

Fundamental Theorem of Matrix Product States

A Formalization Blueprint

Authors

Last update: May 17, 2026

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Chapter 1

Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices $A = (A^i)_{i=0}^{d-1}$, which assigns to each chain length N the vector $|V^{(N)}(A)\rangle$ with coefficients

$$\langle i_1, \dots, i_N | V^{(N)}(A) \rangle = \text{tr}(A^{i_1} \dots A^{i_N}).$$

The main non-periodic route follows the canonical-form and basis-of-normal- tensors argument of [CPGSV16, CPGSV21], with the single-block injective case anchored in the earlier representation theorem of [PGVWC07].

In canonical form one writes

$$A^i = \bigoplus_{j=1}^g \mu_j A_j^i, \quad B^i = \bigoplus_{k=1}^h \nu_k B_k^i,$$

where the displayed blocks are normal tensors. If $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every positive chain length N , then the bases of normal tensors have the same size. After relabelling the blocks of B , each matched pair satisfies

$$B_j^i = \zeta_j X_j A_j^i X_j^{-1}$$

for some invertible matrix X_j and nonzero scalar ζ_j , and the equal case compares the coefficient power sums attached to this matched basis. The block gauges and the weight comparison then assemble into a single gauge conjugacy of the two weighted block-diagonal tensors. For a single injective block this reduces to the familiar conclusion $B^i = X A^i X^{-1}$ for all physical indices i .

Chapters 2–12 contain the aperiodic proof route. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and periodic-chain tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4, 5, 6, and 7 develop the positive-map, multiplicative- domain, Perron–Frobenius, and overlap-decay tools used to control normal blocks. Chapters 8 and 9 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapters 10 and 11 isolate block permutation, block separation, bases of normal tensors, and the power-sum comparison. Chapter 12 records the current conditional non-periodic theorem statements; the remaining BNT matching and fixed-length span inputs are kept as explicit hypotheses until the corresponding source steps are formalized.

The chapters after Chapter 12 collect applications, alternative formulations, and later tensor-network material. Chapter 14 uses the Fundamental Theorem to obtain virtual symmetry gauges,

projective bond representations, and string-order criteria. Chapter 15 develops the direct periodic theorem in irreducible form, together with the compatibility theorems for physical-index rotation (Definition 14.1 in Chapter 14). Chapter 16 gives the fixed-length algebraic Fundamental Theorem of [MGRPG⁺18], and Chapter 17 records the PEPS analogues. The later chapters cover dynamical semigroups, entropy, matrix product density operators, parent Hamiltonians, correlations, and examples; these are downstream of the aperiodic proof rather than part of its dependency chain.

Chapter 2

Matrix Product Vectors

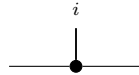
Fix a physical dimension d and a bond dimension D . This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). [Lean](#) [✓ Stmt](#) A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \quad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

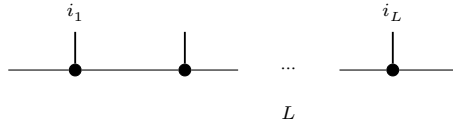


The black node denotes the tensor A , the horizontal legs are virtual, and the upper leg is the physical index i .

Definition 2.2 (Word evaluation). [Lean](#) [✓ Stmt](#) Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^\emptyset = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A , the virtual legs remain open, and the physical legs are labelled by the word $(i_1, \dots, i_L)$.

Lemma 2.3 (Multiplicativity of word evaluation). [Lean](#) [✓ Stmt](#) For any words w_1, w_2 , $A^{w_1 w_2} = A^{w_1} A^{w_2}$.

Proof. [✓ Proof](#) Immediate from the definition of A^w . $\square$

Definition 2.4 (Matrix product vector). [Lean](#) [✓ Stmt](#) The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

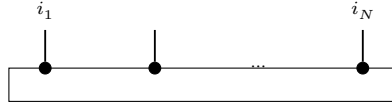
Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w , we also write

$$c_w(A) := \text{tr}(A^w),$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction

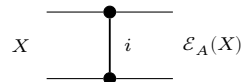


of N copies of the local tensor A , with the outer virtual legs closed by the trace.

Definition 2.5 (Transfer map). [Lean](#) [✓ Stmt](#) The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A , the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.6. [✓ Stmt](#) The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

2.2 Site-dependent periodic chains

Definition 2.7 (Periodic-chain tensor family). [Lean](#) [✓ Stmt](#) For a fixed period m , a *periodic-chain tensor family* is a site-dependent family of local tensors indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D .

Definition 2.8 (Periodic same-state and gauge-equivalence relations). [Lean](#) [Lean](#) [✓ Stmt](#) For periodic-chain tensors A, B at fixed period m :

- equality of the induced periodic-chain states;
- cyclic gauge equivalence.

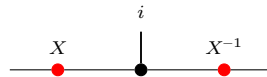
Lemma 2.9 (Equivalence structures for periodic relations). [Lean](#) [Lean](#) [✓ Stmt](#) Both periodic relations are equivalence relations (reflexive, symmetric, transitive).

2.3 Gauge equivalence and same MPV

Definition 2.10 (Gauge equivalence). [Lean](#) [✓ Stmt](#) Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that, for every $i \in \{0, \dots, d-1\}$,

$$B^i = X A^i X^{-1}$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index i , and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.11 (Same MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B (of the same bond dimension) generate the same MPV family if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N .

Definition 2.12 (Same MPV — different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors A and B of possibly different bond dimensions D_1 and D_2 , we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every N .

Definition 2.13 (Positive-length same MPV — different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors A and B of possibly different bond dimensions, we write $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N > 0$.

Remark 2.14. [✓ Stmt](#) Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.15 (Proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B (of possibly different bond dimensions) generate weakly proportional MPV families if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$.

Definition 2.16 (Nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B (of possibly different bond dimensions) generate nonzero proportional MPV families if for every N there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. This is the projective reading of the proportionality hypothesis in [CPGSV16, Theorem II.1].

Definition 2.17 (Eventually nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B are eventually nonzero proportional if, for all sufficiently large N , there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. This auxiliary form is used with the eventual linear-independence statement Lemma 11.3.

Lemma 2.18 (Elementary nonzero proportionality rules). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Nonzero proportionality implies weak proportionality and eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar 1.*

Proof. [✓ Proof](#) Forgetting the nonvanishing condition gives weak proportionality. The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar c_N at each length, and equality is the case $c_N = 1$. $\square$

Lemma 2.19 (Scaling of word evaluation). [Lean](#) [✓ Stmt](#) *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w ,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w .

Proof. [✓ Proof](#) Induction on w : the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ . $\square$

Lemma 2.20 (Scaling of MPV components). [Lean](#) [✓ Stmt](#) *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for every N and every configuration $\sigma = (i_1, \dots, i_N)$,*

$$V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. [✓ Proof](#) Apply Lemma 2.19 to the word $w = (i_1, \dots, i_N)$ of length N , then take the trace; the scalar ζ^N pulls out by linearity. $\square$

Definition 2.21 (Gauge-phase equivalence). [Lean](#) [Lean](#) [✓ Stmt](#) *Two tensors A and B are gauge-phase equivalent if there exist $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that, for every physical index i ,*

$$B^i = \zeta X A^i X^{-1}$$

Proof. [✓ Proof](#) Gauge equivalence is the special case $\zeta = 1$. $\square$

Remark 2.22. [✓ Stmt](#) After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.23 (Gauge covariance of word evaluation). [Lean](#) [✓ Stmt](#) *If $B^i = X A^i X^{-1}$ for all i , then for every word w ,*

$$B^w = X A^w X^{-1}.$$

Proof. [✓ Proof](#) Induction on the word length, using $XX^{-1} = \mathbb{1}$. $\square$

Theorem 2.24 (Gauge invariance of MPVs). [Lean](#) [✓ Stmt](#) *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. [✓ Proof](#) Let $B^i = X A^i X^{-1}$. By Lemma 2.23, $B^w = X A^w X^{-1}$ for every word w . Then $\text{tr}(B^w) = \text{tr}(X A^w X^{-1}) = \text{tr}(A^w)$ by cyclicity of the trace. $\square$

Theorem 2.25 (Gauge-phase scaling of MPVs). [Lean](#) [✓ Stmt](#) *If, for every physical index i ,*

$$B^i = \zeta X A^i X^{-1}$$

then for every system size N and configuration σ one has

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

Proof. [✓ Proof](#) Combine gauge covariance of word evaluation with scalar scaling, then take traces. $\square$

2.4 Injectivity and normality

Definition 2.26 (Injectivity). [Lean](#) [✓ Stmt](#) A tensor A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.27 (L -block injectivity). [Lean](#) [✓ Stmt](#) A tensor A is *L -block injective* if $\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Lemma 2.28 (One-site injectivity gives one-block injectivity). [Lean](#) [✓ Stmt](#) If the one-site matrices $\{A^i\}_{i=0}^{d-1}$ span $M_D(\mathbb{C})$, then the length-one word products span $M_D(\mathbb{C})$, so A is 1-block injective.

Proof. [✓ Proof](#) The length-one words are exactly the letters $i \in \{0, \dots, d-1\}$, and evaluating a length-one word gives the matrix A^i . Thus the length-one word span is the same span as in one-site injectivity. $\square$

Definition 2.29 (Normal tensor). [Lean](#) [✓ Stmt](#) A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0 -block injective.

Remark 2.30. [✓ Stmt](#) The algebraic characterization of normality in Definition 2.29 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map $\mathcal{E}_A$ is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Theorem 8.66, Theorem 8.83, and Theorem 9.16. In particular, every subsequent use of “normal” here refers to this single notion; Definition 9.18 employs the spectral formulation, but it describes the same class of tensors.

2.5 Canonical form

Definition 2.31 (Block-injective canonical form). [Lean](#) [✓ Stmt](#) Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

This block-injective formulation of the canonical form is used in [PGVWC07]; see also [CPGSV21] for the normal canonical form.

Remark 2.32. [✓ Stmt](#) Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.26. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the spectral sense. Chapter 9 treats the normal canonical form needed for the blocking-based multi-block theory, while Chapter 15 develops the separate periodic irreducible-form extension. This section isolates the stronger block-injective form used in the injective and block-injective arguments.

Definition 2.33 (Block projection). [Lean](#) [✓ Stmt](#) For a finite direct sum $\bigoplus_k \mathbb{C}^{D_k}$, the block projection P_j is the matrix that is the identity on the j -th summand and zero on all other summands.

Definition 2.34 (Block-diagonal matrix). [Lean](#) [✓ Stmt](#) A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if it is a direct sum of square matrices, one on each summand.

Lemma 2.35 (Off-block characterization). [Lean](#) [✓ Stmt](#) A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if and only if every entry from one summand to a distinct summand is zero.

Proof. [✓ Proof](#) The forward direction is immediate from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix. $\square$

Lemma 2.36 (Commutation with block projections). [Lean](#) [✓ Stmt](#) If a matrix on $\bigoplus_k \mathbb{C}^{D_k}$ commutes with every block projection P_j , then it is block diagonal.

Proof. [✓ Proof](#) Compare the (i, j) block of the identity $XP_j = P_jX$ for $i \neq j$. The right multiplication by P_j keeps that block, while the left multiplication by P_j kills it. Hence every off-block entry is zero, and Lemma 2.35 applies. $\square$

Definition 2.37 (Block-diagonal tensor from blocks). [Lean](#) [✓ Stmt](#) Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

Lemma 2.38 (Word evaluation of a block-diagonal tensor). [Lean](#) [✓ Stmt](#) For a word w , the word evaluation of a block-diagonal tensor remains block diagonal:

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w.$$

Proof. [✓ Proof](#) Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor μ_k on the k -th block. $\square$

Theorem 2.39 (MPV decomposition). [Lean](#) [✓ Stmt](#) The MPV of a block-diagonal tensor decomposes as

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma.$$

Proof. [✓ Proof](#) By Lemma 2.38, the full word evaluation is block diagonal with blocks $\mu_k^N A_k^w$ for a word w of length N . Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$. $\square$

2.6 Blocking

Definition 2.40 (Blocked tensor). Lean ✓ Stmt Given a tensor A and a blocking length L , the L -blocked tensor is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D , whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}.$$

Lemma 2.41 (Blocked word evaluation). Lean ✓ Stmt If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L -tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\tilde{w}$ of length mL in the original alphabet and one has

$$(A^{[L]})^w = A^{\tilde{w}}.$$

Proof. ✓ Proof Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. $\square$

Lemma 2.42 (Blocking preserves MPV coefficients). Lean ✓ Stmt For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L -tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that

$$V^{(N)}(A^{[L]})_{\sigma} = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.41). $\square$

Theorem 2.43 (Blocking preserves same MPV). Lean ✓ Stmt If A and B generate the same MPV family, then for any blocking length L , the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.

Proof. ✓ Proof By Lemma 2.42, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. $\square$

2.7 MPV overlap

Definition 2.44 (MPV as Hilbert-space vector). Lean ✓ Stmt We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ -component $V^{(N)}(A)_{\sigma}$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.45 (MPV inner product). Lean ✓ Stmt The MPV inner product between two tensors A and B at system size N is

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_{\sigma}} V^{(N)}(B)_{\sigma},$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.46 (MPV overlap). [Lean](#) [✓ Stmt](#) The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.47, it is the complex conjugate of the Hilbert-space inner product from Definition 2.45.

Lemma 2.47 (Overlap equals conjugate inner product). [Lean](#) [✓ Stmt](#) $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}.$

Proof. [✓ Proof](#) The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. $\square$

Lemma 2.48 (Overlap determined by positive-length MPV). [Lean](#) [✓ Stmt](#) If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ , then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N .

Proof. [✓ Proof](#) Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. $\square$

Chapter 3

The Single-Block Fundamental Theorem

This chapter proves the single-block injective Fundamental Theorem of MPS [CPGSV21, Section III.B]: if an injective MPS tensor A generates the same MPV family as another tensor B of the same bond dimension, then there exists an invertible matrix X such that $B^i = XA^iX^{-1}$ for all physical indices i . The proof is purely algebraic and does not require the later overlap or normal-form reductions.

3.1 Trace pairing

Theorem 3.1 (Nondegeneracy of the trace pairing). Lean ✓ Stmt For $M \in M_D(\mathbb{C})$, $\text{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.

Proof. ✓ Proof Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\text{tr}(E_{ji}M) = M_{ij}$. Hence if $\text{tr}(MN) = 0$ for all N , then every entry of M vanishes. $\square$

Lemma 3.2 (Same MPV implies trace agreement). Lean ✓ Stmt If A and B generate the same MPV family, then $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w .

Proof. ✓ Proof Viewing the word w as a configuration of length $|w|$, Definition 2.4 gives the corresponding MPV coefficient as the trace coefficient $c_w(A) = \text{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. $\square$

Definition 3.3 (Trace pairing map). Lean ✓ Stmt For a tensor A , attach to the chosen family $\{A^i\}$ the auxiliary linear map $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ defined by

$$\Phi_A(M)_i = \text{tr}(M \cdot A^i).$$

Theorem 3.4 (Injectivity of the trace pairing map). Lean ✓ Stmt If A is injective, then $\ker \Phi_A = \{0\}$.

Proof. ✓ Proof If $\Phi_A(M) = 0$, then $\text{tr}(M \cdot A^i) = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\text{tr}(MN) = 0$ for all N , so $M = 0$ by Theorem 3.1. $\square$

Lemma 3.5 (Nonvanishing trace on injective tensors). Lean ✓ Stmt *If $D \geq 1$, A is injective, and A and B generate the same MPV family, then B^i cannot be identically zero: $\neg(\forall i, B^i = 0)$.*

Proof. ✓ Proof If $B^i = 0$ for all i , then Lemma 3.2 gives $\text{tr}(A^w) = 0$ for every word w , in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\text{tr}(\mathbb{1}_D) = D \neq 0$. $\square$

3.2 Linear extension

Theorem 3.6 (Existence and uniqueness of the linear extension). Lean ✓ Stmt *If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i .*

Proof. ✓ Proof Write $\ell_A : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). Applying Lemma 3.2 to all words of length 2 shows $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\text{range}(\Phi_A) \subseteq \text{range}(\Phi_B)$. By Theorem 3.4, $\ker \Phi_A = \{0\}$, so rank-nullity gives $\dim(\text{range}(\Phi_A)) = \dim(M_D(\mathbb{C}))$. Hence $\dim(\text{range}(\Phi_B)) \geq \dim(M_D(\mathbb{C}))$, and rank-nullity again gives $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. $\square$

Theorem 3.7 (Multiplicativity of the linear extension). Lean ✓ Stmt *If A is injective and A, B generate the same MPV family, and $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is the unique linear map satisfying $T(A^i) = B^i$ for all i , then $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. ✓ Proof **Step 1** ($\Phi_B \circ T = \Phi_A$). Applying Lemma 3.2 to words of length 2 gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^i A^j) = B^i B^j$). Lemma 3.2 applied to the word (i, j, k) gives $\text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k)$ for all i, j, k . So $\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k$. The same rank-nullity argument used in the proof of Theorem 3.6 gives $\ker \Phi_B = \{0\}$. Hence $T(A^i A^j) = B^i B^j$.

Step 3 (bilinear extension). For fixed j , the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. $\square$

3.3 Simplicity and Skolem–Noether

Theorem 3.8 (Simplicity of $M_D(\mathbb{C})$). Lean ✓ Stmt *A nonzero multiplicative $\mathbb{C}$ -linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. $\square$

Theorem 3.9 (Skolem–Noether). Lean ✓ Stmt *Every $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \text{Aut}_{\mathbb{C}\text{-alg}}(M_D(\mathbb{C}))$, there exists $X \in \text{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M .*

Proof. ✓ Proof Via the canonical identification $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\text{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X . $\square$

Lemma 3.10 (Unit preservation of multiplicative surjective maps). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$ -linear map, then $T(\mathbb{1}) = \mathbb{1}$.*

Proof. ✓ Proof By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$ -algebra homomorphism. $\square$

3.4 Proof of the single-block theorem

Theorem 3.11 (Single-block Fundamental Theorem). Lean ✓ Stmt *Let A be an injective MPV tensor and let B be an MPV tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \text{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i .*

Proof. ✓ Proof By Theorem 3.6, there is a unique linear T with $T(A^i) = B^i$. By Theorem 3.7, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i , contradicting Lemma 3.5. By Theorem 3.8, T is bijective. By Lemma 3.10, T is a $\mathbb{C}$ -algebra automorphism. Theorem 3.9 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. $\square$

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive maps and channels

Definition 4.1 (Positive map). [Lean](#) [✓ Stmt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 4.2. [✓ Stmt](#) We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 4.3 (Trace-preserving map). [Lean](#) [✓ Stmt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *trace-preserving* if $\text{tr}(E(X)) = \text{tr}(X)$ for all X .

Definition 4.4 (Completely positive map). [Lean](#) [✓ Stmt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that, for every $X \in M_D(\mathbb{C})$,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger$$

We use the Kraus characterisation because it matches the later Kadison–Schwarz arguments. By Choi’s theorem, this is equivalent to $E \otimes \text{id}_n$ being positive for all n .

Theorem 4.5 (CP maps are positive). [Lean](#) [✓ Stmt](#) *Every completely positive map is positive.*

Proof. [✓ Proof](#) If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. $\square$

Theorem 4.6 (Channels are positive). [Lean](#) [✓ Stmt](#) *Every quantum channel is positive.*

Proof. [✓ Proof](#) A quantum channel is completely positive by definition, so Theorem 4.5 applies. $\square$

Theorem 4.7 (Positive maps preserve Hermiticity). [Lean](#) [✓ Stmt](#) *If E is positive and $X = X^\dagger$, then $E(X) = E(X)^\dagger$.*

Proof. ✓ Proof Decompose $X = X_+ - X_-$ into its positive and negative parts. Positivity makes $E(X_+)$ and $E(X_-)$ positive semidefinite, hence Hermitian, so their difference $E(X)$ is Hermitian. $\square$

Definition 4.8 (Quantum channel). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 4.9 (Density matrices). Lean Lean ✓ Stmt The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

Theorem 4.10 (Density matrices are compact). Lean ✓ Stmt $\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).

Proof. ✓ Proof The PSD cone is closed (preimage of the closed nonnegative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\text{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are nonnegative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. $\square$

Theorem 4.11 (Density matrices are convex). Lean ✓ Stmt $\mathcal{D}_D$ is convex.

Proof. ✓ Proof Convex combination of PSD matrices is PSD, and trace is linear. $\square$

Theorem 4.12 (Density matrices are nonempty). Lean ✓ Stmt For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.

Proof. ✓ Proof $\frac{1}{D}\mathbb{1}_D$ is a density matrix. $\square$

Theorem 4.13 (Channels preserve density matrices). Lean ✓ Stmt If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.

Proof. ✓ Proof Complete positivity implies positivity (Theorem 4.5), so $E(\rho) \geq 0$; trace preservation gives $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$. $\square$

4.2 Irreducibility

Definition 4.14 (Orthogonal projection). Lean ✓ Stmt A matrix $P \in M_D(\mathbb{C})$ is an *orthogonal projection* if $P = P^\dagger$ and $P^2 = P$.

Definition 4.15 (Irreducible map). Lean ✓ Stmt A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Theorem 6.2, condition (1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 4.16 (Injectivity implies irreducibility). Lean ✓ Stmt If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.

Proof. ✓ Proof If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{1}$. $\square$

Remark 4.17 (Transfer map is CP). Lean ✓ Stmt The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive: it is already written in Kraus form, with Kraus operators $\{A^i\}$.

Theorem 4.18 (Transfer map is a channel). [Lean](#) [✓ Stmt](#) If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).

Proof. [✓ Proof](#) Complete positivity is the observation of Remark 4.17. Trace preservation: $\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}((\sum_i (A^i)^\dagger A^i)X) = \text{tr}(X)$ by cyclicity and the normalization hypothesis. $\square$

4.3 Cesàro fixed-point theorem

Definition 4.19 (Cesàro mean). [Lean](#) [✓ Stmt](#) The *Cesàro mean* of a linear map E at order N , applied to a starting matrix ρ_0 , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 4.20 (Cesàro telescope). [Lean](#) [✓ Stmt](#) $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. [✓ Proof](#) Telescoping: $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. $\square$

Theorem 4.21 (Hermitian fixed-point decomposition). [Lean](#) [✓ Stmt](#) Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\text{tr}(P_+ P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Proposition 6.8].

Proof. [✓ Proof](#) Let Q be the support projection of P_+ . Trace preservation and positivity give $\text{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. $\square$

Theorem 4.22 (Cesàro fixed-point theorem). [Lean](#) [✓ Stmt](#) Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.

Proof. [✓ Proof](#) Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 4.12). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 4.10), extract a convergent subsequence $\sigma_{N_k} \rightarrow \rho$. The telescope identity (Theorem 4.20) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \rightarrow 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. $\square$

4.4 Fixed-point projection

Definition 4.23 (Fixed-point projection). [Lean](#) [✓ Stmt](#) Given a linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 4.24 (Power decomposition). [Lean](#) [✓ Stmt](#) If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. [✓ Proof](#) P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. $\square$

4.5 Peripheral spectrum and primitivity

Definition 4.25 (Peripheral spectrum). [Lean](#) [✓ Stmt](#) The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T .

Definition 4.26 (Peripheral eigenvalues). [Lean](#) [✓ Stmt](#) The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

Note: The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1. For a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Theorem 4.27 (1 is a peripheral eigenvalue). [Lean](#) [✓ Stmt](#) If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E .

Proof. [✓ Proof](#) ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. $\square$

Remark 4.28 (Peripheral eigenvalues lie on the unit circle). [Lean](#) [✓ Stmt](#) Every peripheral eigenvalue λ satisfies $|\lambda| = 1$. This is simply the defining condition in Definition 4.26.

Theorem 4.29 (Finiteness of peripheral eigenvalues). [Lean](#) [✓ Stmt](#) On a finite-dimensional space, the set of peripheral eigenvalues is finite.

Proof. [✓ Proof](#) Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. $\square$

Theorem 4.30 (Power-stable peripheral eigenvalues are roots of unity). [Lean](#) [✓ Stmt](#) Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.

Proof. [✓ Proof](#) Since μ^n is an eigenvalue for every n , the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. $\square$

Theorem 4.31 (Peripheral powers imply root of unity). [Lean](#) [✓ Stmt](#) If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is closed under powers ($\mu \in \text{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \text{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.

Proof. [✓ Proof](#) The closure hypothesis provides $\mu^n \in \text{peripheral}(E)$ for all n , which in particular means μ^n is an eigenvalue. Theorem 4.30 then gives $\mu^p = 1$ for some $p > 0$. $\square$

4.5.1 Periodicity removal by powering

Remark 4.32. ✓ Stmt The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p -blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 4.33 (Common exponent for a finite set of roots of unity). Lean ✓ Stmt *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. ✓ Proof Take p to be the least common multiple of the exponents p_μ . □

Lemma 4.34 (Powering collapses peripheral eigenvalues). Lean ✓ Stmt *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then $\text{peripheral}(E^p) = \{1\}$.*

Proof. ✓ Proof The fixed point ρ gives $1 \in \text{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \text{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \text{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 4.35 (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the “bi-canonical” case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 4.41) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.31. This is the simplest argument, but it requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

4.5.2 Kadison–Schwarz input for peripheral and fixed-point arguments

Definition 4.36 (Kraus maps, adjoints, and normalizations). Lean Lean Lean Lean ✓ Stmt For operators $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$, the associated *Kraus map* and *adjoint Kraus map* are

$$E(X) = \sum_i K_i X K_i^\dagger, \quad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when $\sum_i K_i K_i^\dagger = \mathbb{1}$ and *trace-preserving* when $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Theorem 4.37 (Kadison–Schwarz inequality). Lean ✓ Stmt *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map. Then for every $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. ✓ Proof Apply the unital completely positive map entrywise to the positive 2×2 block matrix $\begin{pmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{pmatrix}$ and take the Schur complement of the lower-right block. □

Theorem 4.38 (Hilbert–Schmidt contraction). [Lean](#) [✓ Stmt](#) *If a Kraus map is both unital and trace-preserving, then*

$$\mathrm{tr}(E(X)^\dagger E(X)) \leq \mathrm{tr}(X^\dagger X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation then identifies the traces of its two terms. $\square$

Theorem 4.39 (KS gap decomposition). [Lean](#) [✓ Stmt](#) *For a unital Kraus map,*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \quad R_i := XK_i^\dagger - K_i^\dagger E(X).$$

Proof. [✓ Proof](#) Expand the right-hand side, distribute the products, and use $\sum_i K_i K_i^\dagger = \mathbb{1}$ to recover the Kadison–Schwarz gap. $\square$

Theorem 4.40 (KS equality implies Kraus commutation). [Lean](#) [✓ Stmt](#) *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then, for every i ,*

$$XK_i^\dagger = K_i^\dagger E(X)$$

Proof. [✓ Proof](#) The Kadison–Schwarz gap is a sum of positive semidefinite terms $\sum_i R_i^\dagger R_i$. If the gap vanishes, each R_i must vanish. $\square$

Theorem 4.41 (KS equality for peripheral eigenvectors). [Lean](#) [✓ Stmt](#) *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

Proof. [✓ Proof](#) The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation and the relation $E(X) = \mu X$ with $|\mu| = 1$ show that its trace is zero, hence the gap vanishes. $\square$

Theorem 4.42 (Left multiplicative identity from KS equality). [Lean](#) [✓ Stmt](#) *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then for every $Y \in M_D(\mathbb{C})$,*

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

Proof. [✓ Proof](#) The commutation relation of Theorem 4.40 lets one move $X^\dagger$ past each Kraus operator inside the sum. $\square$

Definition 4.43 (Multiplicative domain). [Lean](#) [✓ Stmt](#) *The multiplicative domain of a Kraus map E is the set of matrices that are simultaneously left- and right-multiplicative for E .*

Theorem 4.44 (Multiplicative domain is a $*$ -subalgebra). [Lean](#) [✓ Stmt](#) *For a unital Kraus map, the multiplicative domain is a $*$ -subalgebra of $M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix. $\square$

4.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV17a, Appendix A].

Lemma 4.45 (Peripheral closure via adjoint fixed point). Lean ✓ Stmt Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:

1. the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and
2. E is irreducible.

Then $\text{peripheral}(E)$ is closed under powers: if $\mu \in \text{peripheral}(E)$ then $\mu^n \in \text{peripheral}(E)$ for every $n \in \mathbb{N}$.

Proof. ✓ Proof Let $E(X) = \mu X$ with $|\mu| = 1$, and set $G := E(X^\dagger X) - E(X)^\dagger E(X)$. By Kadison–Schwarz, $G \geq 0$. Since $E^*(\rho) = \rho$,

$$\text{tr}(\rho G) = \text{tr}(\rho E(X^\dagger X)) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(\rho X^\dagger X) - |\mu|^2 \text{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because $\rho > 0$ and $G \geq 0$, this forces $G = 0$, so $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 6.4). So X is invertible. By Theorem 4.40 (KS equality implies Kraus commutation), X commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 4.42) gives $E(X^n) = \mu^n X^n$, so μ^n is peripheral. $\square$

Theorem 4.46 (Roots of unity via adjoint fixed point). Lean ✓ Stmt Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital, and assume that the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point and that E is irreducible. Then every peripheral eigenvalue of E is a root of unity.

Proof. ✓ Proof Combine the closure-under-powers result (Lemma 4.45) with Theorem 4.31. $\square$

Remark 4.47. ✓ Stmt Lemma 4.45 and Theorem 4.46 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV17a, Appendix A], which uses a faithful invariant state rather than bi-canonicity.

Definition 4.48 (Channel period). Lean ✓ Stmt The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\text{peripheral}(E)|$.

Definition 4.49 (Primitive map). Lean ✓ Stmt A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\text{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 4.50 (Primitivity equals period one). Lean ✓ Stmt A linear map with a nonzero fixed point is primitive if and only if its period is 1.

Proof. ✓ Proof If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then since $1 \in \text{peripheral}(E)$ (Theorem 4.27), the only element is 1. $\square$

Theorem 4.51 (Primitive maps have a spectral gap on the complement). [Lean](#) [✓ Stmt](#) Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be trace-preserving, and let $\rho \neq 0$ satisfy $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$. Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the fixed-point projection associated to ρ . Assume that:

1. E is primitive;
2. every eigenvalue μ of E satisfies $|\mu| \leq 1$;
3. whenever $E(X) = X$ and $\text{tr}(X) = 0$, one has $X = 0$.

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. [✓ Proof](#) Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\text{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E ; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\text{tr}(X) \neq 0$, trace preservation and $\text{tr}(P(X)) = \text{tr}(X)$ give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu \text{tr}(X) + \text{tr}(P(X)) = \nu \text{tr}(X) + \text{tr}(X),$$

so $\nu \text{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. □

4.6 Representations

The following definitions support the Choi–Jamiołkowski isomorphism (Theorem 4.57) and the representation theorems of [Wol12, Chapter 2].

Definition 4.52 (Partial traces). [Lean](#) [Lean](#) [✓ Stmt](#) Let $X \in M_{d,d'}(\mathbb{C})$ be a bipartite matrix indexed by

$$(\{0, \dots, d-1\} \times \{0, \dots, d'-1\})^2.$$

The *left partial trace* $\text{tr}_A(X)$ and *right partial trace* $\text{tr}_B(X)$ are the $d' \times d'$ and $d \times d$ matrices defined by

$$(\text{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \quad (\text{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

Definition 4.53 (Maximally entangled state). [Lean](#) [✓ Stmt](#) The *maximally entangled state* on $\mathbb{C}^d \otimes \mathbb{C}^d$ is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely, $(|\Omega\rangle\langle\Omega|)_{(i_1,i_2),(j_1,j_2)} = \frac{1}{d} \delta_{i_1 j_1} \delta_{i_2 j_2}$ and $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$.

Definition 4.54 (Tensor product of a map with identity). [Lean](#) [✓ Stmt](#) Given a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *tensor extension* $T \otimes \text{id}$ acts on bipartite matrices $X \in M_{D \times D}(\mathbb{C})$ by applying T to each “slice”:

$$(T \otimes \text{id})(X)_{(i_1,i_2),(j_1,j_2)} = (T(X^{(i_2,j_2)}))_{i_1 j_1},$$

where $X_{ab}^{(i_2,j_2)} = X_{(a,i_2),(b,j_2)}$ is the bipartite slice.

4.7 Choi–Jamiołkowski isomorphism

Definition 4.55 (Choi matrix). [Lean](#) [✓ Stmt](#) The *Choi matrix* of a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is

$$\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|) \in M_{D \times D}(\mathbb{C}).$$

Concretely, $\tau_{(i_1, i_2), (j_1, j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}$ where $E_{i_2 j_2}$ is the matrix unit. This is the Choi–Jamiołkowski convention of [Wol12, Proposition 2.1].

Theorem 4.56 (Choi matrix of the identity map). [Lean](#) [✓ Stmt](#) The identity channel has Choi matrix $|\Omega\rangle\langle\Omega|$.

Proof. [✓ Proof](#) Apply the identity map in the definition of the Choi matrix. $\square$

Theorem 4.57 (Complete positivity iff the Choi matrix is positive semidefinite). [Lean](#) [✓ Stmt](#) A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is completely positive (in the Kraus sense) if and only if its Choi matrix $\tau \geq 0$.

Proof. [✓ Proof](#) ($\Rightarrow$) If $T(X) = \sum_i K_i X K_i^\dagger$, then $\tau = \sum_i |v_i\rangle\langle v_i|$ where $(v_i)_{(a,b)} = c K_i(a, b)$ and $c = 1/\sqrt{D}$. This is a sum of rank-one PSD matrices.

($\Leftarrow$) If $\tau \geq 0$, write $\tau = \sum_m |v_m\rangle\langle v_m|$ by spectral decomposition, define $K_m(a, b) = v_m(a, b)/c$, and recover $T(X) = \sum_m K_m X K_m^\dagger$ by linearity on matrix units. $\square$

Theorem 4.58 (Trace preservation implies the partial-trace condition). [Lean](#) [✓ Stmt](#) If T is trace-preserving, then $\text{tr}_A(\tau) = \frac{1}{D} \mathbb{1}_D$.

Proof. [✓ Proof](#) $(\text{tr}_A(\tau))_{ij} = \text{tr}(T(\Omega^{(ij)})) = \text{tr}(\Omega^{(ij)})$ by trace preservation, where $\Omega^{(ij)} = \frac{1}{D} E_{ij}$ is the (i, j) -slice of $|\Omega\rangle\langle\Omega|$. Taking the trace gives $\frac{1}{D} \delta_{ij}$. $\square$

Theorem 4.59 (Hermiticity preservation and the Choi matrix). [Lean](#) [✓ Stmt](#) The Choi matrix τ is Hermitian if and only if $T(B^\dagger) = T(B)^\dagger$ for all $B \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) The Choi matrix entries satisfy $\tau_{(a,i),(b,j)} = \frac{1}{D} (T(E_{ij}))_{ab}$. Hermiticity of τ says $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$, which unravels to $T(E_{ji})_{ba} = \overline{T(E_{ij})_{ab}}$, i.e., $T(E_{ji}^\dagger) = T(E_{ij})^\dagger$. By linearity this extends to all B . $\square$

Theorem 4.60 (Trace of the Choi matrix of a trace-preserving map). [Lean](#) [✓ Stmt](#) If T is trace-preserving, then $\text{tr}(\tau) = 1$.

Proof. [✓ Proof](#) $\text{tr}(\tau) = \text{tr}(\text{tr}_A(\tau)) = \text{tr}(\frac{1}{D} \mathbb{1}_D) = 1$. $\square$

4.8 Representation corollaries for channel decompositions

The next three corollaries follow from the Choi/Kraus/Stinespring representations already developed above. They correspond to [Wol12, Propositions 2.2–2.4].

Theorem 4.61 (Polarization of sesquilinear sandwiches). [Lean](#) [✓ Stmt](#) For any $A, B, X \in M_D(\mathbb{C})$,

$$4AXB^\dagger = (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger + i(A+iB)X(A+iB)^\dagger - i(A-iB)X(A-iB)^\dagger.$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

Proof. ✓ Proof Working entrywise, reduces to the scalar polarization identity $4\alpha\bar{\delta} = \sum_{k=0}^3 i^k(\alpha + i^k\beta)(\gamma + i^k\delta)$ in $\mathbb{C}$, which holds after substituting $i^2 = -1$. □

Theorem 4.62 (CP decomposition of sandwich sums). Lean ✓ Stmt For any finite Kraus-like families $\{A_i\}, \{B_i\}$, the map $T(X) = \sum_i A_i X B_i^\dagger$ satisfies $4T = T_1 - T_2 + iT_3 - iT_4$ with each T_k a CP map given explicitly by a single-side Kraus sum of the combined families.

Proof. ✓ Proof Apply the polarization identity summand-by-summand. □

Theorem 4.63 (No information without disturbance). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be linear. If $T(|v\rangle\langle v|) = |v\rangle\langle v|$ for every $v \in \mathbb{C}^D$, then $T = \text{id}$. In particular, a quantum channel preserving every pure state equals the identity.

Proof. ✓ Proof Rank-one outer products $|u\rangle\langle v|$ polarize into rank-one self-outer-products: $4|u\rangle\langle v| = \sum_{k=0}^3 i^k |u + i^k v\rangle\langle u + i^k v|$, so T preserves every $|u\rangle\langle v|$. Since the matrix units $E_{ij} = |e_i\rangle\langle e_j|$ span $M_D(\mathbb{C})$, T equals the identity on a spanning set, hence everywhere. □

Definition 4.64 (Pure-state ensemble density). Lean ✓ Stmt Given a finite family $\{\psi_i\}_{i \in \iota}$ of (unnormalized) vectors in $\mathbb{C}^D$, its *pure-ensemble density* is $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$. Weights $p_i \geq 0$ with $\sum p_i = 1$ can be absorbed by replacing $\psi_i \mapsto \sqrt{p_i} \psi_i$.

Theorem 4.65 (Isometric mixing preserves the density). Lean ✓ Stmt If two pure-state ensembles $\{\psi_i\}_{i \in \iota_1}$ and $\{\phi_j\}_{j \in \iota_2}$ are related by an isometric mixing matrix $V \in M_{\iota_1 \times \iota_2}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij} \phi_j$, then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 4.66.

Proof. ✓ Proof Expanding and applying the orthogonality relation $\sum_i V_{ij} \overline{V_{ij'}} = \delta_{jj'}$ from $V^\dagger V = \mathbb{1}$:

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left(\sum_i V_{ij} \overline{V_{ij'}} \right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

□

Theorem 4.66 (Hughston–Jozsa–Wootters converse). Lean ✓ Stmt If two pure-state ensembles $\{\psi_i\}_{i \in \iota_1}$ and $\{\phi_j\}_{j \in \iota_2}$ induce the same pure-ensemble density operator and $|\iota_2| \leq |\iota_1|$, then there exists a tall isometric mixing matrix $V \in M_{\iota_1 \times \iota_2}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $\psi_i = \sum_j V_{ij} \phi_j$. The cardinality hypothesis is what makes V a tall isometry; the symmetric case $|\iota_1| \leq |\iota_2|$ follows by swapping the roles of the ensembles.

Proof. ✓ Proof Embed each vector ψ_i, ϕ_j as the 0-th column of a $D \times D$ matrix with zeros elsewhere; call these $K_i, L_j \in M_D(\mathbb{C})$. A direct entry-wise computation shows that for any $X \in M_D(\mathbb{C})$ both $\sum_i K_i X K_i^\dagger$ and $\sum_j L_j X L_j^\dagger$ collapse to $X_{00} \cdot \rho$, so the density equality $\rho_\psi = \rho_\phi$ forces the two Kraus families to define the same CP map. The rectangular Kraus freedom 4.77 then supplies an isometry V with $V^\dagger V = \mathbb{1}$ and $K_i = \sum_j V_{ij} L_j$; reading the equation off at column 0 recovers the vector relation $\psi_i = \sum_j V_{ij} \phi_j$. □

Theorem 4.67 (Hughston–Jozsa–Wootters equivalence). Lean ✓ Stmt Under the cardinality hypothesis $|\iota_2| \leq |\iota_1|$, two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.

Proof. ✓ Proof Combine the two directions above. □

4.9 Kraus representation theorem

Theorem 4.68 (Trace preservation implies Kraus normalization). [Lean](#) [✓ Stmt](#) If $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving, then $\sum_i K_i^\dagger K_i = \mathbb{1}$.

Proof. [✓ Proof](#) For any N , $\text{tr}((\sum_i K_i^\dagger K_i) N) = \sum_i \text{tr}(K_i^\dagger K_i N) = \sum_i \text{tr}(K_i N K_i^\dagger) = \text{tr}(T(N)) = \text{tr}(N)$. Non-degeneracy of the trace pairing forces $\sum_i K_i^\dagger K_i = \mathbb{1}$. $\square$

Theorem 4.69 (Kraus normalization implies trace preservation). [Lean](#) [✓ Stmt](#) If $\sum_i K_i^\dagger K_i = \mathbb{1}$, then $T(X) = \sum_i K_i X K_i^\dagger$ is trace-preserving.

Proof. [✓ Proof](#) $\text{tr}(T(X)) = \sum_i \text{tr}(K_i X K_i^\dagger) = \sum_i \text{tr}(K_i^\dagger K_i X) = \text{tr}((\sum_i K_i^\dagger K_i) X) = \text{tr}(X)$. $\square$

Theorem 4.70 (Unitality implies Kraus unital normalization). [Lean](#) [✓ Stmt](#) If $T(X) = \sum_i K_i X K_i^\dagger$ satisfies $T(\mathbb{1}) = \mathbb{1}$, then

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

Proof. [✓ Proof](#) Evaluate the Kraus formula at the identity matrix. $\square$

Theorem 4.71 (Unitary freedom in Kraus operators). [Lean](#) [✓ Stmt](#) Let U be a unitary $r \times r$ matrix ($U^\dagger U = \mathbb{1}$) and suppose $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$. Then $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same Kraus map:

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger \quad \forall X \in M_D(\mathbb{C}).$$

Proof. [✓ Proof](#) Substitute the combination formula, expand the triple sum over j, ℓ, ℓ' , swap summation order, and use the entry-wise form of $U^\dagger U = \mathbb{1}$ to collapse to $\ell = \ell'$ diagonal terms. $\square$

Theorem 4.72 (Isometric mixing preserves the Kraus map). [Lean](#) [✓ Stmt](#) Let W be an isometry between two Kraus index spaces, so $W^\dagger W = \mathbb{1}$. If

$$K_j = \sum_\ell W_{j\ell} \tilde{K}_\ell,$$

then the Kraus families $\{K_j\}$ and $\{\tilde{K}_\ell\}$ define the same completely positive map.

Proof. [✓ Proof](#) Expand the Kraus sums, interchange the finite summations, and use $W^\dagger W = \mathbb{1}$ to collapse the coefficient matrix. $\square$

Theorem 4.73 (The transition matrix of orthonormal Kraus families is unitary). [Lean](#) [✓ Stmt](#) Let $\{K_j\}_{j=0}^{r-1}$ and $\{K'_j\}_{j=0}^{r-1}$ be two Hilbert–Schmidt orthonormal Kraus families, and suppose

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell} K'_\ell.$$

Then the transition matrix U is unitary: $U^\dagger U = \mathbb{1}_r$.

Proof. [✓ Proof](#) Expand $\text{tr}(K_j^\dagger K_i)$ using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity $U U^\dagger = \mathbb{1}_r$, hence also $U^\dagger U = \mathbb{1}_r$. $\square$

Theorem 4.74 (Dual map equality from primal map equality). [Lean](#) [✓ Stmt](#) If two Kraus families satisfy $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$ for all X , then $\sum_{\alpha} B_{\alpha}^{\dagger} Y B_{\alpha} = \sum_j A_j^{\dagger} Y A_j$ for all Y .

Proof. [✓ Proof](#) Use the trace pairing: for all X , $\text{tr}(X^{\dagger}(\sum B_{\alpha}^{\dagger} Y B_{\alpha})) = \text{tr}((\sum B_{\alpha} X^{\dagger} B_{\alpha}^{\dagger})Y) = \text{tr}((\sum A_j X^{\dagger} A_j^{\dagger})Y) = \text{tr}(X^{\dagger}(\sum A_j^{\dagger} Y A_j))$. Nondegeneracy of the trace pairing gives the result. $\square$

Theorem 4.75 (Equal Stinespring Gramians). [Lean](#) [✓ Stmt](#) If two Kraus families define the same CPM, then $\sum_{\alpha} B_{\alpha}^{\dagger} B_{\alpha} = \sum_j A_j^{\dagger} A_j$.

Proof. [✓ Proof](#) Specialise Theorem 4.74 at $Y = \mathbb{1}$. $\square$

Theorem 4.76 (Rectangular Kraus freedom). [Lean](#) [✓ Stmt](#) If $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$ for all X and $r_2 \leq r_1$ (where $\{B_{\alpha}\}_{\alpha=0}^{r_1-1}$ is the larger family and $\{A_j\}_{j=0}^{r_2-1}$ the smaller), then there exists a rectangular isometry V ($r_1 \times r_2$, $V^{\dagger} V = \mathbb{1}_{r_2}$) such that $B_{\alpha} = \sum_j V_{\alpha j} A_j$.

Proof. [✓ Proof](#) Pad the smaller family to size r_1 , establish Gram matrix equality $M_B^{\dagger} M_B = M_{A'}^{\dagger} M_{A'}$ from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry V with $V^{\dagger} V = \mathbb{1}_{r_2}$. $\square$

Theorem 4.77 (Rectangular Kraus freedom with general finite index types). [Lean](#) [✓ Stmt](#) Variant of Theorem 4.76 with general finite index sets ι_1, ι_2 in place of standard index sets of cardinalities r_1 and r_2 .

Proof. [✓ Proof](#) Reindex by choosing enumerations of those finite index sets and apply Theorem 4.76. $\square$

Theorem 4.78 (Isometric freedom of Kraus representations). [Lean](#) [✓ Stmt](#) Let $\{B_{\alpha}\}_{\alpha \in \iota_1}$ and $\{A_j\}_{j \in \iota_2}$ be finite Kraus families with $|\iota_2| \leq |\iota_1|$. Then the following are equivalent:

1. $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$ for all $X \in M_D(\mathbb{C})$.
2. There exists a matrix $V = (V_{\alpha j})$ with $V^{\dagger} V = \mathbb{1}$ such that $B_{\alpha} = \sum_j V_{\alpha j} A_j$ for every α .

Proof. [✓ Proof](#) The implication (1) $\Rightarrow$ (2) is Theorem 4.77. Conversely, expand $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger}$ using $B_{\alpha} = \sum_j V_{\alpha j} A_j$, interchange the finite sums, and use $V^{\dagger} V = \mathbb{1}$ to collapse the coefficient matrix to the diagonal. $\square$

Theorem 4.79 (Unitary freedom of Kraus representations). [Lean](#) [✓ Stmt](#) Let $\{B_{\alpha}\}_{\alpha \in \iota}$ and $\{A_j\}_{j \in \iota}$ be two Kraus families with the same finite index set. Then the following are equivalent:

1. $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$ for all $X \in M_D(\mathbb{C})$.
2. There exists a unitary matrix $U = (U_{\alpha j})$ such that $B_{\alpha} = \sum_j U_{\alpha j} A_j$ for every α .

Proof. [✓ Proof](#) Apply Theorem 4.78 with $|\iota| = |\iota|$. In the square case an isometry matrix is unitary, and the converse is immediate. $\square$

Remark 4.80 (Choi rank and minimal Kraus cardinality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) If a completely positive map admits a Kraus representation with exactly r operators, then its Choi matrix is a sum of r rank-one outer products. Consequently,

$$\text{rank}(\tau_E) \leq r.$$

Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly $\text{rank}(\tau_E)$ operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

Proof. ✓ Proof For the forward implication, expand the Choi matrix using the Kraus formula and write $\tau_E = \sum_j v_j v_j^\dagger$ with one vector v_j for each Kraus operator. The column space of this sum lies in the span of the r vectors $\{v_j\}$, so the rank is at most r . For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators. $\square$

4.10 Stinespring dilation

Definition 4.81 (Stinespring isometry). Lean ✓ Stmt Given Kraus operators $\{K_j\}_{j=0}^{r-1}$ with $K_j \in M_D(\mathbb{C})$, the *Stinespring isometry* $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$ is the $(D \cdot r) \times D$ matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements $V = \sum_j K_j \otimes |j\rangle$.

Lemma 4.82 (Entry formula for the Stinespring isometry). Lean ✓ Stmt For indices $i, k \in \{0, \dots, D-1\}$ and $j \in \{0, \dots, r-1\}$,

$$V_{(i,j),k} = (K_j)_{ik}.$$

Theorem 4.83 (Stinespring Gram identity). Lean ✓ Stmt *The Stinespring isometry satisfies*

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

Theorem 4.84 (Stinespring dual representation). Lean ✔ Stmt For Kraus operators $\{K_j\}$ and $A \in M_D(\mathbb{C})$,

$$V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j = T^*(A).$$

This is the Stinespring representation of the dual (Heisenberg picture) map T^* .

Proof. ✓ Proof Compute $(V^\dagger(A \otimes \mathbb{1}_r)V)_{ab}$ by summing over the product index (i, j) ; the δ_{jl} from $\mathbb{1}_r$ collapses the sum over l , giving $\sum_j K_j^\dagger A K_j$. $\square$

Theorem 4.85 (The Stinespring isometry condition iff trace preservation). Lean ✓ Stmt $V^\dagger V = \mathbb{1}_D$ if and only if $\sum_j K_j^\dagger K_j = \mathbb{1}_D$. In particular, V is an isometry precisely when the Kraus map is trace-preserving.

Proof. ✓ Proof $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}} V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}} (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}. \quad \square$

Theorem 4.86 (Stinespring Schrödinger representation). [Lean](#) [✓ Stmt](#) The Kraus map $T(\rho) = \sum_j K_j \rho K_j^\dagger$ equals the partial trace over the dilation space:

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

Proof. [✓ Proof](#) Expand $(V \rho V^\dagger)_{(i,k),(j,k)}$ and sum over k to recover $\sum_k K_k \rho K_k^\dagger$. □

Theorem 4.87 (Existence of a Stinespring dilation). [Lean](#) [✓ Stmt](#) Every completely positive map admits an ancilla dimension r , Kraus operators $\{K_j\}_{j=0}^{r-1}$, and the concrete representation $\pi(A) = A \otimes \mathbb{1}_r$ such that

$$E(A) = V^\dagger \pi(A) V.$$

Proof. [✓ Proof](#) Choose a Kraus representation of E and apply the explicit Heisenberg-picture Stinespring formula. □

Theorem 4.88 (CPTP maps admit isometric Stinespring dilations). [Lean](#) [✓ Stmt](#) Every quantum channel admits a Stinespring dilation whose matrix V is an isometry, equivalently $V^\dagger V = \mathbb{1}$.

Proof. [✓ Proof](#) Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition $V^\dagger V = \mathbb{1}$. □

4.11 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Theorems 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

Definition 4.89 (CP partial order). [Lean](#) [✓ Stmt](#) For linear maps $S, T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, write $T \leq S$ (in the CP order) when $S - T$ is completely positive.

Definition 4.90 (Block-top contraction). [Lean](#) [✓ Stmt](#) For natural numbers r, s , let $C = C_{r,s} \in M(\mathbb{C})(\mathbb{C})$ be the rectangular $r \times (r + s)$ matrix whose rows are the first r rows of the identity on $\mathbb{C}^{r+s}$: $C_{ij} = 1$ if $j = i < r$ and 0 otherwise.

Lemma 4.91 (C is a co-isometry). [Lean](#) [✓ Stmt](#) The block-top matrix satisfies $CC^\dagger = \mathbb{1}_r$.

Proof. [✓ Proof](#) Direct entrywise computation: $(CC^\dagger)_{ii'} = \sum_j C_{ij} \overline{C_{i'j}}$ collapses to $\delta_{ii'}$. □

Lemma 4.92 (C is a contraction). [Lean](#) [✓ Stmt](#) The block-top matrix satisfies $C^\dagger C \leq \mathbb{1}_{r+s}$.

Proof. [✓ Proof](#) The matrix $C^\dagger C$ is diagonal with a 1 on the first r coordinates and a 0 on the last s , so $\mathbb{1} - C^\dagger C$ is positive semidefinite. □

Lemma 4.93 (Intertwining of Stinespring isometries). [Lean](#) [✓ Stmt](#) Let $K : \{0, \dots, r-1\} \rightarrow M_D(\mathbb{C})$ and $L : \{0, \dots, s-1\} \rightarrow M_D(\mathbb{C})$ be two Kraus families. Write $K ++ L$ for their concatenation as a family indexed by $\{0, \dots, r+s-1\}$. Then

$$V_K = (\mathbb{1}_D \otimes C_{r,s}) V_{K++L}.$$

Proof. [✓ Proof](#) Entrywise expansion of both sides: the Kronecker product with $C_{r,s}$ picks out the first r coordinates of the dilation space, matching the action of V_{K++L} on those coordinates, which is precisely V_K . $\square$

Theorem 4.94 (Ordered CP maps via Stinespring contractions). [Lean](#) [✓ Stmt](#) Let $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be CP maps with $T_1 \leq T_2$. Then there exist an ancilla dimension m , Heisenberg-form Stinespring matrices V_1, V_2 realizing T_1, T_2 via $T_i(A) = V_i^\dagger(A \otimes \mathbb{1})V_i$, and a contraction $\tilde{C}$ on the dilation space with $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$ such that

$$V_1 = (\mathbb{1}_D \otimes \tilde{C}) V_2.$$

This is [Wol12, Theorem 2.3].

Proof. [✓ Proof](#) Choose a Heisenberg-form Kraus family K_1 of T_1 with Stinespring matrix $V_1 = V_{K_1}$, and Kraus operators L of the CP map $T_2 - T_1$ in Schrödinger orientation. Let $L^\dagger$ denote the conjugate-transposed family. Set $m = r_1 + s$ and let $K_2 = K_1 ++ L^\dagger$. Then $T_2(A) = T_1(A) + (T_2 - T_1)(A) = \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger$, and the latter equals $\sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger$ which is the Heisenberg form for K_2 , yielding $T_2(A) = V_{K_2}^\dagger(A \otimes \mathbb{1})V_{K_2}$. Taking $\tilde{C} = C_{r_1,s}$ the intertwining $V_{K_1} = (\mathbb{1}_D \otimes \tilde{C})V_{K_2}$ is Lemma 4.93, and $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$ is Lemma 4.92. $\square$

Definition 4.95 (Block-diagonal projectors on the dilation space). [Lean](#) [Lean](#) [✓ Stmt](#) For r, s , let $P_{\text{top}} = C_{r,s}^\dagger C_{r,s}$ and $P_{\text{bot}} = \mathbb{1} - P_{\text{top}}$. Both are PSD (for P_{bot} , by Lemma 4.92) and $P_{\text{top}} + P_{\text{bot}} = \mathbb{1}$.

Theorem 4.96 (Radon–Nikodym theorem for CP maps). [Lean](#) [✓ Stmt](#) Let $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be completely positive. There exist an ancilla dimension m , a Kraus family K with Stinespring matrix $V = V_K$, and positive semidefinite operators $P_1, P_2 : \mathbb{C}^m \rightarrow \mathbb{C}^m$ with $P_1 + P_2 = \mathbb{1}_m$ such that, for every $A \in M_D(\mathbb{C})$,

$$T_i(A) = V^\dagger(A \otimes P_i)V, \quad i = 1, 2.$$

This is [Wol12, Theorem 2.4].

Proof. [✓ Proof](#) Take Heisenberg-form Kraus families K_1 for T_1 and Schrödinger-form L for T_2 , and form $K = K_1 ++ L^\dagger$ on $\mathbb{C}^{r_1+s}$. Choose $P_1 = P_{\text{top}}$, $P_2 = P_{\text{bot}}$ as in Definition 4.95. The Kronecker identity $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger (A \otimes \mathbb{1})(\mathbb{1} \otimes C)$ rewrites $V^\dagger(A \otimes P_{\text{top}})V$ as $((\mathbb{1} \otimes C)V)^\dagger (A \otimes \mathbb{1})((\mathbb{1} \otimes C)V)$, which equals $V_{K_1}^\dagger(A \otimes \mathbb{1})V_{K_1} = T_1(A)$ by the intertwining Lemma 4.93. For the complementary block, $A \otimes P_{\text{bot}} = A \otimes \mathbb{1} - A \otimes P_{\text{top}}$ and sandwiching by V yields $(T_1 + T_2)(A) - T_1(A) = T_2(A)$. $\square$

Theorem 4.97 (Open-system representation of quantum channels). [Lean](#) [✓ Stmt](#) Every quantum channel $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits a Stinespring isometry V on an ancilla space $\mathbb{C}^r$ such that

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

Proof. [✓ Proof](#) Apply the existence of an isometric Stinespring dilation (Theorem 4.88) and the Schrödinger-picture identity (Theorem 4.86). $\square$

Theorem 4.98 (Open-system representation, partial-trace form). [Lean](#) [✓ Stmt](#) Every quantum channel T admits an isometric Stinespring dilation V such that $T(\rho) = \text{tr}_E(V \rho V^\dagger)$.

Proof. [✓ Proof](#) Rewrite the componentwise identity of Theorem 4.97 as a partial trace over the ancilla factor. $\square$

Remark 4.99 (Unitary form). The fully unitary open-system form $T(\rho) = \text{tr}_E(U(\rho \otimes |\phi\rangle\langle\phi|)U^\dagger)$ of [Wol12, Theorem 2.5] follows from the isometric form by extending the Stinespring isometry V to a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^r$; this uses orthonormal basis extension and is recorded as an outstanding formalization item.

4.12 POVMs and Naimark dilation

Definition 4.100 (Positive operator-valued measure). [Lean](#) [✓ Stmt](#) A *positive operator-valued measure* with n outcomes on $\mathbb{C}^D$ is a family $\{E_i\}_{i=1}^n$ of positive semidefinite operators on $\mathbb{C}^D$ satisfying the resolution of identity

$$\sum_{i=1}^n E_i = \mathbb{1}_D.$$

Definition 4.101 (Naimark Kraus square roots). [Lean](#) [Lean](#) [✓ Stmt](#) For a POVM $\{E_i\}$, each effect $E_i \geq 0$ admits a square-root factorisation $E_i = M_i^\dagger M_i$ with $M_i \in M_D(\mathbb{C})$. The operators M_i are the *Naimark Kraus square roots*.

Theorem 4.102 (Naimark square roots satisfy the Kraus normalization). [Lean](#) [✓ Stmt](#) For a POVM $\{E_i\}$ with square roots $E_i = M_i^\dagger M_i$, one has

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

Proof. [✓ Proof](#) Sum the identities $E_i = M_i^\dagger M_i$ and use the defining resolution of the identity for the POVM. $\square$

Definition 4.103 (Naimark isometry). [Lean](#) [✓ Stmt](#) The *Naimark isometry* $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ of a POVM is the Stinespring-type construction

$$V = \sum_{i=1}^n M_i \otimes |i\rangle, \quad V_{(a,i),k} = (M_i)_{a,k}.$$

Definition 4.104 (Naimark projective measurement). [Lean](#) [Lean](#) [✓ Stmt](#) The *Naimark projectors* on $\mathbb{C}^D \otimes \mathbb{C}^n$ are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \quad (P_i)_{(a,b),(c,d)} = \delta_{a,c} \delta_{b,i} \delta_{d,i}.$$

Theorem 4.105 (Naimark isometry condition). [Lean](#) [✓ Stmt](#) The Naimark isometry satisfies $V^\dagger V = \mathbb{1}_D$.

Proof. [✓ Proof](#) The Stinespring Gram identity gives $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$ by the resolution of identity. $\square$

Theorem 4.106 (Naimark projectors form a projective measurement). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Naimark projectors satisfy $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, and $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$.

Proof. [✓ Proof](#) Each identity follows from direct entrywise computation using the delta-function form of $(P_i)_{(a,b),(c,d)}$. $\square$

Theorem 4.107 (Naimark dilation). [Lean](#) [✓ Stmt](#) Every POVM $\{E_i\}_{i=1}^n$ arises as a projective measurement on a dilation: for the isometry V and projectors P_i above,

$$E_i = V^\dagger P_i V.$$

Proof. [✓ Proof](#) Computing entrywise, $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}} (M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$. $\square$

Theorem 4.108 (Existential Naimark dilation). [Lean](#) [✓ Stmt](#) For every POVM $\{E_i\}$ on $\mathbb{C}^D$ there exist a dilation dimension r , an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$, and a projective measurement $\{P_i\}$ on the dilation satisfying $E_i = V^\dagger P_i V$.

Proof. [✓ Proof](#) Take $r = n$ and the explicit witnesses $V = \sum_i M_i \otimes |i\rangle$ and $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. $\square$

Definition 4.109 (Naimark dilation as a structure). [Lean](#) [✓ Stmt](#) A Naimark dilation of a POVM $\{E_i\}$ consists of an isometry V into a larger Hilbert space together with a projective measurement $\{P_i\}$ there such that $E_i = V^\dagger P_i V$ for all i .

Theorem 4.110 (Canonical dilation satisfies the Naimark dilation axioms). [Lean](#) [✓ Stmt](#) The explicit isometry $V = \sum_i M_i \otimes |i\rangle$ and projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ satisfy the defining axioms of a Naimark dilation.

Proof. [✓ Proof](#) Combine the previously proved identities $V^\dagger V = \mathbb{1}_D$, $P_i^2 = P_i$, $P_i^\dagger = P_i$, $P_i P_j = 0$ for $i \neq j$, $\sum_i P_i = \mathbb{1}$, and $V^\dagger P_i V = E_i$. $\square$

Theorem 4.111 (Concrete uniqueness for canonical projectors). [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$ satisfy $V^\dagger P_i V = E_i$ for the canonical projectors $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$. Then there exists an isometry W on the dilated space such that $V = W V_0$, where V_0 is the canonical Naimark isometry of $\{E_i\}$.

Proof. [✓ Proof](#) For each outcome i , the i -th block of V has Gram matrix E_i . Comparing with the canonical square root M_i gives $V_i^\dagger V_i = M_i^\dagger M_i$, hence $V_i = U_i M_i$ for a unitary U_i on $\mathbb{C}^D$. Assemble the U_i block-diagonally to obtain an isometry $W = \bigoplus_i U_i$ on $\mathbb{C}^D \otimes \mathbb{C}^n$, and then $V = W \sum_i M_i \otimes |i\rangle = W V_0$. $\square$

Definition 4.112 (POVM from a PSD resolution of identity on a dilation). [Lean](#) [✓ Stmt](#) Given an isometry $V : \mathbb{C}^D \rightarrow \mathbb{C}^{d'}$ with $V^\dagger V = \mathbb{1}_D$ and a family $\{P_i\}_{i=1}^n$ of positive semidefinite operators on $\mathbb{C}^{d'}$ summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

Definition 4.113 (Quantum instrument). [Lean](#) [✓ Stmt](#) A quantum instrument with n outcomes is a family $\{\Phi_i\}_{i=1}^n$ of completely positive maps on $M_D(\mathbb{C})$ whose sum $\sum_i \Phi_i$ is trace-preserving.

Definition 4.114 (Instrument operations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Associated to an instrument are the total channel $\sum_i \Phi_i$, the unnormalized update $\rho \mapsto \Phi_i(\rho)$ for each outcome i , the outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$, and the normalized posterior state $\Phi_i(\rho)/p_i(\rho)$ whenever $p_i(\rho) \neq 0$.

Theorem 4.115 (Instrument total map is a channel). [Lean](#) [✓ Stmt](#) The total map $\sum_i \Phi_i$ of an instrument is a quantum channel.

Proof. ✓ Proof Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition. $\square$

Theorem 4.116 (Conservation of probability for instruments). Lean ✓ Stmt *For every state $\rho \in M_D(\mathbb{C})$, the outcome probabilities $p_i(\rho) := \text{tr}(\Phi_i(\rho))$ sum to $\text{tr}(\rho)$.*

Proof. ✓ Proof Linearity of trace and trace preservation of $\sum_i \Phi_i$ give $\sum_i \text{tr}(\Phi_i(\rho)) = \text{tr}((\sum_i \Phi_i)(\rho)) = \text{tr}(\rho)$. $\square$

Theorem 4.117 (Non-negativity of instrument probabilities). Lean ✓ Stmt *If $\rho \in M_D(\mathbb{C})$ is positive semidefinite, then each outcome probability $p_i(\rho) = \text{tr}(\Phi_i(\rho))$ is a non-negative real number.*

Proof. ✓ Proof Each Φ_i is completely positive, hence positive, so $\Phi_i(\rho)$ is positive semidefinite and its trace is a non-negative real. $\square$

4.12.1 Trace-pairing expansion in transfer-matrix form

Definition 4.118 (Trace-self-dual basis). Lean ✓ Stmt A basis $\{\sigma_i\}_i$ of $M_D(\mathbb{C})$ is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \text{tr}(\sigma_i X) \sigma_i.$$

Definition 4.119 (Trace-pairing coefficients). Lean ✓ Stmt For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a trace-self-dual basis $\{\sigma_i\}$, define coefficients

$$t_{ij} := \text{tr}(\sigma_i T(\sigma_j)).$$

Theorem 4.120 (Trace-pairing expansion of a linear map). Lean ✓ Stmt *If $\{\sigma_i\}$ is trace-self-dual, then every linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits the expansion*

$$T(\rho) = \sum_i \sum_j t_{ij} \text{tr}(\sigma_j \rho) \sigma_i, \quad t_{ij} = \text{tr}(\sigma_i T(\sigma_j)).$$

Proof. ✓ Proof Expand ρ and each $T(\sigma_j)$ in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula. $\square$

4.12.2 SVD normal form (existence)

[Wol12, Section 2.3] discusses two families of normal forms for the transfer-matrix representation of a quantum channel: the *singular value decomposition* (SVD) and, for qubit channels, the *Lorentz normal form* obtained by $\text{SL}(2, \mathbb{C})$ filtering operations. The SVD normal form is proved below. The Lorentz normal form theorem (Theorem 4.132) requires the compactness and minimisation argument for $\text{SL}(2, \mathbb{C})$ filterings in [Wol12, Propositions 2.9 and 2.11]; see Subsection 4.12.3.

Theorem 4.121 (SVD for positive semi-definite matrices). Lean ✓ Stmt *Every positive semi-definite matrix $M \in M_D(\mathbb{C})$ admits a decomposition*

$$M = U \text{diag}(\sigma) U^\dagger,$$

where $U \in \text{U}(D)$ is unitary and $\sigma : \{1, \dots, D\} \rightarrow \mathbb{R}_{\geq 0}$ has non-negative entries.

Proof. [✓ Proof](#) The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semi-definiteness forces those eigenvalues to be non-negative, and the decomposition is packaged in SVD form with U the eigenvector unitary and σ the eigenvalue sequence. $\square$

Theorem 4.122 (SVD existence for invertible matrices). [Lean](#) [✓ Stmt](#) *Every invertible complex square matrix $M \in M_D(\mathbb{C})$ admits a singular value decomposition*

$$M = U \operatorname{diag}(\sigma) V^\dagger,$$

with $U, V \in U(D)$ unitary and $\sigma_i > 0$ for all i .

Proof. [✓ Proof](#) Apply the spectral theorem to the positive definite matrix $M^\dagger M$ to obtain $M^\dagger M = V \operatorname{diag}(\lambda) V^\dagger$ with $\lambda_i > 0$. Set $\sigma_i := \sqrt{\lambda_i}$ and $\Sigma := \operatorname{diag}(\sigma)$; since Σ is a real positive diagonal, it is self-adjoint and invertible. Define $U := MV\Sigma^{-1}$. Then $U^\dagger U = \Sigma^{-1} V^\dagger (M^\dagger M) V \Sigma^{-1} = \Sigma^{-1} \operatorname{diag}(\lambda) \Sigma^{-1} = I$, so U is unitary, and $U \Sigma V^\dagger = MV(\Sigma^{-1} \Sigma) V^\dagger = M V V^\dagger = M$. $\square$

Theorem 4.123 (SVD representation of a transfer matrix). [Lean](#) [✓ Stmt](#) *Every invertible transfer matrix $\widehat{T}$ of a linear super-operator $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ admits a singular value decomposition $\widehat{T} = U \operatorname{diag}(\sigma) V^\dagger$ with U, V unitary on $\mathbb{C}^D \otimes \mathbb{C}^D$ and $\sigma_i > 0$.*

Proof. [✓ Proof](#) Direct specialisation of Theorem 4.122 to the index type $\{1, \dots, D\} \times \{1, \dots, D\}$ used by the transfer-matrix representation. $\square$

Remark 4.124 (Sorted / unique singular values). Theorem 4.122 produces the singular values as an unordered family, without the usual convention that σ_i are sorted in non-increasing order or the statement that they are uniquely determined by M . Downstream applications in [Wol12, Section 2.3], such as trace-norm identities, polar decomposition, and the Lorentz normal form, will typically require the sorted / unique variant; that refinement is future work.

4.12.3 Lorentz normal form

This subsection states the existence theorems from [Wol12, Section 2.3].

Definition 4.125 (SL-filtering operation). [Lean](#) [✓ Stmt](#) An *SL-filtering* for $D \times D$ matrices is a completely positive map of the form $\Phi(X) = SXS^\dagger$ where $S \in M_D(\mathbb{C})$ satisfies $\det S = 1$. Such maps are invertible and have Kraus rank 1.

Definition 4.126 (Doubly-stochastic map). [Lean](#) [✓ Stmt](#) A linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *doubly-stochastic* if $T(1) \propto 1$ and the reduced density matrix $\operatorname{tr}_1[\tau]$ of its Choi matrix $\tau = (T \otimes \operatorname{id})(|\Omega\rangle\langle\Omega|)$ is proportional to the identity. This is the normal form in [Wol12, Proposition 2.8].

Lemma 4.127 (Infimum attainment for SL-filterings). [Lean](#) *For a positive-definite Choi matrix τ , the infimum of $\operatorname{tr}[(S_2 \otimes_k S_1) \tau (S_2 \otimes_k S_1)^\dagger]$ over $S_1, S_2 \in M_D(\mathbb{C})$ with $\det S_1 = \det S_2 = 1$ is attained. **Not yet proved**—requires compactness of bounded subsets of $\operatorname{SL}(D, \mathbb{C})$ and the extreme-value theorem.*

Theorem 4.128 (Generic normal form for CP maps). [Lean](#) *Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a completely positive map whose Choi matrix is positive-definite (equivalently, T has full Kraus rank). Then there exist SL-filterings Φ_1, Φ_2 such that $\Phi_2 \circ T \circ \Phi_1$ is doubly-stochastic. **Not yet proved**—proof blocked on Lemma 4.127.*

Definition 4.129 (Diagonal Lorentz normal form). [Lean](#) [✓ Stmt](#) For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, write $\widehat{T}'$ for its matrix in the normalized Pauli basis $\{\sigma_0/\sqrt{2}, \sigma_1/\sqrt{2}, \sigma_2/\sqrt{2}, \sigma_3/\sqrt{2}\}$. The channel is in *diagonal Lorentz normal form* when it is unital and every off-diagonal entry of $\widehat{T}'$ is zero.

Definition 4.130 (Non-diagonal Lorentz normal form). [Lean](#) [✓ Stmt](#) For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, write $\widehat{T}'$ in the normalized Pauli basis. The channel is in *non-diagonal Lorentz normal form* when, for some $x \in [0, 1]$,

$$\widehat{T}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x/\sqrt{3} & 0 & 0 \\ 0 & 0 & x/\sqrt{3} & 0 \\ 2/3 & 0 & 0 & 1/3 \end{pmatrix}.$$

Trace preservation supplies the first row of the displayed matrix.

Definition 4.131 (Singular Lorentz normal form). [Lean](#) [✓ Stmt](#) For a completely positive, trace-preserving qubit channel $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, write $\widehat{T}'$ in the normalized Pauli basis. The channel is in *singular Lorentz normal form* when

$$\widehat{T}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

equivalently, every input state is mapped to $(1 + \sigma_3)/2$. Trace preservation supplies the first row of the displayed matrix.

Theorem 4.132 (Lorentz normal form for qubit channels). [Lean](#) For every qubit channel $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, there exist $\text{SL}(2, \mathbb{C})$ -filterings Φ_1, Φ_2 such that the filtered channel $T' = \Phi_2 \circ T \circ \Phi_1$ is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular. *Not yet proved*—proof blocked on Lemma 4.127 and the Lorentz group classification of $\text{SL}(2, \mathbb{C})$ orbits.

4.13 Determinant of a quantum channel

Definition 4.133 (Channel determinant). [Lean](#) [✓ Stmt](#) For a linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the *channel determinant* $\det T$ is the determinant of the matrix of T with respect to the standard matrix-unit basis of $M_D(\mathbb{C})$. This is the quantity studied in [Wol12, Section 6.1.1].

Remark 4.134 (Channel determinant as an operator determinant). [Lean](#) [✓ Stmt](#) The channel determinant agrees with the ordinary determinant of the linear endomorphism $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$.

Definition 4.135 (Unitary channel). [Lean](#) [Lean](#) [✓ Stmt](#) For a unitary matrix $U \in \mathcal{U}(D)$, the *unitary channel* is $T(\rho) = U\rho U^\dagger$. It is automatically a quantum channel.

Theorem 4.136 (Unitary channels have determinant one). [Lean](#) [Lean](#) [✓ Stmt](#) If $T(\rho) = U\rho U^\dagger$ is a unitary channel, then $\det T = 1$ and hence $|\det T| = 1$.

Proof. [✓ Proof](#) Vectorization identifies T with a Kronecker product $\overline{U} \otimes U$, whose determinant is $\overline{\det U}^D (\det U)^D = 1$. $\square$

Theorem 4.137 (Determinant bound for positive trace-preserving maps). [Lean](#) [Lean](#) [✓ Stmt](#) For any positive trace-preserving map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, $|\det T| \leq 1$. This is [Wol12, Theorem 6.1(1)].

Proof. [✓ Proof](#) Every eigenvalue μ of T satisfies $|\mu| \leq 1$ (positivity and trace preservation force spectral radius ≤ 1). Since $\det T$ is the product of the eigenvalues (counted with algebraic multiplicity), the bound $|\det T| = \prod_i |\mu_i| \leq 1$ follows by induction. $\square$

Theorem 4.138 (Determinant one iff the channel is unitary). [Lean](#) [Lean](#) [✓ Stmt](#) For a CPTP map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Theorem 6.1(2)].

Proof. [✓ Proof](#) Reverse direction is Theorem 4.136. Forward direction: $|\det T| = 1$ together with $|\mu| \leq 1$ for every eigenvalue μ (from trace preservation and positivity) forces every eigenvalue to satisfy $|\mu| = 1$. Hence T is bijective. A bijective CPTP map is a $*$ -automorphism: the Heisenberg dual $T^*(Y) = \sum_i K_i^\dagger Y K_i$ is unital (since T is TP), so the Kadison–Schwarz inequality $T^*(Y^\dagger Y) \geq T^*(Y)^\dagger T^*(Y)$ becomes an equality for every Y . Hence T^* , and therefore T , preserves products. By the Skolem–Noether theorem, every $*$ -automorphism of $M_D(\mathbb{C})$ has the form $T(\rho) = U\rho U^\dagger$ for some unitary U . Kraus freedom then gives $K_i = c_i U$ with $\sum_i |c_i|^2 = 1$. $\square$

4.14 Fixed-point algebra

Definition 4.139 (Kraus fixed points). [Lean](#) [✓ Stmt](#) The *fixed-point set* of a Kraus map $E(X) = \sum_i K_i X K_i^\dagger$ is $\text{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$.

Theorem 4.140 (Fixed points lie in the multiplicative domain). [Lean](#) [✓ Stmt](#) Let E be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point $\rho > 0$. Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.

Proof. [✓ Proof](#) The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. $\square$

Theorem 4.141 (Fixed points form a $*$ -subalgebra). [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map whose adjoint map E^* has a positive definite fixed point $\rho > 0$. Then $\text{Fix}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$. This is [Wol12, Theorem 6.12].

Proof. [✓ Proof](#) Closure under addition and scalar multiplication is immediate from linearity. Closure under $\dagger$: if $E(X) = X$ then $E(X^\dagger) = X^\dagger$ because fixed points are stable under adjoint. Closure under multiplication: if $E(X) = X$ and $E(Y) = Y$, the weighted Kadison–Schwarz argument gives $E(X^\dagger X) = E(X)^\dagger E(X)$ and, applied to $X^\dagger$, also $E(X X^\dagger) = E(X) E(X)^\dagger$. Thus X lies in the multiplicative domain, and Theorem 4.42 yields $E(XY) = E(X)E(Y) = XY$. $\square$

Remark 4.142. [✓ Stmt](#) Theorem 4.140 is the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

Theorem 4.143 (Projected elements lie in the multiplicative domain). [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose $E^2 = E$. Then $E(X)$ lies in the multiplicative domain of E for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Idempotence gives $E(E(X)) = E(X)$, so $E(X)$ is a fixed point. By the argument in Theorem 4.141, the Kadison–Schwarz gap for the fixed point $E(X)$ vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under † , so the same argument applied to $E(X)^\dagger$ gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

Therefore Theorem 5.28 places $E(X)$ in the multiplicative domain. □

Theorem 4.144 (Choi–Effros left absorption). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(E(X)E(Y)).$$

Proof. [✓ Proof](#) The previous theorem places $E(X)$ in the multiplicative domain, so $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.145 (Choi–Effros right absorption). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(XE(Y)) = E(E(X)E(Y)).$$

Proof. [✓ Proof](#) The previous theorem places $E(Y)$ in the multiplicative domain, so $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$. Applying the same multiplicativity to $E(X)$ and $E(Y)$ gives $E(E(X)E(Y)) = E(E(X))E(E(Y))$, and idempotence rewrites this as $E(X)E(Y)$. □

Theorem 4.146 (Choi–Effros symmetric absorption). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)Y) = E(XE(Y)).$$

Proof. [✓ Proof](#) Both sides equal $E(E(X)E(Y))$ by the two preceding theorems. □

Theorem 4.147 (Choi–Effros projected product). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all $X, Y \in M_D(\mathbb{C})$,*

$$E(E(X)E(Y)) = E(X)E(Y).$$

Proof. [✓ Proof](#) The previous theorem places $E(X)$ and $E(Y)$ in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since E is idempotent under the standing hypotheses, $E(E(X)) = E(X)$ and $E(E(Y)) = E(Y)$, so

$$E(E(X)E(Y)) = E(X)E(Y).$$

□

Definition 4.148 (Adjoint fixed points). [Lean](#) [✓ Stmt](#) The *adjoint fixed-point set* $\text{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$, where $E^*(X) = \sum_i K_i^\dagger X K_i$.

Theorem 4.149 (Fixed points form a $*$ -subalgebra in the Heisenberg picture). [Lean](#) [Lean](#) [✓ Stmt](#) *If the Kraus map is trace-preserving and has a positive definite fixed point $\rho > 0$ with $E(\rho) = \rho$, then $\text{Fix}(E^*)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$.*

Proof. ✓ Proof Apply Theorem 4.141 to the adjoint family $\{K_i^\dagger\}$. □

Definition 4.150 (Kraus commutant). Lean Lean ✓ Stmt The *Kraus commutant* is $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \ \forall i\}$. It is a $*$ -subalgebra of $M_D(\mathbb{C})$.

Theorem 4.151 (Adjoint fixed points equal the Kraus commutant). Lean ✓ Stmt Under the hypotheses of Theorem 4.149, the adjoint fixed-point $*$ -subalgebra equals the Kraus commutant:

$$\text{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Theorem 6.13].

Proof. ✓ Proof ($\supseteq$): if X commutes with all K_i and $K_i^\dagger$, then $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$ by trace preservation. ($\subseteq$): any $*$ -subalgebra inside $\text{Fix}(E^*)$ lies in the Kraus commutant, because for $X \in \text{Fix}(E^*)$ with $X^\dagger X \in \text{Fix}(E^*)$ the Kadison–Schwarz gap vanishes, which forces $XK_i = K_iX$ and $XK_i^\dagger = K_i^\dagger X$. □

Theorem 4.152 (Kraus commutant inside the adjoint fixed points). Lean ✓ Stmt Under the hypotheses above, the Kraus commutant is the largest $*$ -subalgebra contained in the adjoint fixed-point set.

Proof. ✓ Proof By Theorem 4.151, every $*$ -subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a $*$ -subalgebra. □

4.15 Quantum dynamical semigroups

The one-parameter semigroup theory of Wolf Chapter 7 is treated in Chapter 18. That later chapter develops the exponential representation of norm-continuous semigroups, perturbation bounds, and the GKSL/Lindblad description of generators. We do not duplicate that material here.

Chapter 5

Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory following [Wol12, Chapter 5]. We begin with the Kadison–Schwarz inequality for Kraus maps, then describe the multiplicative domain and its algebraic structure. These two parts are used in the non-periodic Fundamental Theorem through peripheral-eigenvector rigidity and normal-block comparison. We also establish the extension to normal operators for positive subunital maps, the resulting order and spectral contractivity statements, and the standard example of a Schwarz map that is not completely positive. The operator-convexity and Jensen material from [Wol12, Chapter 5] is treated separately in Chapter 19.

5.1 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 4.4). The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map is a Kraus map by construction (Remark 4.17), so these results apply directly to transfer maps.

Definitions 5.1–5.4 below restate the Kraus-map notions from Chapter 4 for the reader’s convenience.

Definition 5.1 (Kraus map). [Lean](#) [✓ Stmt](#) Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 4.4) if and only if it can be written in this form.

Definition 5.2 (Adjoint Kraus map). [Lean](#) [✓ Stmt](#) The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A , this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 5.3 (Unital Kraus map). [Lean](#) [✓ Stmt](#) A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$. In the later gauge language this is the right-canonical condition.

Definition 5.4 (Trace-preserving Kraus map). [Lean](#) [✓ Stmt](#) A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 5.5 (Kadison–Schwarz inequality). [Lean](#) [✓ Stmt](#) Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

Proof. [✓ Proof](#) The 2×2 block matrix $\begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix}$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$ -block maps to $\mathbb{1}$). The Schur complement of the $(2,2)$ -block $\mathbb{1}$ gives the result. $\square$

Theorem 5.6 (Kadison–Schwarz for the adjoint channel). [Lean](#) [✓ Stmt](#) If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.

Proof. [✓ Proof](#) The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 5.5 to $\{K_i^\dagger\}$. $\square$

5.1.1 Two-positive maps and the generalized Schwarz inequality

Definition 5.7 (k -positive map). [Lean](#) [✓ Stmt](#) A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is k -positive if $E \otimes \mathbb{1}_k$ is positive on $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$.

Definition 5.8 (Two-positive map). [Lean](#) [✓ Stmt](#) A linear map is *two-positive* if it is 2-positive.

Theorem 5.9 (Completely positive maps are k -positive). [Lean](#) [✓ Stmt](#) Every completely positive map is k -positive for all $k \geq 1$.

Theorem 5.10 ($(k+1)$ -positive implies k -positive). [Lean](#) [✓ Stmt](#) Positivity under amplification is monotone in k .

Theorem 5.11 (Completely positive implies two-positive). [Lean](#) [✓ Stmt](#) Every completely positive map is two-positive.

Theorem 5.12 (Two-positive implies positive). [Lean](#) [✓ Stmt](#) Every two-positive map is positive.

Definition 5.13 (Unital linear map). [Lean](#) [✓ Stmt](#) A linear map E is unital if $E(\mathbb{1}) = \mathbb{1}$.

Theorem 5.14 (Kadison–Schwarz for unital two-positive maps). [Lean](#) [✓ Stmt](#) If E is unital and two-positive, then

$$E(X^\dagger X) \geq E(X)^\dagger E(X)$$

for all $X \in M_D(\mathbb{C})$.

Proof. Form the 2×2 block matrix $\begin{pmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{pmatrix} \geq 0$. Apply $E \otimes \text{id}_2$ (positive by 2-positivity) and read off the $(2,2)$ -block Schur complement: $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$. $\square$

Theorem 5.15 (Kraus-form Kadison–Schwarz as a special case). [Lean](#) [✓ Stmt](#) The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.

5.1.2 Douglas-type factorization lemmas

Theorem 5.16 (Range inclusion implies right factorization). [Lean](#) [✓ Stmt](#) If $\text{ran}(A) \subseteq \text{ran}(B)$, then there exists C such that $A = BC$.

Proof. In finite dimensions, $\text{ran}(A) \subseteq \text{ran}(B)$ implies $A = B(B^\dagger B)^\dagger B^\dagger A$, where $(B^\dagger B)^\dagger$ denotes the Moore–Penrose pseudoinverse. Set $C = (B^\dagger B)^\dagger B^\dagger A$. $\square$

Theorem 5.17 (Vectorwise range criterion implies factorization). [Lean](#) [✓ Stmt](#) If $Av \in \text{ran}(B)$ for every vector v , then there exists C with $A = BC$.

Theorem 5.18 (Hilbert–Schmidt contraction). [Lean](#) [✓ Stmt](#) If a Kraus map is both unital and trace-preserving, then $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) By Theorem 5.5, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. $\square$

Lemma 5.19 (Trace pairing for Kraus map and adjoint map). [Lean](#) [✓ Stmt](#) For all $X, Y \in M_D(\mathbb{C})$,

$$\text{tr}(X E(Y)) = \text{tr}(E^*(X) Y).$$

Lemma 5.20 (Positive-definite trace test). [Lean](#) [✓ Stmt](#) If $A > 0$, $X \geq 0$, and $\text{tr}(AX) = 0$, then $X = 0$.

Theorem 5.21 (Fixed-point peripheral eigenvectors satisfy KS equality). [Lean](#) [✓ Stmt](#) Let E be unital and let E^* be trace-preserving with a positive definite fixed point $\rho > 0$. If $E(X) = \mu X$ with $|\mu| = 1$, then

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

5.2 Multiplicative domain

Theorem 5.22 (KS gap decomposition). [Lean](#) [✓ Stmt](#) For a unital Kraus map E , the Kadison–Schwarz gap decomposes as

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.

Proof. [✓ Proof](#) Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. $\square$

Theorem 5.23 (KS equality implies Kraus commutation). [Lean](#) [✓ Stmt](#) Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X , then $XK_i^\dagger = K_i^\dagger E(X)$ for all i .

Proof. [✓ Proof](#) By Theorem 5.22, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. $\square$

Theorem 5.24 (KS equality for peripheral eigenvectors). [Lean](#) [✓ Stmt](#) Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the KS gap vanishes: $E(X^\dagger X) = E(X)^\dagger E(X)$.

Proof. ✓ Proof The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 5.5. Its trace satisfies $\text{tr}(G) = \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\text{tr}(E(X)^\dagger E(X)) = |\mu|^2 \text{tr}(X^\dagger X) = \text{tr}(X^\dagger X)$. So $\text{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. $\square$

Theorem 5.25 (Left multiplicative identity from KS equality). Lean ✓ Stmt Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.

Proof. ✓ Proof From Theorem 5.23, $XK_i^\dagger = K_i^\dagger E(X)$ for all i . Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

$\square$

5.2.1 Full multiplicative-domain structure

Definition 5.26 (Right and left multiplicative domains). Lean Lean ✓ Stmt For a Kraus map E , define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(XY) = E(X)E(Y)\}$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(YX) = E(Y)E(X)\}.$$

We follow the convention that $\mathcal{A}_R(E)$ controls right multiplication and $\mathcal{A}_L(E)$ controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Eqs. (5.11)–(5.12)].

Definition 5.27 (Full multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of E is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

Theorem 5.28 (Multiplicative-domain characterization). Lean Lean ✓ Stmt Let E be a unital Kraus map. Then

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Theorem 5.7].

Proof. ✓ Proof If X lies in a one-sided multiplicative domain, evaluate the defining identity at $Y = X^\dagger$. Conversely, the left-domain implication is exactly Theorem 5.25, and the right-domain implication follows by applying Theorem 5.25 to $X^\dagger$. $\square$

Theorem 5.29 (One-sided multiplicative domains are subalgebras). Lean Lean ✓ Stmt For a unital Kraus map E , both $\mathcal{A}_R(E)$ and $\mathcal{A}_L(E)$ are unital subalgebras of $M_D(\mathbb{C})$.

Proof. ✓ Proof Closure under addition follows from linearity of E . Closure under multiplication: if $X, Y \in \mathcal{A}_R(E)$, then $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$ for all Z , using Theorem 5.28 iteratively, and the $\mathcal{A}_L(E)$ case is analogous. Containment of $\mathbb{1}$ follows from unitality of E . $\square$

Theorem 5.30 (Multiplicative domain is a $*$ -subalgebra). [Lean](#) [✓ Stmt](#) For a unital Kraus map E , the full multiplicative domain $\mathcal{A}(E)$ is a $*$ -subalgebra of $M_D(\mathbb{C})$. This is the algebraic content of [Wol12, Theorem 5.7].

Proof. [✓ Proof](#) If $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$, then $E(X^\dagger X) = E(X)^\dagger E(X)$ and $E(XX^\dagger) = E(X)E(X)^\dagger$. The characterization of Theorem 5.28 shows $X^\dagger \in \mathcal{A}(E)$. Combined with Theorem 5.29, this gives a $*$ -subalgebra. $\square$

Theorem 5.31 (Restriction to multiplicative domain is a $*$ -homomorphism). [Lean](#) [✓ Stmt](#) If E is unital, then the restricted map $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \rightarrow M_D(\mathbb{C})$ is a $*$ -algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].

Proof. [✓ Proof](#) Multiplicativity on $\mathcal{A}(E)$ is immediate from the definition. For $X \in \mathcal{A}(E)$, the Kraus commutation relation from Theorem 5.23 gives $E(X^\dagger) = E(X)^\dagger$, so the restriction preserves adjoints. $\square$

5.2.2 Schwarz inequality for normal operators

Theorem 5.32 (CP Schwarz inequality for normal operators). [Lean](#) [Lean](#) [✓ Stmt](#) Let $E^*(X) = \sum_i K_i^\dagger X K_i$ be the adjoint Kraus map of a trace-preserving Kraus family, and let $A \in M_D(\mathbb{C})$ be normal. Then the Schwarz gap

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A).$$

This is the completely positive case of [Wol12, Proposition 5.1].

Proof. [✓ Proof](#) The positivity statement is exactly Theorem 5.6. The second formulation is just the corresponding Loewner-order reformulation. $\square$

Theorem 5.33 (Schwarz inequality for normal operators). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is normal, then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Proposition 5.1].

Proof. [✓ Proof](#) Since A is normal, diagonalize $A = UDU^\dagger$ with D diagonal and U unitary. Express $A^\dagger A = U|D|^2U^\dagger$ as a sum $\sum_{i,j} \bar{d}_i d_j P_i P_j$ where $P_i = U e_i e_i^\dagger U^\dagger$. Each P_i is rank-one PSD, and the images $T(P_i)$ pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown nonneg by a simultaneous-diagonalization argument: positivity of T ensures the scalar PSD matrix $(\langle v_k, T(P_i P_j) v_k \rangle)_{i,j}$ dominates the product $(\langle v_k, T(P_i) v_k \rangle \langle v_k, T(P_j) v_k \rangle)_{i,j}$, and combining over a common eigenbasis of the $T(P_i)$ yields the result. $\square$

Remark 5.34. The proof follows the positivity-on-abelian argument of [Wol12, Proposition 1.6]: a positive map restricted to the commutative $*$ -subalgebra generated by the spectral projectors of A satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

5.2.3 Schwarz inequality for subnormal and commuting-dominant operators

Theorem 5.35 (Schwarz inequality for subnormal operators). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ is subnormal (i.e. there exists a normal operator on a larger space whose compression to $\mathbb{C}^D$ is A), then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [\[Wol12, Theorem 5.5\]](#).

Proof. [✓ Proof](#) Embed A as the northwest corner of a normal operator N on $\mathbb{C}^D \oplus \mathbb{C}^E$. Extend T to a positive subunitary map $\tilde{T}$ on the larger space (acting as T on the northwest block, zero elsewhere). Apply Theorem 5.33 to $\tilde{T}$ and N , then compress the resulting inequality back to $\mathbb{C}^D$. $\square$

Theorem 5.36 (Schwarz inequality for commuting-dominant operators). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $T(\mathbb{1}) \leq \mathbb{1}$. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant $B \geq 0$ with $A^\dagger A \leq B$ and B commuting with A (in the sense $BA = AB$, $BA^\dagger = A^\dagger B$), then

$$T(A^\dagger)T(A) \leq T(B).$$

This is [\[Wol12, Theorem 5.6\]](#).

Proof. [✓ Proof](#) Let $B^{1/2}$ be the positive square root of B . Since B commutes with A and $A^\dagger$, the contraction $C = B^{-1/2}A$ (defined on the range of $B^{1/2}$) is subnormal. An ε -regularisation $B_\varepsilon = B + \varepsilon \mathbb{1}$ makes $B_\varepsilon > 0$ and the contraction C_ε globally defined. Apply Theorem 5.35 to C_ε , multiply both sides by $T(B_\varepsilon)$, use positivity and monotonicity, and take $\varepsilon \rightarrow 0$. $\square$

Theorem 5.37 (CP variant: Kadison–Schwarz for commuting-dominant inputs). [Lean](#) [✓ Stmt](#) Let E^* be the adjoint of a trace-preserving Kraus map. If $A \in M_D(\mathbb{C})$ admits a positive semidefinite dominant B commuting with A , $A^\dagger$, and satisfying $A^\dagger A \leq B$, then $E^*(A^\dagger)E^*(A) \leq E^*(B)$.

Proof. [✓ Proof](#) Apply Theorem 5.36 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis). $\square$

5.2.4 Positive maps preserve order and spectral intervals

Theorem 5.38 (Positive maps are monotone). [Lean](#) [✓ Stmt](#) If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive and $A \leq B$, then $T(A) \leq T(B)$.

Proof. [✓ Proof](#) Since $B - A \geq 0$, positivity gives $T(B - A) \geq 0$. By linearity, this is $T(B) - T(A) \geq 0$. $\square$

Theorem 5.39 (Positive maps preserve adjoints). [Lean](#) [✓ Stmt](#) If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is positive, then $T(A^\dagger) = T(A)^\dagger$ for all $A \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Write $A = B + iC$ with B and C Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints. $\square$

Theorem 5.40 (Order intervals are preserved). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$. If $a \leq 0 \leq b$ and $a\mathbb{1} \leq A \leq b\mathbb{1}$, then

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}.$$

This is the matrix form of [Wol12, Equation (5.21)].

Proof. [✓ Proof](#) By monotonicity, $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$. For the lower bound, $T(a\mathbb{1}) = aT(\mathbb{1})$. Since $a \leq 0$ and $T(\mathbb{1}) \leq \mathbb{1}$, we have $aT(\mathbb{1}) \geq a\mathbb{1}$. For the upper bound, $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$ since $b \geq 0$. Therefore $a\mathbb{1} \leq T(A) \leq b\mathbb{1}$. $\square$

Theorem 5.41 (Spectral interval contractivity). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be positive with $T(\mathbb{1}) \leq \mathbb{1}$, and let $A \in M_D(\mathbb{C})$ be Hermitian. If $\text{spec}(A) \subseteq [a, b]$ with $a \leq 0 \leq b$, then

$$\text{spec}(T(A)) \subseteq [a, b].$$

This is [Wol12, Equation (5.21)].

Proof. [✓ Proof](#) For Hermitian matrices, the inclusion $\text{spec}(A) \subseteq [a, b]$ is equivalent to the order bounds $a\mathbb{1} \leq A \leq b\mathbb{1}$. Apply Theorem 5.40 and translate the resulting inequalities back into spectral bounds. $\square$

5.2.5 A positive Schwarz map that is not completely positive

Example 5.42 (Transpose-trace map on $M_2(\mathbb{C})$). [Lean](#) [✓ Stmt](#) Consider the linear map $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\text{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Ex. 5.3].

Theorem 5.43 (The transpose-trace map is positive). [Lean](#) [✓ Stmt](#) The map from Example 5.42 is positive.

Proof. [✓ Proof](#) It is a nonnegative linear combination of the transpose map and the map $A \mapsto \text{tr}(A)\mathbb{1}$, and both of these maps send positive semidefinite matrices to positive semidefinite matrices. $\square$

Theorem 5.44 (The transpose-trace map satisfies the Schwarz inequality). [Lean](#) [✓ Stmt](#) For every $A \in M_2(\mathbb{C})$,

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0,$$

where T is the map from Example 5.42.

Proof. [✓ Proof](#) Write the Schwarz gap as $F(A)$. One checks that $F(A + \lambda\mathbb{1}) = F(A)$ for every scalar λ , so it is enough to treat the case $\text{tr}(A) = 0$. In that case a direct 2×2 computation shows that $F(A)$ dominates $\frac{1}{2}(A^\dagger A)^T$, hence is positive semidefinite. $\square$

Theorem 5.45 (The transpose-trace map is not completely positive). [Lean](#) [✓ Stmt](#) The map from Example 5.42 is not completely positive.

Proof. [✓ Proof](#) Compute the Choi matrix of T and test it on the antisymmetric vector $|01\rangle - |10\rangle$. The resulting expectation value is negative, so the Choi matrix is not positive semidefinite. $\square$

Chapter 6

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory used in subsequent chapters: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf’s spectral characterization of irreducibility. The conceptual source is Wolf’s treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

Theorem 6.1 (Brouwer fixed point on density matrices). Lean Lean Lean Lean Lean ✓ Stmt
Let $D > 0$. Every continuous map from the compact convex set of density matrices in $M_D(\mathbb{C})$ to itself has a fixed point.

Proof. ✓ Proof The proof starts from Brouwer’s theorem for products of simplices, transfers it to closed cubes, and then to compact retracts of finite-dimensional real normed spaces. The density-matrix set is such a compact retract, using the explicit density retraction. □

6.1 Positive definiteness

Remark 6.2. ✓ Stmt Wolf’s general theory treats irreducible positive maps first ([Wol12, Theorem 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Theorem 6.3]). The injective-tensor version (Theorem 6.3) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 6.4) follows via the support-projection argument of [Wol12, Theorem 6.2].

Theorem 6.3 (PSD fixed point is positive definite). Lean ✓ Stmt *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. ✓ Proof Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is nonneg; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i : the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M . Given any vector w , choose M with $M v = w$. Then $\rho w = 0$ for every w , so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Theorem 6.3, item 2], the same conclusion is reached via the growth condition $(\mathbb{1} + T)^{d-1}(X) > 0$ of [Wol12, Theorem 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. □

Theorem 6.4 (PSD fixed point is positive definite — irreducible version). Lean ✓ Stmt *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. ✓ Proof Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q \mathcal{E}_A(QXQ) Q = \mathcal{E}_A(QXQ)$ for all X . By irreducibility of $\mathcal{E}_A$ (Definition 4.15), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite, contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Theorem 6.2, condition (1)]. $\square$

Theorem 6.5 (Orthogonal trace condition). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\text{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D - 1$ such that*

$$\text{tr}(B \cdot E^t(A)) > 0.$$

Proof. ✓ Proof The growth theorem for irreducible CP maps ([Wol12, Theorem 6.2, item 2]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\text{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\text{tr}\left(B \cdot \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A)\right) > 0.$$

Each term $\text{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\text{tr}(BA) = 0$). Hence at least one $k \in \{1, \dots, D - 1\}$ contributes a strictly positive term. This is [Wol12, Theorem 6.2, item 4]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. $\square$

6.2 Uniqueness

Theorem 6.6 (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. ✓ Proof Both ρ and σ are positive definite by Theorem 6.3. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H . Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 6.3, τ must be zero, giving $\sigma = c_0\rho$.

Relation to Wolf: In [Wol12, Theorem 6.3, item 2], uniqueness is proved via the growth condition of [Wol12, Theorem 6.2]. The same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 6.3) in place of the growth condition. $\square$

Remark 6.7. ✓ Stmt The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 6.8 (Uniqueness of positive eigenvalue for irreducible CP maps). Lean ✓ Stmt *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. ✓ Proof Extract Kraus operators K for E . Irreducibility of E passes to the adjoint transfer map $E^\dagger(\cdot) = \sum_i K_i^\dagger(\cdot) K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 6.18) together with the irreducible positive-definiteness upgrade (Theorem 6.4) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \text{tr}(\tau X) = \text{tr}(\tau \cdot E(X)) = \text{tr}(E^\dagger(\tau) \cdot X) = t \cdot \text{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\text{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. **Relation to Wolf:** This is [Wol12, Theorem 6.3, item 3], establishing non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps. The proof follows Wolf’s dual-map trace argument. $\square$

6.3 Existence and the Perron–Frobenius theorem

Theorem 6.9 (Existence of PSD fixed point). Lean ✓ Stmt Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.

Proof. ✓ Proof Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 4.18). The Cesàro fixed-point theorem (Theorem 4.22) then provides the fixed point; injectivity is not needed for this step. $\square$

Theorem 6.10 (Quantum Perron–Frobenius). Lean ✓ Stmt Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.

Proof. ✓ Proof Combine Theorems 6.9, 6.3, and 6.6. $\square$

6.4 Right- and left-canonical gauges

Theorem 6.11 (Right-canonical (unital) gauge). Lean ✓ Stmt Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization used in subsequent chapters.

Proof. ✓ Proof Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_{\rho} (A^i)^\dagger (S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i \rho (A^i)^\dagger = \rho$ gives $S^{-1} \rho (S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. $\square$

Theorem 6.12 (Left-canonical (trace-preserving) gauge). Lean ✓ Stmt Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization used in subsequent chapters.

Proof. ✓ Proof Expand each summand: $(SA^iS^{-1})^\dagger (SA^iS^{-1}) = (S^\dagger)^{-1} (A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^i S^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1} \sigma S^{-1} = (S^\dagger)^{-1} S^\dagger S S^{-1} = \mathbb{1}$. $\square$

Remark 6.13. ✓ Stmt The right-canonical gauge of Theorem 6.11 and the left-canonical gauge of Theorem 6.12 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

6.5 Similarity preserves irreducibility

Lemma 6.14 (Similarity transform preserves irreducibility). [Lean](#) [✓ Stmt](#) *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) = c \cdot C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. [✓ Proof](#) Suppose Q is an invariant projection for E' : $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1}-Q)E'(QXQ)Q = 0$ for all X . Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E . Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Proposition 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. $\square$

Lemma 6.15 (Similarity is an involution). [Lean](#) [✓ Stmt](#) *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transforms by C and C^{-1} compose to the identity:*

$$X \mapsto C(C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}) C^\dagger = E(X).$$

Proof. [✓ Proof](#) Direct computation: the inner conjugation $C \cdot (C^{-1} X (C^\dagger)^{-1}) \cdot C^\dagger$ collapses to X using $CC^{-1} = \mathbb{1}$ and $(C^\dagger)^{-1}C^\dagger = \mathbb{1}$, and the outer wrapping by $C(\cdot)C^\dagger$ then cancels the inner $C^{-1}(\cdot)(C^\dagger)^{-1}$ on $E(X)$ for the same reason. $\square$

Lemma 6.16 (Irreducibility is invariant under similarity). [Lean](#) [✓ Stmt](#) *For any invertible $C \in M_D(\mathbb{C})$ and any linear map E on $M_D(\mathbb{C})$, the similarity transform $X \mapsto C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}$ is irreducible if and only if E is.*

Proof. [✓ Proof](#) The forward direction follows from Lemma 6.14 applied with the inverse C^{-1} (whose determinant is also nonzero): if the similarity transform of E by C is irreducible, then so is its similarity transform by C^{-1} , which equals E by Lemma 6.15. The reverse direction is Lemma 6.14 itself (with $c = 1$). $\square$

Theorem 6.17 (Scaling preserves irreducibility). [Lean](#) [✓ Stmt](#) *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Proposition 6.6].*

Proof. [✓ Proof](#) An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). $\square$

6.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV17a, Appendix A].

Theorem 6.18 (PSD eigenvector for positive maps). [Lean](#) [✓ Stmt](#) *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ . Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Theorem 6.5]; see also [EHK78]). The proof applies Brouwer’s fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\text{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 6.19 (Adjoint transfer map is nonzero). [Lean](#) [✓ Stmt](#) If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.

Proof. [✓ Proof](#) If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. $\square$

Theorem 6.20 (Positive-definite adjoint eigenvector for irreducible tensors). [Lean](#) [✓ Stmt](#) Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.

This combines the general PF existence ([Wol12, Theorem 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3, item 2]); the application to the adjoint transfer map follows [CPGSV17a, Appendix A].

Proof. [✓ Proof](#) By Theorem 6.19, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map. More precisely: if P is invariant for $\mathcal{E}_A$ (i.e. $(\mathbb{1} - P)K_i P = 0$ for all Kraus operators K_i), then taking adjoints gives $PK_i^\dagger(\mathbb{1} - P) = 0$, which means $\mathbb{1} - P$ is invariant for $\mathcal{E}_A^\dagger$. Hence $\mathcal{E}_A$ irreducible $\iff \mathcal{E}_A^\dagger$ irreducible, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ . Hence Theorem 6.18 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by Theorem 6.17, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 6.4) gives σ_0 positive definite. $\square$

Theorem 6.21 (TP-gauge existence for irreducible tensors). [Lean](#) [✓ Stmt](#) Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r , a positive definite matrix σ , and a gauge-equivalent tensor B such that

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the “spectral rescaling + TP gauge” step of [CPGSV17a, Appendix A]. The similarity is generally non-unitary.

Proof. [✓ Proof](#) By Theorem 6.20, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the TP gauge construction (Theorem 6.12), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. $\square$

6.7 Exponential positivity for irreducible CP maps

Theorem 6.22 (Exponential truncation is positive definite). [Lean](#) [✓ Stmt](#) Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then the finite exponential truncation

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is the finite-sum core of [Wol12, Theorem 6.2, item (3)].

Proof. ✓ Proof If a nonzero vector v annihilated this quadratic form, then every term $E^k(A)$ with $0 \leq k \leq D-1$ would annihilate v . The growth theorem for irreducible CP maps (that is, $(\mathbb{1} + E)^{D-1}(A) > 0$ for irreducible E and nonzero PSD A , as established in the proof of Theorem 6.5) then forces $(\mathbb{1} + E)^{D-1}(A)$ to annihilate v , contradicting its positive definiteness. $\square$

Theorem 6.23 (Exponential positivity for irreducible CP maps). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $A \geq 0$ be nonzero, and let $t > 0$. Then*

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Theorem 6.2, item (3)].

Proof. ✓ Proof By Theorem 6.22, the first D terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail. $\square$

6.8 Ergodicity of irreducible channels

Theorem 6.24 (Unique density-matrix fixed point for an irreducible channel). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$. Then there exists a unique density matrix σ such that $E(\sigma) = \sigma$. Moreover σ is positive definite. This is the qualitative form of [Wol12, Corollary 6.3].*

Proof. ✓ Proof Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1. $\square$

Theorem 6.25 (Cesàro convergence for irreducible channels). Lean ✓ Stmt *Let E be an irreducible quantum channel on $M_D(\mathbb{C})$ with $D > 0$, and let ρ be any density matrix. Then the Cesàro means*

$$\frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of E . This is [Wol12, Corollary 6.3].

Proof. ✓ Proof Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 6.24, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit. $\square$

6.9 Spectral radius at the Perron eigenvalue

Theorem 6.26 (Spectral radius is the Perron eigenvalue). Lean ✓ Stmt *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho > 0$ and $E(\rho) = r\rho$ with $r > 0$. Then the spectral radius of E is r . This is [Wol12, Theorem 6.3, item (4)].*

Proof. ✓ Proof Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by r^{-1} . The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving

map F satisfies $\rho(F) \leq 1$, since $\text{tr}(F^n(X)) = \text{tr}(X)$ for all n bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of E . $\square$

Theorem 6.27 (Real spectral-radius identity). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 6.26, the real-valued spectral radius of E is also equal to r .*

Proof. [✓ Proof](#) This is the real-valued reformulation of Theorem 6.26. $\square$

6.10 Spectral characterization of irreducibility

Definition 6.28 (Spectral properties). [Lean](#) [✓ Stmt](#) A completely positive map E on $M_D(\mathbb{C})$ has the *spectral properties* of [Wol12, Theorem 6.4] if there exist a positive real number r , a positive definite right eigenvector ρ , and a positive definite left eigenvector σ for the adjoint map such that $E(\rho) = r\rho$, $E^\dagger(\sigma) = r\sigma$, every positive semidefinite right eigenvector for eigenvalue r is a scalar multiple of ρ , and the spectral radius of E is equal to r .

Theorem 6.29 (Irreducibility gives the spectral properties). [Lean](#) [✓ Stmt](#) *Every nonzero irreducible completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.28.*

Proof. [✓ Proof](#) Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 6.26. $\square$

Theorem 6.30 (Spectral properties imply irreducibility). [Lean](#) [✓ Stmt](#) *If a completely positive map on $M_D(\mathbb{C})$ has the spectral properties of Definition 6.28, then it is irreducible.*

Proof. [✓ Proof](#) Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 6.28. $\square$

Theorem 6.31 (Irreducibility iff the spectral properties hold). [Lean](#) [✓ Stmt](#) *Let E be a nonzero completely positive map on $M_D(\mathbb{C})$. Then E is irreducible if and only if it has the spectral properties of Definition 6.28. This is [Wol12, Theorem 6.4].*

Proof. [✓ Proof](#) Combine Theorems 6.29 and 6.30. $\square$

Chapter 7

Spectral Gap and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary spectral-gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

7.1 Mixed transfer operator

Definition 7.1 (Mixed transfer operator). [Lean](#) [✓ Stmt](#) The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 7.2 (Rectangular mixed transfer operator). [Lean](#) [✓ Stmt](#) For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\text{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Remark 7.3 (Self-transfer). [Lean](#) [✓ Stmt](#) Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.

Theorem 7.4 (Iterated mixed transfer). [Lean](#) [✓ Stmt](#) For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. ✓ Proof Induction on N : the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. $\square$

Theorem 7.5 (Matrix trace of iterated mixed transfer at the identity). Lean ✓ Stmt For $N \geq 0$,

$$\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^\sigma (B^\sigma)^\dagger).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not be confused with the operator trace $\text{Tr}(F_{AB}^N)$; its relation to the MPV overlap is proved in the next section.

Proof. ✓ Proof Apply the matrix trace to each summand in Theorem 7.4 with $X = \mathbb{1}$: $\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_{\sigma} \text{tr}(A^\sigma (B^\sigma)^\dagger)$. $\square$

7.2 MPV overlap as transfer trace

Recall from Definition 2.46 that

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 7.6 (Trace expansion over matrix units). Lean ✓ Stmt For any linear endomorphism T on $M_D(\mathbb{C})$, the operator trace can be expanded as

$$\text{Tr}(T) = \sum_{p,q=0}^{D-1} (T(E_{pq}))_{pq},$$

where E_{pq} is the matrix with 1 in position (p, q) and 0 elsewhere.

Proof. ✓ Proof The operator trace $\text{Tr}(T)$ equals the sum of $\text{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p, q) , and $\text{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

Theorem 7.7 (Overlap equals transfer trace). Lean ✓ Stmt For tensors A, B of bond dimension $D \geq 1$,

$$O_{AB}(N) = \text{Tr}(F_{AB}^N).$$

Proof. ✓ Proof Expand the operator trace using Lemma 7.6 as a sum over matrix units E_{pq} . Apply Theorem 7.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p, q) -entry, and reassemble. The resulting double sum over (p, q) and σ equals $\sum_{\sigma} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. $\square$

Theorem 7.8 (Rectangular overlap as transfer trace). Lean ✓ Stmt For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,

$$O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N).$$

Proof. ✓ Proof Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Theorem 7.7. $\square$

Lemma 7.9 (Word trace as entrywise inner product). Lean ✓ Stmt For a word $\sigma \in \{0, \dots, d-1\}^N$,

$$\text{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \overline{(B^\sigma)_{jk}}.$$

Proof. ✓ Proof Expand the matrix trace and the (j, j) -entry of $A^\sigma (B^\sigma)^\dagger$. $\square$

7.3 Frobenius norm estimates

The spectral-gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write $\|\cdot\|_F$ for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm $\|\cdot\|_{\text{HS}}$ on finite-dimensional spaces. We use the squared version and an isometric embedding into Euclidean space.

Definition 7.10 (Frobenius norm squared). [Lean](#) [✓ Stmt](#) For a (possibly rectangular) matrix $X \in M_{m \times n}(\mathbb{C})$, the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

Lemma 7.11 (Trace formula for Frobenius norm). [Lean](#) [✓ Stmt](#) $\|X\|_F^2 = \text{Re tr}(X^\dagger X)$.

Proof. [✓ Proof](#) Expand the matrix product and trace entry by entry. □

Definition 7.12 (Euclidean-space embedding). [Lean](#) [✓ Stmt](#) The map $\text{vec} : M_{m \times n}(\mathbb{C}) \rightarrow \mathbb{C}^{mn}$ that flattens matrix entries is an isometric embedding with respect to the Frobenius norm: $\|\text{vec}(X)\|^2 = \|X\|_F^2$.

Lemma 7.13 (Norm of embedded matrix). [Lean](#) [✓ Stmt](#) $\|\text{vec}(X)\|^2 = \|X\|_F^2$.

Proof. [✓ Proof](#) Expand both sides as sums over entries and use $|z|^2 = \bar{z}z$. □

7.4 Eigenvalue bound and spectral gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization (cf. [Wol12, Proposition 6.1] for the general operator-norm version via Russo–Dye). For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument ([Wol12, Theorem 5.3]) to a block embedding of a mixed-transfer eigenvector.

Theorem 7.14 (Eigenvalue bound). [Lean](#) [✓ Stmt](#) Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \leq 1$.

Proof. [✓ Proof](#) Expand $F_{AB}^n(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$ for all n , where C_D depends only on D . If $F_{AB}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n \|X\|_{\text{HS}} = \|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$ for every n . Hence $|\mu| \leq 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. □

Theorem 7.15 (Spectral radius bound). [Lean](#) [✓ Stmt](#) $\rho(F_{AB}) \leq 1$ for normalized tensors.

Proof. [✓ Proof](#) The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 7.14. □

Theorem 7.16 (Spectral radius ≥ 1 implies gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let A, B be injective normalized MPS tensors. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.

Proof. ✓ Proof Since $\rho(F_{AB}) \leq 1$ (Theorem 7.15) and $\rho(F_{AB}) \geq 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \neq 0$: $F_{AB}(X) = \mu X$. The proof proceeds in five steps.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A, ρ_B of the transfer maps $\mathcal{E}_A, \mathcal{E}_B$ (Theorem 6.10) and gauge each tensor separately to a unital Kraus family (Theorem 6.11): set $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B . Let $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$ denote the correspondingly gauged eigenvector.

Step 2 (Block Kraus family and off-diagonal embedding). Form the block Kraus operators $C^i = \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}$. These constitute a unital Kraus family on $M_{2D}(\mathbb{C})$. The off-diagonal matrix $Y = \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}$ satisfies $E_C(Y) = \mu Y$.

Step 3 (KS equality and Kraus commutation). Because $|\mu| = 1$ and E_C is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 5.24) gives $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$. By the Kraus commutation relation (Theorem 5.23), Y commutes with every block Kraus operator: $C^i Y = Y C^i$ for all i .

Step 4 (Intertwining identity and invertibility). Expanding the commutation relation block by block gives $A'^i X' = X' B'^i$ for every i . Since $\{A'^i\}$ span $M_D(\mathbb{C})$ (injectivity of A), X' is a nonzero intertwiner from $M_D(\mathbb{C})$ to $M_D(\mathbb{C})$. The left multiplicative identity (Theorem 5.25) gives $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$, so $X'^\dagger X'$ is a positive semi-definite fixed point of $\mathcal{E}_{B'}$, positive definite by injectivity. Hence X' is invertible.

Step 5 (Skolem–Noether conclusion). The map $M \mapsto X' M (X')^{-1}$ is a $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ sending B'^i to A'^i . By Skolem–Noether (Theorem 3.9) this is an inner automorphism. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for all i , establishing gauge-phase equivalence (Definition 2.21).

The crucial point is that one gauges the individual transfer maps to unital form; one does not require the mixed transfer map itself to be simultaneously unital and trace-preserving. $\square$

Theorem 7.17 (Gauged peripheral eigenvector yields Kraus intertwining). Lean ✓ Stmt *Let A and B be normalized tensors, and let $X \neq 0$ satisfy $F_{AB}(X) = \mu X$ with $|\mu| = 1$. Choose positive-definite fixed points ρ_A, ρ_B of the transfer maps and gauge to unital families $A'^i = S_A^{-1} A^i S_A$ and $B'^i = S_B^{-1} B^i S_B$ with $S_A S_A^\dagger = \rho_A$ and $S_B S_B^\dagger = \rho_B$. Then the transported matrix $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and for every i one has*

$$X' (B'^i)^\dagger = \mu (A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i.$$

Proof. ✓ Proof Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure. $\square$

Theorem 7.18 (Intertwining yields gauge-phase equivalence). Lean ✓ Stmt *Let A and B be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families A' and B' , an invertible matrix X' , and a scalar μ with $|\mu| = 1$ such that, for every physical index i ,*

$$A'^i X' = \mu X' B'^i$$

Then A and B are gauge-phase equivalent.

Proof. ✓ Proof Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating A to a scalar multiple of B . $\square$

Remark 7.19 (Two gauge conventions). ✓ Stmt The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a unital Kraus family ($\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$). A different convention,

$\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$, makes the family *trace-preserving* ($\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the spectral-gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 9).

Theorem 7.20 (Strict spectral gap). [Lean](#) [Stmnt](#) *Let A, B be injective normalized MPS tensors that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. [Proof](#) $\rho(F_{AB}) \leq 1$ by Theorem 7.15. If $\rho(F_{AB}) = 1$, Theorem 7.16 gives gauge-phase equivalence, contradicting the hypothesis. $\square$

7.5 Spectral gap — rectangular case

Theorem 7.21 (Rectangular intertwining forces equal bond dimension). [Lean](#) [Stmnt](#) *Let A and B be injective tensors of bond dimensions D_1 and D_2 . If there exist $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that, for every physical index i ,*

$$X(B^i)^\dagger = \mu (A^i)^\dagger X, \quad A^i X = \mu X B^i$$

then $D_1 = D_2$.

Proof. [Proof](#) The intertwining relation shows that $\ker(X)$ is invariant under all generators of B . Since B is injective, its generators span the full matrix algebra, so $\ker(X)$ is trivial and $D_2 \leq D_1$. Applying the same argument to $X^\dagger$ gives $D_1 \leq D_2$. $\square$

Theorem 7.22 (Rectangular spectral gap). [Lean](#) [Stmnt](#) *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2 , with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\text{rect}}) < 1$.*

Proof. [Proof](#) The same word-expansion argument as in Theorem 7.14 gives $|\mu| \leq 1$ for every eigenvalue of F_{AB}^{rect} . If $\rho(F_{AB}^{\text{rect}}) = 1$, choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A and B separately to unital Kraus families and apply the block-embedding Kadison–Schwarz argument as in the square case. The resulting intertwining relations satisfy the hypotheses of Theorem 7.21, so $D_1 = D_2$, contradicting the hypothesis. $\square$

Theorem 7.23 (Rectangular overlap decay). [Lean](#) [Stmnt](#) *If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [Proof](#) The spectral gap $\rho(F_{AB}^{\text{rect}}) < 1$ (Theorem 7.22) implies $(F_{AB}^{\text{rect}})^N \rightarrow 0$, so the operator trace converges to 0. By Theorem 7.8, $O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N) \rightarrow 0$. $\square$

7.6 MPV overlap decay

Theorem 7.24 (Transfer powers converge to zero). [Lean](#) [Stmnt](#) *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. [Proof](#) The spectral gap $\rho(F_{AB}) < 1$ (Theorem 7.20) implies $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$ by the Gelfand formula, so $F_{AB}^n(X) \rightarrow 0$ for every X . $\square$

Theorem 7.25 (Overlap decay). [Lean](#) [Stmnt](#) *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [Proof](#) By Theorem 7.7, $O_{AB}(N) = \text{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 7.6): $\text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$. By Theorem 7.24, each $F_{AB}^N(E_{pq}) \rightarrow 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \rightarrow 0$. $\square$

7.7 Block separation

These asymptotic facts are the ingredients used later in the canonical-form separation arguments.

Remark 7.26 (Cross-correlation decay). [Lean](#) [✓ Stmt](#) If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ as $N \rightarrow \infty$. This is the scalar trace consequence of the operator convergence $F_{AB}^N(X) \rightarrow 0$ from Theorem 7.24.

Remark 7.27 (Self-correlation persists). [Lean](#) [✓ Stmt](#) If ρ is a fixed point of $\mathcal{E}_A$, then $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 6.10.

7.8 Spectral gap under irreducible-TP hypotheses

The preceding spectral-gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with trace-preserving normalization ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$), following Cirac et al. [CPGSV17b] and the irreducibility theory of [Wol12, Section 6.2]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

Theorem 7.28 (Gauge-equivalence from irreducible-TP spectral radius). [Lean](#) [✓ Stmt](#) Let A, B be irreducible trace-preserving normalized MPS tensors of bond dimension D . If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.

Proof. [✓ Proof](#) Choose a modulus-one eigenvector as in Theorem 7.16. The proof follows the five-step structure of Theorem 7.16, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 6.10) and the right-canonical gauge (Theorem 6.11). The change is in Step 4: the Kadison–Schwarz equality gives $X'^\dagger X'$ as a nonzero positive semidefinite fixed point of $\mathcal{E}_{B'}$. Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 6.4 upgrades it to positive definite, giving invertibility of X' . Formally, Theorem 7.17 combines the separately gauged eigenvector into the required intertwining relations. Once X' is invertible, Theorem 7.18 upgrades the intertwining to gauge-phase equivalence. $\square$

Theorem 7.29 (Strict spectral gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) Let A, B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.

Proof. [✓ Proof](#) Contrapositive of Theorem 7.28, combined with $\rho(F_{AB}) \leq 1$ (Theorem 7.15). $\square$

Theorem 7.30 (Transfer power decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 7.29, $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) By the Gelfand formula, $\rho(F_{AB}) < 1$ implies $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$. $\square$

Theorem 7.31 (Overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) Let A, B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.

Proof. [✓ Proof](#) Combine the spectral gap (Theorem 7.29) with the overlap–trace identity (Theorem 7.7). $\square$

Theorem 7.32 (Rectangular spectral gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) Let A (bond dimension D_1) and B (bond dimension D_2) be irreducible trace-preserving normalized tensors with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\text{rect}}) < 1$.

Proof. [✓ Proof](#) Assume for contradiction that $\rho = 1$. Choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C})$. The Cauchy–Schwarz rigidity argument produces an intertwiner X' with $A'^i X' = X' B'^i$ for all i . Since $\ker(X')$ is invariant under all B'^i , irreducibility forces $\ker(X') = \{0\}$; similarly for $X'^\dagger$. This gives $D_1 = D_2$, contradicting the hypothesis. $\square$

Theorem 7.33 (Rectangular overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.

Proof. [✓ Proof](#) Combine Theorem 7.32 with Theorem 7.8. $\square$

7.9 Conditional expectation from a faithful fixed point

This section follows [Wol12, Section 6.4]; the conditional expectation is built from the fixed-point algebra of [Wol12, Theorem 6.14] via [Wol12, (1.40)]. The scalar conditional expectation E_σ is the special case where the fixed-point $*$ -subalgebra is the scalar algebra $\mathbb{C} \cdot \mathbb{1}$ (the primitive-channel case). The general irreducible case with nontrivial period requires the Wedderburn block decomposition of [Wol12, Theorem 6.14].

Definition 7.34 (Conditional expectation). [Lean](#) [✓ Stmt](#) Given a $*$ -subalgebra $S \subseteq M_D(\mathbb{C})$, a linear map $P : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a conditional expectation onto S if it is idempotent, unital, has range contained in S , and satisfies $P(X) = X$ for all $X \in S$.

Definition 7.35 (Scalar conditional expectation). [Lean](#) [✓ Stmt](#) For $\sigma \geq 0$ with $\text{tr}(\sigma) \neq 0$, define

$$E_\sigma(X) := \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}.$$

Lemma 7.36 (Evaluation formula). [Lean](#) [✓ Stmt](#) $E_\sigma(X) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}$.

Theorem 7.37 (Unitality). [Lean](#) [✓ Stmt](#) $E_\sigma(\mathbb{1}) = \mathbb{1}$.

Theorem 7.38 (Idempotence). [Lean](#) [✓ Stmt](#) $E_\sigma(E_\sigma(X)) = E_\sigma(X)$ for all X .

Lemma 7.39 (Range is scalar). [Lean](#) [✓ Stmt](#) For every X , there exists $c \in \mathbb{C}$ with $E_\sigma(X) = c \mathbb{1}$.

Lemma 7.40 (Fixes scalar operators). [Lean](#) [✓ Stmt](#) For $c \in \mathbb{C}$, $E_\sigma(c \mathbb{1}) = c \mathbb{1}$.

Theorem 7.41 (Absorbs the adjoint map). [Lean](#) [✓ Stmt](#) If $T(\sigma) = \sigma$, then $E_\sigma \circ T^* = E_\sigma$.

Theorem 7.42 (Adjoint map absorbs scalar projection). [Lean](#) [✓ Stmt](#) If T is trace-preserving, then $T^* \circ E_\sigma = E_\sigma$.

Lemma 7.43 (Image lies in adjoint fixed points). [Lean](#) [✓ Stmt](#) For all X , $E_\sigma(X)$ is fixed by T^* .

Theorem 7.44 (Scalar map is a conditional expectation). [Lean](#) [✓ Stmt](#) Let K be trace-preserving, let $\rho > 0$ be a positive-definite fixed point of the associated channel $T(\rho) = \sum_i K_i \rho K_i^\dagger$ (so $T(\rho) = \rho$), and assume every adjoint fixed point of T^* is a scalar multiple of $\mathbb{1}$. Then E_ρ is a conditional expectation onto the adjoint fixed-point $*$ -subalgebra.

Theorem 7.45 (Complete positivity of the scalar conditional expectation). [Lean](#) [✓ Stmt](#) For any positive semidefinite σ , the scalar conditional expectation E_σ is a completely positive map.

Proof. [✓ Proof](#) Write $S = \sqrt{\sigma}$ and define the Kraus family $K_{j,i} = |j\rangle\langle i| S$ indexed by $(j, i) \in \{1, \dots, D\}^2$. Using $|j\rangle\langle i| M |i\rangle\langle j| = M_{ii} |j\rangle\langle j|$ and summing first over i and then over j ,

$$\sum_{j,i} K_{j,i} X K_{j,i}^\dagger = \sum_j |j\rangle\langle j| \operatorname{tr}(SXS) = \operatorname{tr}(\sigma X) \mathbb{1},$$

where we used cyclicity of the trace and $S^2 = \sigma$. Hence $E_\sigma = (\operatorname{tr} \sigma)^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$ is the Kraus map scaled by the nonnegative real number $(\operatorname{tr} \sigma)^{-1}$, and therefore completely positive. $\square$

Theorem 7.46 (σ -trace preservation of the scalar conditional expectation). [Lean](#) [✓ Stmt](#) If $\operatorname{tr} \sigma \neq 0$, then for every X

$$\operatorname{tr}(\sigma E_\sigma(X)) = \operatorname{tr}(\sigma X).$$

Proof. [✓ Proof](#) Direct computation: $\operatorname{tr}(\sigma \cdot \frac{\operatorname{tr}(\sigma X)}{\operatorname{tr} \sigma} \mathbb{1}) = \frac{\operatorname{tr}(\sigma X)}{\operatorname{tr} \sigma} \operatorname{tr} \sigma = \operatorname{tr}(\sigma X)$. $\square$

7.10 Stationary support

This section develops the support-projection theory behind stationary states, following [Wol12, Section 6.4, Propositions 6.10–6.11].

Lemma 7.47 (Lower-triangular Kraus condition implies corner invariance). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor and let P be an orthogonal projection. If, for every physical index i ,

$$(\mathbb{1} - P)A^i P = 0$$

then

$$P \mathcal{E}_A(PXP) P = \mathcal{E}_A(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Expand the transfer map and use $(\mathbb{1} - P)A^i P = 0$ to reduce each Kraus term to the P -corner. $\square$

Theorem 7.48 (Support projection invariance). [Lean](#) [✓ Stmt](#) Let E be a quantum channel and $\rho \geq 0$ a positive semidefinite fixed point ($E(\rho) = \rho$). Let P denote the support projection of ρ . Then

$$P E(PXP) P = E(PXP)$$

for all $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Write E in Kraus form $E(X) = \sum_i K_i X K_i^\dagger$. From $E(\rho) = \rho$ and $\rho \geq 0$ one deduces $(\mathbb{1} - P)K_i P = 0$ for all i . Apply Lemma 7.47. $\square$

Definition 7.49 (Stationary state). [Lean](#) [✓ Stmt](#) For an irreducible channel E on $M_D(\mathbb{C})$ (with $D \geq 1$), the *stationary state* is the unique density-matrix fixed point ρ_∞ satisfying $E(\rho_\infty) = \rho_\infty$, $\rho_\infty > 0$, and $\operatorname{tr}(\rho_\infty) = 1$. Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 6.10).

Definition 7.50 (Stationary support). [Lean](#) [✓ Stmt](#) The *stationary support* of an irreducible channel E is the support projection of the stationary state (Definition 7.49).

Theorem 7.51 (Stationary support is full). [Lean](#) [✓ Stmt](#) For an irreducible channel E , the stationary support equals the identity: $\text{supp}(\rho_\infty) = \mathbb{1}$.

Proof. [✓ Proof](#) The support projection P of ρ_∞ is an orthogonal projection satisfying $PE(PXP)P = E(PXP)$ for all X (Theorem 7.48). By irreducibility, $P = 0$ or $P = \mathbb{1}$. Since $\rho_\infty \neq 0$, we have $P \neq 0$, so $P = \mathbb{1}$. $\square$

Remark 7.52 (Stationary-support formulations). The non-trivial formulations needed here are the following results from Wolf:

- the converse direction of [Wol12, Proposition 6.10]: if the support projection of every positive semidefinite fixed point is full, then the channel is irreducible;
- [Wol12, Proposition 6.11]: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are not used elsewhere in this chapter.

7.10.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra, $*$ -structure, and compression of a PSD fixed point onto its support sector needed for [Wol12, Corollary 6.6]: the corner carries a $\mathbb{C}$ -algebra and $*$ -structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

Lemma 7.53 (Support projection annihilates the kernel). [Lean](#) [✓ Stmt](#) Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \text{supp}(\rho)$. For every $v \in \mathbb{C}^D$, if $\rho v = 0$, then $Pv = 0$.

Proof. [✓ Proof](#) Diagonalize $\rho = U \text{diag}(\lambda) U^\dagger$ with $\lambda_j \geq 0$. Writing $w := U^\dagger v$, the hypothesis $\rho v = 0$ gives $\lambda_j w_j = 0$ for every j ; hence $w_j = 0$ whenever $\lambda_j > 0$. The support projection acts as $\text{diag}(\chi_{\lambda_j > 0})$ in the eigenbasis, so $Pv = U \text{diag}(\chi_{\lambda_j > 0}) w = 0$. $\square$

Definition 7.54 (Corner submodule / idempotent-corner identification). [Lean](#) [✓ Stmt](#) For an idempotent $P \in M_D(\mathbb{C})$, the subspace $\{X \mid PXP = X\}$ and the corner $\{PXP \mid X \in M_D(\mathbb{C})\}$ are identified as $\mathbb{C}$ -linear spaces by the identity on underlying matrices.

Remark 7.55 ($\mathbb{C}$ -algebra and $*$ -structure on the corner). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The corner algebra associated to an idempotent already carries a ring structure with unit P . In the matrix case it also has $\mathbb{C}$ -module and $\mathbb{C}$ -algebra structures and, under the additional assumption $P^\dagger = P$, star, star ring, and star module structures over $\mathbb{C}$.

Theorem 7.56 (Faithful compression onto the support). [Lean](#) [✓ Stmt](#) Let $\rho \succeq 0$ on $\mathbb{C}^D$ with support projection $P = \text{supp}(\rho)$, and let $V : \mathbb{C}^D \rightarrow \mathbb{C}^k$ be a compression isometry satisfying $VV^\dagger = \mathbb{1}_k$ and $V^\dagger V = P$. Then the compression

$$V \rho V^\dagger \in M_k(\mathbb{C})$$

is positive definite.

Proof. [✓ Proof](#) Hermiticity follows from Hermiticity of ρ and the product-star identity. For strict positivity, let $w \neq 0$, set $u := V^\dagger w$, and note that $VV^\dagger = \mathbb{1}_k$ forces $u \neq 0$, while $V^\dagger V = P$ forces $Pu = u$. The quadratic form computes as $w^\dagger (V \rho V^\dagger) w = u^\dagger \rho u$, which is nonnegative by positive semidefiniteness. If it vanishes, then $\rho u = 0$, and by Lemma 7.53 also $Pu = 0$, contradicting $Pu = u \neq 0$. $\square$

7.11 Wedderburn decomposition of the fixed-point algebra

This section proves the structure theorem for the fixed-point algebra of a trace-preserving Kraus map, following [Wol12, Theorems 6.12–6.14]. The proof uses the Jacobson radical characterization of semisimplicity together with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.

Theorem 7.57 (Fixed-point algebra is semisimple). [Lean](#) [✓ Stmt](#) *The adjoint-fixed-point $*$ -subalgebra of a trace-preserving Kraus map on $M_D(\mathbb{C})$ is a semisimple ring.*

Proof. [✓ Proof](#) The subalgebra is finite-dimensional (as a subalgebra of $M_D(\mathbb{C})$), hence Artinian. By [Wol12, Theorem 6.14], it suffices to show the Jacobson radical J vanishes. If $x \in J$, then $x^*x \in J$ (left-ideal closure). The radical of an Artinian ring is nilpotent, so x^*x is nilpotent. A self-adjoint nilpotent matrix satisfies $\|x^*x\|^{2^k} = \|(x^*x)^{2^k}\| = 0$, hence $x^*x = 0$, and the C^* -identity gives $x = 0$. $\square$

Theorem 7.58 (Abstract Wedderburn–Artin decomposition). [Lean](#) [✓ Stmt](#) *There exist $n \in \mathbb{N}$ and dimensions $d_1, \dots, d_n \geq 1$ such that the adjoint-fixed-point $*$ -subalgebra is $\mathbb{C}$ -algebra isomorphic to $\prod_{k=1}^n M_{d_k}(\mathbb{C})$.*

Proof. [✓ Proof](#) Combine semisimplicity (Theorem 7.57) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field. $\square$

Theorem 7.59 (Fixed-point algebra decomposition with multiplicities). [Lean](#) [✓ Stmt](#) *There exist integers n , block sizes $d_1, \dots, d_n \geq 1$, multiplicities $m_1, \dots, m_n \geq 1$, and a $\mathbb{C}$ -algebra isomorphism from the adjoint-fixed-point $*$ -subalgebra to*

$$\prod_{k=1}^n M_{d_k}(\mathbb{C})$$

such that

$$\sum_{k=1}^n d_k m_k \leq D.$$

Proof. [✓ Proof](#) Apply the abstract Wedderburn decomposition (Theorem 7.58). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra $M_D(\mathbb{C})$, this realization also yields multiplicities m_k and the bound $\sum_k d_k m_k \leq D$. $\square$

Remark 7.60 (Block form of the fixed-point algebra). Wolf’s block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with

$$0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form $M_{d_k}(\mathbb{C}) \otimes \rho_k$. The theorem above gives the product decomposition, multiplicities, and dimension bound needed from this block form.

Theorem 7.61 (Weighted fixed points form a $*$ -subalgebra). [Lean](#) [✓ Stmt](#) *Let $T(X) = \sum_i K_i X K_i^\dagger$ be trace-preserving, and let $\rho > 0$ satisfy $T(\rho) = \rho$. If*

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

then $\rho^{-1/2} F_T \rho^{-1/2}$ is a $$ -subalgebra of $M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) Set $B_i = \rho^{-1/2} K_i \rho^{1/2}$. The equation $T(\rho) = \rho$ implies that the family $\{B_i\}$ is unital, and trace preservation of T implies that ρ is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 4.141 to B . The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with $\rho^{-1/2} F_T \rho^{-1/2}$. □

7.12 Further irreducibility and primitivity equivalences

This section collects several criteria and equivalences from [Wol12, Theorems 6.2, 6.7, 6.8] that are used later.

Theorem 7.62 (Exponential positivity condition). [Lean](#) [Lean](#) [✓ Stmt](#) Let E be an irreducible completely positive map, $A \geq 0$ with $A \neq 0$, and $t > 0$. Then $\exp(tE)(A)$ is positive definite. Conversely, if $\exp(tE)(A)$ is positive definite for every such t, A , then E is irreducible (full equivalence [Lean](#)).

Theorem 7.63 (Kraus-span equivalence). [Lean](#) [✓ Stmt](#) Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.

Theorem 7.64 (Combined primitivity equivalence). [Lean](#) [✓ Stmt](#) Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.

Remark 7.65 (Pairwise primitivity equivalences). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) These are the pairwise equivalences in [Wol12, Theorem 6.8].

Theorem 7.66 (Scalar fixed-point algebra case). [Lean](#) [✓ Stmt](#) In the scalar fixed-point algebra setting, the weighted map $X \mapsto \frac{\text{tr}(\rho X)}{\text{tr}(\rho)} \mathbb{1}$ is a conditional expectation onto the adjoint fixed-point $*$ -subalgebra.

7.13 Multi-cycle block-permutation structure

This section develops the block-permutation structure underlying the cycle decomposition of [Wol12, Theorem 6.16]. The multi-cycle decomposition refines the single-cycle case to handle arbitrary peripheral periods.

Definition 7.67 (Block-permutation structure). [Lean](#) [✓ Stmt](#) Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$. A *block-permutation structure* for T consists of a finite index set ι , a family of orthogonal projections $P_k \in M_D(\mathbb{C})$ for $k \in \iota$, and a permutation $\sigma \in \text{Sym}(\iota)$ (not necessarily a single cycle — in general σ may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k) T(X), \quad T(X P_k) = T(X) T(P_k)$$

for every $k \in \iota$ and $X \in M_D(\mathbb{C})$.

Definition 7.68 (Multi-cycle block-permutation structure). [Lean](#) [✓ Stmt](#) A *multi-cycle decomposition* of $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ consists of a finite cycle index ι , a per-cycle period $m : \iota \rightarrow \mathbb{N}_{>0}$,

a projection family $P_{c,k} \in M_D(\mathbb{C})$ for $c \in \iota$ and $k \in \{0, \dots, m(c)-1\}$ consisting of orthogonal projections, together with the per-cycle cyclic action

$$T(P_{c,k+1}) = P_{c,k} \quad (k \in \{0, \dots, m(c)-1\}, c \in \iota)$$

and the multiplicative-domain factorisations $T(P_{c,k} X) = T(P_{c,k}) T(X)$ and $T(X P_{c,k}) = T(X) T(P_{c,k})$.

Theorem 7.69 (Per-cycle corner preservation). [Lean](#) [✓ Stmt](#) *For every cycle $c \in \iota$ and index $k \in \{0, \dots, m(c)-1\}$, the iterate $T^{m(c)}$ preserves the corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$.*

Proof. [✓ Proof](#) By induction on n , the n -th iterate T^n sends $P_{c,k+n} X P_{c,k+n}$ to $P_{c,k} T^n(X) P_{c,k}$, using the multiplicative-domain factorisations and the cyclic action $T(P_{c,j+1}) = P_{c,j}$ at each induction step. Specialising to $n = m(c)$ and using that the indices $k+m(c) = k$ in $\{0, \dots, m(c)-1\}$ yields corner preservation of $T^{m(c)}$. $\square$

Theorem 7.70 (Common-period corner preservation). [Lean](#) [✓ Stmt](#) *If $N \in \mathbb{N}$ is divisible by every per-cycle period $m(c)$, then T^N preserves every corner $P_{c,k} M_D(\mathbb{C}) P_{c,k}$ of the decomposition.*

Proof. [✓ Proof](#) Write $N = m(c)q$. Corner preservation is stable under taking powers, so $T^N = (T^{m(c)})^q$ preserves each corner by Theorem 7.69. $\square$

Definition 7.71 (Flattening to a single-index block-permutation structure). [Lean](#) [✓ Stmt](#) A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index $\Sigma_{c \in \iota} \{0, \dots, m(c)-1\}$: the permutation is the sigma-product of per-cycle cyclic shifts $k \mapsto k+1$ (whose cycle decomposition has one cycle per $c \in \iota$), the projections are $(c, k) \mapsto P_{c,k}$, and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.

7.14 Peripheral eigenvalue group structure

This section proves the peripheral spectrum structure of [Wol12, Theorem 6.6] for irreducible Schwarz maps: the peripheral eigenvalues form a finite cyclic group. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 7.75.

Lemma 7.72 (Peripheral eigenvectors are invertible). [Lean](#) [✓ Stmt](#) *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If $X \neq 0$ satisfies $E(X) = \mu X$ with $|\mu| = 1$, then X is invertible.*

Proof. [✓ Proof](#) The KS equality gives $E(X^\dagger X) = X^\dagger X$, so $X^\dagger X$ is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so $\det(X) \neq 0$. $\square$

Lemma 7.73 (Peripheral eigenvalues: product closure). [Lean](#) [✓ Stmt](#) *Let E be an irreducible unital Kraus map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. If μ, ν are peripheral eigenvalues of E , then so is $\mu\nu$.*

Proof. [✓ Proof](#) Take eigenvectors X, Y for μ, ν . The KS equality at X puts X in the multiplicative domain, so $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$. Since X, Y are units (Lemma 7.72), $YX \neq 0$ and $|\mu\nu| = 1$. $\square$

Theorem 7.74 (Peripheral eigenvalues: cyclic structure). [Lean](#) [✓ Stmt](#) *Let E be an irreducible unital Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point (trace-preservation is not required). Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof The peripheral eigenvalues form a finite subset of $\mathbb{C}$, contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of $\mathbb{C}^\times$. Any finite subgroup of $\mathbb{C}^\times$ is cyclic. The generator γ has order $m = |\text{peripheral spectrum}|$. $\square$

Theorem 7.75 (Peripheral eigenvalues form a cyclic group). Lean ✓ Stmt *Let E be an irreducible unital trace-preserving Schwarz map on $M_D(\mathbb{C})$ with a positive-definite adjoint fixed point. Then there exist $m \geq 1$ and a primitive m -th root of unity γ such that $m \mid D$ and the peripheral eigenvalues of E are exactly $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$.*

Proof. ✓ Proof Apply Theorem 7.74 for the cyclic structure. Divisibility $m \mid D$ follows from the cyclic decomposition: m mutually orthogonal projections summing to $\mathbb{1}$ each have equal trace (by trace preservation), so $m \cdot \text{tr}(P_0) = D$ with $\text{tr}(P_0) \in \mathbb{N}$. $\square$

Theorem 7.76 (Period divides bond dimension). Lean ✓ Stmt *Let E be as in Theorem 7.75, and let γ be a primitive m -th root of unity generating the peripheral eigenvalues. Then $m \mid D$.*

Proof. ✓ Proof Follows from the cyclic projection decomposition and trace preservation, as in the final step of Theorem 7.75. $\square$

7.14.1 Cyclic decomposition of irreducible Schwarz maps

Definition 7.77 (Corner preservation). Lean ✓ Stmt For an orthogonal projection P and linear map E , the corner $PM_D(\mathbb{C})P$ is preserved if $E(PXP) = PE(PXP)P$ for all X .

Definition 7.78 (Corner subspace). Lean ✓ Stmt The corner subspace associated with P is $\{PXP : X \in M_D(\mathbb{C})\}$.

Definition 7.79 (Restricted corner map). Lean ✓ Stmt The restriction of E to an invariant corner $PM_D(\mathbb{C})P$.

Definition 7.80 (Irreducible restriction on a corner). Lean ✓ Stmt Irreducibility of the corner-restricted map.

Definition 7.81 (Rank of an orthogonal projection). Lean ✓ Stmt The rank of an orthogonal projection $P \in M_D(\mathbb{C})$, i.e. the dimension of the corner algebra $PM_D(\mathbb{C})P$.

Theorem 7.82 (Compression isometry for an orthogonal projection). Lean ✓ Stmt *Let $P \in M_D(\mathbb{C})$ be an orthogonal projection of rank n . The corner algebra $PM_D(\mathbb{C})P$ is linearly isomorphic to the full matrix algebra $M_n(\mathbb{C})$. The isomorphism is built from the spectral diagonalisation of P by conjugating the top-left $n \times n$ block by the unitary diagonalising P .*

Theorem 7.83 (Rank equals trace for an orthogonal projection). Lean ✓ Stmt *The rank of an orthogonal projection P equals its trace: $n = \text{tr } P$.*

Theorem 7.84 (Peripheral unitary eigenvector). Lean ✓ Stmt *For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.*

Theorem 7.85 (Powers on a peripheral-unitary orbit). Lean ✓ Stmt *If $E(U) = \mu U$ with $|\mu| = 1$ and U unitary, then $E(U^n) = \mu^n U^n$ for all n .*

Theorem 7.86 (Normalized peripheral unitary). Lean ✓ Stmt *Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.*

Theorem 7.87 (Cyclic projections from a peripheral unitary). Lean ✓ Stmt *A primitive m -th peripheral root yields m orthogonal projections summing to $\mathbb{1}$ and cyclically permuted by the channel.*

Theorem 7.88 (Cyclic decomposition for irreducible Schwarz maps). [Lean](#) [✓ Stmt](#) *Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.*

Theorem 7.89 (Power maps preserve each cyclic sector). [Lean](#) [✓ Stmt](#) *Appropriate powers of the channel preserve each sector corner.*

Theorem 7.90 (Irreducibility of restricted sector dynamics). [Lean](#) [✓ Stmt](#) *Each sector restriction is irreducible.*

Theorem 7.91 (Primitivity of restricted sector dynamics). [Lean](#) [✓ Stmt](#) *Each sector restriction is primitive.*

Theorem 7.92 (Corner invariance at the period). [Lean](#) [✓ Stmt](#) *The channel raised to the period preserves every cyclic corner.*

7.14.2 Group structure of the peripheral spectrum

Theorem 7.93 (Peripheral product closure). [Lean](#) [✓ Stmt](#) *The product of two peripheral eigenvalues is peripheral.*

Theorem 7.94 (Peripheral inverse closure). [Lean](#) [✓ Stmt](#) *The inverse of a peripheral eigenvalue is peripheral.*

Theorem 7.95 (Peripheral cyclic group with period divisibility). [Lean](#) [✓ Stmt](#) *The peripheral eigenvalues form a finite cyclic group whose order divides the dimension.*

Theorem 7.96 (Peripheral eigenvalue multiplicity one). [Lean](#) [✓ Stmt](#) *Each peripheral eigenvalue has algebraic multiplicity one.*

Proof. [✓ Proof](#) Given a peripheral unitary eigenvector U for γ , the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X . By the scalar fixed-point lemma for irreducible unital maps, $XU^\dagger = c \cdot I$, hence $X = c \cdot U$ and the eigenspace is one-dimensional. $\square$

7.15 Primitive overlap convergence (spectral-gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument, corresponding to the primitive-case specialization of [Wol12, Theorem 6.7] where T_ϕ is a rank-one projection. For primitive normalized transfer maps, Chapter 4 supplies this spectral-gap input under explicit hypotheses.

Theorem 7.97 (Complementary spectral gap implies trace convergence to 1). [Lean](#) [✓ Stmt](#) *Let E be trace-preserving with fixed point ρ ($\text{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 4.23). If $\rho_{\text{spec}}(E - P) < 1$, then $\text{Tr}(E^n) \rightarrow 1$ as $n \rightarrow \infty$.*

Proof. [✓ Proof](#) By the power decomposition (Theorem 4.24), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\text{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \rightarrow 0$, so $\text{Tr}(E^n) \rightarrow \text{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\rho$, we have $\text{Tr}(P) = \frac{\text{tr}(\rho)}{\text{tr}(\rho)} = 1$. $\square$

Theorem 7.98 (Self-overlap convergence from a complementary spectral gap). [Lean](#) [✓ Stmt](#) *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \rightarrow 1$ as $N \rightarrow \infty$.*

Proof. ✓ Proof By Theorem 7.7 (with $A = B$), $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$. By Theorem 7.97, $\text{Tr}(\mathcal{E}_A^N) \rightarrow 1$. □

Lemma 7.99 (Norm of self-overlap convergence). Lean ✓ Stmt *If A is an MPS tensor and $O_{AA}(N) \rightarrow 1$ in $\mathbb{C}$ as $N \rightarrow \infty$, then $\|O_{AA}(N)\| \rightarrow 1$ as $N \rightarrow \infty$.*

Proof. ✓ Proof Follows from the continuity of the norm: $O_{AA}(N) \rightarrow 1$ implies $\|O_{AA}(N)\| \rightarrow \|1\| = 1$. □

Chapter 8

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D , the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

8.1 Cumulative span

Definition 8.1 (Word span). [Lean](#) [✓ Stmt](#) The *word span* at length n is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}} \{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Section II].

Definition 8.2 (Cumulative span). [Lean](#) [✓ Stmt](#) The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \dots + S_n(A).$$

Lemma 8.3 (Monotonicity of cumulative span). [Lean](#) [✓ Stmt](#) $T_n(A) \leq T_{n+1}(A)$ for all n .

Proof. [✓ Proof](#) T_{n+1} includes all words of length $\leq n+1$, which contains all words of length $\leq n$. $\square$

Lemma 8.4 (Dimension bound). [Lean](#) [✓ Stmt](#) $\dim T_n(A) \leq D^2$ for all n .

Proof. [✓ Proof](#) $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. $\square$

Theorem 8.5 (Cumulative span stabilisation). [Lean](#) [✓ Stmt](#) If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.

Proof. [✓ Proof](#) If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2} , so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. $\square$

Theorem 8.6 (Dimension growth). [Lean](#) [✓ Stmt](#) If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.

Proof. [✓ Proof](#) $T_n \leq T_{n+1}$ (Lemma 8.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. $\square$

Lemma 8.7 (Word span characterises block injectivity). [Lean](#) [✓ Stmt](#) $S_L(A) = M_D(\mathbb{C})$ if and only if A is L -block injective.

Proof. [✓ Proof](#) Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. $\square$

Lemma 8.8 (Two-sided span from one nonzero matrix). [Lean](#) [✓ Stmt](#) If $C \in M_D(\mathbb{C})$ is nonzero, then

$$\text{span}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}).$$

This is the matrix-factorization input used in [\[PGVWC07, Lemma lem1\]](#).

Proof. [✓ Proof](#) Choose a nonzero entry C_{pq} . Then every matrix unit E_{ij} is a scalar multiple of $E_{ip}CE_{qj}$, so the displayed span contains the standard matrix basis. $\square$

Lemma 8.9 (Exact words around a nonzero middle matrix). [Lean](#) [✓ Stmt](#) If A is L -block injective and $X \neq 0$, then

$$\text{span}\{A_u X A_v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

Proof. [✓ Proof](#) By L -block injectivity, the length- L word products span $M_D(\mathbb{C})$. Bilinearity of $(R, S) \mapsto RXS$ reduces the preceding lemma to products with R and S chosen among those word products. $\square$

8.2 Fitting decomposition

Definition 8.10 (Fitting decomposition). [Lean](#) [✓ Stmt](#) A *Fitting decomposition* of a linear endomorphism $f : V \rightarrow V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ -eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V ,
4. the generalized eigenspaces are linearly independent.

Theorem 8.11 (Fitting decomposition exists). [Lean](#) [✓ Stmt](#) Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.

Proof. [✓ Proof](#) Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [\[SPGWC10\]](#) and [\[Wol12, Lemma 6.3\(b\)\]](#), the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. $\square$

Theorem 8.12 (Nilpotent power bound). [Lean](#) [✓ Stmt](#) If f is nilpotent on a space of dimension n , then $f^n = 0$.

Proof. [✓ Proof](#) A nilpotent endomorphism on an n -dimensional space has nilpotency index $\leq n$. $\square$

8.3 Nonzero trace product

Theorem 8.13 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). Lean ✓ Stmt *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 8.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 8.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0 , hence $T_{L_0} = M_D(\mathbb{C})$). $\square$

Theorem 8.14 (Cumulative span reaches $M_D(\mathbb{C})$). Lean ✓ Stmt *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Take n from Theorem 8.13; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. $\square$

Theorem 8.15 (Nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2$ such that $\text{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is $D \geq 1$; when $D = 0$ the normality hypothesis is vacuous.*

Proof. ✓ Proof By Theorem 8.14, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. In the nontrivial case $D \geq 1$, at least one has nonzero trace (since $\text{tr}(\mathbb{1}) = D \neq 0$). $\square$

Theorem 8.16 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). Lean ✓ Stmt *If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \text{kr}(A)$. This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches $M_D(\mathbb{C})$ by index $D^2 - d' + 1 \leq D^2$.*

Proof. ✓ Proof The key observation is that if $T_n(A) \neq T_{n+1}(A)$, the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, we have $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. $\square$

Theorem 8.17 (Sharp nonzero trace product). Lean ✓ Stmt *If A is normal, then there exists a word w of length $\leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1] with the sharp length bound.*

Proof. ✓ Proof By Theorem 8.16, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$, hence is a linear combination of word products of length $\leq D^2 - d' + 1$. At least one such product has nonzero trace. $\square$

Theorem 8.18 (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If A is normal and $D \geq 2$, then there exists a word w of positive length $1 \leq |w| \leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This strengthens Theorem 8.17 by requiring $|w| \geq 1$, a feature needed for the blocking argument in the general Wielandt bound below.*

Proof. ✓ Proof Define the *positive-level* cumulative span $V_n = \text{span}\{A^w : 1 \leq |w| \leq n\}$. For $D \geq 2$, the span $S_0(A) = \text{span}\{\mathbb{1}\}$ has dimension 1, so $V_1 \supseteq S_1(A)$ has dimension $\geq d' \geq 1$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left-multiplication by each A^i , so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for all $N \geq 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbb{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since $\text{tr}(\mathbb{1}) = D \geq 2 \neq 0$). $\square$

8.4 Eigenvector extraction

Theorem 8.19 (Nonzero trace implies eigenvector). [Lean](#) [✓ Stmt](#) If $M \in M_D(\mathbb{C})$ has $\text{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.

Proof. [✓ Proof](#) A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 8.11) ensures that the corresponding generalized eigenspace is nontrivial. On that generalized eigenspace, $M - \mu\mathbb{1}$ is nilpotent, so its kernel is nontrivial. Any nonzero vector in $\ker(M - \mu\mathbb{1})$ is therefore a genuine eigenvector with eigenvalue μ . $\square$

Theorem 8.20 (Eigenvalue extraction). [Lean](#) [✓ Stmt](#) If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.

Proof. [✓ Proof](#) By Theorem 8.15, there is a word w_0 with $\text{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem 8.19, A^{w_0} has a nonzero eigenvector. $\square$

8.5 Eigenvector spreading

Definition 8.21 (Cumulative vector span). [Lean](#) [✓ Stmt](#) Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 8.22 (Vector span stabilisation). [Lean](#) [✓ Stmt](#) If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \geq n$.

Proof. [✓ Proof](#) Same argument as Theorem 8.5: left-multiplication by A^i cannot escape the stabilised span. $\square$

Lemma 8.23 (Vector span from matrix span). [Lean](#) [✓ Stmt](#) If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.

Proof. [✓ Proof](#) Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N . Applying such matrices to φ recovers all of $\mathbb{C}^D$. $\square$

Theorem 8.24 (Eigenvector spreading). [Lean](#) [✓ Stmt](#) Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].

Proof. [✓ Proof](#) The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 8.22). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 8.14, so Lemma 8.23 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. $\square$

8.6 Proof of the cumulative Wielandt bound

Theorem 8.25 (Wielandt chain). [Lean](#) [✓ Stmt](#) Let A be a normal MPS tensor with bond dimension $D \geq 1$. Then the following hold simultaneously:

1. $T_{D^2}(A) = M_D(\mathbb{C})$ (cumulative span stabilises);

2. there exists a word w_0 with $|w_0| \leq D^2$ and $\text{tr}(A^{w_0}) \neq 0$;
3. there exist $w_0, \mu \neq 0, \varphi \neq 0$ with $|w_0| \leq D^2$ and $A^{w_0}\varphi = \mu\varphi$;
4. for every $\varphi \neq 0$, $K_{D^2}(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Items (1)–(3) follow from Theorems 8.14 and 8.20. Item (4) combines these with the eigenvector spreading bound (Theorem 8.24). □

Theorem 8.26 (Wielandt bound). Lean ✓ Stmt If A is a normal MPS tensor with bond dimension D , then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2 . This is the cumulative-span part of [SPGWC10, Theorem 1]. The cited theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion $S_N(A) = M_D(\mathbb{C})$, stated below as a separate statement.

Proof. ✓ Proof Immediate from Theorem 8.14. □

Remark 8.27 (Explicit blocking-length estimates). ✓ Stmt Theorem 8.26 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d + 1) \cdot D^2$ (Theorem 1 case (1) of [SPGWC10]) is proved in Theorem 8.40 below. The sharp blocking-length estimates for injectivity (D^4) and bi-canonical form ($3D^5$) from [SPGWC10] are not proved here.

Definition 8.28 (One-step augmentation). Lean ✓ Stmt Let A be an MPS tensor with Kraus family $(A^i)_{i=0}^{d-1}$, and let $X \in M_D(\mathbb{C})$. The *one-step augmentation* of A by X is the tensor $\tilde{A}$ with physical dimension $d + 1$ whose first Kraus operator is X and whose remaining Kraus operators are those of A :

$$\tilde{A}^0 = X, \quad \tilde{A}^{i+1} = A^i.$$

Lemma 8.29 (Kraus operators lie in the one-step span). Lean ✓ Stmt Every Kraus operator A^i belongs to the one-step span $S_1(A)$.

Proof. ✓ Proof This is the length-one word with letter i . □

Lemma 8.30 (Redundant one-step augmentation). Lean Lean Lean ✓ Stmt Let $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . Then

$$S_n(\tilde{A}) = S_n(A)$$

for every $n \geq 0$.

Proof. ✓ Proof The new generator X is already in $S_1(A)$, and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length. □

Lemma 8.31 (Normality and Kraus rank under one-step augmentation). Lean Lean ✓ Stmt Let $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . If A is normal, then $\tilde{A}$ is normal, and

$$\text{kr}(\tilde{A}) = \text{kr}(A).$$

Proof. ✓ Proof Since all exact word spans are unchanged by Lemma 8.30, eventual full word span for A gives eventual full word span for $\tilde{A}$. The equality of Kraus ranks is the equality of the dimensions of the common one-step span. □

Theorem 8.32 (Wielandt case (2), invertible generator). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a normal MPS tensor with bond dimension D , and suppose that some Kraus operator A^{i_0} is invertible. Then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

Proof. [✓ Proof](#) The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument applied at all later lengths would contradict normality. Since $\dim S_1(A) = \text{kr}(A)$ and $\dim M_D(\mathbb{C}) = D^2$, the full algebra is reached by length $D^2 - \text{kr}(A) + 1$. The index bound follows from the definition of $\iota(A)$. $\square$

Theorem 8.33 (Wielandt case (2), one-step invertible element). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If $S_1(A)$ contains an invertible matrix X , then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}).$$

In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].

Proof. [✓ Proof](#) Adjoin X as an extra first Kraus operator, obtaining $\tilde{A}$. Primitivity with a positive-definite fixed point and normalization give normality of A , hence normality of $\tilde{A}$; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of $\tilde{A}$ is invertible, so Theorem 8.32 gives $S_{D^2 - \text{kr}(\tilde{A}) + 1}(\tilde{A}) = M_D(\mathbb{C})$. Transfer this equality back to A . $\square$

Corollary 8.34 (Wielandt case (2), index bound). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 8.33,

$$\iota(A) \leq D^2 - \text{kr}(A) + 1,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$.

Proof. [✓ Proof](#) The displayed word-span equality gives an admissible value in the defining minimum for $\iota(A)$. $\square$

Corollary 8.35 (Wielandt case (2), invertible Kraus operator). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is invertible, then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

Proof. [✓ Proof](#) Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.29. $\square$

Lemma 8.36 (Rank-one extraction from a non-invertible eigenvector). [Lean](#) [✓ Stmt](#) Let A be a normal MPS tensor with bond dimension $D \geq 1$. Suppose that a Kraus operator A^{i_0} is non-invertible and that there exists a vector $\varphi \in \mathbb{C}^D$ and a nonzero scalar μ such that $A^{i_0}\varphi = \mu\varphi$. Then, for every $\psi \in \mathbb{C}^D$,

$$|\varphi\rangle\langle\psi| \in S_{D^2 - D + 1}(A).$$

Proof. [✓ Proof](#) Decompose A^{i_0} into its nilpotent and nonzero generalized eigenspaces. On the nonzero part, multiplication by A^{i_0} is injective, so the rectangular spans grow strictly until they fill the whole range. The nilpotent index contributes the remaining length. Since the eigenvalue on φ is nonzero, eigenvector padding moves the resulting cumulative membership to the exact length $D^2 - D + 1$. $\square$

Theorem 8.37 (Wielandt case (3), one-step non-invertible element). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If $S_1(A)$ contains a non-invertible matrix X with a nonzero eigenvalue, then

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

More explicitly, it is enough to have $X\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. In the terminology of [SPGWC10], this is the case where the first application space $S_1(A)$ contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].

Proof. [✓ Proof](#) Adjoin X as an extra first Kraus operator. The augmented tensor is normal and has the same exact word spans as A . The eigenvector relation for X is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length $D - 1$, and Lemma 8.36 gives the rank-one operators $|\varphi\rangle\langle\psi|$ in length $D^2 - D + 1$. Multiplying these two fixed-length spans gives S_{D^2} for the augmented tensor, hence for A . $\square$

Corollary 8.38 (Wielandt case (3), index bound). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 8.37,

$$\iota(A) \leq D^2.$$

Proof. [✓ Proof](#) The equality $S_{D^2}(A) = M_D(\mathbb{C})$ gives an admissible value in the defining minimum for $\iota(A)$. $\square$

Corollary 8.39 (Wielandt case (3), non-invertible Kraus operator). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive with a positive-definite fixed point. If some Kraus operator A^{i_0} is non-invertible and has an eigenvector $\varphi \neq 0$ with nonzero eigenvalue μ , then

$$S_{D^2}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2.$$

Proof. [✓ Proof](#) Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma 8.29. $\square$

Theorem 8.40 (Wielandt's inequality — general bound). [Lean](#) [✓ Stmt](#) Let A be a normalized MPS tensor ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) that is primitive with a positive-definite fixed point. Then

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1) \cdot D^2,$$

where $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

Proof. [✓ Proof](#)

1. $D = 1$: In this case $\iota(A) = 0$, so the bound holds trivially.
2. $D \geq 2$: By Theorem 8.18, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\text{tr}(A^w) \neq 0$, where $d' = \text{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ (Theorem 8.19). Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal (Theorem 8.51), and the encoding of w as a single block index gives $B^{i_0} = A^w$.

- (a) If A^w is invertible (invertible case), $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

- (b) If A^w is non-invertible (non-invertible case), the eigenvector φ spans a proper subspace, and the rank-one extraction argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$.

In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$ where $k_B = \text{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate qualitative fixed-length conclusion stated below does not supply this explicit bound. In both cases $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$. $\square$

Remark 8.41 (Quantitative and qualitative Wielandt bounds). ✓ Stmt The quantitative bound $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$ is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint's own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 8.57: there exists N with $S_N(A) = M_D(\mathbb{C})$. The quantitative bound is not on the fundamental-theorem critical path.

8.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 8.42 (Fixed-length vector span). Lean ✓ Stmt The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \text{span}_{\mathbb{C}} \{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 8.43 (Word span products). Lean ✓ Stmt $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$: the product of word spans at lengths m and n is contained in the word span at length $m + n$.

Proof. ✓ Proof Every product $A^{w_1} A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1 w_2}$ with $|w_1 w_2| = m + n$. $\square$

Lemma 8.44 (Eigenvector padding). Lean ✓ Stmt Suppose $A^{i_0} \varphi = \mu \varphi$ with $\mu \neq 0$. Then for all n ,

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0 . The eigenvector relation gives $A^{w \cdot i_0^k} \varphi = \mu^k \cdot A^w \varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n . The reverse inclusion $H_n \leq K_n$ is immediate. $\square$

Theorem 8.45 (Fixed-length vector spanning implies matrix spanning). Lean ✓ Stmt Assume:

1. $H_n(A, \varphi) = \mathbb{C}^D$ (length- n word products applied to φ span all of $\mathbb{C}^D$), and
2. for each standard basis vector e_j , the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof Fix a matrix unit E_{ij} . By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i \varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i \varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 8.43. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 8.46 (Blocking transfer). Lean ✓ Stmt *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof Each length- n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length- nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. $\square$

Theorem 8.47 (Bounded injective blocking for left-canonical normal tensors). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then there is a positive blocking length $L \leq D^4$ such that the blocked tensor $A^{[L]}$ is one-site injective.*

This is the blocked-injectivity form of the quantum Wielandt input used in the non-periodic Fundamental Theorem chain. It corresponds to the statement in [CPGSV16, Section II, line 332] that every normal tensor becomes injective after at most D^4 blockings, using the index estimate of [SPGWC10, Theorem 1]. The Lean statement includes the left-canonical/trace-preserving hypothesis because that is the canonical-form context in which the blocked-injectivity input is consumed.

Proof. ✓ Proof The proof uses the full-Kraus-rank index bound from the quantum Wielandt inequality. Since A is normal, the index is attained; the one-step Kraus rank is nonzero in the left-canonical case, so the bound $(D^2 - \text{rank } S_1(A) + 1)D^2$ is at most D^4 . The passage from exact word-span fullness to injectivity of the blocked tensor is Lemma 8.46. $\square$

Corollary 8.48 (Uniform injective blocking for normal BNT families). Lean Lean ✓ Stmt *Let $(A_x)_{x \in X}$ be a finite normal-canonical BNT family whose blocks have positive virtual bond dimension. Then there is a positive integer L such that every blocked tensor $A_x^{[L]}$ is one-site injective. The same conclusion holds simultaneously for two finite normal-canonical BNT families whose blocks have positive virtual bond dimension.*

Proof. ✓ Proof Apply Theorem 8.47 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking. $\square$

Remark 8.49 (Relation with the rank-one extraction lemma). ✓ Stmt In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 8.45 is the second step. Theorem 8.56 supplies the rank-one hypothesis in blocked form, and Theorem 8.57 then gives a blocked, fixed-length word-span saturation result.

Remark 8.50 (Use of the quantum Wielandt theorem in the MPS route). The source papers use the quantum Wielandt theorem as an input converting normality into injectivity after blocking. In the notation of this chapter, the input has two distinct conclusions. First, Theorem 8.26 gives cumulative saturation $T_{D^2}(A) = M_D(\mathbb{C})$. This alone is not the injectivity statement used in canonical form, because injectivity requires products of one fixed length. Second, Lemma 2(b) of [SPGWC10], formalized here through Theorem 8.57, supplies a length N for which $S_N(A) = M_D(\mathbb{C})$. This is the statement that matches the injectivity of Γ_N and can be transported through

blocking by Lemma 8.46. Thus any later use of Wielandt in the Fundamental Theorem must specify whether it is using cumulative span, exact-length span, or the blocked version of the exact-length conclusion.

8.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor* $A^{[L]}$ and develops the “rectangular span” used to produce rank-one elements.

Theorem 8.51 (Blocking preserves normality). Lean ✓ Stmt *If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

Proof. ✓ Proof Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A)$$

by Lemma 8.43. Iterating this inclusion gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$; taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma 8.52, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

Lemma 8.52 (Chunking: word span embeds into blocked word span). Lean ✓ Stmt $S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L .

Proof. ✓ Proof Every length- nL word in the original alphabet can be split into n consecutive chunks of length L , each of which is a single blocked Kraus operator. By induction on n , using the word span product lemma (Lemma 8.43). $\square$

Theorem 8.53 (Word eigenvector gives blocked eigenvector). Lean ✓ Stmt *If $A^{w_0} \varphi = \mu \varphi$ for a word w_0 of length L , then $(A^{[L]})^{j_0} \varphi = \mu \varphi$, where j_0 is the encoded blocked index corresponding to w_0 .*

Proof. ✓ Proof By definition, $(A^{[L]})^{j_0} = A^{w_0}$. $\square$

Theorem 8.54 (Blocked fixed-length matrix spanning). Lean ✓ Stmt *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0} \varphi = \mu \varphi$ ($\mu \neq 0, \varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T \psi = \nu \psi$ ($\nu \neq 0, \psi \neq 0$),*
3. *a rank-one element $\varphi \psi^T \in S_m(A^{[L]})$ for some m .*

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.

Remark. The proof gives the explicit bound $N = ((D-1) + m + (D-1)) \cdot L$; the statement above gives only the existential conclusion.

Proof. ✓ Proof Apply Theorem 8.45 to the blocked tensor $A^{[L]}$, which is normal by Theorem 8.51. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 8.53. Transfer back to the original tensor via Lemma 8.46. $\square$

Theorem 8.55 (Blocked fixed-length matrix spanning from word eigenvectors). Lean ✓ Stmt *Suppose A is normal and $L > 0$. If a length- L word has a nonzero eigenvector and a length- L word has a nonzero transpose eigenvector, then there exists N such that $S_N(A) = M_D(\mathbb{C})$.*

Proof. ✓ Proof The rank-one extraction theorem supplies the blocked rank-one element required by Theorem 8.54; apply the blocked assembly theorem. $\square$

Theorem 8.56 (Rank-one extraction for blocked tensor). Lean ✓ Stmt Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0 , nonzero vectors φ, ψ , nonzero scalars μ, ν , and $m \in \mathbb{N}$ such that:

1. $A^{\sigma_0} \varphi = \mu \varphi$,
2. $(A^{\tau_0})^T \psi = \nu \psi$,
3. $\varphi \psi^T \in S_m(A^{[N_0]})$.

Proof. ✓ Proof Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, we have $S_1(B) = M_D(\mathbb{C})$ by definition of the blocked tensor. Hence $\mathbb{1} \in S_1(B)$, so in the nontrivial case $D \geq 1$ not all blocked generators can have trace zero. Choose j_0 with $\text{tr}(B^{j_0}) \neq 0$. By Theorem 8.19, B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0} \varphi = \mu \varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0 , a nonzero vector ψ , and $\nu \neq 0$ with $(A^{\tau_0})^T \psi = \nu \psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem 8.51. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular $\varphi \psi^T \in S_M(B)$. Taking $m = M$ proves (3). $\square$

Theorem 8.57 (Fixed-length word-span saturation via blocking). Lean ✓ Stmt If A is a normal MPS tensor with bond dimension D , then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].

Proof. ✓ Proof By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 8.56 produces eigenvectors and a rank-one element $\varphi \psi^T \in S_m(A^{[N_0]})$. Theorem 8.54 then gives $S_{((D-1)+m+(D-1)) \cdot N_0}(A) = M_D(\mathbb{C})$. $\square$

Remark 8.58 (Blocking argument for fixed-length spanning). ✓ Stmt Theorem 8.57 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 8.56) and then applying the fixed-length matrix-spanning theorem (Theorem 8.54). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

8.9 Primitive MPS tensors

Definition 8.59 (Primitive MPS tensor). Lean ✓ Stmt Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \rho \geq 0, \quad \rho \neq 0, \quad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ , then $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This is weaker than the primitive condition of [SPGWC10], which requires $\rho > 0$; see Remark 8.65. The stronger positive-definite hypothesis is imposed separately in Theorems 8.66 and 8.84.

Theorem 8.60 (Primitive overlap convergence). Lean ✓ Stmt If A is a primitive MPS tensor, then $O_{AA}(N) \rightarrow 1$.

Proof. ✓ Proof Immediate from the self-overlap convergence under a complementary spectral gap (Theorem 7.98). $\square$

8.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 8.59: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1 .

Theorem 8.61 (Primitive MPS tensors have nonzero word products). Lean ✓ Stmt *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length exactly n with $A^w \neq 0$.*

Proof. ✓ Proof The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. $\square$

Theorem 8.62 (Iterated transfer does not vanish). Lean ✓ Stmt *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. ✓ Proof By Theorem 8.61, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. $\square$

8.10.1 From primitivity to normality

The primitivity condition of Definition 8.59 combines TP normalization, a nonzero PSD fixed point, and a complementary spectral gap. It is connected to normality through the following results.

Theorem 8.63 (Fixed-point uniqueness from spectral gap). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ , then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ : $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$.*

Proof. ✓ Proof Set $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$. Then $\text{tr}(\sigma') = 0$ and $(E - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $E - P_\rho$, contradicting the spectral gap $\rho_{\text{spec}}(E - P_\rho) < 1$. $\square$

Theorem 8.64 (Complement powers tend to zero). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ , then $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$ in operator norm.*

Proof. ✓ Proof The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the spectral-gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. $\square$

Remark 8.65 (Primitivity: spectral gap vs. positive definiteness). ✓ Stmt The primitivity condition used in §8.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ , and the spectral gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of “primitive” in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the spectral-gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \text{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Theorem 8.66 gives irreducibility. Together with the Burnside argument below (§8.10.3) and Theorem 8.82, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \text{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 8.77. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies that primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

8.10.2 Primitive implies irreducible

Theorem 8.66 (Primitive tensors with positive-definite fixed point are irreducible). [Lean](#)

[✓ Stmt](#) If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV17a, Section 2.3]).

Proof. [✓ Proof](#) By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1} - P)A^iP = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1} - P)A^iP = 0$ implies $(\mathbb{1} - P) \cdot \mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).
2. *Convergence:* By Theorem 8.64, $\mathcal{E}_A^n(\sigma) \rightarrow \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\text{tr}(P\rho P) > 0$, so $\frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.
4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

□

8.10.3 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 9.11) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 8.67 (Algebra span). [Lean](#) [✓ Stmt](#) For an MPS tensor A , let $\text{alg}(A)$ be the unital $\mathbb{C}$ -subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}$:

$$\text{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C}).$$

Definition 8.68 (Invariant submodule). [Lean](#) [✓ Stmt](#) A subspace $W \leq \mathbb{C}^D$ is A -invariant if $A^iW \subseteq W$ for every i .

Definition 8.69 (Irreducible action). [Lean](#) [✓ Stmt](#) The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A -invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 8.70 (Irreducible tensor implies irreducible action). [Lean](#) [✓ Stmt](#) If A is irreducible as a tensor (Definition 9.11), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 8.69).

Proof. [✓ Proof](#) Let $W \leq \mathbb{C}^D$ be an A -invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^iP = 0$ for all i . Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. □

Theorem 8.71 (Burnside's theorem for Kraus algebras). [Lean](#) [✓ Stmt](#) Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:

$$\text{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].

Proof. ✓ Proof Transport $\text{alg}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. $\square$

Remark 8.72. ✓ Stmt This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

Theorem 8.73 (Irreducible action implies eventual cumulative spanning). Lean ✓ Stmt Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof By Theorem 8.71, $\text{alg}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some n . Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \leq T_1(A) \leq \dots$ stabilizes, so one can choose a uniform N with $T_N(A) = M_D(\mathbb{C})$. $\square$

8.10.4 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect the spectral-gap primitivity of Definition 8.59 directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

Theorem 8.74 (Primitive tensors with positive-definite fixed point have irreducible transfer map). Lean ✓ Stmt If A is a primitive MPS tensor with positive-definite fixed point ρ , then the transfer map $\mathcal{E}_A$ is irreducible.

The proof uses fixed-point uniqueness (Theorem 8.63): every PSD fixed point is proportional to ρ , so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

Proof. ✓ Proof By Theorem 8.63, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\text{tr } \sigma}{\text{tr } \rho} \rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. $\square$

Theorem 8.75 (Primitive tensors with positive-definite fixed point are strongly irreducible). Lean ✓ Stmt If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$.

Proof. ✓ Proof Combine Theorem 8.74 (irreducibility) with the peripheral-spectrum primitivity from the spectral-gap hypothesis (Theorem 8.60). $\square$

Theorem 8.76 (Primitive MPS with positive-definite fixed point is normal). Lean ✓ Stmt If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.

This is the key connection between the spectral-gap primitivity of Definition 8.59 and normality; it removes the need for the aperiodicity-based argument.

Proof. ✓ Proof By Theorem 8.75, A is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 7.64 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank). $\square$

8.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). An aperiodicity hypothesis upgrades cumulative spanning to a single-length word span.

Remark 8.77 (Aperiodicity condition). ✓ Stmt We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the primitive condition with positive-definite fixed point (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 7.64) without passing through aperiodicity.

Remark 8.78 (Counterexample: cumulative spanning does not imply normality). ✓ Stmt In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\text{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n . Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n , and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 8.79 (Word span monotonicity from aperiodicity). Lean ✓ Stmt If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all n .

Proof. ✓ Proof For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 8.43. □

Theorem 8.80 (Cumulative span equals word span under aperiodicity). Lean ✓ Stmt If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all n .

Proof. ✓ Proof By Theorem 8.79, the word spans are monotone: $S_0 \leq S_1 \leq \dots$. The supremum $T_n = S_0 + \dots + S_n$ therefore equals the largest term S_n . □

Theorem 8.81 (Cumulative spanning plus aperiodicity implies normality). Lean ✓ Stmt If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.

Proof. ✓ Proof By Theorem 8.80, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. □

Theorem 8.82 (Algebra span plus aperiodicity implies normality). Lean ✓ Stmt If $\text{alg}(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.

Proof. ✓ Proof Since $\text{alg}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 8.81. □

Theorem 8.83 (Irreducible tensor plus aperiodicity implies normality). Lean ✓ Stmt If A is an irreducible tensor, $D > 0$, and $\mathbb{1} \in S_1(A)$, then A is normal. This assembles the full chain:

$$\text{irreducible tensor} \xrightarrow{8.70} \text{irreducible action} \xrightarrow{\text{Burnside}} \text{alg}(A) = M_D(\mathbb{C}) \xrightarrow{8.82} \text{normal}.$$

This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. The Burnside step is supplied by Theorems 8.70 and 8.71.

Proof. ✓ Proof By Theorem 8.70, A acts irreducibly on $\mathbb{C}^D$. By Theorem 8.71, $\text{alg}(A) = M_D(\mathbb{C})$. By Theorem 8.82 with the aperiodicity hypothesis, A is normal. □

Theorem 8.84 (Primitive + positive definite fixed point imply normality). [Lean](#) [✓ Stmt](#) *If A is a primitive MPS tensor with fixed point ρ and ρ is positive definite, then A is normal. No aperiodicity assumption is needed.*

Proof. [✓ Proof](#) By Theorem 8.75, A is strongly irreducible, i.e. primitive in Wolf’s sense. Under normalization, Theorem 7.64 therefore gives normality directly, hence eventual full Kraus rank. $\square$

Remark 8.85 (Normal canonical form avoids the aperiodicity step). The aperiodicity results in §8.11 are needed only for the block-injective argument through Definition 2.29, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 9–11 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV17a]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

8.12 Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9)

Theorem 8.86 (TP-primitive irreducible tensor is injective after blocking). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then some blocking $A^{[L]}$ is one-site injective.*

Proof. [✓ Proof](#) The normality theorem (Theorem 9.32) gives eventual block injectivity, and the blocked-chain equivalence identifies this with one-site injectivity of the corresponding blocked tensor. $\square$

Remark 8.87. See Theorem 8.51 for the formalized statement that blocking preserves normality.

Chapter 9

Canonical Form Reduction

This chapter reduces a general MPS tensor to block-diagonal data via iterated invariant-subspace decomposition. Two reduction schemes are relevant. The block-injective reduction aims at injective TP-normalized blocks and the canonical form conditions of Chapter 10. The normal-canonical reduction uses a weighted family of irreducible TP-normalized blocks with primitive transfer maps and pairwise distinct nonzero weight moduli, in the sense of [CPGSV17a]. Thus the reductions recorded here are conditional primitive-block results. They should not be read as the full translation-invariant canonical-form theorem of [PGVWC07, Theorem Th:TIcanonical, lines 742–763]. The normalization convention is also dual to the one used in the cited theorem: the blocks are written in the unital orientation $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ for $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$ [PGVWC07, Theorem Th:TIcanonical, lines 754–758], whereas this chapter uses the trace-preserving, or left-canonical, orientation $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. Passing to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$ exchanges the two orientations: the transfer map of K is the Frobenius adjoint of $\mathcal{E}_A$ (formalized in `MPS/CanonicalForm/BlockingViaAdjoint.lean`). The Perron–Frobenius eigenvector existence and TP-gauge construction are in Chapter 6; this chapter uses them to produce left-canonical normalization data and the normal canonical form data. The reduction draws on [PGVWC07], [CPGSV21], and [Wol12, Chapter 6].

9.1 Normal tensors, canonical forms, and bases of normal tensors

This section records the unstrengthened definitions of *normal tensor*, *canonical form*, and *basis of normal tensors* from [CPGSV17a]. The stronger predicates used elsewhere in this chapter (Definition 9.18 and the canonical-form BNT predicate of Chapter 11) bundle additional structure—left-canonical normalization, strict modulus ordering, one-copy-per-sector—which is not present in the literal source-paper definitions stated below.

Definition 9.1 (CPSV normal tensor (NT)). [Lean] [✓ Stmt] A tensor $A : \text{MPSTensor } d D$ is a *normal tensor* if (i) A admits no nontrivial invariant orthogonal projection (`IsIrreducibleTensor`), and (ii) its associated CPM $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ has a unique eigenvalue of magnitude equal to its spectral radius which is equal to one (`IsPrimitive (transferMap A)`, after the spectral-radius normalization in [CPGSV17a, paragraph preceding the NT definition]).

Definition 9.2 (CPSV canonical form (CF)). [Lean] [✓ Stmt] A tensor A is in *canonical form* if there is a finite family of normal tensors $A_k : \text{MPSTensor } d D_k$ and weights $\mu_k \in \mathbb{C}$ ($k = 1, \dots, r$)

with

$$A^i \cong \bigoplus_{k=1}^r \mu_k A_k^i$$

in the sense that A and the block-assembled tensor `toTensorFromBlocks` $\mu A_\bullet$ generate the same MPV family at every length, and the weights satisfy the normalization $|\mu_k| \leq 1$ for all k and $|\mu_k| = 1$ for at least one k .

Definition 9.3 (CPSV basis of normal tensors (BNT)). Lean ✓ Stmt A finite family $\{A_j\}_{j=1}^g$ of normal tensors is a *basis of normal tensors* for A if:

1. each A_j is normal in the sense of Definition 9.1;
2. for every $N \geq 1$ there exist coefficients $c_j^{(N)} \in \mathbb{C}$ with

$$V^{(N)}(A) = \sum_{j=1}^g c_j^{(N)} V^{(N)}(A_j);$$

3. there exists N_0 such that for all $N > N_0$ the family $(V^{(N)}(A_j))_{j=1}^g$ is linearly independent.

Remark 9.4 (Relation to the strong predicates). A tensor A which is both irreducible (Definition 9.11) and has primitive transfer map is normal in the sense of Definition 9.1. The reverse implications from the stronger predicates of Definition 9.18 and of Chapter 11 require either the fundamental-theorem step or the algebraic-to-spectral normality equivalence; these appear in downstream chapters.

9.2 Transfer map normalization

Remark 9.5 (Scalar normalization conventions). ✓ Stmt The reductions in this chapter use four routine normalization facts: scaling preserves injectivity, the transfer map under a scalar ζ rescales by $|\zeta|^2$, unit-modulus phase rescaling preserves the left-canonical condition $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and the MPV of a weighted block-diagonal tensor factorizes as $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$ with $\eta_k := \mu_k / |\mu_k|$. These are scalar identities, formalized in the Lean module `SharedInfra/Scaling.lean`.

Without first placing each block in the left-canonical normalization, the implication $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k, \mathcal{V}(A_k) = \mathcal{V}(B_k)$ fails even for two scalar blocks with distinct nonzero weights: for $\mu = (1, 2)$ and $A = (1, 3/2)$, $B = (3, 1/2)$, the two direct sums generate identical MPV families but the first blocks differ at $N = 1$. The canonical-form hypotheses (Definition 9.2) remove this ambiguity by combining left-canonical normalization with the strict ordering $|\mu_1| > \dots > |\mu_r|$.

9.3 Invariant subspace decomposition

Theorem 9.6 (Unitary conjugation preserves the MPV family). Lean ✓ Stmt If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.

Proof. ✓ Proof By cyclicity of the trace: $\text{tr}((U^\dagger A^{i_1} U) \dots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \dots A^{i_N})$. □

Definition 9.7 (Support projection). Lean ✓ Stmt For a positive semi-definite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ . It is constructed via the spectral decomposition: if $\rho = U \text{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$, then $P = U \text{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger$, where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 9.8 (Invariant projection predicate). [Lean] ✓ Stmt An MPS tensor A has an *invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^iP = 0$ for all i . A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 9.11).

Theorem 9.9 (PSD fixed point gives invariant projection). [Lean] ✓ Stmt Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be the support projection of ρ . Then for all i , $(\mathbb{1} - P)A^iP = 0$: the range of P is invariant under all A^i .

Proof. ✓ Proof From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i\rho(A^i)^\dagger$, $(\mathbb{1} - P)A^i\rho(A^i)^\dagger(\mathbb{1} - P) = 0$ for each i . Since ρ is supported on the range of P , this forces $(\mathbb{1} - P)A^iP = 0$.

This is the finite-dimensional version of [Wol12, Proposition 6.10] (“Stationary subspaces”: the support projection of any stationary state is sub-harmonic, i.e. invariant under the channel). The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Theorem 6.2]; see also [CPGSV21, Section IV.A.1]. $\square$

Theorem 9.10 (Two-block decomposition). [Lean] ✓ Stmt If the transfer map has a PSD fixed point ρ that is not positive definite (i.e. ρ has a nontrivial kernel), then there exist MPS tensors A_1, A_2 of smaller bond dimensions such that the two-block tensor $A_1 \oplus A_2$ generates the same MPV family as A : $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.

Proof. ✓ Proof The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not PD). Theorem 9.9 gives $(\mathbb{1} - P)A^iP = 0$ for all i . A unitary change of basis that block-diagonalizes P puts A^i in upper-triangular block form, with diagonal blocks B_{11}^i and B_{22}^i . Theorem 9.6 ensures the MPV is preserved by the conjugation. The trace of the upper-triangular product equals the sum of the diagonal-block traces, so the block-diagonal tensor obtained by discarding the off-diagonal blocks has the same MPV coefficients as the conjugated tensor, and hence as A . $\square$

9.4 Iterated reduction

Definition 9.11 (Irreducible tensor). [Lean] ✓ Stmt An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 9.12 (Irreducible block decomposition). [Lean] ✓ Stmt Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ where each A_k is irreducible and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$.

Proof. ✓ Proof By strong induction on the bond dimension D . If A is already irreducible, take $r = 1$. Otherwise, by Definition 9.11, A has a nontrivial invariant projection P . Diagonalizing P yields the same two-block upper-triangular splitting used in Theorem 9.10; the block-diagonal tensor obtained by discarding the off-diagonal part has the same MPV family as A , and the two diagonal blocks have strictly smaller bond dimensions. Iterating this splitting produces a block-diagonal tensor $\bigoplus_k A_k$ whose blocks are irreducible. Apply the induction hypothesis to each block and concatenate. (If one prefers a positive-eigenvector argument without assuming TP normalization, the correct replacement for the channel-specific Cesàro theorem is Theorem 6.18; the induction itself needs only the invariant projection.)

Both [PGVWC07, Theorem 4] and [CPGSV21, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting used here. An alternative derivation, developed in [Wol12, Theorem 6.14], characterises the fixed-point set of a quantum channel as a $*$ -algebra and then applies the Wedderburn–Artin decomposition ([Wol12, Equation (1.39)]) to read off the block structure directly. The block decomposition is obtained by iterating the invariant-projection splitting and using strong induction on D to ensure termination. $\square$

Theorem 9.13 (Unitary diagonal positive fixed point in TP gauge). [Lean] [Stmt] Let A be an irreducible MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. [Proof] The TP hypothesis makes $\mathcal{E}_A$ a channel, so it has a nonzero PSD fixed point. The irreducible tensor condition implies that $\mathcal{E}_A$ is irreducible as a CP map, and Theorem 6.4 upgrades the fixed point to a positive definite one. The spectral theorem diagonalizes this matrix by a unitary conjugation, and unitary conjugation preserves the TP normalization. $\square$

The Burnside-style passage from irreducible tensors to full algebra spanning is developed alongside the Wielandt material in Chapter 8; see in particular Theorem 8.73.

9.5 Canonical form from primitivity

The canonical form conditions (Definition 9.2) include an explicit hypothesis that the self-overlap $O_{A_k A_k}(N) \rightarrow 1$ for each block. The single source-faithful result of this section is Theorem 9.16: blocking an irreducible TP block by the least common multiple of its peripheral periods makes the resulting transfer map primitive, removing periodicity.

Remark 9.14 (Two routes to the canonical-form predicate from normal blocks). [Stmt] Given a family of injective, left-canonical, strictly-ordered nonzero-weight blocks (μ_k, A_k) , the canonical-form conditions of Definition 9.2 can be discharged from either of two spectral hypotheses on the per-block transfer map $\mathcal{E}_{A_k}$: a positive-definite primitive fixed point, or the peripheral-spectrum condition $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$. In both cases the overlap clause $O_{A_k A_k}(N) \rightarrow 1$ follows from the spectral-gap criterion of Chapter 7. Both statements assume the blocks are already in the required normal form; they are not the arbitrary-input canonical-form existence theorem of [PGVWC07, Theorem Th:TIcanonical].

Remark 9.15. [Stmt] The remark above shows that the overlap convergence hypothesis in the canonical form conditions is redundant once peripheral-spectrum primitivity is known. The argument factors through the spectral-gap formulation of Chapter 7: peripheral-spectrum primitivity is first converted to that spectral-gap condition, and the overlap limit is then deduced from it. In [CPGSV21], primitivity is obtained from irreducibility plus periodicity removal by blocking to period one. Theorem 9.16 is precisely this periodicity-removal step under irreducibility and TP normalization.

Theorem 9.16 (Blocking yields a peripheral-spectrum primitive transfer map). [Lean] [Stmt] Let A be an irreducible MPS tensor with $D > 0$. Assume that A satisfies the TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that the transfer map $\mathcal{E}_{A^{[p]}}$ of the blocked tensor $A^{[p]}$ is primitive, i.e. its peripheral spectrum is $\{1\}$.

Proof. [Proof] The period- p blocking step is the diagrammatic move of grouping the chain into consecutive blocks of length p . Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$. The TP normalization of A makes K unital, and irreducibility of A gives irreducibility of the associated Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint map of K , so Theorem 4.46 shows that every peripheral eigenvalue is a root of unity. Since $\text{peripheral}(\mathcal{E}_A)$ is finite (Theorem 4.29), Lemma 4.33 gives a common exponent p such that $\mu^p = 1$ for all $\mu \in \text{peripheral}(\mathcal{E}_A)$. Lemma 4.34 then implies $\text{peripheral}(\mathcal{E}_A^p) = \{1\}$. Finally, blocking corresponds to taking a power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. $\square$

Remark 9.17. ✓ Stmt Theorem 9.16 is the “periodicity removal by blocking” step in [CPGSV17a, Appendix A] for an irreducible block already placed in TP gauge. The later existence theorems then reuse this step.

9.6 Normal canonical form

Definition 9.18 (Normal canonical form conditions). Lean ✓ Stmt A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here “normal” is used in the spectral sense of [CPGSV17a]: irreducible blocks whose TP-normalized transfer maps are primitive. As noted in the remark after Definition 2.29, this is equivalent to the eventual block-injectivity formulation of Definition 2.29; see [SPGWC10, Proposition 3].

Theorem 9.19 (Self-overlap is derived in normal canonical form). Lean ✓ Stmt If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \rightarrow 1$ for every block k .

Proof. ✓ Proof Peripheral-spectrum primitivity is part of the data. Together with irreducibility and TP normalization, it yields the spectral-gap primitive theorem needed for overlap convergence, so the self-overlap condition follows from the other hypotheses rather than appearing as a separate assumption. □

Remark 9.20. See Theorem 7.28 for the formalized modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

Lemma 9.21 (Mixed-transfer spectral radius from unit-modulus overlap). Lean ✓ Stmt Let A and B be MPS tensors of the same bond dimension. If $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$ as $N \rightarrow \infty$, then the mixed transfer map has spectral radius at least 1.

Proof. ✓ Proof If the mixed transfer spectral radius were strictly less than 1, the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace, and via the trace identity $\text{tr}(F_{AB}^N) = \langle V^{(N)}(A), V^{(N)}(B) \rangle$ the overlap itself would tend to 0, contradicting the hypothesis. □

Theorem 9.22 (Gauge-phase equivalence from unit-modulus overlap). Lean ✓ Stmt Let A and B be irreducible MPS tensors of the same bond dimension satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$, then A and B are gauge-phase equivalent.

Proof. ✓ Proof By Lemma 9.21, the mixed transfer spectral radius is at least 1. Theorem 7.28 then gives gauge-phase equivalence. □

9.7 Steps toward canonical form existence

This section combines the components of the canonical-form construction following [CPGSV17a, Section 2.3 and Appendix A]. The approach here requires blocking to remove non-trivial periodicity. A separate reduction for periodic blocks without blocking is developed in [DICCSPG17]; see Chapter 15 for that direct periodic extension.

Theorem 9.23 (Left-canonical normalization for irreducible TP blocks). Lean ✓ Stmt *Let A be an irreducible MPS tensor already in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that:*

1. $B^i := U^\dagger A^i U$ is TP,
2. $\mathcal{E}_B(\Lambda) = \Lambda$,
3. Λ is diagonal and positive definite,
4. A and B generate the same MPV family.

This is the left-canonical normalization step (Canonical Form II) of [CPGSV17a, Appendix A]. The unitary conjugation occurs only after TP normalization has already been achieved; it is a second step inside TP gauge, not a unitary normalization of an arbitrary irreducible block.

Proof. ✓ Proof Apply Theorem 9.13 to obtain the unitary and the diagonal PD fixed point. Unitary conjugation preserves trace-preservation and the MPV family (Theorem 9.6). □

Theorem 9.24 (Irreducible block decomposition with left-canonical normalization). Lean ✓ Stmt *For any MPS tensor A , there exist irreducible blocks $(A_k)_{k=1}^r$ that generate the same MPV family as A . Moreover, for each block k , if that block has already been placed in TP gauge and $D_k > 0$, one obtains a left-canonical normalization (unitary, diagonal PD fixed point, MPV preservation). This yields the left-canonical normalization; the further reduction to normal canonical form is carried out in the subsequent spectral analysis.*

Proof. ✓ Proof Combine Theorems 9.12 and 9.23. □

Theorem 9.25 (Normal canonical form from a primitive block decomposition). Lean ✓ Stmt *Suppose A generates the same MPV family as a weighted block-diagonal tensor $\bigoplus_k \mu_k A_k$, and assume that:*

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli $|\mu_k|$ are pairwise distinct,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Then there exist a blocking length $p > 0$, weights (ν_k) , and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions.

Proof. ✓ Proof No new blocking is needed here: the given blocks are already primitive, so one may take $p = 1$. The theorem is stated with a blocking length because earlier reduction steps may require a common blocking period; under these hypotheses that period can be chosen to be 1. Since the moduli $|\mu_k|$ are pairwise distinct, there is a reordering of the blocks for which they become strictly decreasing. After this reordering, set $\nu_k := \mu_k$ and $B_k := A_k$ in the new order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are unchanged by the reordering, so the reordered family satisfies Definition 9.18 and generates the same MPV family as A . $\square$

Remark 9.26 (Scope of Theorem 9.25). Theorem 9.25 assumes all six block hypotheses—irreducibility, TP normalization, primitivity, distinct nonzero weight moduli, and positive bond dimensions—in advance, and produces the normal canonical form by reordering. It is *not* the arbitrary-input canonical-form existence theorem of [CPGSV17a, lines 237–250], which starts from an arbitrary tensor and derives the primitive block structure. That arbitrary-input direction is the open formalization gap documented in Section 9.12.2.

9.8 TP-gauge reduction for arbitrary tensors

Theorem 9.27 (TP-gauge reduction for arbitrary tensors with zero-block separation). Lean

✓ Stmt For any MPS tensor A , there exist:

1. a zero-tail bond dimension D_0 ,
2. nontrivial blocks $(B_k)_{k=1}^r$ with nonzero weights $(\mu_k)_{k=1}^r$, each block irreducible and left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), with positive bond dimension,

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d, D_0})_\sigma + V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

This is the furthest unconditional reduction step available for arbitrary tensors before periodicity removal.

Proof. ✓ Proof Apply Theorem 13.3 to separate the trivial block and nontrivial blocks. Then apply the blockwise TP-gauge theorem to the nontrivial blocks. The MPV identity chains through both steps. $\square$

9.9 Blocking and primitivity reduction

The intermediate constructions that transport MPV equalities, weights, transfer maps, and physical-alphabet casts through blocking by an integer p are collected in the Lean modules `MPS/Core/BlockingInfrastructure.lean` and `MPS/Core/BlockingTransfer.lean`. The single source-facing consequence needed downstream is the existence of a common blocking period under which every transfer map in a finite family becomes primitive.

Theorem 9.28 (Common blocking period for a block family). Lean ✓ Stmt Let $(B_k)_{k=1}^r$ be a finite family of MPS tensors with positive bond dimensions, each admitting a blocking period p_k making its transfer map primitive. Then $p = \text{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k .

Proof. ✓ Proof Each p_k divides the least common multiple p , so primitivity of $\mathcal{E}_{B_k^{[p_k]}}$ transports to primitivity of $\mathcal{E}_{B_k^{[p]}}$ via the standing divisibility identity for transfer-map powers. $\square$

9.10 Period removal (cyclic-sector decomposition)

In the non-periodic route of [CPGSV17a] the only required step is periodicity removal by blocking: after blocking each irreducible TP block by the least common multiple of its peripheral periods, the resulting tensor has no nontrivial p -periodic vectors [CPGSV17a, Section 2.3]. The cyclic-sector decomposition itself comes from the peripheral spectrum of the adjoint transfer map [Wol12, Theorem 6.6]. Three source-facing statements summarize this step: existence of the cyclic decomposition after blocking, the version specialized to irreducible TP tensors, and the strengthened version where the resulting sector blocks are themselves primitive and irreducible.

Theorem 9.29 (Period removal by blocking (1606.00608, §2.3)). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor of period m . Then the blocked tensor $A^{[m]}$ admits:*

1. *left-canonical sector tensors $(C_k)_{k=1}^m$,*
2. *orthogonal projections $(P_k)_{k=1}^m$ summing to $\mathbb{1}$ and satisfying $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,*
3. *compression linear equivalences $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,*
4. *a direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^m C_k$.*

This is the period-removal step: after blocking by m , the result admits a unit-weight decomposition with no non-trivial p -periodic vectors. The cyclic projections P_k come from the peripheral spectrum of the adjoint transfer map ([Wol12, Theorem 6.6]).

Proof. ✓ Proof The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating the cyclic relation m times shows $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and the blocked-transfer identity $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$ identifies these as the adjoint transfer map of the blocked tensor. Apply the block decomposition from adjoint-fixed projections (formalized in MPS/CanonicalForm/CyclicSectors/FixedAdjoint.lean). □

Theorem 9.30 (Period removal for irreducible TP tensors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there exist $m \geq 1$ and left-canonical blocks $(C_k)_{k=1}^m$ such that the m -blocked tensor $A^{[m]}$ admits a unit-weight decomposition into the C_k . This is the period-removal step of [CPGSV17a, Section 2.3].*

Proof. ✓ Proof Left-canonicity gives $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so the conjugate-transposed family $K_i := (A^i)^\dagger$ is unital; irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point. The cyclic peripheral-spectrum theorem then provides a period m and the cyclic spectral data, and Theorem 9.29 produces the decomposition. □

Theorem 9.31 (Period removal with primitive irreducible sectors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period $m \geq 1$ and a unit-weight decomposition of $A^{[m]}$ into sector blocks $(C_k)_{k=1}^m$ such that every C_k is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension.*

Equivalently: the cyclic-sector decomposition of $A^{[m]}$ produces unit blocks that already satisfy the trace-preservation, primitivity, and irreducibility hypotheses needed for the common-blocking step (Lemma 9.50).

Proof. ✓ Proof The proof repeats the cyclic-sector construction of Theorem 9.30, keeps the projection and compression data, and applies Theorem 13.30 to the resulting sector blocks. □

9.11 Reduction to primitive blocks

The structural after-blocking statements that take an arbitrary tensor or a TP-primitive-irreducible block to a primitive block decomposition appear in Chapter 13, and the block-injectivity statement in Chapter 8. The single statement recorded in this chapter is the normality of a TP-primitive irreducible block.

Theorem 9.32 (TP-primitive irreducible tensor is normal). [Lean](#) [✓ Stmt](#) *Let A be a left-canonical MPS tensor with $D > 0$, irreducible transfer map, and peripheral spectrum $\{1\}$. Then A is normal.*

Proof. [✓ Proof](#) Peripheral primitivity and irreducibility yield the spectral-gap condition. The PSD fixed point is positive definite by Theorem 6.4. Apply Theorem 8.76. $\square$

9.11.1 Basis of Normal Tensors

Theorem 9.33 (Sorting by strictly decreasing weight norms). *Let $(\mu_k)_{k=1}^r$ be nonzero complex weights whose moduli are pairwise distinct. Then there exists a permutation σ of $\{1, \dots, r\}$ such that*

$$|\mu_{\sigma(1)}| > |\mu_{\sigma(2)}| > \dots > |\mu_{\sigma(r)}| > 0.$$

Theorem 9.34 (Sorted decomposition preserves the represented MPS family). *Under the same hypothesis, with σ as above and $\tilde{\mu}_k := \mu_{\sigma(k)}$, $\tilde{B}_k := B_{\sigma(k)}$,*

$$\mathcal{V}(\bigoplus_k \mu_k B_k) = \mathcal{V}(\bigoplus_k \tilde{\mu}_k \tilde{B}_k), \quad |\tilde{\mu}_1| > \dots > |\tilde{\mu}_r|.$$

Theorem 9.35 (Normal canonical form after sorting). *Suppose $(\mu_k, B_k)_{k=1}^r$ satisfies $\mu_k \neq 0$, $|\mu_k|$ pairwise distinct, and each B_k is irreducible, left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), with primitive transfer map and $\dim B_k > 0$. Then after sorting by strictly decreasing $|\mu_{\sigma(k)}|$, the resulting family $(\tilde{\mu}_k, \tilde{B}_k)$ satisfies the normal canonical form conditions of Definition 9.18.*

Definition 9.36 (Granular sector decomposition). [Lean](#) [✓ Stmt](#) Given nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(B_k)_{k=1}^r$, the *granular sector decomposition* has r sectors, one basis block B_k per sector, multiplicity 1 in each sector, and sector weight μ_k on the k th sector. Equivalently, as a finite sum,

$$\sum_{k=1}^r \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma.$$

Lemma 9.37 (Granular sector decomposition preserves the represented family). [Lean](#) [✓ Stmt](#) *The granular sector decomposition associated to $(\mu_k, B_k)_k$ represents the same MPS family as the original weighted block sum $\bigoplus_k \mu_k B_k$.*

Proof. [✓ Proof](#) Expanding the sector decomposition gives the sum $\sum_k \mu_k^N V^{(N)}(B_k)_\sigma$, which is exactly the finite block-sum expression for $\bigoplus_k \mu_k B_k$. $\square$

Theorem 9.38 (Trivial sector decomposition for the sorted distinct-norm case). *If $\mu_k \neq 0$ for all k and $|\mu_1| > \dots > |\mu_r|$, then the granular sector decomposition $P = \text{trivialSectorDecomp}(\mu, B)$ satisfies $V^{(N)}(P) = V^{(N)}(\bigoplus_k \mu_k B_k)$ for every $N \geq 1$ and has exactly one copy per sector.*

Definition 9.39 (Norm-class partition). Let $(\mu_k)_{k=1}^r$ be a finite family of complex weights. A *norm-class partition* consists of: a number of classes $s \leq r$; representative values $\lambda_1 > \dots > \lambda_s >$

0; multiplicities $(m_j)_{j=1}^s$ with $\sum_j m_j = r$; for each j an enumeration $\iota_j : \{1, \dots, m_j\} \hookrightarrow \{1, \dots, r\}$ of the indices k with $|\mu_k| = \lambda_j$; and the regrouping identity

$$\sum_{k=1}^r f(k) = \sum_{j=1}^s \sum_{q=1}^{m_j} f(\iota_j(q))$$

for every function f .

Definition 9.40 (Norm-class enumeration in decreasing order). For any finite family $(\mu_k)_{k=1}^r$ of complex weights, sorting the distinct moduli $\{|\mu_k| : 1 \leq k \leq r, \mu_k \neq 0\}$ in strictly decreasing order yields a norm-class partition (in the sense of Definition 9.39) with representative values $\lambda_1 > \dots > \lambda_s$.

9.12 Open directions

Remark 9.41 (Canonical-form reduction before the equal-case comparison). ✓ Stmt The canonical-form proof in the MPS literature first separates the nilpotent part (which contributes only to the length-zero trace) from the nonzero part, decomposes the nonzero part into irreducible blocks, places each block in TP gauge, blocks each period- m block over m sites, and restricts to its cyclic spectral sectors. The sector tensors are trace-preserving, primitive, irreducible, and have positive bond dimension. For a finite nonzero block family one chooses a common multiple of all periods so that every sector is expressed over the same blocked physical alphabet [CPGSV17a, Section 2.3 and Appendix A].

The common-alphabet cyclic-sector family and the after-blocking zero-tail and common-sector transport statements are stated and proved in Chapter 13; the present chapter records only Theorems 9.29, 9.30, and 9.31. The sorted normal-canonical-form theorem (Theorem 9.35) assumes pairwise distinct nonzero weight norms; equal-norm families require the basis-of-normal-tensors and multiplicity comparison developed in Chapter 11.

9.12.1 Material from PGVWC06 not covered by this chapter

This chapter follows the presentation of [CPGSV17a] and [CPGSV21], which works with site-independent translation-invariant tensors, transfer-map primitivity, and CPM fixed-point normalization. The 2007 preprint [PGVWC07] contains eight named theorems in its canonical-form section §III (lines 399–1177 of `MPSarchive.tex`). Theorem `Th:TIcanonical` has a partial, conditional counterpart here; the opening paragraphs of this chapter make the scope limitation explicit. The remaining seven theorems of [PGVWC07] §III are not covered:

- **Thm. OBC-Vidal (lines 431–443): completeness and OBC canonical form.** Any pure state admits an OBC-MPS representation of bond dimension $D \leq d^{\lfloor N/2 \rfloor}$ with simultaneous left-normalization and a positive full-rank diagonal interleaving matrix $\Lambda^{[m]}$. *Not covered:* the chapter starts from a site-independent TI tensor; constructing the finite-OBC Schmidt canonical form requires a site-dependent Chain layer not present in this development.
- **Thm. free-OBC (lines 466–486): OBC representation freedom.** Every OBC-MPS $\{B_i^{[m]}\}$ is related to the canonical one by site-local invertible gauge matrices Y_j, Z_j with $Y_j Z_j = \mathbf{1}$. *Not covered:* same reason as OBC-Vidal; local-gauge analysis for site-dependent open-boundary matrices is outside the TI-MPS scope.

- **Thm. “Site-independent matrices” (lines 620–630).** Every TI pure state with PBC on a finite ring of N sites has a site-independent MPS representation; a bond- D OBC representation lifts to one at bond dimension at most ND . *Not covered:* the chapter assumes a site-independent MPV tensor as input; the block-circulant construction producing it is not covered.
- **Thm. Th:periodic (lines 849–858): periodic decomposition.** If the transfer map of a single-block TI canonical MPS has p eigenvalues of modulus 1, the state is a superposition of p p -periodic states of bond dimension D , and vanishes when $p \nmid N$. *Partially formalized:* the cyclic-sector machinery of Section 9.10 (Theorem 9.29) removes the period and yields primitive sector tensors, but the explicit state-vector decomposition into p -periodic summands is not separately stated.
- **Thm. “Interpretation of Λ ” (lines 987–993).** Under condition C2, the eigenvalues of the half-chain reduced density operator converge as $N \rightarrow \infty$ to $\Lambda \otimes \Lambda$ with Λ from Theorem Th:TIcanonical. *Not covered:* Theorem 9.23 produces Λ , but the asymptotic eigenvalue statement requires correlation-length and entropy material not addressed here.
- **Uniqueness of the canonical form ([PGVWC07, lines 1002–1015]).** Under condition C1 with unique OBC canonical form, two TI canonical D -MPS representations of the same state satisfy $B_i = UC_iU^\dagger$ for a global unitary U . *Not covered in this chapter.* A modern analogue is proved in Chapter 12 (global-gauge equality theorem); the original finite-ring uniqueness of [PGVWC07] under C1/C2 and $N > 2L_0 + D^4$ is not covered here.
- **Thm. “Obtaining the TI canonical form” (lines 1154–1165).** The site-independent $D \times D$ matrices can be recovered from an OBC canonical form by solving an N -independent system (S) of $O(D^8)$ quadratic equations; every solution is related to the canonical form by a global unitary. *Not covered:* this chapter addresses existence and uniqueness abstractly; numerical and algorithmic recovery is not formalized.

9.12.2 Building a BNT from normal blocks with normalized weights

The BNT predicate (Definition 9.3, formalized as `IsBNTCanonicalForm` in Chapter 11) bundles the basis-of-normal-tensors structure of [CPGSV17a] with multiplicity and span fields. The following theorem starts from a family of normalized block data and produces a sector decomposition satisfying that predicate.

Theorem 9.42 (BNT from normal blocks with normalized weights). Lean ✓ Stmt *Let $r \in \mathbb{N}$, let $\dim : \text{Fin } r \rightarrow \mathbb{N}$, let $\mu : \text{Fin } r \rightarrow \mathbb{C}$, and let $\text{blocks}_k : \text{MPSTensor } d(\dim k)$ for $k < r$. Assume:*

1. **positive bond dimensions:** $0 < \dim k$ for all k ,
2. **TP normalization:** each block satisfies $\sum_i (\text{blocks}_k^i)^\dagger \text{blocks}_k^i = \mathbb{1}$ (*IsLeftCanonical*),
3. **primitivity:** the transfer map $\mathcal{E}_{\text{blocks}_k}$ is primitive for each k ,
4. **irreducibility:** each block satisfies *IsIrreducibleTensor*,
5. **injectivity:** each block satisfies *IsInjective*,
6. **nonzero weights:** $\mu_k \neq 0$ for all k ,
7. **bounded weights:** $|\mu_k| \leq 1$ for all k ,

8. **global unit witness:** $\exists k, |\mu_k| = 1$.

Then there exists a *SectorDecomposition* P such that

- $P.toTensor$ and $toTensorFromBlocks$ μ blocks generate the same MPV family at every length (*SameMPV₂*), and
- P satisfies *IsBNTCanonicalForm*.

Proof. ✓ Proof The phase-class quotient collapses gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars (derived from the TP + primitive + irreducible hypotheses via the MPV phase-class infrastructure). The resulting sector decomposition inherits the BNT data fields from `exists_bnt_sectorDecomp_of_tp_primitive_irr_blocks`; the bounded-weight and global-unit hypotheses close the weight-norm fields of *IsBNTCanonicalForm* directly. $\square$

Remark 9.43 (Arbitrary-input direction: open formalization gap). Theorem 9.42 assumes eight pre-normalized hypotheses on the block data; it does not produce a sector decomposition starting from an arbitrary MPS tensor. Three independent obstacles separate it from the arbitrary-input statement.

1. **Zero-tail at $N = 0$.** The same-MPV equivalence used in the BNT predicate quantifies over all $N \geq 0$, so the $N = 0$ clause forces the bond dimension of the sector decomposition to equal the bond dimension of the input. The arbitrary-input reduction naturally yields equality only for $N \geq 1$ together with a separate length-zero dimension identity; matching the zero-tail bond dimension fails once all-zero blocks are removed by Theorem 13.3.
2. **Global weight normalization.** The BNT predicate requires $|\mu_k| \leq 1$ for all k and $|\mu_k| = 1$ for some k . These are normalization conventions, not consequences of an arbitrary tensor. If a left-canonical block C has unit weight, the scaled input $A = 2 \cdot C$ has weight of modulus 2, violating the bound. The hypotheses must therefore be supplied externally, or the statement weakened to a rescaled form.
3. **Per-block unit-modulus witness.** The BNT predicate requires $\forall j, \exists q, |\text{weight } j q| = 1$. For a decaying sector such as $C \oplus \frac{1}{2}D$, this fails: no weight of the D sector has unit modulus. The condition is therefore an explicit hypothesis rather than a consequence of the block decomposition.

Closing the gap for arbitrary tensors remains open.

9.12.3 Auxiliary lemmas

The lemmas below collect the small structural facts about transfer maps, blocking, and cyclic-sector reblocking that are used in the after-blocking constructions of Chapter 12 and Chapter 13 and in the periodic extension of Chapter 15.

Lemma 9.44 (Canonical form from peripheral primitivity). *Let $(\mu_k, A_k)_{k=1}^r$ be a family of nonzero weights and left-canonical irreducible blocks with positive bond dimensions, with strictly decreasing moduli $|\mu_1| > \dots > |\mu_r| > 0$ and $|\mu_1| = 1$, and assume each transfer map satisfies $\sigma_{\partial}(\mathcal{E}_{A_k}) = \{1\}$ (peripheral-spectrum primitivity). Then $(\mu_k, A_k)_{k=1}^r$ satisfies the canonical-form conditions of Definition 10.16. In particular, $O_{A_k A_k}(N) \rightarrow 1$ for every k .*

Lemma 9.45 (Adjoint primitivity equivalence). Lean ✓ Stmt Let $\mathcal{E}$ be a finite-dimensional completely positive map. Then $\mathcal{E}$ is primitive if and only if its Frobenius adjoint $\mathcal{E}^\dagger$ is primitive. Equivalently, $\sigma_\partial(\mathcal{E}) = \{1\}$ iff $\sigma_\partial(\mathcal{E}^\dagger) = \{1\}$, since the peripheral spectrum of $\mathcal{E}^\dagger$ consists of the complex conjugates of the peripheral spectrum of $\mathcal{E}$.

Lemma 9.46 (Primitivity under blocking by a multiple). Lean ✓ Stmt Let A be an MPS tensor with positive bond dimension and let $p_0, p \geq 1$ with $p_0 \mid p$. If $\mathcal{E}_{A[p_0]}$ is primitive, then $\mathcal{E}_{A[p]}$ is primitive.

Lemma 9.47 (Blocking the assembled weighted block sum). Lean ✓ Stmt For nonzero weights $(\mu_k)_{k=1}^r$, blocks $(B_k)_{k=1}^r$, and any blocking length $p \geq 1$, the blocked assembled tensor agrees with the assembled tensor of the blocked blocks with powered weights:

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]},$$

where $\sim$ denotes generation of the same MPV family at every length.

Lemma 9.48 (Nonzero weights survive blocking). Lean ✓ Stmt If $\mu \in \mathbb{C}$ is nonzero and $p \geq 1$, then $\mu^p \neq 0$. In particular, blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.

Lemma 9.49 (Primitive cyclic sectors of irreducible TP tensors). Lean ✓ Stmt Let A be a left-canonical irreducible MPS tensor with $D > 0$. Then there exist a period $m \geq 1$ and a family $(C_k)_{k=1}^m$ of left-canonical blocks with primitive transfer maps, tensor-irreducible structure, and positive bond dimensions such that $A^{[m]}$ generates the same MPV family as the unit-weight assembled tensor $\bigoplus_{k=1}^m C_k$.

Lemma 9.50 (Common reblocking of cyclic-sector families). Lean ✓ Stmt Given a finite family $(B_k)_{k=1}^r$ of left-canonical irreducible TP tensors with positive bond dimensions and per-block period-removal lengths $(m_k)_{k=1}^r$, there exists a common blocking length p such that, for every k , the blocked tensor $B_k^{[p]}$ admits a unit-weight assembled decomposition over a single common blocked alphabet $d^{[p]}$ into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.

Lemma 9.51 (Nonzero unit weight of the common reblocked family). Lean ✓ Stmt The unit weight on each sector of the common reblocked cyclic-sector family of Lemma 9.50 is nonzero.

Lemma 9.52 (Sector irreducibility for the adjoint blocked map). Lean ✓ Stmt Let A be a left-canonical irreducible MPS tensor with $D > 0$, let m be the period of the cyclic-sector decomposition of A , and let $(P_u)_{u=1}^m$ be the cyclic projections of the adjoint transfer map satisfying $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ and $\sum_u P_u = \mathbb{1}$. Then the restriction of $(\mathcal{E}_{A^\dagger})^m$ to each corner $P_u M_D(\mathbb{C}) P_u$ is irreducible.

Lemma 9.53 (Iterated blocking at the transfer-map level). Lean ✓ Stmt For any MPS tensor A and integers $m, n \geq 1$,

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

Definition 9.54 (Primitive irreducible cyclic-sector data). Lean ✓ Stmt A has primitive irreducible cyclic sectors of period $m \geq 1$ if there exists a family $(C_k)_{k=1}^m$ of MPS tensors such that $A^{[m]}$ generates the same MPV family as $\bigoplus_{k=1}^m C_k$ (unit weights), and each C_k is left-canonical, has primitive transfer map, is tensor-irreducible, and has positive bond dimension.

Lemma 9.55 (Common reblocking at a prescribed multiple). Lean ✓ Stmt With hypotheses as in Lemma 9.50, and given any integer p that is a common multiple of the per-block period-removal lengths (m_k) , the common reblocking can be performed at the prescribed length p rather than only at some sufficiently large length.

Lemma 9.56 (Structural properties of the common reblocked sectors). Lean ✓ Stmt Each sector tensor in the common reblocked cyclic-sector family of Lemma 9.50 is trace-preserving ($\sum_i (B^i)^\dagger B^i = 1$), has primitive transfer map ($\sigma_\partial(\mathcal{E}_B) = \{1\}$), is tensor-irreducible, and has positive bond dimension.

Lemma 9.57 (Common-alphabet flattening). Lean ✓ Stmt With $p = m_k e_k$ for each k , the iterated blocked tensor $(B_k^{[m_k]})^{[e_k]}$ agrees with the directly blocked tensor $B_k^{[p]}$ in the same-MPV sense, after the canonical identification of the p -blocked alphabet $d^{[p]}$ with the e_k -fold tensor power of the m_k -blocked alphabet $d^{[m_k]}$.

Lemma 9.58 (Reindexed nonzero-part transport). Lean ✓ Stmt The nonzero-part decomposition of a weighted block sum $\bigoplus_k \mu_k B_k$ transports through the common-alphabet identification of Lemma 9.57: the common-alphabet sectors have the same nonzero-part decomposition as the direct p -blocked tensors.

Definition 9.59 (Coordinate-grouping predicate). Lean ✓ Stmt The canonical identification of the common-alphabet sectors with the iterated alphabet $(d^{[m_k]})^{e_k}$ agrees with grouping any direct word of length $p = m_k e_k$ into e_k consecutive words of length m_k .

Lemma 9.60 (Digit-level flattening cast). Lean ✓ Stmt For $p = mn$, the canonical word-flattening cast identifies a p -letter word with the concatenation of n consecutive m -letter words: the two encodings of a word of length p over $d^{[p]}$ and over $(d^{[m]})^n$ are equal under the canonical alphabet identification.

Lemma 9.61 (Direct vs. iterated blocking comparison). Lean ✓ Stmt Direct blocking at length $p = m_k e_k$ and iterated blocking $(B_k^{[m_k]})^{[e_k]}$ produce, after the canonical alphabet identification, the same weighted finite sum: for every $N \geq 0$ and every blocked word σ ,

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}((B_k^{[m_k]})^{[e_k]})_{\sigma'},$$

where σ' is the image of σ under the coordinate-grouping identification.

Chapter 10

Block Permutation and Separation

This chapter develops the algebra of block permutations, the product-algebra maps built from per-block same-MPV data, the invariant-projection splitting step that reduces a tensor to smaller blocks, and the block-separation argument that recovers the individual blocks from a block-diagonal same-MPV hypothesis. The block-separation results in this chapter are conditional same-structure lemmas: they assume blockwise normalization and separation hypotheses, including strict ordering of the weight moduli when required below. They should not be read as a formalization of the translation-invariant canonical-form uniqueness theorem of [PGVWC07, Theorem thm-uniq, lines 1002–1015], which requires injectivity at some finite blocking length, uniqueness of the OBC canonical representation, and a finite-size lower bound. The source theorem is stated in the cited PGVWC07 passage; the project-level scope distinction is recorded in the accompanying paper-gap note.¹

10.1 Block permutation algebra

Definition 10.1 (Block ideal). Lean ✓ Stmt Let $\{R_k\}_{k=1}^r$ be simple rings. The k -th *block ideal* in $\prod_{j=1}^r R_j$ is

$$I_k := \{x \in \prod_{j=1}^r R_j : x_j = 0 \text{ for } j \neq k\}.$$

Theorem 10.2 (Each block ideal is an atom). Lean ✓ Stmt Each block ideal I_k is an atom in the lattice of two-sided ideals of $\prod_{j=1}^r R_j$.

Proof. ✓ Proof Under the order isomorphism between two-sided ideals of $\prod_{j=1}^r R_j$ and products of two-sided ideal lattices, I_k corresponds to the tuple with $\top$ in the k -th coordinate and $\perp$ elsewhere. This tuple is an atom, hence so is I_k . $\square$

Theorem 10.3 (Ring isomorphisms permute block ideals). Lean ✓ Stmt Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \rightarrow \prod_k R_k$ permutes the block ideals: there exists a permutation π such that T maps the k -th block ideal to the $\pi(k)$ -th block ideal.

¹docs/paper-gaps/pgvwc07_ti_uniqueness_scope.tex.

Proof. ✓ Proof By Theorem 10.2, each block ideal is an atom in the lattice of two-sided ideals of $\prod_k R_k$. A ring isomorphism induces an order automorphism of this lattice, so it permutes the atoms. Since the atoms are exactly the block ideals, this gives the required permutation π . $\square$

Remark 10.4 (Ring level versus algebra level). Theorem 10.3 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorem 10.6 stays at this ring level: once the block ideals are permuted, one can extract the corresponding single-block maps without using scalar multiplication. Theorems 10.5 and 10.7 require an upgrade to $\mathbb{C}$ -algebra automorphisms. The reason is threefold: dimension preservation compares the blocks as $\mathbb{C}$ -vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$ -algebras; and the final per-block maps must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

Theorem 10.5 (Dimension preservation). Lean ✓ Stmt Assume $D_k \geq 1$ for all k . If T is an algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and π is the induced permutation, then $D_{\pi(k)} = D_k$ for all k .

Proof. ✓ Proof The restriction of T to the k -th block ideal induces a $\mathbb{C}$ -algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$, which forces $D_k = D_{\pi(k)}$ by comparing dimensions as $\mathbb{C}$ -vector spaces. $\square$

Theorem 10.6 (Per-block component maps are bijections). Lean ✓ Stmt Assume $D_k \geq 1$ for all k . Let T be a ring automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and let π be the induced permutation from Theorem 10.3. For each block index k , the map

$$M \mapsto (T(0, \dots, 0, M, 0, \dots, 0))_{\pi(k)}$$

from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ is bijective.

Proof. ✓ Proof Theorem 10.3 forces a single-support element in the k -th block ideal to remain supported on the $\pi(k)$ -th block after applying T . Injectivity follows from injectivity of T . For surjectivity, apply the same argument to T^{-1} and the inverse permutation. $\square$

Theorem 10.7 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$). Lean ✓ Stmt Assume $D_k \geq 1$ for all k . Every $\mathbb{C}$ -algebra automorphism T of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$ decomposes as a block permutation π followed by per-block inner automorphisms: there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that, for every tuple $M = (M_1, \dots, M_r)$, $T(M)_k = X_k M_{\pi(k)} X_k^{-1}$, where $T(M)_k$ denotes the k -th block component of $T(M)$.

Proof. ✓ Proof Theorem 10.3 gives the permutation π . Theorem 10.5 identifies the matrix sizes $D_{\pi(k)} = D_k$. For each k , Theorem 10.6 gives a bijective multiplicative map from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ extracted from the elements supported on one factor of T ; after reindexing by the dimension equality, this is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$. Skolem–Noether (Theorem 3.9) makes each such automorphism inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, Section IV.A] for the MPS context. $\square$

10.2 Product algebra linear extension

Definition 10.8 (Per-block linear extension). Lean ✓ Stmt Given injective tensors A_k, B_k of bond dimension D_k with the same MPV for each block k , the *per-block linear extension* T_k is the unique linear map $M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for all i . It exists (and is unique) because $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$ by injectivity.

Definition 10.9 (Product algebra automorphism). [Lean](#) [✓ Stmt](#) The *product algebra automorphism* is the componentwise map $T : \prod_k M_{D_k}(\mathbb{C}) \rightarrow \prod_k M_{D_k}(\mathbb{C})$ defined by $T(M)_k := T_k(M_k)$. Theorem 10.10 shows that each T_k is a $\mathbb{C}$ -algebra homomorphism and that this map is a $\mathbb{C}$ -algebra automorphism.

Theorem 10.10 (Per-block linear extensions define a product algebra automorphism). [Lean](#) [✓ Stmt](#) Let A_k and B_k be injective tensors of bond dimension D_k for $k = 1, \dots, r$, and suppose the per-block same-MPV condition holds. Then the per-block linear extensions $T_k : A_k^i \mapsto B_k^i$ define an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$, which decomposes as a permutation π with $D_{\pi(k)} = D_k$ and per-block conjugation matrices X_k .

Proof. [✓ Proof](#) Each T_k exists uniquely and is multiplicative by Theorems 3.6 and 3.7. Promotion to algebra homomorphisms (Lemma 3.10) and componentwise composition give an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$. Theorem 10.7 then decomposes this automorphism as a block permutation together with per-block inner automorphisms. $\square$

Lemma 10.11 (Per-block same MPV gives global gauge equivalence). [Lean](#) [✓ Stmt](#) If all block tensors A_k are injective, and B_k are block tensors of the same bond dimensions such that per-block same-MPV holds, then each pair (A_k, B_k) is gauge equivalent, and the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are globally gauge equivalent.

Proof. [✓ Proof](#) Per-block gauge equivalence follows from the single-block FT (Theorem 3.11) applied to each block. Global gauge equivalence follows by placing the per-block gauge matrices on the block diagonal (Theorem 12.56). $\square$

10.3 Burnside and invariant projections

Lemma 10.12 (The algebra span reaches a finite cumulative span). [Lean](#) [✓ Stmt](#) If the generated unital algebra satisfies $\text{alg}(A) = M_D(\mathbb{C})$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Every element of $\text{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n \geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \dots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. $\square$

Lemma 10.13 (Irreducible action gives an irreducible tensor). [Lean](#) [✓ Stmt](#) If the letters of A act irreducibly on $\mathbb{C}^D$, then A has no nontrivial invariant orthogonal projection.

Proof. [✓ Proof](#) If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A . Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. $\square$

Theorem 10.14 (Invariant projection gives a two-block decomposition). [Lean](#) [✓ Stmt](#) Suppose P is an orthogonal projection satisfying, for every physical index i ,

$$(\mathbb{1} - P)A^iP = 0.$$

Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. ✓ **Proof** Diagonalize P by a unitary U . Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem 9.6. In this basis the relation $(\mathbb{1} - P)A^i P = 0$ makes each letter upper block triangular. Discarding the off-diagonal blocks does not change the trace of any word product, so the resulting block-diagonal tensor has the same MPV coefficients. Reindex the two diagonal blocks by $\{0, \dots, n-1\}$ and $\{0, \dots, m-1\}$. $\square$

Theorem 10.15 (Nontrivial invariant projection gives strict two-block reduction). Lean ✓ **Stmt** Under the hypotheses of Theorem 10.14, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2 , such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.12.

Proof. ✓ **Proof** Apply Theorem 10.14. In the diagonal form of P , the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. $\square$

10.4 Block separation

Recall from Definition 2.46 and Theorem 7.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \text{Tr}(F_{AB}^N).$$

We use this notation throughout the block-separation argument.

Definition 10.16 (Canonical form conditions). Lean ✓ **Stmt** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form conditions* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \rightarrow 1$ for each k .

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Theorem 6.7, item 3]), $T^n(\rho) \rightarrow \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Theorem 6.8] for the completely-positive characterisation and Chapter 7 for the spectral-gap proof.

Remark 10.17 (Canonical-form theorem versus local predicates). The translation-invariant canonical-form theorem of [PGVWC07, Theorem Th:TIcanonical, lines 742–763] starts from an arbitrary TI-MPS representation and constructs a block canonical form with positive weights, unital block normalization, full-rank diagonal invariant densities, and uniqueness of the identity fixed point. Definition 10.16 is a local predicate used after such structure has already been supplied or separately proved. In particular, it assumes injectivity and the self-overlap limit, and the block-separation lemmas below additionally require strict separation of the weight moduli. These results are therefore conditional consequences, not the source canonical-form existence or uniqueness theorems.

Remark 10.18 (Normalization convention). The convention here differs from [PGVWC07, Theorem 4], which uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$. This chapter instead works in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides a gauge transformation between these two normalizations (Theorem 6.12).

Remark 10.19 (Overlap hypothesis). In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum = $\{1\}$). Theorem 9.44 gives this implication: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form conditions retain condition (5) explicitly because it can be verified independently of primitivity.

Remark 10.20 (Normal-canonical companion). Definition 10.16 gives the block-injective conditions. The normal-canonical analogue instead uses Definition 9.18: injectivity and the explicit self-overlap hypothesis are replaced by irreducibility, TP normalization, and primitive transfer maps. Theorem 9.19 then supplies the self-overlap limit automatically.

Lemma 10.21 (Repeated-word evaluation). Lean Stmt For any word w and any $L \geq 0$, the evaluation of A on the L -fold concatenation w^L is $(A^w)^L$.

Proof. Proof By induction on L , using the multiplicativity of word evaluation under concatenation. $\square$

Lemma 10.22 (Block separation core). Lean Stmt Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. Assume moreover that the weight moduli are strictly ordered:

$$|\mu_1| > |\mu_2| > \dots > |\mu_r|.$$

If $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all N, σ , then each pair (A_k, B_k) generates the same MPV family [CPGSV16, Appendix AppendixMPV, lines 1054–1192].

Proof. Proof By induction on the number of blocks r . In the inductive step, take bilinear overlaps with the leading block A_0 : from $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ one obtains

$$\sum_k \bar{\mu}_k^{-N} (O_{A_0 A_k}(N) - O_{A_0 B_k}(N)) = 0.$$

After division by $\bar{\mu}_0^N$, the contributions from $k \geq 1$ decay because $|\mu_k/\mu_0| < 1$ (strict weight ordering) and the bilinear overlaps $O_{A_0 A_k}(N)$ and $O_{A_0 B_k}(N)$ are bounded (Theorems 7.23 and 7.25 supply the relevant bounds), leaving $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \rightarrow 0$. Condition (5) gives $O_{A_0 A_0}(N) \rightarrow 1$, hence $O_{A_0 B_0}(N) \rightarrow 1$. Since $O_{A_0 B_0}(N) \rightarrow 1 \neq 0$, the bond dimensions must match: $D_{A_0} = D_{B_0}$; otherwise Theorem 7.23 would give $O_{A_0 B_0}(N) \rightarrow 0$, a contradiction. With equal bond dimensions established, the contrapositive of Theorem 7.25 forces A_0 and B_0 to be gauge-phase equivalent; the limit $O_{A_0 B_0}(N) \rightarrow 1$ then makes the phase trivial, so A_0 and B_0 generate the same MPV family. Peeling off the leading block and dividing by μ_0^N , the induction hypothesis applies. $\square$

Lemma 10.23 (Block separation from canonical form). Lean Stmt Let $((\mu_k), (A_k))$ satisfy the canonical form conditions, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. Assume moreover that the weight moduli are strictly ordered:

$$|\mu_1| > |\mu_2| > \dots > |\mu_r|.$$

If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family [CPGSV16, Appendix AppendixMPV, lines 1054–1192].

Proof. ✓ Proof Apply Lemma 10.22 to the identity obtained from the equality of the two block-diagonal MPV families. The induction peels off the dominant block and treats the remainder as a decaying tail. Concretely, one isolates the leading term $\mu_1^N V(A_1)$ and bounds the remaining contribution by $O(|\mu_2/\mu_1|^N)$ after dividing by μ_1^N . The canonical form conditions supply injectivity, TP normalization, strict nonzero weights, and self-overlap convergence for (A_k) , while the hypotheses on (B_k) supply the comparison injectivity and TP normalization. $\square$

Lemma 10.24 (Block separation from normal canonical form). Lean ✓ Stmt *Let $((\mu_k), (A_k))$ satisfy the normal canonical form conditions, and let (B_k) be irreducible tensors of the same bond dimensions, satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. Assume moreover that the weight moduli are strictly ordered:*

$$|\mu_1| > |\mu_2| > \dots > |\mu_r|.$$

If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family [CPGSV16, Appendix AppendixMPV, lines 1054–1192].

Proof. ✓ Proof The proof follows the same peeling argument as Lemma 10.23, with Theorem 7.28 (modulus-one eigenvalue rigidity for irreducible tensors) replacing the injective overlap-decay argument at the identification step, since the blocks are irreducible rather than injective. Theorem 9.19 replaces the explicit self-overlap hypothesis on the leading block. Since both block families already use the same dimension function, the conclusion is direct per-block same MPV. $\square$

Lemma 10.25 (Canonical-form gauge comparison, same-structure case). Lean ✓ Stmt *Under the hypotheses of Lemma 10.23, including strict ordering of the weight moduli, each pair (A_k, B_k) is gauge equivalent. Hence the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent. This is a same-structure local comparison theorem, not the full PGVWC07 uniqueness theorem [PGVWC07, Theorem thm-uniq, lines 1002–1015].*

Proof. ✓ Proof Lemma 10.23 gives per-block same MPV. Lemma 10.11 then converts this to per-block gauge equivalence and then to global gauge equivalence of the block-diagonal tensors. $\square$

Remark 10.26 (Relation to the literature). The block-permutation part follows the decomposition of automorphisms of semisimple matrix algebras used in [CPGSV21, Section IV.A]. The separation part is the local strict-weight peeling argument used in [CPGSV16, Appendix AppendixMPV, lines 1054–1192]; the checked proof here formulates it as a same-structure block-separation lemma using overlap bounds and mixed-transfer decay.

Chapter 11

Basis of Normal Tensors

This chapter introduces the *basis of normal tensors* (BNT) notion, the canonical-form variants with BNT separation, the cross-overlap decay theorem, and the permutation-rigidity arguments used in the proof of the Fundamental Theorem. The chapter concludes with auxiliary results on coefficient convergence and Newton–Girard identities. The main references are [PGVWC07], [CPGSV16], and [CPGSV21].

11.1 Basis of normal tensors

Definition 11.1 (Basis of Normal Tensors (BNT)). [Lean](#) [✓ Stmt](#) A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each A_j is normal (Definition 2.29),
2. at each system size N , the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\rangle_{j=1}^g$,
3. for sufficiently large N , the block MPV vectors $\{|V^{(N)}(A_j)\rangle\rangle_{j=1}^g$ are linearly independent.

This is [CPGSV21, Definition 4.2]. In the canonical-form characterization of [CPGSV16, Section 2.3 and Appendix A], a BNT is a minimal family of representatives for the relation

$$\tilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$$

among the normal blocks in the canonical form.

Lemma 11.2 (Finite-index overlap-orthonormality implies eventual linear independence). [Lean](#) [✓ Stmt](#) Let I be a finite index set and let $(A_i)_{i \in I}$ be a finite family of tensors such that $O_{A_i A_i}(N) \rightarrow 1$ for each i , and $O_{A_i A_j}(N) \rightarrow 0$ whenever $i \neq j$. Then the MPV states $\{|V^{(N)}(A_i)\rangle\rangle_{i \in I}$ are linearly independent for all sufficiently large N .

Proof. [✓ Proof](#) The Gram matrix of the family $\{|V^{(N)}(A_i)\rangle\rangle_{i \in I}$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. $\square$

Lemma 11.3 (Overlap-orthonormality implies eventual linear independence). [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^g$ be a finite family of tensors such that $O_{A_j A_j}(N) \rightarrow 1$ for each j and $O_{A_i A_j}(N) \rightarrow 0$ for $i \neq j$. Then the MPV states $\{|V^{(N)}(A_j)\rangle\rangle_{j=1}^g$ are linearly independent for all sufficiently large N .

Proof. [✓ Proof](#) Apply Lemma 11.2 with $I = \{1, \dots, g\}$. □

Lemma 11.4 (Eventual coefficient extraction from linear independence). [Lean](#) [✓ Stmt](#) *If a finite family of vectors is linearly independent for all sufficiently large N , and two finite linear combinations of that family are equal for all sufficiently large N , then the corresponding coefficients are equal for all sufficiently large N .*

Proof. [✓ Proof](#) Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length. □

Lemma 11.5 (Overlap-orthonormality for two families). [Lean](#) [✓ Stmt](#) *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap $O_{A_j B_k}(N)$ tends to 0. Then the combined MPV family*

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large N .

Proof. [✓ Proof](#) Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks. □

Definition 11.6 (No gauge-phase-equivalent distinct blocks). [Lean](#) [✓ Stmt](#) A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

Definition 11.7 (Canonical form with BNT separation). [Lean](#) [✓ Stmt](#) A family is in *canonical form with BNT separation* if it satisfies Definition 10.16 and, in addition:

1. the moduli are *strictly* decreasing: $|\mu_1| > |\mu_2| > \dots > |\mu_r|$ (strengthened from the non-increasing ordering of Definition 10.16),
2. distinct blocks of the same bond dimension are *not* gauge-phase equivalent.

This is an already grouped special case. In the source BNT expansion of [CPGSV16, Section 2.3], repeated copies of a selected basis tensor appear as

$$A^i = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1}.$$

The coefficients $\mu_{j,q}$, multiplicities r_j , and gauges $X_{j,q}$ are not part of this restricted single-representative definition. This definition therefore records only strict weight ordering and the absence of phase-gauge equivalence among the selected blocks. It is not the source notion of block-injective canonical form, which is the fixed-length direct-sum span condition discussed in Chapter 8 and Chapter 12.

Theorem 11.8 (Distinct bond dimensions force BNT separation). [Lean](#) [✓ Stmt](#) *If a canonical-form family has strictly decreasing weight moduli and pairwise distinct bond dimensions, then it is in canonical form with BNT separation.*

Proof. [✓ Proof](#) The only new clause in Definition 11.7 is the prohibition on gauge-phase equivalence between distinct blocks of the same bond dimension. When the bond dimensions are pairwise distinct, this clause is vacuous. □

Definition 11.9 (Normal canonical form with BNT separation). [Lean](#) [✓ Stmt](#) A family is in *normal canonical form with BNT separation* if it satisfies Definition 9.18 and, in addition:

1. the moduli are strictly decreasing (strengthened from non-increasing),
2. distinct blocks of the same bond dimension are not gauge-phase equivalent,
3. the dominant block has unit modulus, $\|\mu_1\| = 1$ ([CPGSV16, paragraph after eq:II_CF1]).

As in Definition 11.7, the strict ordering describes an already selected or grouped representative family. It is not the source BNT expansion

$$A^i = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1},$$

where repeated copies of the same basis tensor remain visible through their coefficients, multiplicities, and gauges.

Lemma 11.10 (Restricted norm-class collapse). *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family with $\mu_k \neq 0$ for every k . Assume that whenever $\|\mu_j\| = \|\mu_k\|$, the tensors A_j and A_k generate the same MPV family. Then one can regroup the blocks into a sector decomposition whose associated total tensor generates the same MPV family as $\bigoplus_k \mu_k A_k$, and whose sector-weight norms are strictly decreasing. This is a special-case regrouping lemma, not the full BNT characterization of [CPGSV16, Section 2.3]: the hypothesis that equal-norm blocks already generate the same MPV family is an assumption here, while the source BNT construction chooses a minimal family of representatives for the phase-gauge equivalence classes of normal blocks.*

Proof. Order the distinct norms of the weights in decreasing order, choose one representative block from each norm class, and retain the multiplicities inside each class as sector copies. The equal-MPV hypothesis lets every block in a class be replaced by the chosen representative, so the regrouped decomposition has the same MPV family. By construction, the sector-weight norms are strictly decreasing. $\square$

Theorem 11.11 (Cross-overlap decay from explicit BNT hypotheses). [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^r$ be injective and trace-preserving normalized. Assume that distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \rightarrow 0$ for every pair $j \neq k$.

Proof. [✓ Proof](#) If the bond dimensions differ, Theorem 7.23 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 7.25 gives the decay. $\square$

Theorem 11.12 (Cross-overlap decay for BNT-separated blocks). [Lean](#) [✓ Stmt](#) If $((\mu_k), (A_k))$ is in canonical form with BNT separation and $j \neq k$, then $O_{A_j A_k}(N) \rightarrow 0$.

Proof. [✓ Proof](#) Definition 11.7 supplies injectivity, left-canonical normalization, and the BNT separation hypothesis, so Theorem 11.11 applies directly. $\square$

Lemma 11.13 (BNT from explicit blockwise hypotheses). [Lean](#) [✓ Stmt](#) Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume that the blocks are injective and trace-preserving normalized, that $O_{A_k A_k}(N) \rightarrow 1$ for each k , and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.

Proof. ✓ Proof Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 11.11 supplies the off-diagonal decay. Therefore Lemma 11.3 gives the eventual linear independence required in Definition 11.1. $\square$

Lemma 11.14 (Restricted CF-BNT hypotheses yield a basis of normal tensors). Lean ✓ Stmt *If $((\mu_k), (A_k))$ is in canonical form with BNT separation, then the block tensors form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. ✓ Proof Definition 11.7 provides injective blocks, left-canonical normalization, normalized self-overlaps, and BNT separation. Hence Lemma 11.13 applies. $\square$

Remark 11.15 (Comparing the two normality conditions). Definition 11.9 references the normal canonical form conditions (Definition 9.18), which encode normality via the spectral characterization (primitive transfer map). Definition 11.1 requires normality via the algebraic characterization (eventual block injectivity, Definition 2.29). These are equivalent by Remark 2.20, but the equivalence must be invoked explicitly: the Wielandt theory of Chapter 8 provides the implication from primitivity to eventual block injectivity needed here.

Lemma 11.16 (Restricted normal-CF-BNT hypotheses give a BNT). Lean ✓ Stmt *If a family is in normal canonical form with BNT separation and each block is also known to be normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor.*

Proof. ✓ Proof The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 11.1. $\square$

11.2 Permutation rigidity for bases of normal tensors

Lemma 11.17 (Invertible change of basis). Lean ✓ Stmt *Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \text{GL}_g(\mathbb{C})$ such that*

$$w_j = \sum_{i=1}^g U_{ij} v_i$$

for every j .

Proof. ✓ Proof Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^g$; these coefficients form the columns of U . If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. $\square$

Lemma 11.18 (Eventual change of basis from equal spans). Lean ✓ Stmt *If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size N , then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.*

Proof. ✓ Proof By Lemma 11.3, both families are linearly independent for all sufficiently large N . At such an N , the span equality lets us apply Lemma 11.17. $\square$

Theorem 11.19 (Permutation rigidity for equal block counts). [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be injective families satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size N , then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.

Proof. [✓ Proof](#) For large N , Lemma 11.18 gives an invertible matrix U_N relating the two MPV families. Fix j . If every mixed overlap $O_{A_i B_j}(N)$ tended to 0, then taking inner products of the relation with the A_i and using that the Gram matrix of the A -family tends to the identity would force the j th column of U_N to vanish asymptotically, contradicting the self-overlap limit of B_j . Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 7.23 and 7.25 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B -family makes f injective, hence bijective, and its inverse is the required permutation. $\square$

Lemma 11.20 (Permutation rigidity with asymptotically orthonormal overlaps). [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size N ,

$$\text{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \text{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j .

Proof. [✓ Proof](#) By Lemma 11.3, both families are linearly independent for all sufficiently large N . At such an N , the common span has the same dimension when computed from the A -family or the B -family, so $g_A = g_B$. After identifying the two index sets, Theorem 11.19 yields the permutation conclusion. $\square$

Theorem 11.21 (Non-equivalent irreducible TP blocks are eventually non-proportional). [Lean](#) [✓ Stmt](#) Let A and B be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound N_0 there is $N \geq N_0$ such that $V^{(N)}(A)$ is not proportional to $V^{(N)}(B)$. Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.

Proof. [✓ Proof](#) If the two MPV states were proportional for all sufficiently large N , the irreducible trace-preserving overlap decay theorem would force gauge-phase equivalence. This contradicts the hypothesis. $\square$

11.3 Coefficient convergence

Lemma 11.22 (Strict-dominance coefficient ratios). [Not ready](#) Let $(\mu_k)_k$ be a strictly ordered nonzero weight family with μ_0 dominant. Then the normalized powers $(\mu_k/\mu_0)^N$ converge to 1 for the dominant block and to 0 for every other block.

Proof. [Not ready](#) The dominant ratio is 1. For every other block, strict ordering gives $|\mu_k/\mu_0| < 1$, so its powers converge to 0. $\square$

Lemma 11.23 (Dominant-weight normalization). Not ready Factoring the dominant weight from a block-diagonal tensor rewrites its MPV as μ_0^N times the MPV of the normalized block family. Hence a proportionality relation between two such tensors induces the corresponding proportionality relation between their normalized block families, with the scalar adjusted by the dominant-weight ratio.

Proof. Not ready Expand the MPV of the block-diagonal tensor as the sum over blocks and factor μ_0^N from each coefficient. The proportionality statement follows by dividing by the nonzero dominant power. $\square$

Remark 11.24 (Strict-dominance regime). Not ready The coefficient-ratio decay auxiliary applies when one modulus is strictly dominant: $|\mu_0| > |\mu_k|$ for $k \geq 1$. In that restricted situation the ratios $(\mu_k/\mu_0)^N$ tend to zero, and the MPV of the normalized block family gives the corresponding proportional comparison after factoring out the leading weight. In the more general periodic setting of [CPGSV21, Equation (72a)], the coefficient attached to block j can take the form $\sum_q \mu_{j,q}^N$, where the $\mu_{j,q}$ are the individual weights contributed by that block. The oscillatory case with $|\mu_{j,q}| = 1$ is treated separately in the periodic comparison theorem.

11.4 Newton–Girard identities

Theorem 11.25 (Newton–Girard trace recursion). Lean Lean ✓ Stmt If $A, B \in M_n(\mathbb{C})$ satisfy $\text{tr}(A^k) = \text{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial. It is enough to assume this equality for $1 \leq k \leq n$.

Proof. ✓ Proof Let e_m denote the m th elementary symmetric polynomial in the eigenvalues, with $e_0 = 1$. The Newton–Girard recursion

$$me_m = \sum_{i=1}^m (-1)^{i-1} e_{m-i} \text{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces $\text{tr}(A^i)$. Equal power sums for A and B therefore give equal coefficients, hence equal characteristic polynomials. $\square$

Remark 11.26 (How the power-sum equality is used). The equal-MPV corollary in the source compares scalar power sums only after the BNT representatives have already been matched. If the matched basis tensors satisfy $B_j = e^{i\phi_j} Y_j A_j Y_j^{-1}$, then for every positive length N the coefficient $\sum_q \mu_{j,q}^N$ of $|V^{(N)}(A_j)\rangle$ in the BNT expansion of A equals the coefficient $\sum_q (\nu_{j,q} e^{i\phi_j})^N$ coming from B . This scalar equality is the input to the Newton–Girard step.

Thus the power-sum lemma determines the multiset of weights with multiplicity, but it is not a replacement for the BNT matching argument. The block permutation and the phase gauges must first identify which basis tensor is being compared. Once that matching is available, the power-sum lemma supplies the multiplicity and weight equality needed to identify the weighted block-diagonal gauge.

The finite-range statement below includes the same-cardinality form, the nonzero-entry unequal-cardinality form, and the positive-power conclusion after deleting zeros. The nonzero-entry form pads the shorter family by zeros, applies the same-cardinality result at length $\max(m, n)$, and then uses the nonzero-entry hypothesis to count the padded zeros. The Appendix lemma in the source fixes an ordering convention for the two families; here the conclusion is stated as multiset equality, so no sorting hypothesis is needed. If equality at exponent 0 is also assumed, the nonzero-entry hypothesis can be omitted because $\sum_k \lambda_{a,k}^0 = x_a$ and $\sum_k \lambda_{b,k}^0 = x_b$.

Theorem 11.27 (Finite-range equal power sums imply equal multisets). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for $1 \leq k \leq n$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal. More generally, if $\alpha : \{0, \dots, m-1\} \rightarrow \mathbb{C}$ and $\beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$ have no zero entries and their power sums agree for $1 \leq k \leq \max(m, n)$, then $m = n$ and the two multisets are equal. There is also an exponent-zero list form: for lists $(\lambda_{a,k})_{k=1}^{x_a}$ and $(\lambda_{b,k})_{k=1}^{x_b}$, the hypothesis $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$ for $1 \leq N \leq \max(x_a, x_b)$ implies

$$\{\lambda_{a,k} \mid \lambda_{a,k} \neq 0\}_{k=1}^{x_a} = \{\lambda_{b,k} \mid \lambda_{b,k} \neq 0\}_{k=1}^{x_b}$$

as multisets. If the equality is also assumed at $N = 0$, then the original lists have the same multiset.

Proof. [✓ Proof](#) Apply the finite-range Newton–Girard theorem to $\text{diag}(\alpha)$ and $\text{diag}(\beta)$, then extract the roots of $\prod_i (X - \alpha_i)$ and $\prod_i (X - \beta_i)$. For unequal cardinalities, pad both lists with zeros to length $\max(m, n)$. The equal-root conclusion compares the padded multisets, and the nonzero-entry hypothesis makes the number of padded zeros exactly $\max(m, n) - m$ and $\max(m, n) - n$, forcing $m = n$. Without a nonzero-entry hypothesis, first delete the zero entries; for positive N , this does not change $\sum_k \lambda_k^N$. The nonzero-entry form then compares the remaining multisets. In the exponent-zero list form, the equality at $N = 0$ gives $x_a = x_b$, so the same-cardinality finite-range theorem applies directly. $\square$

Theorem 11.28 (Equal power sums imply equal multisets). [Lean](#) [✓ Stmt](#) Let $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.

Proof. [✓ Proof](#) Construct diagonal matrices $\text{diag}(\alpha)$ and $\text{diag}(\beta)$. The power-sum hypothesis gives $\text{tr}(\text{diag}(\alpha)^k) = \text{tr}(\text{diag}(\beta)^k)$. By Theorem 11.25, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. $\square$

Remark 11.29. Theorems 11.25 and 11.28 remain relevant for the equal-MPV source corollary discussed in Section 12.2, where the paper uses a power-sum step to recover the multiplicities r_j and the weights $\mu_{j,q}$. The BNT multiplicity lemmas handle this recovery directly rather than through a separate explicit-coefficient equal-case theorem.

11.5 The BNT canonical form on a sector decomposition

This section introduces the BNT canonical-form predicate following [CPGSV17a, §II, lines 287–301] and [CPGSV21, Definition 4.3, lines 1846–1884]. The predicate is defined on a sector decomposition with raw sector weights $\mu_{j,q}$ and sector coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$; no equal-modulus factorization is assumed.

Definition 11.30 (BNT canonical form). [Lean](#) [✓ Stmt](#) Given a sector decomposition P , the BNT canonical-form predicate records the minimal BNT data of [CPGSV17a, CGPSV21] without any equal-modulus or strict-ordering condition on the raw sector weights. It asserts that each basis block has positive bond dimension, is injective, irreducible, and left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent in the cast-compatible shape; and the [CPGSV17a, §II.A, line 246] normalization holds,

$$\|\mu_{j,q}\| \leq 1 \text{ for every } (j, q), \quad \exists (j_*, q_*) : \|\mu_{j_*, q_*}\| = 1.$$

The MPV expansion takes the raw two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^g \left(\sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma,$$

matching [CPGSV17a, § II, lines 271–301] and [CPGSV21, Definition 4.3, lines 1846–1884]. The line-246 normalization admits every example of the source paper (in particular $C \oplus D$, $C \oplus (-C)$, $C \oplus (1/2)C$, and $C \oplus (-C) \oplus (1/2)C$) and is not derivable from the per-block normality data alone.

Lemma 11.31 (Raw two-layer sector coefficient identity). [Lean] [✓ Stmt] *For every sector decomposition P , every length N , and every basis index j , the sector coefficient is the raw power sum*

$$c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

This is [CPGSV17a, eq. II, lines 287–301] and [CPGSV21, Definition 4.3, lines 1846–1884].

Proof. [✓ Proof] Definitional unfolding of the sector coefficient. □

Lemma 11.32 (Sector coefficient bound by multiplicity). [Lean] [✓ Stmt] *Under the optional global modulus normalization [CPGSV17a, §II, lines 217–246], every raw weight satisfies $\|\mu_{j,q}\| \leq 1$, and the triangle inequality bounds the norm of the sector coefficient by the multiplicity:*

$$\|c_N^{(j)}\| \leq r_j.$$

Proof. [✓ Proof] Apply the triangle inequality to the raw power sum from Lemma 11.31 and bound each term by 1 using $\|\mu_{j,q}\| \leq 1$. □

Lemma 11.33 (Cross-overlap decay between distinct basis blocks). [Lean] [✓ Stmt] *If P is a BNT canonical form and $j \neq k$, then $O_{P_j P_k}(N) \rightarrow 0$ as $N \rightarrow \infty$, where P_j and P_k are the basis blocks at indices j and k . The argument follows the normal-tensor overlap dichotomy of [CPGSV17a, lines 1080–1091]: differing bond dimensions force decay, and matching bond dimensions are ruled out by the gauge-phase distinctness clause of [CPGSV17a, lines 264–279].*

Proof. [✓ Proof] Case on whether the bond dimensions match. If they differ, apply the dimension-mismatch overlap-decay theorem. If they agree, the basis-distinctness clause of the BNT canonical form rules out gauge-phase equivalence in the cast-compatible shape, so the equal-dimension overlap-decay theorem applies. □

Lemma 11.34 (Combined-family eventual linear independence). [Lean] [✓ Stmt] *Let P and Q each be a BNT canonical form, and suppose every mixed overlap $O_{P_j Q_k}(N)$ tends to 0. Then the union family*

$$\{|V^{(N)}(P_j)\rangle\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\rangle\}_{k=1}^{g_Q}$$

is linearly independent for all sufficiently large N . This is the corollary at [CPGSV17a, lines 1121–1132], matching the BNT linear-independence input recorded at [CPGSV21, line 1850].

Proof. [✓ Proof] Feed the per-family normalized self-overlaps (from the canonical-form data), the within-family cross-overlap decay (Lemma 11.33), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma 11.5). □

Lemma 11.35 (Sector coefficient bound from the line-246 structural field). [Lean] [Stmt] Under the BNT canonical form, the [CPGSV17a, § II.A, line 246] normalization yields directly

$$\|c_N^{(j)}\| \leq r_j$$

for every N and basis index j , without an additional explicit modulus-bound hypothesis at the call site.

Proof. [Proof] Feed the global modulus normalization field into Lemma 11.32. □

Lemma 11.36 (Unit-modulus weight witness from the line-246 structural field). [Lean] [Stmt] Under the BNT canonical form, the [CPGSV17a, § II.A, line 246] unit-modulus existential provides indices (j_*, q_*) with $\|\mu_{j_*, q_*}\| = 1$.

Proof. [Proof] Direct projection onto the unit-modulus existential field of the canonical-form predicate. □

Example 11.37 ($C \oplus (1/2)C$, unequal-modulus admissibility). [Lean] [Stmt] The single-sector two-copy decomposition with raw weights $(1, 1/2)$ demonstrates that the BNT canonical form admits unequal-modulus copies: the modulus bound holds because $\|1\| \leq 1$ and $\|1/2\| = 1/2 \leq 1$, and the unit-modulus existential is witnessed by the first copy with weight 1. This is the [CPGSV17a, § II.A, line 246] normalization in positive form, applied to the $C \oplus (1/2)C$ example.

11.6 Cesàro non-decay and unit-block coefficient

This section formalizes the analytic input that supplies, for any sector j_0 carrying a unit-modulus weight, the “sector- j_0 coefficient does not asymptotically vanish” ingredient used in the matching step of [CPGSV17a, § II, lines 1181–1188]. The reduction is to a pure-analytic statement: a finite sum of N -th powers of complex numbers in the closed unit disk, with at least one unit-modulus summand, cannot tend to zero.

The proof is the elementary Cesàro-mean identity: $M_T := T^{-1} \sum_{N=0}^{T-1} |c(N)|^2 = \sum_{p,q} T^{-1} \sum_{N=0}^{T-1} (\mu_p \bar{\mu}_q)^N$ splits into: T^{-1} times the partial geometric sum for each pair (p, q) ; each pair with $\mu_p \bar{\mu}_q = 1$ contributes the limit 1, while each pair with $\mu_p \bar{\mu}_q \neq 1$ contributes the limit 0 (since $|(z^T - 1)/(z - 1)| \leq 2/|z - 1|$ divided by $T \rightarrow 0$). The diagonal pair (q^*, q^*) at a unit-modulus index q^* contributes, so $M_T \rightarrow \#\{\text{matching pairs}\} \geq 1 > 0$, contradicting $c(N) \rightarrow 0 \Rightarrow |c(N)|^2 \rightarrow 0 \Rightarrow M_T \rightarrow 0$.

Lemma 11.38 (Cesàro non-decay of a finite power sum). [Lean] [Stmt] Let $\mu : \{0, \dots, r-1\} \rightarrow \mathbb{C}$ satisfy $|\mu_q| \leq 1$ for every q and let some q^* satisfy $|\mu_{q^*}| = 1$. Then the sequence $N \mapsto \sum_{q=0}^{r-1} (\mu_q)^N$ does not tend to 0 as $N \rightarrow \infty$. This is the analytic content of the unit-block non-decay step at [CPGSV17a, § II.A, lines 246, 1181–1188].

Proof. [Proof] Cesàro-mean argument as described above. □

Lemma 11.39 (Unit-block coefficient non-decay). [Lean] [Stmt] Let P be a BNT canonical form and let j_0 be a sector index such that some copy of sector j_0 carries a unit-modulus weight, that is, there exists q with $\|\mu_{j_0, q}\| = 1$. Then the sector- j_0 coefficient $c_N^{(j_0)} = \sum_q \mu_{j_0, q}^N$ does not tend to 0 as $N \rightarrow \infty$.

This reads the [CPGSV17a, § II.A, lines 246 and 1244] normalization sector by sector. The unit-modulus witness is supplied externally (as the hypothesis above) because the source statement is global (“at least one of them equals one” over the two-layer index (j, q)) and does not pin the witness to a specific sector.

Proof. ✓ **Proof** Apply Lemma 11.38 to the family $q \mapsto \mu_{j_0, q}$, using the modulus bound $\|\mu_{j_0, q}\| \leq 1$ from the canonical-form normalization together with the per-sector unit-modulus hypothesis. $\square$

Remark 11.40. Used inside the matching theorem of [CPGSV17a, § II, lines 1181–1188] this lemma supplies the non-decay step on $c_N^{(j_0)}$. The matching theorem takes the unit-modulus sector and witness as explicit parameters, mirroring the global character of the source normalization at [CPGSV17a, § II.A, line 246].

11.6.1 Matched-sector weight multiset equality

Given a matched basis-sector pair $(j_0, \tilde{k}_0)$ and a nonzero scalar $\zeta \in \mathbb{C} \setminus \{0\}$ such that the matched coefficient identity $c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q)$ holds for every N past some threshold N_0 , this subsection records the two algebraic conclusions: the *matched-sector weight multiset equality* derived from the coefficient identity by power-sum recovery (Theorem 11.28), and the *matched-sector weight-permutation extractor* upgrading the multiset equality to an explicit per-copy indexing. Together they realise the $\mu_{j, q} = \nu_{j, q} \cdot e^{i\varphi_j}$ recovery of [CPGSV17a, § II, lines 1184–1188]. The coefficient identity itself is the linear-independence projection step of [CPGSV17a, § II, lines 1170–1192] and is supplied by the substitution argument of §11.7.2 below.

Lemma 11.41 (Matched-sector weight multiset equality). Lean ✓ **Stmt** *Let P, Q be sector decompositions with weights $\mu_{j, q}$ and $\nu_{\tilde{k}, q'}$, and fix indices $j_0, \tilde{k}_0$, and a nonzero scalar $\zeta \in \mathbb{C} \setminus \{0\}$. Suppose that for every $N > N_0$,*

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q).$$

Then the per-sector multiplicities agree, $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$, and the rescaled per-copy weight multisets coincide. This is [CPGSV17a, § II, lines 1184–1188] ($\mu_{j, q} = \nu_{j, q} \cdot e^{i\varphi_j}$ and matching multiplicities $r_{a, j} = r_{b, j}$) read on a single matched sector.

Proof. ✓ **Proof** Unfold both sides of the coefficient identity to per-copy power sums $\sum_q \mu_{j_0, q}^N$ and $\sum_{q'} \nu_{\tilde{k}_0, q'}^N$, and distribute ζ^N through the right-hand sum to obtain, for every $N > N_0$,

$$\sum_q \mu_{j_0, q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0, q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 11.27), using $\zeta \neq 0$ together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality. $\square$

Lemma 11.42 (Matched-sector weight-permutation extractor). Lean ✓ **Stmt** *Under the hypotheses of Lemma 11.41, the per-sector multiplicities agree and there is a permutation τ of the per-copy index set identifying the individual weights up to the gauge phase: for every q ,*

$$\nu_{\tilde{k}_0, \tau(q)} = \zeta^{-1} \cdot \mu_{j_0, q}.$$

This is the per-copy realisation of [CPGSV17a, § II, line 1188] ($\mu_{j, q} = \nu_{j, q} \cdot e^{i\varphi_j}$): the multiset equality from Lemma 11.41 is upgraded to a concrete indexing between the matched P - and Q -copies.

Proof. ✓ **Proof** Invoke Lemma 11.41 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality cast and cancel ζ via $\zeta \neq 0$ to convert the gauge factor on the right of the pointwise identity to ζ^{-1} on the left. $\square$

11.7 Strong existential matching of unit-modulus sectors

This section lifts the weak existential matching $\exists j, \exists k$ to the strong existential $\forall k, \exists j$ on the original pair of BNT canonical forms, without recursion, partial sector dropping, or any combined-linear-independence obligation on a partial union. The result is a single $\forall k, \exists j$ statement on the original (P, Q) , iterated externally over each Q -side unit-modulus sector.

The unit-modulus restriction on k matches the source convention: [CPGSV17a, § II.A, line 246] records the modulus-one hypothesis as a single *global* existential over the two-layer (j, q) index of the BNT display, not a per-block one. Sectors whose weights all have modulus strictly less than one have coefficients $c_N^{(k)}(Q) = \sum_q \nu_{k,q}^N$ that decay exponentially to zero, contribute nothing to the thermodynamic state, and are not constrained by the matching ([CPGSV17a, § II.C, lines 1244–1248] returns to this point through the transfer-power fixed-point characterisation). The bijective-matching theorem below applies the strong existential theorem twice (with (P, Q) swapped) to derive the cardinality identity and the per-pair bijection / gauge data on the full basis (every sector is a unit-modulus block under the per-block normalization).

Theorem 11.43 (Strong existential matching, full-basis form). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. For every sector k there exist a sector j with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence between the cast basis tensors, and a non-decaying cross-overlap.*

Proof. ✓ Proof The contrapositive of [CPGSV17a, § II.C, lines 1182–1186], combined with linear independence on the full family $\{P_j\}_j \cup \{Q_k\}_k$, is iterated externally over each sector k . Each iteration is discharged by the full-basis weak existential matching theorem with the roles of P and Q swapped: supply the canonical-form data in reversed order, the basis non-emptiness witness from the retained global unit-modulus witness, and the proportional-MPV hypothesis through its pointwise-symmetric flip. The resulting conclusion is re-oriented by symmetry of gauge-phase equivalence across a bond-dimension cast and symmetry of overlap convergence. $\square$

Remark 11.44 (Linear independence on the full family). Theorem 11.43 consumes *only* the weak existential matching theorem, which itself routes through the linear-independence step (Lemma 11.34) on the *full* family $\{P_j\}_j \sqcup \{Q_k\}_k$, the corollary input of [CPGSV17a, § II.C, lines 1131–1133], not on any partial union.

11.7.1 Bijective matching by symmetry

This subsection derives the symmetry argument of [CPGSV17a, § II.C, lines 1184–1186]: $g_a \geq g_b$ and $g_b \geq g_a$ together imply $g_a = g_b$.

The strong existential matching theorem (Theorem 11.43) gives one direction of the matching on the unit-modulus subset. Applying the same theorem once more with the roles of P and Q swapped (and the trivially symmetric flip of the proportional-MPV hypothesis) gives the other direction. The two matchings, combined with the basis-distinctness clause of the canonical-form predicate [CPGSV17a, § II, lines 264–279], force injectivity in both directions.

Scope. The literal source conclusion “ $g_a = g_b$ and a relabelling $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$ ” ([CPGSV17a, § II.C, lines 1184–1186]) reads as a bijection between the unit-modulus subsets of the two BNT canonical forms. The theorem below delivers the symmetric injective matching (the source’s $g_a \geq g_b \wedge g_b \geq g_a$ step) on the cleanest surface: two injective embeddings, one in each direction, with codomains in the full basis of P and Q respectively. Upgrading the codomains to the

unit-modulus subsets (and packaging as a bijection) requires the additional per-sector weight-equivariance argument of [CPGSV17a, § II.C, lines 1148–1167] combined with the gauge-phase identity on the weighted sectors $\sum_q \mu_{j,q}^N |V^{(N)}(P_j)\rangle$; the non-decay output of the strong existential theorem is on the basis MPV states, not on the weighted sectors, and so does not by itself force the matched j to carry a unit-modulus weight. That upgrade is left to the substitution argument of §11.7.2.

Theorem 11.45 (Bijective matching by symmetry). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family. Write*

$$J_1(P) := \{j \in J(P) : \exists q, |\mu_{j,q}| = 1\}$$

and let $J'_1(Q)$ denote the analogous unit-modulus subset of Q -sectors. Then there exist injective embeddings

$$\varphi : J'_1(Q) \hookrightarrow J(P), \quad \psi : J_1(P) \hookrightarrow J(Q),$$

such that for every $k \in J'_1(Q)$ the bond dimensions match, the basis tensors $P_{\varphi(k)}$ and Q_k are gauge-phase equivalent in the cast-left shape, and the basis cross-overlap does not tend to zero; and symmetrically for every $j \in J_1(P)$ through ψ .

Proof. ✓ Proof Apply Theorem 11.43 first with (P, Q) and the original hypothesis to obtain a raw matched-sector function via classical choice, and then again with (Q, P) and the symmetric flipped hypothesis to obtain the symmetric raw function.

Forward injectivity. Suppose $\varphi_0(k_1) = \varphi_0(k_2) = j$. The two matching witnesses give a common centre P_j with two gauge-phase equivalences (after a bond-dimension cast) to Q_{k_1} and Q_{k_2} . Composing through this centre (cast-aware transitivity of gauge-phase equivalence) yields a gauge-phase equivalence between Q_{k_1} and Q_{k_2} under the dimension equality $\dim_Q(k_1) = \dim_Q(k_2)$. The basis-distinctness clause of the canonical-form predicate on Q ([CPGSV17a, § II, lines 264–279]) then forces $k_1 = k_2$. Backward injectivity is symmetric.

Injective embeddings. The injective functions, paired with the injectivity proofs, are exactly the embedding pair stated in the theorem. The matching data are extracted by re-applying the classical-choice specification at each subtype index. $\square$

11.7.2 Substitution argument for the unit-modulus subset

Given the matched gauges produced by the strong existential theorem (Theorem 11.43) encoded as the injection from §11.7.1, the per-block coefficient identity of [CPGSV17a, § II.C, lines 1187–1188] follows by substituting the per-pair gauge identity $V^{(N)}(Q_k)_\sigma = \zeta_k^N \cdot V^{(N)}(P_{\varphi(k)})_\sigma$ into the proportional-MPV hypothesis and projecting the resulting overlap onto each basis tensor P_{j_0} .

Scope. The literal form of the matched-pair clause is the eventual exact equality $c_N^{(\varphi(k))}(P) = \zeta_k^N \cdot c_N^{(k)}(Q)$ for all sufficiently large N . Deriving the eventual exact equality from the partial-bijection data (with non-unit Q -sectors unconstrained) requires linear independence on the partial union of bases, which is not naturally available from the canonical-form predicate alone. The theorem below delivers the asymptotic-difference form $c_N^{(\varphi(k))}(P) - \zeta_k^N \cdot c_N^{(k)}(Q) \rightarrow 0$ as $N \rightarrow \infty$, which is the natural output of the overlap-projection argument; the upgrade to the eventual-exact form is provided downstream by the multiset-rigidity Theorem 11.28, the Newton–Girard input behind [CPGSV17a, § II.C, lines 1187–1188]. The non-matched decay clause is delivered in full.

Lemma 11.46 (Coefficient decay for subnormal sectors). Lean ✓ Stmt Let P be a BNT canonical form and let j be a sector for which every copy weight is strictly subnormal, that is, $\|\mu_{j,q}\| < 1$ for every q . Then $\lim_{N \rightarrow \infty} c_N^{(j)} = 0$.

Proof. ✓ Proof The coefficient $c_N^{(j)} = \sum_q \mu_{j,q}^N$ is a finite sum. Each summand $\mu_{j,q}^N$ tends to 0 as $N \rightarrow \infty$ by the strict subnormality hypothesis. A finite sum of null sequences is null. $\square$

Theorem 11.47 (Substitution argument for the unit-modulus subset). Lean ✓ Stmt Let P, Q be two BNT canonical forms generating proportional MPVs. Let

$$J'_1(Q) := \{k \in J(Q) \mid \exists q, \|\nu_{k,q}\| = 1\}$$

be the unit-modulus subset of Q -sectors ([CPGSV17a, § II.A, line 246] normalization), and let $\varphi : J'_1(Q) \hookrightarrow J(P)$ be an injection equipped with per-pair matched-gauge data for every $k \in J'_1(Q)$. Then:

1. Matched-pair asymptotic identity. For every $k \in J'_1(Q)$ there exists a phase $\zeta_k \in \mathbb{C}$ with $\|\zeta_k\| = 1$ such that the asymptotic difference of coefficient sequences vanishes,

$$\lim_{N \rightarrow \infty} (c_N^{(\varphi(k))}(P) - \zeta_k^N \cdot c_N^{(k)}(Q)) = 0.$$

2. Non-matched decay. For every $j_0 \in J(P)$ outside the range of φ ,

$$\lim_{N \rightarrow \infty} c_N^{(j_0)}(P) = 0.$$

Proof. ✓ Proof Fix j_0 and project the proportional-MPV identity onto the overlap with P_{j_0} :

$$\sum_j c_N^{(j)}(P) \cdot \langle V^{(N)}(P_j) | V^{(N)}(P_{j_0}) \rangle = \sum_k c_N^{(k)}(Q) \cdot \langle V^{(N)}(Q_k) | V^{(N)}(P_{j_0}) \rangle.$$

On the left-hand side, the off-diagonal terms $j \neq j_0$ decay by Lemma 11.33 ([CPGSV17a, lines 1080–1091]), and the diagonal term $c_N^{(j_0)}(P) \cdot (\langle V^{(N)}(P_{j_0}) | V^{(N)}(P_{j_0}) \rangle - 1)$ decays by the normalized self-overlap ([CPGSV21, line 1818]). Hence $\text{LHS} - c_N^{(j_0)}(P) \rightarrow 0$.

On the right-hand side, split k into $J'_1(Q)$ and its complement.

- For $k \in J'_1(Q)$, substitute via the gauge identity $V^{(N)}(Q_k)_\sigma = \zeta_k^N \cdot V^{(N)}(P_{\varphi(k)})_\sigma$ of [CPGSV17a, line 1186]; the unit-modulus closure $\|\zeta_k\| = 1$ follows from the norm-from-overlap lemma and the convergence of both per-block normalized self-overlaps to 1. If $\varphi(k) = j_0$, the diagonal self-overlap contributes $c_N^{(k)}(Q) \cdot \zeta_k^N + o(1)$; otherwise the cross-overlap decay gives $o(1)$.
- For $k \notin J'_1(Q)$, every copy weight has strict modulus < 1 , so $c_N^{(k)}(Q) \rightarrow 0$ by Lemma 11.46; combined with the uniform overlap bound from left-canonical normalization ([CPGSV21, lines 1815–1837]), the contribution decays to 0.

Combining the per- k contributions yields

$$\text{RHS} - \sum_{k \in J'_1(Q), \varphi(k)=j_0} \zeta_k^N \cdot c_N^{(k)}(Q) \rightarrow 0.$$

Since φ is an embedding the inner sum has at most one term. With $\text{LHS} = \text{RHS}$ (the projection identity), this gives $c_N^{(j_0)}(P) - (\text{matched term}) \rightarrow 0$.

Specialising: if $j_0 = \varphi(k_0)$ for some $k_0 \in J'_1(Q)$ the matched term is $\zeta_{k_0}^N \cdot c_N^{(k_0)}(Q)$, giving clause (1); if $j_0 \notin \text{range}(\varphi)$ the matched term is 0, giving clause (2). $\square$

Remark 11.48. Theorem 11.47 realises the substitution route of [CPGSV17a, § II.C, lines 1187–1188]: substitute the per-pair gauge phases into the proportional-MPV identity, reduce to the P -only relation, and extract per- j coefficient identities from cross-overlap decay alone. The result handles the $J'_1(Q)$ -restricted bijection setting; the full bijection $J(P) \simeq J(Q)$ is not delivered by the bijective-matching theorem (see the scope discussion above).

Chapter 12

Proof of the Fundamental Theorem

This chapter records the aperiodic comparison results used in the proof of the Fundamental Theorem of Matrix Product States. We first state strict special cases for canonical-form tensors with already selected BNT representatives, and then state the sector-decomposition theorem that keeps the repeated-copy multiplicities appearing in the CPSV BNT comparison. We then treat the equal-MPV case, including the common-block-structure specialization that deduces gauge equivalence. The results combine the canonical form reduction (Chapter 9), block separation (Chapter 10), spectral gap (Chapter 7), and the BNT permutation arguments of Chapter 11.

The theorems in this chapter follow the approach of [CPGSV17a]: they first block the tensor to remove non-trivial periodicity, then apply the aperiodic fundamental theorem. Chapter 15 develops the stronger irreducible-form framework of [DCCSPG17], which gives a fundamental theorem valid for periodic tensors directly, with an extra Z -gauge degree of freedom.

12.1 Homogeneous Pair-Span Padding

The pair-span padding lemmas below are shared algebraic infrastructure for the aperiodic Fundamental Theorem route. They are used to pass from the Wielandt/block-injective homogeneous span information of [CPGSV17a] to the fixed-length separation statements used by BNT comparison. Parent-Hamiltonian arguments later reuse the same predicates, but the source role of the identity-pair lemma is the Fundamental Theorem pair-span step.

Lemma 12.1 (Identity pair from full homogeneous pair-word span). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Stmnt](#) Let A and B be MPS tensors of bond dimensions D_1 and D_2 .

1. If the homogeneous pair-word span is the full product algebra at some length n , i.e.

$$\text{span}\{(A^w, B^w) \mid |w| = n\} = M_{D_1}(\mathbb{C}) \times M_{D_2}(\mathbb{C}),$$

then the simultaneous identity pair (I_{D_1}, I_{D_2}) belongs to that same homogeneous span.

2. If the span is full at one positive length $S > 0$, then the identity pair lies in the homogeneous pair-word span at every sufficiently large length:

$$\exists L \in \mathbb{N}, \forall n \geq L, (I_{D_1}, I_{D_2}) \in \text{span}\{(A^w, B^w) \mid |w| = n\}.$$

Proof. ✓ Proof Statement (1) is immediate: if the span is already the whole product algebra, then it contains the identity pair. For (2), fullness at a positive length propagates by multiplying a word of length T by a one-letter word, so induction gives fullness at every length $S + k$. Applying (1) at each such length gives the conclusion with $L = S$. $\square$

12.2 Equal-MPV Comparisons

The source corollary for equal MPVs says that if two canonical-form tensors generate the same MPV family for all system sizes, then their matrix dimensions coincide and the full tensors are related by one invertible gauge [CPGSV17a, Corollary II.2]. In the proof, this corollary is obtained by matching the BNT blocks and then comparing the sector-weight power sums.

The block-matching conclusion (block-count, bond-dimension, and gauge-phase identification of corresponding sectors) and the global gauge identification are formalised on the BNT canonical form of Chapter 11. The coordinate packaging of [CPGSV17a, § II.C, lines 1184–1192] is the in-flight follow-up.

Theorem 12.2 (Bond dimension equality from unit overlap). Lean ✓ Stmt *Let A and B be irreducible trace-preserving blocks, possibly with different bond dimensions. If*

$$\lim_{N \rightarrow \infty} |\langle V^{(N)}(B) | V^{(N)}(A) \rangle| = 1,$$

then the two bond dimensions are equal.

Proof. ✓ Proof The proof is the contrapositive of rectangular overlap decay: if the two bond dimensions differ, the mixed overlap tends to 0, contradicting the assumed convergence of its modulus to 1. $\square$

Lemma 12.3 (Common-block equal-MPV comparison). Lean ✓ Stmt *Fix a common block structure: the same number of blocks g , the same bond dimensions $(D_k)_{k=1}^g$, and the same weights $(\mu_k)_{k=1}^g$. Let $(\mu_k, A_k)_{k=1}^g$ and $(\mu_k, B_k)_{k=1}^g$ be two families in canonical form with BNT separation on this common structure. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) is gauge equivalent, and hence the corresponding block-diagonal tensors are gauge equivalent.*

Proof. ✓ Proof On the BNT canonical form this is the specialization of the global-gauge equality theorem (Chapter 11) to the case where P and Q share a common (g, D_k, μ_k) block structure: the basis bijection β is trivial, the per-block gauge matrices X_k furnish the per-pair gauge equivalences, and the flattened direct-sum gauge X assembles them into the global conjugation on $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$. $\square$

12.2.1 Sector multiplicities and weight recovery

This subsection records the coefficient comparison used when several canonical-form blocks are phase and gauge transforms of the same basis normal tensor. If the j th basis block occurs with nonzero weights $\mu_{j,1}, \dots, \mu_{j,r_j}$, its length- N contribution is governed by the power sum

$$c_{S,j}^{(N)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

This is the scalar part of the source decomposition

$$A^i = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1}.$$

The sector tensor used below is the representative form after the gauge conjugations $X_{j,q}$ have already been absorbed into the choice of basis block and gauge phases. Thus BNT linear independence separates the coefficient arrays $c_{S,j}^{(N)}$, and elementary power-sum algebra recovers both the multiplicity r_j and the multiset of weights attached to the basis block. In the later PEPS application this information is not presented as a separate chain of MPS comparison theorems: Theorem IV.13 of [CPGSV17a] uses structure coefficients

$$c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L), \quad m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta,$$

and the appendix writes the vertical canonical forms as $\bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ and $\bigoplus_\beta \nu_\beta \otimes A_\beta$, with $m_\alpha = \text{tr}(\mu_\alpha)$ and $n_\beta = \text{tr}(\nu_\beta)$. Here μ_α and ν_β are diagonal positive matrices in the source paper; only their traces are the scalar coefficients compared in this subsection. The one-index coefficients $c_{S,j}^{(N)}$ below are the corresponding power-sum comparison after the basis blocks have already been matched. The geometric extrapolation and Newton–Girard steps are external finite-sequence facts used to recover the lists of weights from those power sums.

Definition 12.4 (Sector weights and sector decomposition). [Lean](#) [Lean](#) [✓ Stmt](#) A sector-weight tuple over g basis blocks consists of multiplicities $r_j > 0$, weights $\mu_{j,q} \neq 0$ for $q \in \{1, \dots, r_j\}$, and the *sector coefficients*

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N$$

for the tuple S . A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with these multiplicities and weights.

Lemma 12.5 (Sector decomposition formula). [Lean](#) [✓ Stmt](#) For a sector decomposition S with associated tensor A_{tot} , the MPV satisfies

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^g c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Lemma 12.6 (Eventual coefficient comparison from BNT linear independence). [Lean](#) [✓ Stmt](#) If two sector-weight tuples S, T over the same basis, with coefficient arrays $c_{S,j}^{(N)}$ and $c_{T,j}^{(N)}$, produce equal total MPVs and the basis is eventually linearly independent (the BNT property), then there is N_0 such that

$$\forall N > N_0, \forall j, \quad c_{S,j}^{(N)} = c_{T,j}^{(N)}.$$

Lemma 12.7 (BNT recovery of copy counts and sector weights). [Lean](#) [✓ Stmt](#) Let S and T be two sector-weight tuples over the same basis, without assuming their multiplicities agree. If the basis is eventually linearly independent and the total MPVs of S and T agree, then there are copy-count equalities $r_j^S = r_j^T$ for every basis block j , and, after transporting along these equalities, the sector weight multisets are equal for every j :

$$\{\mu_{j,q}^S : 1 \leq q \leq r_j^S\} = \{\mu_{j,q}^T : 1 \leq q \leq r_j^T\}.$$

Proof. [✓ Proof](#) BNT linear independence and equality of total MPVs give eventual coefficient equality by Lemma 12.6. Thus, for each basis sector j , the two finite nonzero weight families satisfy

$$\sum_{q=1}^{r_j^S} (\mu_{j,q}^S)^N = \sum_{q=1}^{r_j^T} (\mu_{j,q}^T)^N$$

for all sufficiently large N . The signed-geometric-sequence argument extends this to every positive exponent. Applying the finite-range unequal-cardinality power-sum theorem (Theorem 11.27) gives both $r_j^S = r_j^T$ and equality of the sector weight multisets. $\square$

Lemma 12.8 (Phase matching recovers copy counts and sector weights). [Lean](#) [✓ Stmt](#) *Let P and Q be sector decompositions. Assume there is a permutation π matching the basis blocks and, for each j , a nonzero complex number ζ_j such that*

$$V^{(N)}(Q_{\pi(j)})_{\sigma} = \zeta_j^N V^{(N)}(P_j)_{\sigma}$$

for every N and σ . If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then the copy counts satisfy $r_j^P = r_{\pi(j)}^Q$, and after multiplying the weights of $Q_{\pi(j)}$ by ζ_j the sector weight multisets agree.

Proof. [✓ Proof](#) Absorb the factors ζ_j into the sector weights of Q and keep the basis blocks of P . The phase-matching hypothesis and the sector decomposition formula show that the resulting decomposition has the same total MPV family as Q , hence also as P . One is then in the shared-basis situation of Lemma 12.7, which recovers both the copy-count equalities and the claimed equality of weight multisets. $\square$

Lemma 12.9 (Two-basis sector comparison after phase matching). [Lean](#) [✓ Stmt](#) *Let P and Q be sector decompositions. Assume there is a permutation π matching the basis blocks, the multiplicities satisfy $r_j^P = r_{\pi(j)}^Q$, and for each j there is a nonzero complex number ζ_j such that*

$$V^{(N)}(Q_{\pi(j)})_{\sigma} = \zeta_j^N V^{(N)}(P_j)_{\sigma}$$

for every N and σ . If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ by multiplication by ζ_j .

Proof. [✓ Proof](#) Lemma 12.8 gives copy-count equalities and the corresponding weight-multiset equality. The recovered copy-count equalities coincide with the assumed ones, so the weight-multiset equality gives the conclusion. $\square$

Lemma 12.10 (Gauge-phase paired bases imply phase-matched sector comparison). [Lean](#) [✓ Stmt](#) *Let P and Q be sector decompositions. Assume there is a permutation π matching the basis blocks, the multiplicities satisfy $r_j^P = r_{\pi(j)}^Q$, and each matched pair of basis blocks is gauge-phase equivalent. If the total MPV families of P and Q agree and the basis blocks of P are eventually linearly independent, then for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ after multiplication by the corresponding gauge phase.*

Proof. [✓ Proof](#) Gauge-phase equivalence gives the required phase-scaling identity for the MPVs by Theorem 2.25. The conclusion then follows immediately from Lemma 12.9. $\square$

Definition 12.11 (Sector basis matching witness). [Lean](#) [✓ Stmt](#) *A sector basis matching between two sector decompositions P and Q consists of*

1. a basis permutation (bijection) $\pi : \{1, \dots, g_P\} \simeq \{1, \dots, g_Q\}$,
2. per-block multiplicity equalities $r_j^P = r_{\pi(j)}^Q$,
3. per-block bond-dimension equalities $D_j^P = D_{\pi(j)}^Q$, and
4. per-block gauge-phase equivalences $P_j \simeq_{\text{gp}} Q_{\pi(j)}$ after transporting P_j across the bond-dimension equality.

The hypotheses consumed by Lemma 12.10 are exactly the fields of this matching tuple.

Definition 12.12 (Sector basis pre-matching). Lean ✓ Stmt A *sector basis pre-matching* between sector decompositions P and Q consists of a permutation of their basis blocks, equality of the matched bond dimensions, and gauge-phase equivalence of each matched pair. It does not include the copy-count equality; that equality is recovered from total MPV equality and BNT linear independence in Lemma 12.16.

Definition 12.13 (Single-family overlap-orthogonality hypotheses for sector bases). Lean ✓ Stmt For one sector decomposition P , this condition states positive basis dimensions, trace-preserving normalization of all basis blocks, self-overlap limits equal to 1, and off-overlap limits equal to 0.

Definition 12.14 (Primitive overlap-span hypotheses for sector bases). Lean ✓ Stmt For sector decompositions P and Q , this condition states the primitive overlap-rigidity hypotheses for their basis blocks: nonzero bond dimensions, injectivity, trace-preserving normalization, self-overlap limits equal to 1, off-overlap limits equal to 0, and equality of the finite-length spans of the basis MPV states for every system size.

Lemma 12.15 (Single-family overlap hypotheses plus span give overlap-span hypotheses). Lean ✓ Stmt *If two sector bases each satisfy the single-family overlap-orthogonality hypotheses, and one also supplies one-site injectivity of all their basis blocks together with equality of the finite-length MPV spans, then the two bases satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof This is just collecting the one-family fields on the left and right, and adding the remaining injectivity and span-comparison hypotheses. □

Lemma 12.16 (A pre-matching gives a sector basis matching). Lean ✓ Stmt *Let M be a sector basis pre-matching between P and Q . If the total MPV families agree and the basis MPVs of P are eventually linearly independent, then the copy-count equalities recovered by the phase-matched BNT coefficient comparison turn M into a full sector basis matching.*

Proof. ✓ Proof Gauge-phase equivalence gives the required phase-scaling identities for the corresponding basis MPVs:

$$V^{(N)}(Q_{\pi(j)})_{\sigma} = \zeta_j^N V^{(N)}(P_j)_{\sigma}.$$

Lemma 12.8 then recovers the missing copy-count equalities, which are precisely the extra fields needed for Definition 12.11. □

Theorem 12.17 (Overlap rigidity gives a sector basis matching). Lean ✓ Stmt *Assume the basis blocks of two sector decompositions have nonzero bond dimensions, are trace-preserving and injective, have asymptotically orthonormal overlaps within each family, and span the same MPV subspace at every system size. If the total MPV families also agree, then there exists a sector basis matching between the two decompositions.*

Proof. [✓ Proof](#) The permutation-rigidity lemma for bases of normal tensors gives a permutation, the matched bond dimensions, and the gauge-phase equivalences of the basis blocks, hence a sector basis pre-matching. Asymptotic orthonormality gives the BNT linear-independence condition, and Lemma 12.16 recovers the copy-count equalities from equality of the total MPV families. $\square$

Lemma 12.18 (Primitive overlap-span hypotheses give a sector basis matching). [Lean](#) [✓ Stmt](#) *If two sector decompositions satisfy the primitive overlap-span hypotheses and their total MPV families agree, then there exists a sector basis matching between them.*

Proof. [✓ Proof](#) The collected hypotheses are exactly the nonzero-dimension, injectivity, normalization, overlap, and span assumptions of Theorem 12.17. Applying that theorem gives the matching witness. $\square$

Theorem 12.19 (Span equality for representative sector bases). [Lean](#) [✓ Stmt](#) *Let two nonzero-weight block families be trace-preserving, primitive, irreducible, injective, and carry nonzero weights. If their finite-length MPV spans agree for every system size, then there exist sector decompositions P and Q whose total tensors have the same MPVs as the two weighted nonzero-block tensors, whose bases satisfy BNT linear independence, and whose sector bases satisfy the primitive overlap-span hypotheses.*

Proof. [✓ Proof](#) Apply the phase-class construction on both sides and expose the representative overlap/span data internally. The one-sided positive-dimension, injectivity, normalization, and overlap fields fill all fields of Definition 12.14 except the two-family span equality. That span equality follows by transporting the assumed nonzero-block span equality through the two representative-span identities. $\square$

Definition 12.20 (MPV phase equivalence for nonzero-weight blocks). [Lean](#) [✓ Stmt](#) Two blocks A and B are MPV-phase equivalent if there is a nonzero scalar ζ such that, for every system size N and every physical word, the MPV coefficient of B equals ζ^N times the MPV coefficient of A . The two bond dimensions are allowed to differ.

Definition 12.21 (Common MPV phase cover). [Lean](#) [✓ Stmt](#) For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

Lemma 12.22 (Common MPV phase cover gives nonzero-block span equality). [Lean](#) [✓ Stmt](#) *Let two finite nonzero-weight block families map onto a third finite family of common blocks. Suppose both maps are surjective, and each block has finite-length MPV vectors equal to a nonzero scalar power times the MPV vectors of its common block. Then, for every system size, the two nonzero-weight block families span the same MPV subspace.*

Proof. [✓ Proof](#) Each nonzero-block MPV vector lies in the span of the common family because it is a scalar multiple of its common block's MPV vector. Surjectivity gives the reverse inclusion, so each nonzero-block span equals the common span; transitivity gives the equality of the two nonzero-block spans. $\square$

Lemma 12.23 (Common phase cover gives nonzero-block span equality). [Lean](#) [✓ Stmt](#) *A common MPV phase cover for two nonzero-weight block families gives equality of their finite-length MPV spans at every system size.*

Proof. ✓ Proof The common cover specifies the common family, the two surjective class maps, and the MPV-phase equivalences. The span lemma then applies directly. $\square$

Lemma 12.24 (Common phase cover from a BNT block-matching conclusion). Lean ✓ Stmt *If a BNT block-matching comparison has produced a permutation of the two basis families together with matched bond dimensions and gauge-phase equivalences, then the two basis families have a common MPV phase cover.*

Proof. ✓ Proof Take the left basis family as the common family. The comparison permutation and its inverse give the two class maps, while each gauge-phase equivalence gives the required MPV-phase equivalence. $\square$

Lemma 12.25 (Common phase cover from a bijective phase matching). Lean ✓ Stmt *Let $(A_s)_{s \in S}$ and $(B_t)_{t \in T}$ be two finite block families. Suppose there is a bijection $\sigma : S \rightarrow T$ and, for each $s \in S$, a scalar $\omega_s \in \mathbb{C}^\times$ such that*

$$|V^{(N)}(B_{\sigma(s)})\rangle = \omega_s^N |V^{(N)}(A_s)\rangle$$

for every length $N \geq 0$, with $|V^{(0)}(C)\rangle$ interpreted as the bond dimension of C . For each matched pair, the case $N = 0$ records equality of bond dimensions. Then the two block families admit a common MPV phase cover.

Proof. ✓ Proof Use the first family as the common family. The two covering maps are the identity on S and the inverse of σ . The displayed phase relation provides the required phase factor for the corresponding block of the second family. $\square$

Lemma 12.26 (Span equality from a BNT block-matching comparison). Lean ✓ Stmt *A BNT block-matching comparison gives equality, for every system size, of the finite-length MPV spans of the two nonzero-weight block families.*

Proof. ✓ Proof First form the common MPV phase cover from the block-matching comparison. The common phase-cover span lemma then gives the stated span equality. $\square$

Lemma 12.27 (Span equality from a bijective phase matching). Lean ✓ Stmt *A bijective phase matching gives equality, for every system size, of the finite-length MPV spans of the two block families.*

Proof. ✓ Proof First form the common MPV phase cover from the bijective phase matching. The common phase-cover span lemma then gives the stated span equality. $\square$

Theorem 12.28 (Overlap-span hypotheses from a common phase cover). Lean ✓ Stmt *Let two block families be trace-preserving, primitive, irreducible, injective, and carry nonzero weights. If they have a common MPV phase cover, then there exist BNT sector decompositions P and Q for the two weighted nonzero parts, and the sector bases satisfy the primitive overlap-span hypotheses.*

Proof. ✓ Proof The common cover gives finite-length span equality for the two nonzero-weight block families. The phase-class representative construction then transports this span equality to the BNT sector bases and supplies the remaining one-family overlap hypotheses. $\square$

Theorem 12.29 (Sector comparison via a sector-basis matching). Lean ✓ Stmt *Given two sector decompositions P and Q , a sector basis matching (Definition 12.11) between them, eventual linear independence of the basis MPV states of P , and equality of the total MPV families of P and Q , one obtains nonzero phases $(\zeta_j)_j$ such that for each j the sector weight multiset of P_j equals the multiset obtained from that of $Q_{\pi(j)}$ by multiplication by ζ_j .*

Proof. ✓ Proof The sector-basis matching contains exactly the permutation, copy, and gauge-phase hypotheses of Lemma 12.10, so the conclusion follows directly from that lemma. $\square$

Lemma 12.30 (Cancellation of leftover zero blocks). Lean ✓ Stmt *Let A and B have the same MPV family, each decomposed as an all-zero leftover block plus a nonzero part. If the leftover zero-block dimensions agree, then the nonzero parts themselves have the same MPV family, including at length zero.*

Proof. ✓ Proof At positive lengths the all-zero leftover block vanishes, so the nonzero parts agree. Theorem 13.2 gives the exact value

$$V^{(0)}(\mathbf{0}_{d, D_0})_{\emptyset} = D_0,$$

so equality of the two leftover zero-block dimensions gives cancellation at length zero as well. $\square$

Theorem 12.31 (Sector comparison from leftover zero blocks and span hypotheses). Lean ✓ Stmt *Let A and B have the same MPV family. Fix $p > 0$ and suppose $A^{[p]}$ and $B^{[p]}$ each decompose as an all-zero leftover block plus a weighted nonzero part, with all blocks in the nonzero part trace-preserving, primitive, irreducible, injective, and with nonzero weights. If the two leftover zero-block dimensions agree and the two nonzero-weight block families have equal finite-length MPV spans at every system size, then there exist a period $p' > 0$ and sector decompositions P, Q such that $A^{[p']}$ and $B^{[p']}$ agree with the corresponding sector tensors at every positive length, P and Q have the same full MPV family, both have BNT bases, and their sector weights satisfy the matched sector-weight conclusion.*

Proof. ✓ Proof The finite-length span equality gives the primitive overlap-span hypotheses for the BNT sector bases of the two nonzero parts. Cancellation of leftover zero blocks identifies the two nonzero parts as full MPV families, while the leftover zero-block summands vanish at positive lengths, giving the positive-length comparison of $A^{[p]}$ and $B^{[p]}$ with the two sector decompositions. The sector-basis matching and two-basis comparison theorems then give the sector-weight conclusion. $\square$

The following common-primitive hypotheses are not separate statements in arXiv:1606.00608. They name the extra formal assumptions used here before the zero-block sector comparison: span equality, phase-cover hypotheses, and block-matching conclusions for the common primitive sectors.

Definition 12.32 (Span hypotheses for common primitive sectors). Lean ✓ Stmt For two common primitive nonzero-sector families at one blocking length, this condition consists of the hypotheses needed before applying the zero-block sector comparison: equality of the two leftover zero-block dimensions, one-site injectivity of all blocks on both sides, and equality of the finite-length MPV spans of the two block families for every system size.

Theorem 12.33 (Phase covers supply span hypotheses). Lean ✓ Stmt *If the two common primitive nonzero-sector families have equal leftover zero-block dimensions, are one-site injective, and are equipped with a common MPV phase cover, then they satisfy the span hypotheses of Definition 12.32.*

Proof. ✓ Proof The zero-block dimension equality and injectivity fields are exactly the supplied hypotheses. The common phase cover gives the finite-length span equality by Lemma 12.23. $\square$

Definition 12.34 (Phase-cover hypotheses for common primitive sectors). [Lean](#) [✓ Stmt](#) For two common primitive nonzero-sector families at one blocking length, this condition is the phase-cover analogue of Definition 12.32: equality of the two leftover zero-block dimensions, one-site injectivity of all blocks on both sides, and a common MPV phase cover for the two block families.

Theorem 12.35 (Phase-cover hypotheses imply span hypotheses). [Lean](#) [✓ Stmt](#) *If two common primitive nonzero-sector families satisfy the phase-cover hypotheses of Definition 12.34, then they satisfy the span hypotheses of Definition 12.32.*

Proof. [✓ Proof](#) The zero-block dimension equality and the two injectivity assertions are already part of the phase-cover hypotheses. Apply Theorem 12.33 to a common MPV phase cover for the two block families. $\square$

Definition 12.36 (Block-matching hypotheses for common primitive sectors). [Lean](#) [✓ Stmt](#) For two common primitive nonzero-sector families at one blocking length, this condition is the block-matching analogue of Definition 12.34: equality of the two leftover zero-block dimensions, one-site injectivity of all blocks on both sides, and a BNT block-matching conclusion for the two block families (a permutation, equal bond dimensions, and gauge-phase equivalences).

Theorem 12.37 (Block-matching hypotheses imply phase-cover hypotheses). [Lean](#) [✓ Stmt](#) *If the block-matching hypotheses of Definition 12.36 hold, then the phase-cover hypotheses of Definition 12.34 hold.*

Proof. [✓ Proof](#) The zero-block dimension equality and injectivity assumptions transfer directly. The common MPV phase cover is provided by the general block-matching lemma applied to the two block families. $\square$

Theorem 12.38 (Block-matching hypotheses imply span hypotheses). [Lean](#) [✓ Stmt](#) *If the block-matching hypotheses hold, then the span hypotheses hold by passing through the phase-cover hypotheses.*

Proof. [✓ Proof](#) The block-matching hypotheses give the phase-cover hypotheses via the preceding theorem, and the phase-cover hypotheses imply the span hypotheses. $\square$

Definition 12.39 (Per-block block-matching gauge-phase data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The block-matching conclusion can be unpacked into explicit per-block data: a permutation matching the two block families, equalities of matched bond dimensions, gauges X_k , and nonzero phases ζ_k such that

$$B_{\sigma(k)}^i = \zeta_k X_k A_k^i X_k^{-1}$$

after casting the source block along the dimension equality. The auxiliary projections name the reindexed target family, the cast source family, and the dependent and flattened block-diagonal gauges assembled from the X_k .

Theorem 12.40 (Per-block gauge phases assemble to an explicit global gauge). [Lean](#) [Lean](#) [✓ Stmt](#) *If the block weights absorb the per-block phases, $\mu_A(k) = \mu_B(\sigma(k))\zeta_k$, then the weighted direct sum of the permuted right blocks is conjugate to the weighted direct sum of the cast left blocks by the explicit flattened gauge $X = \bigoplus_k X_k$. Consequently this global matrix witnesses the gauge equivalence of the two assembled tensors.*

Proof. [✓ Proof](#) The scalar identity moves each phase ζ_k from the tensor relation into the corresponding block weight. After this absorption, every weighted block satisfies an ordinary conjugation identity by X_k ; the block-diagonal conjugation formula assembles those identities into the displayed global gauge equivalence. $\square$

Theorem 12.41 (Common primitive sectors with span hypotheses). [Lean](#) [✓ Stmt](#) Assume A and B have the same MPV family and that the canonically blocked nonzero parts agree with the same families written after identifying blocked physical words. The common primitive theorem then produces one positive blocking length with $A^{[p]} \sim \mathbf{0}_{d^p, z_A} \oplus A_{\text{nz}}$ and $B^{[p]} \sim \mathbf{0}_{d^p, z_B} \oplus B_{\text{nz}}$, and trace-preserving primitive irreducible nonzero-sector families with positive bond dimensions and nonzero weights. If these produced families also satisfy Definition 12.32, then the leftover zero-block sector comparison conclusion holds: after a positive blocking, there are sector decompositions P, Q agreeing with A and B at all positive lengths, with P and Q generating the same full MPV family, with BNT bases, and with their sector weights matched up to a permutation and nonzero phases.

Proof. [✓ Proof](#) Apply the common primitive theorem to obtain the nonzero-sector families. The span hypotheses provide $z_A = z_B$, one-site injectivity, and finite-length nonzero-sector span equality. These are exactly the additional hypotheses required by the leftover-zero-block comparison theorem. $\square$

Theorem 12.42 (Common primitive irreducible sectors with conditional block-span consequences). [Lean](#) [✓ Stmt](#) Let A and B have the same MPV family and assume the blocked-word identification for the common-sector families. The common primitive irreducible-block theorem supplies one positive blocking length and trace-preserving, primitive, irreducible nonzero-sector families on both sides. This theorem records that, for exactly those families, either a common MPV phase cover or a BNT block-matching conclusion supplies the finite-length block-span equality required by Theorem 12.31. Thus the remaining mathematical inputs – the blocked-word identification and the later common-phase or BNT matching comparison – are kept explicit.

Definition 12.43 (Blocked-word identification for common sectors). [Lean](#) [✓ Stmt](#) This is the one-sided assertion that the canonically blocked weighted nonzero part agrees, as an MPV family, with the same common cyclic-sector family after identifying blocked physical words. It isolates the coordinate equality between direct and iterated blocking. The coordinate-grouping theorem below supplies this condition in the theorems that take it as a hypothesis, so it is not a new paper-level mathematical hypothesis.

Definition 12.44 (Global coordinate-grouping hypothesis for common sectors). [Lean](#) [✓ Stmt](#) This is the corresponding global assertion at the level of blocked words: every common blocked cyclic-sector family satisfies the coordinate-grouping condition for each original nonzero block. Equivalently, the canonical identification with each iterated blocked alphabet agrees with grouping the direct blocked word into consecutive words of the period-removal length. The next lemma proves this assertion from the concrete word-cast construction.

Lemma 12.45 (Canonical coordinate grouping). [Lean](#) [✓ Stmt](#) The canonical coordinate grouping holds for every common blocked cyclic-sector family. The proof identifies each grouped block with the corresponding direct blocked word.

Proof. [✓ Proof](#) Apply the grouped-block cast comparison to each original nonzero block. $\square$

Lemma 12.46 (Coordinate grouping gives the word identification). [Lean](#) [✓ Stmt](#) If the global coordinate-grouping hypothesis holds, then the blocked-word identification used by the common-sector structural theorem holds. The proof is exactly the weighted finite-sum comparison for direct and iterated blocking applied to each common cyclic-sector family.

Proof. [✓ Proof](#) Apply the coordinate-grouping comparison to each one-sided family and sum over the original nonzero blocks with the transported powers of the weights. $\square$

Theorem 12.47 (Unconditional common primitive block decompositions). Lean ✓ Stmt *Let A and B have the same MPV family. Then there is a positive blocking length at which both blocked tensors decompose into a leftover all-zero block and a weighted direct sum of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two nonzero-sector families have the same positive-length MPV family, and the leftover zero blocks satisfy the length-zero identity.*

Proof. ✓ Proof The direct digit computation gives the global coordinate-grouping hypothesis. This implies the blocked-word identification for common sectors. Applying the common primitive decomposition theorem with this identification gives the unconditional decomposition. $\square$

Theorem 12.48 (Common sectors from phase covers). Lean ✓ Stmt *Let A and B have the same MPV family. Assume the blocked-word identification for the common-sector families. The common-sector structural theorem then produces a common blocking length with $A^{[p]} \sim \mathbf{0}_{d^p, z_A} \oplus A_{\text{nz}}$ and $B^{[p]} \sim \mathbf{0}_{d^p, z_B} \oplus B_{\text{nz}}$, and trace-preserving, primitive, irreducible nonzero-sector families on both sides. If those families satisfy the phase-cover hypotheses of Definition 12.34, then the sector decompositions obtained from the two sides have BNT bases, agree as full MPV families, and their sector weights satisfy the matched sector-weight conclusion.*

Proof. ✓ Proof Use Theorem 12.35 to form the span hypotheses of Definition 12.32. Then apply the common-sector comparison theorem with those span hypotheses. $\square$

Lemma 12.49 (After-blocking comparison from supplied BNT matching hypotheses). Not ready *Let A and B generate the same MPV family. Assume the after-blocking BNT matching hypotheses (blocking to normal tensors, equality of the residual zero-block dimensions, and BNT comparison of the nonzero summands). Then there is a positive blocking length with sector decompositions P, Q carrying BNT bases, generating the same full MPV family, and with sector weights matched up to a permutation and nonzero phases.*

Proof. Not ready Convert the BNT matching hypotheses to common phase-cover hypotheses and apply the common-sector comparison theorem. $\square$

Remark 12.50 (Equal-case comparison after canonical reduction). The equal-case proof in the source starts after both tensors have been put in canonical form and after a basis of normal tensors has been chosen. In that notation the BNT expansion has the form

$$|V^{(N)}(A)\rangle = \sum_{j=1}^g \left(\sum_{q=1}^{r_j^A} (\mu_{j,q}^A)^N \right) |V^{(N)}(A_j)\rangle,$$

and similarly for B . The proportional theorem of [CPGSV16, CPGSV21] first matches the BNT representatives: after a relabelling,

$$B_j^i = e^{i\phi_j} Y_j A_j^i Y_j^{-1}.$$

Only after this representative matching does the equal case compare the scalar coefficient power sums

$$\sum_{q=1}^{r_j^A} (\mu_{j,q}^A)^N = \sum_{q=1}^{r_j^B} (\mu_{j,q}^B e^{i\phi_j})^N \quad (j = 1, \dots, g; N \geq 1).$$

Newton–Girard then recovers the multiplicities and weight multisets, and the block gauges Y_j are placed on the diagonal to give the global conjugating matrix. This is the equal-case corollary

[CPGSV17a, Corollary II.2, lines 354–361 and appendix 1172–1192], restated in [CPGSV21, Corollary IV.5].

The formalization of this route is split into two theorems. The first packages the BNT basis bijection β , the per-block gauge-phase witnesses, copy-count equality, and the scalar coefficient comparison $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau(q)}$ matching the source power-sum identity displayed above. The second assembles the per-block gauge matrices $Y_j = X_k$ on the flattened direct sum to produce the global conjugation X , matching the appendix construction at [CPGSV17a, lines 1189–1192].

The comparison theorems above formalize only the part of this route after the common blocked nonzero sectors and their BNT matching data have been supplied. The structural reduction supplies a common positive blocking length and decompositions

$$A^{[p]} \sim \mathbf{0}_{d^p, z_A} \oplus A_{\text{nz}}, \quad B^{[p]} \sim \mathbf{0}_{d^p, z_B} \oplus B_{\text{nz}}, \quad z_A = z_B,$$

with trace-preserving, primitive, irreducible nonzero sectors. The remaining inputs are the one-site injectivity of those sectors and the BNT block-matching comparison for the representative families. These are genuine upstream assumptions of the conditional statements here; they are not proved from equality of MPV families in this chapter.

Common phase covers are an auxiliary way to express the finite-length span equality needed by the sector-comparison theorem. They are not a separate form of the source theorem. Lemma 12.23 turns a common phase cover into span equality, while Lemma 12.24 constructs such covers from BNT block-matching data. Establishing the missing representative BNT comparison and exact-length block-injective span remains the open source-faithful passage from equality of total MPV vectors to the componentwise power-sum identities.

12.3 Auxiliary constructions relocated from Chapter 9

The following auxiliary results were previously presented in Chapter 9. They are reduction steps of the fundamental-theorem proof rather than statements about canonical-form data, so they are collected here.

12.3.1 Block-diagonal corollaries of the single-block theorem

Remark 12.51. See Definition 2.37 for the formalized block-diagonal tensor construction.

Theorem 12.52 (Per-block single-block FT). [Lean] [✓ Stmt] *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. [✓ Proof] Apply the single-block fundamental theorem (Theorem 3.11) to each block independently. □

Remark 12.53 (Per-block restatement, not the multi-block paper theorem). This is a block-sum reformulation of the per-block injective single-block FT: the block pairing (A_k, B_k) is assumed already given and the blocks already matched. It is not the multi-block fundamental theorem of [CPGSV17a, Theorem II.1, lines 349–352], which derives the block matching and permutation from only the global same-MPV hypothesis.

Theorem 12.54 (Per-block same MPV implies global same MPV). [Lean] [✓ Stmt] *If A_k and B_k generate the same MPV family for each block k , then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. ✓ Proof By the MPV decomposition (Theorem 2.39), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. $\square$

Remark 12.55 (Direct-sum congruence). This is a direct-sum congruence statement: per-block MPV equality lifts to global MPV equality via the block-diagonal decomposition formula.

Theorem 12.56 (Global multi-block fundamental theorem). Lean ✓ Stmt *If each block A_k is injective and each pair (A_k, B_k) generates the same MPV family, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. ✓ Proof Apply the per-block single-block fundamental theorem (Theorem 12.52) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors. $\square$

Remark 12.57 (Block-diagonal gauge assembly, not the FT block-matching theorem). This combines per-block gauges into a block-diagonal gauge for the global assembled tensor, given pre-matched same-index blocks. It is not the FT block-matching theorem: it does not derive the block pairing or permutation from a global same-MPV hypothesis; the block correspondence is assumed in advance.

12.3.2 Proportional single-block ingredient

Theorem 12.58 (Proportional MPV implies gauge-phase equivalence). Lean ✓ Stmt *Let A and B be injective MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If A and B generate proportional MPV families $(V^{(N)}(A))_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$ and $O_{AA}(N) \rightarrow 1$, $O_{BB}(N) \rightarrow 1$, then A and B are gauge-phase equivalent.*

Proof. ✓ Proof By contradiction. If A and B are not gauge-phase equivalent, then $O_{AB}(N) \rightarrow 0$ (Theorem 7.25). From the proportionality hypothesis, $O_{AB}(N) = c_N O_{BB}(N)$. Since $O_{BB}(N) \rightarrow 1$, it follows that $c_N \rightarrow 0$. But $O_{AA}(N) = |c_N|^2 O_{BB}(N) \rightarrow 0$, contradicting $O_{AA}(N) \rightarrow 1$.

In [PGVWC07], the proportional case is handled as part of the canonical form argument. Here it is obtained directly from the overlap-decay theorem. $\square$

12.3.3 Equal-norm primitive irreducible blocks (BNT input)

Theorem 12.59 (Gauge phases have unit modulus for primitive irreducible blocks). Lean ✓ Stmt *For trace-preserving primitive irreducible blocks, every gauge-phase equivalence has a phase factor of modulus 1.*

Proof. ✓ Proof Primitivity forces the peripheral spectrum to consist of $\{1\}$ alone, so the gauge phase must have modulus one. $\square$

Remark 12.60. See Lemma 11.10 for the formalized restricted norm-class collapse statement.

Theorem 12.61 (Existential BNT independence from overlap limits). Lean ✓ Stmt *If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.*

Proof. ✓ Proof This is Lemma 11.3, restated with an explicit threshold N_0 . $\square$

Theorem 12.62 (Granular sector decomposition from eventual BNT independence). Lean ✓ Stmt *If the MPV states of the blocks are linearly independent for all sufficiently large system sizes and all weights are nonzero, then the granular sector decomposition represents the same MPS family and satisfies the BNT linear-independence condition.*

Proof.  **Proof** The equality of represented families is Lemma 9.37. The assumed eventual linear independence is exactly the BNT condition for the basis blocks of the granular sector decomposition. $\square$

Definition 12.63 (MPV phase class data).     **Stmt** For a finite family of blocks, put $j \sim k$ when there is a nonzero scalar factor ζ such that $|V^{(N)}(B_k)\rangle = \zeta^N |V^{(N)}(B_j)\rangle$ for every length N . The associated structure enumerates the finite quotient, chooses one representative per class, records the scalar relating the representative to each member, and shows that distinct representatives are not gauge-phase equivalent.

Theorem 12.64 (Phase-class BNT sector construction).   **Stmt** For trace-preserving primitive irreducible blocks of positive bond dimension with nonzero weights, there is a sector decomposition representing the same weighted block sum and satisfying the BNT linear-independence condition. The construction chooses one representative per MPV phase class and absorbs the nonzero scalar factors into the sector weights.

Proof.  **Proof** Finite phase classes partition the original block indices. Expanding the sector decomposition gives the original finite sum after replacing each class member by its representative and moving the factor ζ^N into $(\zeta\mu_k)^N$. If two chosen representatives were gauge-phase equivalent, they would lie in the same MPV phase class, contradicting their being distinct classes. Therefore the representatives satisfy the separated primitive-normal hypothesis, and the overlap-limit criterion (Theorem 12.61) gives the BNT linear-independence condition. $\square$

Chapter 13

Canonical-form data after blocking

This chapter collects the canonical-form bridge data consumed by Chapter 12 after the initial blocking and zero-tail reductions. The statements form an interface between the canonical-form chapter and the Fundamental-Theorem comparison: they package trace-preserving gauges, primitive irreducible sector data, common blocked alphabets, and zero-tail bookkeeping.

Conceptually, this material belongs to the blocking part of the canonical-form construction in [CPGSV16, Section 2.3] and to the normal/BNT setup surrounding [CPGSV21, Section IV.A]. The individual blueprint statements below are implementation bridge lemmas rather than paper-numbered theorems.

13.1 Zero-tail separation (relocated from Chapter 9)

In the irreducible block decomposition some blocks may have all-zero Kraus operators. Such blocks contribute to the MPV only at length $N = 0$, where the contribution equals their bond dimension. The three statements below separate the all-zero blocks from the nonzero blocks; they are consumed throughout this chapter (zero-tail bookkeeping) and in Chapter 12.

Definition 13.1 (Zero MPS tensor). [Lean](#) [✓ Stmt](#) The *zero MPS tensor* of physical dimension d and bond dimension D , denoted $\mathbf{0}_{d,D}$, is the tensor whose Kraus operators are all zero.

Theorem 13.2 (MPV of zero tensor). [Lean](#) [✓ Stmt](#) For any spin configuration σ of length N ,

$$V^{(N)}(\mathbf{0}_{d,D})_{\sigma} = \begin{cases} D & \text{if } N = 0, \\ 0 & \text{if } N \geq 1. \end{cases}$$

Proof. [✓ Proof](#) For $N = 0$, $(\mathbf{0}_{d,D})^{\emptyset} = \mathbf{1}_D$, so the MPV equals $\text{tr}(\mathbf{1}_D) = D$. For $N \geq 1$, every word product of a zero tensor is zero. $\square$

Theorem 13.3 (Zero-block separation). [Lean](#) [✓ Stmt](#) Every MPS tensor A admits a block decomposition into:

1. a trivial block of bond dimension D_0 with all Kraus operators zero (accumulating all all-zero irreducible blocks), and

2. a finite family of nontrivial blocks $(B_k)_{k=1}^r$, each irreducible, each with at least one nonzero Kraus operator, and each with positive bond dimension,

such that

$$V^{(N)}(A)_\sigma = V^{(N)}(\mathbf{0}_{d,D_0})_\sigma + V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma$$

for every spin configuration σ of length N . At $N = 0$ this reads $D = D_0 + \sum_{k=1}^r D_k$. For $N \geq 1$ the zero-tail term vanishes.

Proof. ✓ Proof Apply the irreducible block decomposition (Theorem 9.12) and partition the blocks by whether each has a nonzero Kraus operator. The all-zero blocks satisfy a dimension bound via irreducibility; their total bond dimension is D_0 . The MPV identity follows from the block structure. □

13.2 Reduction and structural after-blocking results

Theorem 13.4 (Blockwise TP-gauge normalization of irreducible blocks). Lean ✓ Stmt Suppose A generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^r B_k$ with each block B_k irreducible and containing at least one nonzero Kraus operator. Then, after rescaling the sector weights, one obtains blocks (C_k) such that $A \sim \bigoplus_k \mu_k C_k$, each C_k is irreducible, each C_k is left-canonical, all $\mu_k \neq 0$, and all bond dimensions are positive.

Proof. ✓ Proof The Perron–Frobenius TP-gauge construction is applied separately to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the new weights. □

Theorem 13.5 (Primitive nonzero blocks are normal). Lean ✓ Stmt A left-canonical irreducible tensor with primitive transfer map and positive bond dimension is normal.

Proof. ✓ Proof Apply Theorem 9.32. □

Theorem 13.6 (Blocking preserves irreducibility for primitive irreducible blocks). Lean ✓ Stmt If A is left-canonical, irreducible, primitive, and has positive bond dimension, then every positive blocking length P preserves irreducibility: $A^{[P]}$ is again irreducible.

Proof. ✓ Proof A positive-definite fixed point of $\mathcal{E}_A$ remains fixed by the blocked transfer map. Primitivity gives uniqueness of positive semidefinite fixed points for the blocked map, hence the blocked transfer map is irreducible; this is exactly the irreducibility criterion for the blocked tensor. □

Theorem 13.7 (Extra blocking of primitive irreducible sector blocks). Lean ✓ Stmt If a sector block is already trace-preserving, has primitive transfer map, and is tensor-irreducible, then every later positive blocking length k preserves all three properties. This k is the subsequent common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.

Proof. ✓ Proof Trace preservation follows from blocking a trace-preserving tensor, primitivity from taking a positive power of a primitive transfer map, and irreducibility from Theorem 13.6. □

Theorem 13.8 (Sorted TP primitive irreducible data form a normal canonical form). Lean ✓ Stmt Let (μ_k, A_k) be a finite block family with strictly decreasing norms $|\mu_1| > \dots > |\mu_r|$. If every block A_k is left-canonical, primitive, irreducible, has positive bond dimension, and every $\mu_k \neq 0$, then (μ_k, A_k) is a normal canonical form.

Proof. ✓ Proof This is the defining constructor for normal canonical form with all required data supplied by the hypotheses. □

Theorem 13.9 (Normal canonical form from an already primitive block decomposition). Lean

✓ Stmt *If A already admits a block decomposition into irreducible left-canonical primitive blocks with pairwise distinct weight norms, nonzero weights, and positive bond dimensions, then some blocked tensor $A^{[p]}$ is MPV-equivalent to a normal canonical form.*

Proof. ✓ Proof Apply Theorem 9.25 to the given primitive block decomposition. □

Theorem 13.10 (Common blocking period for two primitive normal tensors). Lean ✓ Stmt

Let A and B be left-canonical normal tensors of positive bond dimensions. Suppose $A^{[p_A]}$ and $B^{[p_B]}$ already have primitive transfer maps for some positive periods p_A, p_B . Then there exists a common positive period p such that both $A^{[p]}$ and $B^{[p]}$ are left-canonical, primitive, and normal.

Proof. ✓ Proof Take p to be the least common multiple of p_A and p_B . Primitivity follows from divisibility, and left-canonical normalization and normality are both preserved under blocking. □

Theorem 13.11 (Structural after-blocking decomposition). Lean ✓ Stmt *If two tensors generate the same MPV family, then each tensor admits some blocking after which it decomposes into a zero-tail tensor plus nonzero left-canonical blocks with primitive transfer maps and positive bond dimensions. Moreover, at every blocked system size and every blocked word, the MPV of the blocked tensor is the sum of the zero-tail MPV and the MPV of the weighted nonzero-block tensor:*

$$V^{(N)}(A^{[p_A]})_\sigma = V^{(N)}(Z_{p_A, z_A})_\sigma + V^{(N)}\left(\bigoplus_k \mu_k^A A_k\right)_\sigma,$$

and likewise for B . The blocked tensors also preserve the original MPV equality:

$$A^{[p_A]} \sim B^{[p_A]}, \quad A^{[p_B]} \sim B^{[p_B]}.$$

Proof. ✓ Proof Apply Theorem 13.31 separately to the two tensors. □

Remark 13.12 (Structural after-blocking decomposition; not the FT block-matching theorem). This theorem produces, for each of the two same-MPV tensors independently, a TP/primitive/left-canonical block decomposition after blocking. It does *not* derive any block matching, gauge equivalence, or phases between the two decompositions; those are produced by the subsequent common-sector construction in this chapter. The statement is a preliminary step for the fundamental-theorem proof, not the multi-block fundamental theorem of [CPGSV17a, Theorem II.1, lines 349–352].

Theorem 13.13 (Zero-tail decomposition after blocking). Lean ✓ Stmt

Suppose A is expressed as a zero-tail tensor plus a nonzero part L at all system sizes. For every positive blocking length p , the blocked tensor $A^{[p]}$ is expressed as the same zero-tail bond dimension plus $L^{[p]}$. The zero-tail term still contributes only at blocked length 0.

Proof. ✓ Proof Flatten a blocked word of length N to an original word of length Np . Since $p > 0$, this length is zero exactly when $N = 0$, so the zero-tail MPV is unchanged by the reblocking convention. □

Theorem 13.14 (Zero-tail weighted block decomposition after blocking). [Lean](#) [✓ Stmt](#) Suppose A is expressed as a zero-tail tensor plus the weighted nonzero-block tensor $\bigoplus_k \mu_k B_k$. For every positive blocking length p , $A^{[p]}$ is expressed as the same zero-tail contribution plus the weighted blocked nonzero tensor $\bigoplus_k \mu_k^p B_k^{[p]}$.

Proof. [✓ Proof](#) Apply Theorem 13.13 to the assembled weighted nonzero tensor and rewrite the blocked assembled tensor using Theorem 9.47. $\square$

Theorem 13.15 (Length-zero identity after blocking weighted blocks). [Lean](#) [✓ Stmt](#) If the length-zero zero-tail identity holds for two assembled weighted block tensors, then after any blocking length $p \geq 0$ the same identity holds for the blocked families with weights transported to the powers μ_k^p .

Proof. [✓ Proof](#) At blocked length zero the flattened configuration is empty. The blocked assembled tensors are rewritten by Theorem 9.47, and the original length-zero identity is applied to the empty configuration. $\square$

Theorem 13.16 (Zero-tail identity for nonzero block tensors). [Lean](#) [✓ Stmt](#) Suppose two tensors have the same MPV family and each is expressed as a zero-tail tensor plus a weighted nonzero-block tensor. Then the two nonzero-block tensors agree on every positive system size, and at size 0 the equality is exactly the equality of the two zero-tail contributions plus the two nonzero bond-dimension traces.

Proof. [✓ Proof](#) For $N > 0$, the MPV of a zero-tail tensor is zero, so the two decompositions can be compared directly through the common MPV family. For $N = 0$, the zero-tail MPV is its bond dimension, giving the stated length-zero identity. $\square$

Theorem 13.17 (Blocked nonzero parts and length zero). [Lean](#) [✓ Stmt](#) In the setting of Theorem 13.16, after any positive common blocking length p , the powered weighted nonzero-block tensors agree on all positive blocked system sizes. At blocked length 0, the same zero-tail identity holds with weights transported from μ_k to μ_k^p .

Proof. [✓ Proof](#) Use Theorem 13.14 on both zero-tail decompositions, block the original MPV equality, and apply Theorem 13.16 at the blocked physical dimension. $\square$

Theorem 13.18 (Equal zero tails give full nonzero-part MPV equality). [Lean](#) [✓ Stmt](#) In the setting of Theorem 13.16, if the two zero-tail dimensions are equal, then the two nonzero-block tensors have the same MPV family, including the length-zero case.

Proof. [✓ Proof](#) The positive-length cases are exactly the first conclusion of Theorem 13.16. At $N = 0$, substitute the equality of zero-tail dimensions into the length-zero identity and cancel the common scalar term. $\square$

Theorem 13.19 (Cyclic sectors after the zero-tail split). [Lean](#) [✓ Stmt](#) If two tensors generate the same MPV family, then after the zero-tail split and trace-preserving gauge each side has irreducible nonzero-weight blocks. The nonzero-block tensors agree at all positive system sizes, while the length-zero case is included as the exact zero-tail identity. Moreover each nonzero-weight block has primitive irreducible cyclic sectors. The period-removal length for those sectors is kept separate from any later common blocking or injectivity length.

Proof. [✓ Proof](#) Apply the arbitrary-tensor TP-gauge theorem separately to A and B . The zero-tail identity theorem compares the two resulting nonzero parts at positive lengths and gives the $N = 0$ identity. Finally apply Theorem 9.49 to every trace-preserving irreducible nonzero-weight block. $\square$

Theorem 13.20 (Common blocking for cyclic sectors). [Lean](#) [✓ Stmt](#) *If two tensors generate the same MPV family, then the zero-tail/TP-gauge reduction for nonzero-weight blocks can be combined with a common reblocking of all per-block cyclic sectors. Each side obtains a finite flattened sector family at one physical blocking length, with trace preservation, primitive transfer maps, tensor irreducibility, positive bond dimensions, and nonzero unit weights. The theorem keeps the positive-length equality of the nonzero parts and the exact $N = 0$ zero-tail identity for the original nonzero-block tensors.*

Proof. [✓ Proof](#) Apply Theorem 13.19. Then apply Theorem 9.50 to the per-block cyclic-sector data on each side. The nonzero unit-weight conclusion is Theorem 9.51. $\square$

The next results put the cyclic-sector decompositions on a common blocked alphabet. They compare direct blocking with iterated cyclic-sector blocking and transport the zero-tail identity through this comparison.

Theorem 13.21 (Common cyclic sectors after reblocking). [Lean](#) [✓ Stmt](#) *If two tensors generate the same MPV family, then after reblocking the nonzero blocks one obtains cyclic-sector families over a common alphabet. These families preserve the MPV of the reblocked nonzero part, with nonzero transported weights μ_k^p , and retain the zero-tail identity.*

Proof. [✓ Proof](#) Start from the common reblocked cyclic-sector families on the two sides. Apply the zero-tail reblocking theorem to the original nonzero-block decomposition, then use the common-alphabet sector decomposition and the derived structural properties of the common-sector family. $\square$

Theorem 13.22 (Common blocking length for cyclic sectors). [Lean](#) [✓ Stmt](#) *If two tensors generate the same MPV family, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length. At this length the nonzero parts still agree at positive system sizes, and the length-zero identity is exactly the zero-tail identity.*

Proof. [✓ Proof](#) Start from the per-block cyclic-sector theorem. Let p_A and p_B be the least common multiples of the period-removal lengths on the two sides, and use $p = \text{lcm}(p_A, p_B)$. The prescribed-common-multiple theorem constructs both one-sided sector families at this same length. The zero-tail and positive-length equalities are the existing reblocking statements, and the sector conclusions are the common-alphabet MPV equalities. $\square$

Theorem 13.23 (Length-zero identity under compatible blocking). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Assume a zero-tail/nonzero decomposition on one side. If direct blocking of each nonzero block agrees with the corresponding iterated cyclic-sector blocking, then the zero-tail identity can be rewritten using the common cyclic-sector family. The coordinate-grouping condition supplies this agreement at any prescribed common blocking length.*

Proof. [✓ Proof](#) Transport the zero-tail/nonzero decomposition to the common blocking length. The blockwise comparison then replaces the directly blocked nonzero part by the weighted common-sector tensor. $\square$

Theorem 13.24 (Length-zero identity under word identification). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Assume a zero-tail/nonzero decomposition on one side. If the blocked nonzero tensor agrees, after identifying blocked words, with the common cyclic-sector family, then the same zero-tail identity can be written in that common alphabet. At positive system sizes this says that the blocked tensor and the weighted common-sector tensor have the same MPV coefficients.*

Proof. ✓ **Proof** Transport the zero-tail/nonzero decomposition to the common blocking length. The word-identification hypothesis then replaces the blocked nonzero tensor by the weighted common-sector tensor. At positive lengths the zero-tail tensor has zero MPV coefficients. $\square$

Theorem 13.25 (Primitive block decompositions at a common length). Lean ✓ **Stmt** *Assume that the nonzero part can be identified with the common blocked alphabet. Then two tensors with the same MPV family admit one positive blocking length at which both nonzero parts are weighted sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights.*

Proof. ✓ **Proof** Apply Theorem 13.22 to choose the common blocking length and the two common cyclic-sector families. The assumed word identification converts the blocked nonzero part into the weighted common-sector family on each side. The zero-tail equations then give the displayed decompositions. The structural properties and nonzero weights come from the common-sector families. $\square$

Theorem 13.26 (Transport of MPV and length-zero identities). Lean ✓ **Stmt** *Let L_A, L_B be two nonzero parts whose MPV families agree at positive lengths and whose zero-tail dimensions satisfy the length-zero identity. If L'_A and L'_B generate the same MPV families as L_A and L_B , respectively, then L'_A and L'_B satisfy the same positive-length equality and the same length-zero zero-tail identity.*

Proof. ✓ **Proof** Use the two MPV equalities to replace L'_A by L_A and L_B by L'_B in the positive-length equality and in the $N = 0$ identity. $\square$

Theorem 13.27 (Length-zero identity for common sectors). Lean ✓ **Stmt** *If the blocked nonzero part agrees with the common-sector tensor, then the zero-tail identity can be written with that common-sector tensor. Nonzero original weights remain nonzero after the blocking power.*

Proof. ✓ **Proof** Combine Theorem 13.24 with the common-sector MPV equality and the nonzero-weight statement. $\square$

Theorem 13.28 (Common sectors from grouped coordinates). Lean ✓ **Stmt** *Assume a zero-tail/nonzero decomposition with nonzero weights. If the common blocked alphabet is identified with each iterated alphabet by grouping consecutive words, then the zero-tail identity can be written with the weighted common-sector family.*

Proof. ✓ **Proof** The coordinate-grouping comparison identifies the directly blocked nonzero part with the weighted common-sector family. Combine this with the zero-tail identity and the nonzero-weight statement. $\square$

Theorem 13.29 (Common sectors at one blocking length). Lean ✓ **Stmt** *If two tensors generate the same MPV family, choose a common blocking length for their cyclic-sector decompositions. If each blocked nonzero part agrees with its common-sector form after identifying words, then all zero-tail identities and positive-length MPV equalities can be stated using the common sector families.*

Proof. ✓ **Proof** Apply Theorem 13.22. The two word-identification hypotheses identify the blocked nonzero parts with the weighted common-sector tensors. Substituting these MPV equalities into the zero-tail equations gives the rewritten zero-tail identities, the positive-length MPV equalities, and the $N = 0$ identity. $\square$

13.3 Primitive-block reduction (relocated from Chapter 9)

The following two statements were previously presented in Chapter 9. They describe primitive blocked sectors and a primitive-block decomposition starting from an arbitrary tensor; both are after-blocking structural results, so they are collected here.

Theorem 13.30 (Unconditional primitive blocked sectors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive m -th root of unity γ . Let $(C_u)_{u=1}^m$, $(P_u)_{u=1}^m$, and $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$ be the blocked cyclic-sector data. Then every blocked sector tensor C_u has primitive transfer map and is irreducible.*

Proof. ✓ Proof Theorem 9.52 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u . The blocked cyclic-sector data provide compression equivalences φ_u intertwining the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The same cyclic peripheral-spectrum argument that produces the sector decomposition shows that each corner restriction is primitive. Transport primitivity and irreducibility across φ_u , then use Theorem 9.45 to pass from the adjoint blocked map to the primal transfer map of C_u . $\square$

Theorem 13.31 (Arbitrary tensor to primitive block decomposition). Lean ✓ Stmt *For any MPS tensor A , there exist a period $p \geq 1$, a zero-tail bond dimension D_0 , nonzero weights $(\mu_k)_{k=1}^r$, and left-canonical blocks $(B_k)_{k=1}^r$ with primitive transfer maps and positive bond dimensions, such that*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\mathbf{0}_{d[p], D_0})_\sigma + V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma.$$

Proof. ✓ Proof Apply Theorem 9.27 to get the trivial block and left-canonical nontrivial blocks. Apply Theorem 9.28 to find a common blocking period p making all transfer maps primitive. Lemma 9.48 keeps the blocked weights nonzero, and Theorem 13.14 transports the zero-tail/nonzero MPV identity through the blocking step. $\square$

Chapter 14

Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem [PGVWC07], following the symmetry analysis of [PGWS⁺08]. For injective tensors, physical on-site symmetries determine virtual gauges, those gauges define projective representations on the bond space, and the same virtual data controls string-order phenomena.

14.1 Physical Symmetries Give Virtual Gauges

For periodic tensors we use the following physical-index rotation, both in this chapter and in Chapter 15.

Definition 14.1 (Physical-index rotation). Lean ✓ Stmt Given a matrix $u \in M_d(\mathbb{C})$ and an MPS tensor A , the *physical-index rotation* is the tensor $(A_u)^i := \sum_j u_{ij} A^j$.

The compatibility theorems for this definition — Theorems 15.4, 15.5, 15.6, 15.7, 15.8, 15.9, and 15.10 — are proved in Chapter 15 (which follows this chapter).

Corollary 14.2 (Periodic symmetry yields gauge equivalence up to a root of unity). Lean ✓ Stmt Let A be in irreducible form II and let u be a matrix such that $B := A_u$ is also in irreducible form II and $\mathcal{V}(A) = \mathcal{V}(B)$. Then there exists $m > 0$ such that A and B are gauge equivalent up to an m -th root of unity.

Proof. ✓ Proof Direct application of the periodic equal-case fundamental theorem (Theorem 15.24) to A and B . □

14.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

Definition 14.3 (Twisted tensor). Lean ✓ Stmt Given a group representation $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g -twisted tensor* is defined, for each $i \in \{0, \dots, d-1\}$, by

$$\tilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij} A^j$$

Definition 14.4 (On-site symmetry). [Lean](#) [✓ Stmt](#) A tensor A is *on-site symmetric* under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$ for every $g \in G$.

Lemma 14.5 (Functoriality of twisting). [Lean](#) [✓ Stmt](#) The twisted tensor satisfies the composition law

$$\tilde{A}_{gh} = \widetilde{(\tilde{A}_h)_g},$$

i.e. twisting by gh is the same as first twisting by h and then by g .

Proof. [✓ Proof](#) Expanding both sides using $U(gh) = U(g)U(h)$, then swapping the order of summation. $\square$

Lemma 14.6 (Identity twist). [Lean](#) [✓ Stmt](#) Twisting by the identity group element is trivial: $\tilde{A}_1 = A$.

Proof. [✓ Proof](#) Since $U(1) = I_d$, each component reduces to $\sum_j \delta_{ij} A^j = A^i$. $\square$

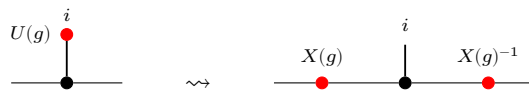
Lemma 14.7 (Symmetric twists are gauge equivalent). [Lean](#) [✓ Stmt](#) If A is injective and on-site symmetric under U , then for every $g \in G$ the twisted tensor $\tilde{A}_g$ is gauge equivalent to A .

Proof. [✓ Proof](#) On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$, and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence. $\square$

Remark 14.8 (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from string-order rigidity. The on-site symmetry condition first identifies A and $\tilde{A}_g$ as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 14.7. Gauge uniqueness and the twist composition law then assemble these gauges into Theorem 14.15. The string-order results later in this chapter use this virtual representation as an input.

Theorem 14.9 (Virtual symmetry equation). [Lean](#) [✓ Stmt](#) If A is injective and on-site symmetric under U , then for every $g \in G$ there exist an invertible matrix $X(g) \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that $\tilde{A}_g^i = \varphi(g) X(g) A^i X(g)^{-1}$ for all i .

Proof. [✓ Proof](#) The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 14.7 to obtain $X(g)$, then set $\varphi(g) = 1$. $\square$

Lemma 14.10 (Gauge uniqueness up to scalar). [Lean](#) [✓ Stmt](#) Let A be an injective tensor. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.

Proof. [✓ Proof](#) The matrix $Z = Y^{-1}X$ commutes with every A^i . Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 16.12, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. $\square$

Theorem 14.11 (Gauge uniqueness from equality of MPV families). [Lean](#) [✓ Stmt](#) Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.

Proof. [✓ Proof](#) This is exactly Lemma 14.10; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced. $\square$

Definition 14.12 (Scalar 2-cochain). [Lean](#) [✓ Stmt](#) A scalar 2-cochain on a group G is a function $\omega : G \times G \rightarrow \mathbb{C}^\times$.

Definition 14.13 (Projective representation). [Lean](#) [✓ Stmt](#) Given a scalar 2-cochain ω , a projective representation of G on $\mathbb{C}^D$ is a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh)$$

Lemma 14.14 (Associativity forces the cocycle condition). [Lean](#) [✓ Stmt](#) If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity:

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k).$$

Proof. [✓ Proof](#) Expanding $\rho(g) (\rho(h) \rho(k))$ and $(\rho(g) \rho(h)) \rho(k)$ using the projective multiplication law and equating via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor. $\square$

Theorem 14.15 (Virtual representation theorem for injective MPS). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. Then there exist:

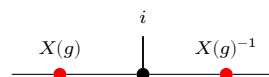
1. a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$, and
2. a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh)$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has, for every $g \in G$ and $i \in \{0, \dots, d-1\}$,

$$\tilde{A}_g^i = X(g) A^i X(g)^{-1}$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. [✓ Proof](#) Since A is injective and on-site symmetric, each twisted tensor $\tilde{A}_g$ is gauge equivalent to A (Lemma 14.7). Choose $X(g) \in \text{GL}_D(\mathbb{C})$ with $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$ for all i .

By functoriality (Lemma 14.5), the product $X(h) X(g)$ also intertwines A with $\tilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 14.10) gives a nonzero scalar $\omega'(h, g)$ with $X(h) X(g) = \omega'(h, g) X(gh)$.

Defining $\rho(g) := X(g^{-1})$ converts this to the standard law: $\rho(g) \rho(h) = \omega(g, h) \rho(gh)$ where $\omega(g, h) := \omega'(h^{-1}, g^{-1})$. $\square$

Corollary 14.16 (2-cocycle identity for the factor system). [Lean](#) [✓ Stmt](#) If $D > 0$, the factor system ω from Theorem 14.15 satisfies the 2-cocycle identity for all $g, h, k \in G$:

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k)$$

Proof. [✓ Proof](#) Theorem 14.15 gives ρ and ω . By Lemma 14.14, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices. $\square$

14.3 Cohomology Classes of Virtual Symmetry Data

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient $H^2(G, \mathbb{C}^\times)$.

Definition 14.17 (Coboundary). [Lean](#) [✓ Stmt](#) A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega(g, h) = \varphi(g) \varphi(h) \varphi(gh)^{-1}$$

Definition 14.18 (Cohomologous cocycles). [Lean](#) [✓ Stmt](#) Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega_1(g, h) = \varphi(g) \varphi(h) \varphi(gh)^{-1} \omega_2(g, h)$$

Theorem 14.19 (Cohomologous-to is an equivalence relation). [Lean](#) [✓ Stmt](#) The relation “cohomologous to” is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.

Proof. [✓ Proof](#) Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1} . Transitivity: compose the witnesses φ and ψ pointwise. $\square$

Lemma 14.20 (Coboundary iff cohomologous to the trivial cocycle). [Lean](#) [✓ Stmt](#) A cocycle ω is a coboundary if and only if $\omega \sim \mathbf{1}$, where $\mathbf{1}(g, h) = 1$ for all g, h .

Proof. [✓ Proof](#) Both directions follow by absorbing or introducing the factor $\mathbf{1}(g, h) = 1$ via $x \cdot 1 = x$. $\square$

Theorem 14.21 (Gauge independence of the cocycle class). [Lean](#) [✓ Stmt](#) If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U , then $\omega_1 \sim \omega_2$.

Proof. [✓ Proof](#) For each g , gauge uniqueness gives a nonzero scalar $f(g)$ with $X_2(g^{-1}) = f(g) X_1(g^{-1})$. Equivalently, for each fixed g the two local gauges

$$\begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \\ X_1(g^{-1}) \quad X_1(g^{-1})^{-1} \end{array} \quad \text{and} \quad \begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \\ X_2(g^{-1}) \quad X_2(g^{-1})^{-1} \end{array}$$

describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling $X_1(g \cdot h)$ yields $\omega_2(g, h) = f(g) f(h) f(gh)^{-1} \omega_1(g, h)$. $\square$

Definition 14.22 (Multiplicative 2-cocycle condition). [Lean](#) [✓ Stmt](#) A scalar 2-cochain $\omega: G \times G \rightarrow \mathbb{C}^\times$ is a *2-cocycle* if it satisfies, for all $g, h, k \in G$,

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k)$$

Definition 14.23 (Second cohomology quotient). [Lean](#) [✓ Stmt](#) The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times) / \sim,$$

where $\sim$ is the coboundary-equivalence relation.

Definition 14.24 (Projective equivalence). [Lean](#) [✓ Stmt](#) Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

Theorem 14.25 (Projective equivalence and cohomologous cocycles). [Lean](#) [✓ Stmt](#) Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2 . Then

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

Proof. [✓ Proof](#) This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems. $\square$

14.4 Permutation-Based Physical Symmetry

The linear-combination twist $\tilde{A}_g^i = \sum_j U(g)_{ij} A^j$ is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is a formal convenience for symmetries that act by reordering basis states rather than by a general linear map.

Definition 14.26 (Permutation symmetry data). [Lean](#) [✓ Stmt](#) An *on-site symmetry datum* for a group G on a physical space of dimension d consists of a matrix-valued map $u : G \rightarrow M_d(\mathbb{C})$ and a physical-index action $\sigma : G \rightarrow \text{End}(\{0, \dots, d-1\})$.

Definition 14.27 (Permutation-twisted tensor). [Lean](#) [✓ Stmt](#) Given on-site symmetry data (u, σ) , the *permutation-twisted tensor* at group element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:

Lemma 14.28 (Identity permutation twist). [Lean](#) [✓ Stmt](#) If $\sigma(1) = \text{id}$, then $(A^{\sigma_1})^i = A^i$ for all i .

Proof. [✓ Proof](#) Immediate from $\sigma(1)(i) = i$ for all i . $\square$

Composition order. The linear-combination twist (Lemma 14.5) composes as $\widetilde{(\tilde{A}_h)_g} = \tilde{A}_{gh}$, applying h first then g . The permutation twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner permutation $\sigma(h)$ acts first on the index.

Lemma 14.29 (Composition of permutation twists). [Lean](#) [✓ Stmt](#) If $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$, then

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

Proof. [✓ Proof](#) Expanding both sides: $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

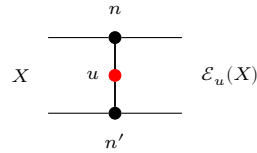
14.5 String Order and Local Symmetry Equivalence

The string order parameter [\[PGWS⁺08\]](#) detects whether an on-site unitary u acts as a local symmetry on the state generated by an injective MPS tensor.

Definition 14.30 (Twisted transfer map). [Lean](#) [✓ Stmt](#) Given an MPS tensor A and a physical-index matrix u , the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n' | u | n \rangle A_n X A_{n'}^\dagger.$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

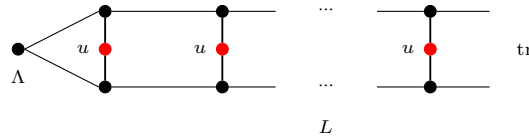


The upper and lower black nodes denote A_n and $A_{n'}^\dagger$, and the red node labelled u couples the physical legs.

Definition 14.31 (String order parameter). [Lean](#) [✓ Stmt](#) The *string order parameter* for twist u and stationary state Λ at length L is

$$R_L(u) := \text{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length- L doubled chain with a copy of u inserted on each physical rung:



The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 14.32 (Boundary string order parameter). [Lean](#) [✓ Stmt](#) For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *boundary string order parameter* is

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The ordinary string order parameter is the special case $X = Y = \mathbb{1}$.

Definition 14.33 (Local symmetry). [Lean](#) [✓ Stmt](#) An MPS tensor A has *local symmetry* under a unitary u with respect to a boundary state Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger.$$

The unitary V intertwines the physical action of u with a virtual conjugation, and the boundary state Λ is invariant under V .

Definition 14.34 (String order). [Lean](#) [✓ Stmt](#) String order *exists* for u with respect to a boundary state Λ if there exist boundary matrices X, Y and a positive constant $c > 0$ such that $c \leq \|R_L(u; X, Y)\|$ for all L , where $R_L(u; X, Y)$ is the boundary-refined string-order parameter. Informally, this means the scalar string-order parameter $R_L(u)$ does not decay to zero.

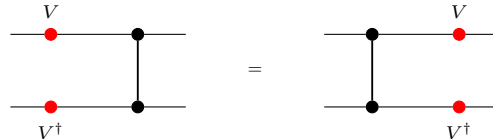
Lemma 14.35 (String order prevents decay to zero). [Lean](#) [✓ Stmt](#) If A has string order for u with respect to a boundary state Λ , then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \rightarrow \infty$.

Proof. [✓ Proof](#) A uniform lower bound $c \leq \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. $\square$

Definition 14.36 (Local physical intertwining). [Lean](#) [✓ Stmt](#) The local physical intertwining relation is $\sum_j u_{ij} A^j = V A^i V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



Definition 14.37 (Transfer-map covariance). [Lean](#) [✓ Stmt](#) Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation: $\mathcal{E}(V X V^\dagger) = V \mathcal{E}(X) V^\dagger$. In doubled-layer notation, the virtual operators slide through the local transfer cell:

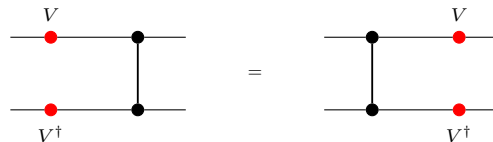


Definition 14.38 (Commutation in the doubled transfer picture). [Lean](#) [✓ Stmt](#) In the transfer-map language, the commutation of the doubled transfer matrix $E = \sum_i A_i \otimes \bar{A}_i$ with $V \otimes \bar{V}$ is

$$V \mathcal{E}_1(X) V^\dagger = \mathcal{E}_1(V X V^\dagger).$$

Theorem 14.39 (Transfer-map covariance and doubled commutation are equivalent). [Lean](#) [✓ Stmt](#) Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V .

Proof. [✓ Proof](#) Both sides express the same local picture with opposite orientation:



The two formulas are literally the same identity read from opposite sides. $\square$

Lemma 14.40 (Unitary Kraus mixing). [Lean](#) [✓ Stmt](#) *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel: $\sum_i (\sum_j u_{ij} A_j) Y (\sum_j u_{ij} A_j)^\dagger = \sum_i A_i Y A_i^\dagger$.*

Proof. [✓ Proof](#) This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 4.71. $\square$

Theorem 14.41 (Local physical intertwining implies transfer-map covariance). [Lean](#) [✓ Stmt](#) *If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

Proof. [✓ Proof](#) Starting from

one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u -boxes disappear by the unitary Kraus-mixing identity. $\square$

Theorem 14.42 (Transfer-map covariance implies local physical intertwining). [Lean](#) [✓ Stmt](#) *For an injective MPS, if V is unitary and the transfer map is covariant, then there exists a unitary u satisfying the local physical intertwining relation.*

Proof. [✓ Proof](#) Set $B^i := V A^i V^\dagger$. The transfer-map covariance relation

implies that the two Kraus families B and A define the same transfer map. The missing step — supplied by Kraus freedom for equal channels — converts equality of transfer maps $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. $\square$

Definition 14.43 (Twisted companion tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor A and a physical-index matrix u , define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}} A^{n'}.$$

Lemma 14.44 (Twisted transfer as a mixed transfer operator). [Lean](#) [✓ Stmt](#) *The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u) :*

$$\mathcal{E}_u = F_{A, B_u}.$$

Proof. [✓ Proof](#) Expand both definitions and rearrange the double sum. $\square$

Theorem 14.45 (Twisted-transfer eigenvalues have modulus at most one). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \leq 1$.

Proof. [✓ Proof](#) By Lemma 14.44, the twisted transfer map is the mixed transfer operator F_{A,B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common TP gauge. Theorem 7.14 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ . $\square$

Theorem 14.46 (Modulus-one twisted eigenvalue implies gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u .

Proof. [✓ Proof](#) By Lemma 14.44, the twisted transfer map is the mixed transfer operator F_{A,B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common TP gauge. In that gauge the eigenvalue equation becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 7.28 yields gauge-phase equivalence. Undoing the common TP gauge gives the claim for A and B_u . $\square$

Lemma 14.47 (Spectral radius < 1 forces boundary-string decay). [Lean](#) [✓ Stmt](#) If $\rho(\mathcal{E}_u) < 1$, then for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has

$$R_L(u; X, Y) \longrightarrow 0 \quad \text{as } L \rightarrow \infty.$$

Proof. [✓ Proof](#) In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \rightarrow 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \text{tr}(\Lambda X Z)$ gives the stated scalar convergence. $\square$

Theorem 14.48 (String order gives gauge-phase equivalence with the companion tensor). [Lean](#) [✓ Stmt](#) Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ , then A is gauge-phase equivalent to the companion tensor B_u .

Proof. [✓ Proof](#) If $\rho(\mathcal{E}_u) < 1$, Lemma 14.47 forces every boundary string-order sequence to decay to zero, contradicting Definition 14.34. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 14.45 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 14.46 then gives the gauge-phase equivalence. $\square$

Theorem 14.49 (Gauge-phase equivalence with the companion tensor yields a virtual unitary). [Lean](#) [✓ Stmt](#) Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u , then there exist a unitary V and a phase μ with $|\mu| = 1$ such that, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

Proof. ✓ Proof Start from an invertible intertwiner X between A and B_u . The positive matrix $Q := XX^\dagger$ is a positive fixed point of $\mathcal{E}_A$. By uniqueness of positive fixed points for an injective normalized tensor, Q is a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V , and the original intertwining relation becomes the phased covariance equation above, with $|\mu| = 1$. $\square$

Theorem 14.50 (Virtual unitary gives a twisted-transfer eigenmatrix). Lean ✓ Stmt Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and that, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

with V unitary. Then V is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V.$$

Proof. ✓ Proof Expand $\mathcal{E}_u(V)$, insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. $\square$

Theorem 14.51 (Boundary-state invariance of the virtual unitary). Lean ✓ Stmt Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

then

$$V^\dagger \Lambda V = \Lambda.$$

Proof. ✓ Proof The covariance relation shows that $V^\dagger \Lambda V$ is another positive fixed point of the adjoint transfer map. Since Λ is the normalized positive fixed point, uniqueness forces $V^\dagger \Lambda V = \Lambda$. $\square$

Lemma 14.52 (Local symmetry implies string order). Lean ✓ Stmt Assume $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If u is a local symmetry of A with respect to Λ , then string order exists for u .

Proof. ✓ Proof Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the local-symmetry datum. By Theorem 14.50, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L \text{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. $\square$

Theorem 14.53 (Local symmetry iff a unitary peripheral eigenmatrix exists). Lean ✓ Stmt Let A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then u is a local symmetry if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that

$$\mathcal{E}_u(V) = \mu V.$$

By Theorem 14.45, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. ✓ Proof The forward direction is Theorem 14.50. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 14.46 turns it into a gauge-phase equivalence with the companion tensor; then Theorem 14.49 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 14.51 gives the boundary-state invariance needed to assemble the local-symmetry data. $\square$

Theorem 14.54 (String order $\Leftrightarrow$ local symmetry). [Lean](#) [✓ Stmt](#) For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order for u exists iff u is a local symmetry.

Proof. [✓ Proof](#) If string order exists, Theorem 14.48 gives a gauge-phase equivalence with the companion tensor. Theorem 14.49 extracts a virtual unitary and phase from that equivalence, and Theorem 14.51 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 14.52 turns local symmetry into a non-decaying boundary string-order sequence. $\square$

Theorem 14.55 (Virtual unitary from string order). [Lean](#) [✓ Stmt](#) For an injective MPS, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If string order exists for u with boundary parameter Λ , then there exists a virtual unitary V and a unit-modulus scalar μ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all i . Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:

Proof. [✓ Proof](#) String order first gives gauge-phase equivalence with the companion tensor by Theorem 14.48. Theorem 14.49 then normalizes the resulting intertwiner to a unitary and produces the phase μ . $\square$

14.6 SPT Phase Classification

Two injective symmetric tensors belong to the same symmetry-protected topological phase when their virtual representation cocycles are cohomologous [CGW11].

Definition 14.56 (Same SPT phase). [Lean](#) [✓ Stmt](#) Injective tensors A and B , both symmetric under an on-site representation $U: G \rightarrow \text{GL}_d(\mathbb{C})$, are in the *same SPT phase* if there exist virtual cocycles ω_A, ω_B with corresponding projective representations that intertwine the respective twisted tensors and satisfy $\omega_A \sim \omega_B$ (cohomologous).

Theorem 14.57 (String order holds for injective symmetric MPS). [Lean](#) [✓ Stmt](#) Let A be an injective tensor, symmetric under a unitary representation U . Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then string order exists for every group element g .

Proof. [✓ Proof](#) The virtual representation theorem produces $\rho(g^{-1}) \in \text{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 14.46 upgrades this to gauge-phase equivalence with the companion tensor, and Theorem 14.49 extracts a virtual unitary and phase. By Theorem 14.51, this unitary preserves the boundary state, so Lemma 14.52 gives string order. $\square$

Theorem 14.58 (String order is an SPT invariant). [Lean](#) [✓ Stmt](#) If A and B are injective, both on-site symmetric under a unitary representation U , and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite, $\text{tr}(\Lambda_A) = 1$, $\text{tr}(\Lambda_B) = 1$, $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, string order exists for A with twist $U(g)$ if and only if it exists for B . In particular, since Definition 14.56 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase.

Proof.  **Proof** Both directions follow from Theorem [14.57](#), which shows that string order holds universally for injective symmetric tensors with canonical normalisations. $\square$

Chapter 15

Periodic MPS: irreducible form and the periodic Fundamental Theorem

This chapter develops the periodic equal-case Fundamental Theorem of MPS (Theorem 15.24) following [DICCSPG17], and gives its two applications: the symmetry corollary lifting a physical on-site symmetry of a tensor in irreducible form II to a virtual Z -gauge, and the equivalence between p -refinability of the matrix-product-vector family and p -divisibility of the transfer map.

15.1 Periodic irreducible form and physical-index rotation

Definition 15.1 (Periodic tensor). [Lean](#) [✓ Stmt](#) An irreducible MPS tensor A of bond dimension D is m -periodic (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the m -th roots of unity $\{e^{2\pi i k/m} : 0 \leq k < m\}$. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 4.49).

Definition 15.2 (Irreducible form). [Lean](#) [✓ Stmt](#) A tensor A is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j A_j^i,$$

where each block A_j is an m_j -periodic tensor, each weight $\mu_j \neq 0$, and the blocks are pairwise non-repeated (no two distinct j, j' have A_j gauge-equivalent to $A_{j'}$). This is the standard form introduced in [DICCSPG17, Section II]. It extends the normal canonical form (Definition 9.18), which requires the stronger condition that every block is 1-periodic.

Remark 15.3 (Relation to normal canonical form). [Not ready](#) Every tensor in normal canonical form (Definition 9.18) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic.

Physical-index rotation was introduced in Definition 14.1 (Chapter 14). The following compatibility theorems are proved in the present chapter.

Theorem 15.4 (Transfer map preserved by physical rotation). [Lean](#) [✓ Stmt](#) If $uu^\dagger = I$, then the transfer map is invariant under physical-index rotation: $\mathcal{E}_{A_u} = \mathcal{E}_A$.

Proof. ✓ Proof This is the unitary freedom of Kraus representations from [Wol12, Theorem 2.18]. □

Theorem 15.5 (Left-canonical preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is left-canonical, then A_u is left-canonical.*

Proof. ✓ Proof Expand $\sum_i B_i^\dagger B_i$ where $B_i = \sum_j u_{ij} A_j$, swap sums, apply orthogonality $u^\dagger u = I$, and collapse to $\sum_j A_j^\dagger A_j = I$. □

Theorem 15.6 (Irreducibility preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is irreducible, then A_u is irreducible.*

Proof. ✓ Proof Contrapositive: any nontrivial invariant projection P for the rotated tensor yields $(1 - P) A_k P = 0$ for all k via the orthogonality of u , contradicting irreducibility of A . □

Theorem 15.7 (Periodicity preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is periodic with period m , then A_u is periodic with the same period.*

Proof. ✓ Proof Assembles transfer-map invariance, left-canonical preservation, and irreducibility preservation. □

Theorem 15.8 (Equality of MPV families preserved by physical rotation). Lean ✓ Stmt *If $\mathcal{V}(A) = \mathcal{V}(B)$, then $\mathcal{V}(A_u) = \mathcal{V}(B_u)$.*

Proof. ✓ Proof Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation. □

Theorem 15.9 (Physical rotation distributes over block-diagonal composition). Lean ✓ Stmt *The rotated block-diagonal tensor satisfies $(\bigoplus_k \mu_k A_k)_u = \bigoplus_k \mu_k (A_k)_u$.*

Proof. ✓ Proof Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing. □

Theorem 15.10 (Irreducible form preserved by physical rotation). Lean ✓ Stmt *If $uu^\dagger = I$ and A is in irreducible form, then A_u is in irreducible form with the same block structure and rotated blocks.*

Proof. ✓ Proof Each block remains periodic by Theorem 15.7, equality of MPV families transfers via Theorem 15.8, and block-diagonal compatibility follows from Theorem 15.9. □

15.1.1 The Z-gauge degree of freedom

Definition 15.11 (Entrywise periodic Z-gauge ratio). Lean ✓ Stmt Given complex weights μ, ν , define the entrywise Z-gauge ratio by

$$z(\mu, \nu) := \mu / \nu.$$

Lemma 15.12 (Matched powers imply root-of-unity ratio). Lean ✓ Stmt *If $\mu^m = \nu^m$ and $\nu \neq 0$, then*

$$z(\mu, \nu)^m = 1.$$

Lemma 15.13 (Ratio rescales denominator back to numerator). Lean ✓ Stmt *If $\nu \neq 0$, then*

$$z(\mu, \nu) \nu = \mu.$$

Definition 15.14 (Diagonal periodic Z -gauge matrix). [Lean](#) [✓ Stmt](#) For matched weight families $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix

$$Z := \text{diag}(z(\mu_i, \nu_i))_i.$$

Lemma 15.15 (Diagonal Z -gauge has finite order under matched powers). [Lean](#) [✓ Stmt](#) If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all i , then

$$Z^m = \mathbb{1}.$$

Lemma 15.16 (Diagonal Z -gauge transports diagonal weights). [Lean](#) [✓ Stmt](#) Under $\nu_i \neq 0$ for all i ,

$$Z \text{ diag}(\nu) = \text{diag}(\mu).$$

Definition 15.17 (Z -gauge equivalence). [Lean](#) [✓ Stmt](#) Let A be an m -periodic irreducible tensor of bond dimension D . A Z -gauge transformation of A is a diagonal unitary matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. Z commutes with every Kraus operator: $[A^i, Z] = 0$ for all physical indices i , and
2. $Z^m = \mathbb{1}_D$ (all diagonal entries are m -th roots of unity).

Replacing A by ZA (i.e. $A^i \mapsto ZA^i$) leaves the MPV family invariant for lengths N that are multiples of m , because the factor $\text{tr}(Z^N) = \text{tr}(\mathbb{1}) = D$ for $N \equiv 0 \pmod{m}$. Two tensors A, B are Z -gauge equivalent if there exists an invertible Y and a Z -gauge transformation Z for A such that $ZA^i = YB^iY^{-1}$ for all i .

Remark 15.18 (Z -gauge is trivial for period-1 blocks). [Lean](#) [✓ Stmt](#) When $m = 1$ the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so Z -gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The Z -gauge is therefore a genuine new degree of freedom that appears only for $m > 1$.

Theorem 15.19 (Blocking a $\mathbb{Z}_m$ -gauge gives an ordinary gauge). [Lean](#) [✓ Stmt](#) If A and B are $\mathbb{Z}_m$ -gauge equivalent, then their m -blocked tensors $A^{[m]}$ and $B^{[m]}$ are gauge equivalent in the ordinary sense.

Proof. [✓ Proof](#) Write the $\mathbb{Z}_m$ -gauge relation as $ZA^i = YB^iY^{-1}$ with $Z^m = \mathbb{1}$ and $[A^i, Z] = 0$. For a blocked letter, commute one factor of Z past each physical letter in the associated length- m word. The total prefactor becomes $Z^m = \mathbb{1}$, so the entire blocked word is related by the same ordinary gauge Y . $\square$

Corollary 15.20 (Blocking a $\mathbb{Z}_m$ -gauge preserves the MPV family). [Lean](#) [✓ Stmt](#) If A and B are $\mathbb{Z}_m$ -gauge equivalent, then the blocked tensors $A^{[m]}$ and $B^{[m]}$ have the same MPV family.

Proof. [✓ Proof](#) Apply Theorem 15.19 and then the gauge invariance of MPV families. $\square$

15.2 Periodic Fundamental Theorem

The theorems in Chapter 12 all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly $\{1\}$. This is achieved by blocking the original tensor by a common period. [DICCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

15.2.1 Common-period blocking lemmas

Definition 15.21 (LCM period for a finite block family). [Lean](#) [✓ Stmt](#) Given a finite family of blocking periods (p_i) indexed by a finite set, their least common multiple serves as the shared blocking period.

Definition 15.22 (Common-period blocked family). [Lean](#) [✓ Stmt](#) Given a finite family of tensors (B_i) with designated periods (p_i) , block each B_i to the shared period $\text{lcm}(p_i)$, so every block lives in the same physical space.

Theorem 15.23 (Common-period blocking preserves transfer-map primitivity). [Lean](#) [✓ Stmt](#) *If each tensor B_i has a primitive transfer map after blocking by its own positive period p_i , then blocking the entire family to the common LCM period preserves primitivity of each member's transfer map.*

Proof. [✓ Proof](#) Each p_i divides $\text{lcm}(p_i)$. Apply the divisibility monotonicity theorem for primitive blocked transfer maps to each family member separately. $\square$

15.2.2 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 (“equal case”) of [DICCSPG17]. The statements recorded here are conditional components of that theorem: the final comparison still assumes periodic-overlap and power-equality hypotheses, so the literature theorem has not yet been obtained here as a single unconditional result.

Theorem 15.24 (Periodic Fundamental Theorem). [Not ready](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) *Let A and B be MPS tensors in irreducible form, with bases of periodic blocks $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ respectively. Suppose that $\mathcal{V}(A) = \mathcal{V}(B)$ (equal MPV families for all lengths N).*

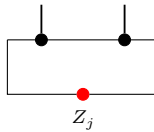
Then $|J| = |K|$ and there is a bijection $\pi : J \rightarrow K$ such that A_j and $B_{\pi(j)}$ are m_j -periodic for the same period m_j . Moreover, for each $j \in J$ there exists:

1. *an invertible matrix Y_j of size $D_j \times D_{\pi(j)}$, and*
2. *a diagonal unitary matrix Z_j of size $D_j \times D_j$ satisfying $Z_j^{m_j} = \mathbb{1}$ and $[A_j^i, Z_j] = 0$ for all i ,*

such that, for every physical index i ,

$$Z_j A_j^i = Y_j B_{\pi(j)}^i Y_j^{-1}$$

For each periodic block, the extra freedom is encoded by



rather than by an ordinary aperiodic gauge alone: the operator Z_j sits on the closing virtual bond and satisfies $Z_j^{m_j} = \mathbb{1}$. Equivalently, setting $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$ and $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$, one has $ZA^i = YB^iY^{-1}$ for all i , with $[A^i, Z] = 0$ and $\mathcal{V}(A) = \mathcal{V}(ZA)$.

Remark. The proof is conditional on the periodic-overlap dichotomy [DICCSPG17, Proposition 3.3], which is left here as a separate hypothesis.

15.2.3 Periodic overlap dichotomy

Definition 15.25 (Cyclic sector decomposition). Lean ✓ Stmt A family of compressed sector tensors $(C_k)_{k=0}^{m-1}$ on corner bond spaces forms a *cyclic sector decomposition* of the blocked tensor $A^{(m)}$ if there exist orthogonal projections $P_0, \dots, P_{m-1}$ on the ambient bond space such that $\sum_k P_k = \mathbb{1}$, the projections satisfy the cyclic-shift relation $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$ with the index $k+1$ taken cyclically modulo m , each P_k commutes with every blocked letter $P_k A_i^{(m)} = A_i^{(m)} P_k$, and $V^{(N)}(C_k)_\sigma = \text{tr}(P_k (A^{(m)})^\sigma)$ for every word σ , and each block carries a $*$ -algebra isomorphism $\varphi_k : M_{\dim C_k}(\mathbb{C}) \rightarrow P_k \cdot M_D(\mathbb{C}) \cdot P_k$ (multiplicative and adjoint-preserving) intertwining the compressed adjoint transfer map with the sector adjoint transfer map on the corner of P_k . This is the inverse cyclic indexing of the off-diagonal convention in [DICCSPG17, Appendix A]. There the same adjoint transfer map is written E^* , and the source projectors satisfy $E^*(P_u) = P_{u+1}$ together with $A^i = \sum_u P_u A^i P_{u+1}$. Here P_k corresponds to the source sector $u = -k \pmod{m}$, which turns the source shift into the displayed adjoint-transfer relation.

Theorem 15.26 (Self-overlap along period multiples). Lean Not ready If A is m -periodic, then the self-overlap along multiples of m converges to m (the period, viewed as a scalar).

Theorem 15.27 (Different periods imply asymptotic orthogonality). Lean ✓ Stmt If A and B are periodic with different periods, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$.

Proof. ✓ Proof Split on whether the bond dimensions agree. If not, the claim follows from Theorem 15.33. If $D_1 = D_2$, suppose for contradiction that A and B are gauge-phase equivalent; Perron–Frobenius eigenvalue uniqueness [Wol12, Theorem 6.3] then forces equal periods, contradicting the hypothesis. Hence the tensors are not gauge-phase equivalent, and the overlap decays. $\square$

Lemma 15.28 (Sector-block normality for periodic tensors). Lean ✓ Stmt For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.

Theorem 15.29 (No sector match implies asymptotic orthogonality). Lean ✓ Stmt If two periodic tensors with the same period admit no gauge-phase matching sector pair after blocking, then their overlap tends to zero.

Lemma 15.30 (Sector-match propagation under translation). Lean Not ready A single gauge-phase match between blocked sectors propagates to all cyclic translates of that sector pair.

Lemma 15.31 (Per-site proportionality from blocked sector match). Lean Not ready If all aligned blocked sectors match up to gauge and phase, then the original tensors are related by repeated-block proportionality.

Theorem 15.32 (Sector match yields repeated-block equivalence). Lean Not ready For periodic tensors of the same period, existence of one matching sector pair implies repeated-block equivalence.

Theorem 15.33 (Different bond dimensions imply asymptotic orthogonality). Lean ✓ Stmt If two periodic tensors have different bond dimensions, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$.

Theorem 15.34 (Periodic overlap dichotomy). Lean Not ready Let A and B be periodic tensors. Then at least one of the following alternatives holds:

1. $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$;
2. A and B are equivalent up to repeated blocks.

Theorem 15.35 (Eventual linear independence of a periodic basis). Lean Not ready *For a finite family of pairwise non-equivalent periodic tensors whose periods divide a common integer p , there is N_0 such that for every $N \geq N_0$ the vectors $\{|V^{(pN)}(A_j)\rangle\}_j$ are linearly independent.*

Remark 15.36 (Comparison with the blocking approach). Not ready Theorem 15.24 and the equal-MPV source corollary discussed in Section 12.2 both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Section 12.2, following [CPGSV17a]): Block both A and B by a common period p so that all blocks of $A^{[p]}$ and $B^{[p]}$ are primitive (1-periodic). Apply the aperiodic fundamental theorem to the blocked tensors. The conclusion is that $A^{[p]}$ and $B^{[p]}$ are gauge equivalent. The Z -gauge freedom does not appear because blocking absorbs the periodicity.
2. *Periodic approach* (Theorem 15.24, following [DICCSPG17]): Work directly with the periodic blocks of A and B . The conclusion includes an extra Z -gauge transformation reflecting the non-trivial phase ambiguity that arises for periodic blocks. The Z matrix satisfies $Z^m = \mathbb{1}$ and commutes with all Kraus operators; it is invisible to the MPV family because it only introduces m -th roots of unity in the trace.

Chapters 9 and 12 follow only the blocking argument: remove periodicity by blocking, then apply the aperiodic theory. The periodic extension treats periodic blocks directly.

Theorem 15.37 (Peripheral proportional case from exact MPV equality). Lean Not ready *Let A and B be periodic tensors with the same matrix product vectors. Then their bond dimensions agree, and after identifying the bond spaces, A and B are equivalent up to repeated blocks.*

Proof. Not ready By Theorem 15.34, either the overlap $\langle V^{(N)}(A) | V^{(N)}(B) \rangle$ tends to 0 or the bond dimensions agree and A and B are equivalent up to repeated blocks. In the decay case, equality of the MPV families identifies the cross-overlap with the self-overlap of A for every N . But Theorem 15.26 gives $\langle V^{(m_a N)}(A) | V^{(m_a N)}(A) \rangle \rightarrow m_a$ along multiples of the period m_a , contradiction. $\square$

Theorem 15.38 (Peripheral proportional case from phase rescaling). Lean Not ready *Assume that whenever two periodic tensors have proportional MPV families, one can rescale one tensor by a unit-modulus phase so that the MPV families agree exactly. Then proportional periodic MPV families already force the tensors to agree up to repeated blocks.*

Proof. Not ready Apply the phase-rescaling hypothesis to replace A by a periodic tensor A' with the same MPV family as B . Theorem 15.37 then yields repeated-block equivalence between A' and B . Since A' differs from A only by a unit-modulus phase, A and A' are themselves equivalent up to repeated blocks, and transitivity gives the claim. $\square$

Remark 15.39 (Proportional periodic FT). Not ready There is also a proportional periodic Fundamental Theorem (Theorem 3.4, “proportional case”, in [DICCSPG17]). The former formulation of this theorem assumed externally supplied coefficient arrays with nonzero limits as hypotheses. That is not the hypothesis set of the source theorem. A faithful statement must instead derive the nondecaying cross-overlap witnesses from the proportional MPV hypothesis and the periodic BNT structure.

15.3 Physical symmetries on periodic MPS

Theorem 15.40 (Physical symmetry yields a virtual Z -gauge). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor in irreducible form II and let $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ be a representation of a group G on the physical leg, acting unitarily ($U(g)U(g)^\dagger = \mathbb{1}$). Assume moreover the periodic equal-case Fundamental Theorem of MPS (Theorem 15.24) as an explicit hypothesis on the pair (d, D) , expressed by the hypothesis [Lean](#) dD . If A is on-site symmetric under U — that is, the twisted tensor $\tilde{A}_g$ has the same MPV family as A for every $g \in G$ — then for each $g \in G$ there exists a positive period m_g and a $\mathbb{Z}_{m_g}$ -gauge equivalence between A and $\tilde{A}_g$.

Concretely there are matrices Z_g and Y_g , with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$, satisfying, for every physical index i ,

$$Z_g A^i = Y_g \tilde{A}_g^i Y_g^{-1}$$

Proof. [✓ Proof](#) The twisted tensor $\tilde{A}_g$ coincides with the physical-index rotation $A_{U(g)}$ of Definition 14.1. By Theorem 15.10 the rotated tensor remains in irreducible form II, and the on-site symmetry hypothesis supplies the equality $\mathcal{V}(A) = \mathcal{V}(A_{U(g)})$. Apply the periodic equal-case Fundamental Theorem (Theorem 15.24) to obtain the asserted $\mathbb{Z}_{m_g}$ -gauge equivalence. $\square$

Definition 15.41 (Periodic projective-rigidity hypothesis). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with physical dimension d and bond dimension D , and assume it is on-site symmetric under a group action $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$. The tensor A satisfies the *periodic projective-rigidity hypothesis* if one may choose a family of virtual gauges $Y : G \rightarrow \mathrm{GL}_D(\mathbb{C})$ and a scalar 2-cochain $u : G \times G \rightarrow \mathbb{C}^\times$ such that

1. every Y_g realises the intertwining from [DICCSPG17, Corollary 4.1] with the twisted tensor $\tilde{A}_g$ for some accompanying Z_g (with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$); and
2. the family obeys the projective multiplication law on the bond space: $Y_h Y_g = u(g, h) Y_{gh}$.

The injective case reduces the rigidity to scalar uniqueness of gauges (cf. §14.2); in the genuinely periodic setting the bond-space commutant of A is generated by the Z_g matrices, so coherence of the factor system u is an independent analytic assumption.

Theorem 15.42 (Projective-representation upgrade). [Lean](#) [✓ Stmt](#) Under the hypotheses of Corollary 15.40 together with the periodic projective-rigidity hypothesis (Definition 15.41), the Y_g family defines a projective representation of G on the bond space: there exist a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$ and virtual gauges $X : G \rightarrow \mathrm{GL}_D(\mathbb{C})$ satisfying

$$X(g) X(h) = \omega(g, h) X(gh),$$

while for each g the matrix $X(g^{-1})$ still realises the $\mathbb{Z}_{m_g}$ -intertwining from [DICCSPG17, Corollary 4.1] between A and the twisted tensor $\tilde{A}_g$.

Proof. [✓ Proof](#) Unpack the rigidity hypothesis to extract the family (Y, u) together with the intertwiners from [DICCSPG17, Corollary 4.1]. Set $X(g) := Y(g^{-1})$ and $\omega(g, h) := u(h^{-1}, g^{-1})$. The projective multiplication law $X(g) X(h) = \omega(g, h) X(gh)$ is then exactly the (h^{-1}, g^{-1}) -instance of the factor-system identity, using $(gh)^{-1} = h^{-1}g^{-1}$. The intertwining relation between A and $\tilde{A}_g$ transports unchanged. $\square$

Theorem 15.43 (Cocycle identity for the upgraded factor system). [Lean](#) [✓ Stmt](#) If the bond dimension D is positive, the factor system ω produced by Theorem 15.42 satisfies the multiplicative 2-cocycle identity $\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k)$.

Proof. ✓ Proof Associativity of matrix multiplication applied to $X(g)X(h)X(k)$, combined with the projective multiplication law, gives

$$\omega(g, h)\omega(gh, k)X((gh)k) = \omega(g, hk)\omega(h, k)X(g(hk)).$$

Since $(gh)k = g(hk)$, both sides are scalar multiples of the same matrix $X((gh)k)$. The matrix lies in $\text{GL}_D(\mathbb{C})$ and is therefore invertible, but the scalar cancellation leading from an equality of scalar multiples of a matrix to equality of the scalar coefficients additionally requires $D > 0$ so that the underlying index set is inhabited. Cancelling then yields $\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k)$ in $\mathbb{C}^\times$, hence in $\text{Units}(\mathbb{C})$. $\square$

Remark 15.44 (SPT classification). The cohomology class of ω in $H^2(G, U(1))$ classifies the SPT phase of the symmetric MPS (cf. Chen–Gu–Wen and Pérez-García et al.). This classification statement is not established here; only the underlying 2-cocycle identity of Theorem 15.43 is available.

Theorem 15.45 (Projective representation and cocycle from symmetry). Lean ✓ Stmt *Theorems 15.42 and 15.43 combine into a single statement: from the periodic projective-rigidity hypothesis (Definition 15.41), one obtains a projective representation of G on the bond space whose factor system is a genuine 2-cocycle whenever $D > 0$.*

15.4 Refinement and divisibility

The next theorem characterises when the matrix-product-vector family of a periodic tensor admits a finer description in terms of a smaller-period tensor whose blocks have been merged. This was the original motivation for the irreducible-form framework in [DICCSPG17].

Definition 15.46 (p -divisible CP map). Lean ✓ Stmt A linear endomorphism $\mathcal{E}$ of $M_D(\mathbb{C})$ is p -divisible if there exists a CPTP map $\mathcal{E}'$ of $M_D(\mathbb{C})$ such that $\mathcal{E} = (\mathcal{E}')^p$.

Definition 15.47 (p -refinable MPS tensor). Lean ✓ Stmt An MPS tensor B with physical dimension d and bond dimension D is p -refinable if there exist another such tensor A and an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$ (encoded as a matrix $W \in M_{d^p \times d}(\mathbb{C})$ with $W^\dagger W = \mathbb{1}$) such that the p -blocked tensor $A^{[p]}$ matches the W -image of B at the level of MPV coefficients:

$$V^{(N)}(A^{[p]})_\tau = \sum_{\sigma \in \{0, \dots, d-1\}^N} \left(\prod_k W_{\tau(k), \sigma(k)} \right) V^{(N)}(B)_\sigma$$

for every $N \in \mathbb{N}$ and every $\tau \in \{0, \dots, d^p - 1\}^N$.

In Dirac notation, this encodes $|V_{pN}(A)\rangle = W^{\otimes N} |V_N(B)\rangle$ for all N .

Theorem 15.48 (Pullback of a refinement witness). Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II. If B is p -refinable, then there exist a tensor A , an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$, and the pullback tensor*

$$C^\alpha := \sum_{i=0}^{d-1} W_{\alpha, i} B^i \quad (\alpha \in \{0, \dots, d^p - 1\})$$

such that C is again in irreducible form II, $\mathcal{E}_C = \mathcal{E}_B$, and C has the same MPV family as $A^{[p]}$.

Proof. [✓ Proof](#) Choose (A, W) from the refinement data. The pullback construction gives $\mathcal{E}_C = \mathcal{E}_B$ and equality of MPV families between C and $A^{[p]}$. To see that C is still in irreducible form II, transport each periodic block B_j of B through the same physical isometry W . The left-canonical normalization is preserved because $W^\dagger W = \mathbb{1}$, while irreducibility and the peripheral spectrum are preserved because the transfer map of each transported block agrees with that of B_j . Reassembling these transported blocks with the same weights gives an irreducible-form decomposition of C . $\square$

Definition 15.49 (Blocked equal-case Z-gauge extraction hypothesis). [Lean](#) [✓ Stmt](#) Fix (d, D, p) . This hypothesis says that whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, there exists an integer $m \geq 1$ and a $\mathbb{Z}_m$ -gauge equivalence between C and $A^{[p]}$.

Definition 15.50 (Blocked-to-root reconstruction hypothesis from Z-gauge). [Lean](#) [✓ Stmt](#) Fix (d, D, p) . This hypothesis says that if $m \geq 1$ and a blocked tensor C in irreducible form II is $\mathbb{Z}_m$ -gauge equivalent to $A^{[p]}$, then there exists a tensor A' with $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$ and $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Definition 15.51 (Blocked equal-case root hypothesis). [Lean](#) [✓ Stmt](#) Fix (d, D, p) . This states the combined root-existence statement: whenever a blocked tensor C in irreducible form II has the same MPV family as a blocked root $A^{[p]}$, one can replace A by a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$.

Theorem 15.52 (Blocked equal-case root from Z-gauge extraction). [Lean](#) [✓ Stmt](#) *If the blocked equal-case Z-gauge extraction hypothesis and the blocked-to-root reconstruction hypothesis both hold, then the combined blocked equal-case root hypothesis holds.*

Proof. [✓ Proof](#) Apply the Z-gauge extraction hypothesis to obtain a blocked $\mathbb{Z}_m$ -gauge witness between C and $A^{[p]}$, and then apply the reconstruction hypothesis to that witness. $\square$

Theorem 15.53 (Reduction to the blocked equal-case root hypothesis). [Lean](#) [✓ Stmt](#) *Assume the blocked equal-case root hypothesis. Then every p -refinable tensor B in irreducible form II admits a left-canonical root A' with $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$.*

Proof. [✓ Proof](#) Apply Theorem 15.48 to the refinement witness of B . This produces a blocked tensor C in irreducible form II with $\mathcal{E}_C = \mathcal{E}_B$ and the same MPV family as $A^{[p]}$. The blocked equal-case root hypothesis then gives a left-canonical tensor A' with $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$. Combining the two transfer-map identities yields $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$. $\square$

Theorem 15.54 (Forward direction from the blocked equal-case root hypothesis). [Lean](#) [✓ Stmt](#) *Assume the blocked equal-case root hypothesis. Then for every tensor B in irreducible form II, p -refinability of B implies p -divisibility of $\mathcal{E}_B$.*

Proof. [✓ Proof](#) Combine Theorem 15.53 with the witness-level forward implication: once $\mathcal{E}_B$ is identified with $\mathcal{E}_{A'^{[p]}}$ for a left-canonical A' , the blocking identity gives $\mathcal{E}_B = (\mathcal{E}_{A'})^p$. $\square$

The following conditional statements isolate the refinability–divisibility equivalence from [DCCSPG17, Theorem 4.1].

Theorem 15.55 (p -refinability iff transfer-map p -divisibility). [Lean](#) [✓ Stmt](#) *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the two root hypotheses used in the one-way statements: the forward left-canonical root hypothesis for refinability $\Rightarrow$ divisibility and the reverse transfer-map root hypothesis for divisibility $\Rightarrow$ refinability. Then B is p -refinable iff its transfer map $\mathcal{E}_B$ is p -divisible.*

Proof. ✓ Proof Bundle the forward implication (Theorem 15.56) with the reverse implication (Theorem 15.59) into a single biconditional. The forward direction assumes the left-canonical root hypothesis; the reverse direction assumes only the transfer-map root hypothesis. Together they give the full equivalence under those hypotheses. $\square$

Theorem 15.56 (*p*-refinability implies transfer-map *p*-divisibility). Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II. Assume a left-canonical root hypothesis: every refinement witness gives a tensor A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. Then *p*-refinability of B implies *p*-divisibility of $\mathcal{E}_B$.*

Proof. ✓ Proof The left-canonical root hypothesis supplies a witness A with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (left-canonical) and $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. The left-canonical property makes $\mathcal{E}_A$ a CPTP channel, and the blocking identity $\mathcal{E}_{A^{[p]}} = (\mathcal{E}_A)^p$ then identifies $\mathcal{E}_B$ with the *p*-th power of the channel $\mathcal{E}_A$. $\square$

Definition 15.57 (Blocked-to-root channel-root hypothesis from Z-gauge). Lean ✓ Stmt Fix (d, D, p) . This sharpened reconstruction hypothesis asks for a CPTP map $\mathcal{E}'$ such that $\mathcal{E}_C = (\mathcal{E}')^p$ and the Choi/Kraus rank of $\mathcal{E}'$ is at most d , whenever C is related to $A^{[p]}$ by a blocked $\mathbb{Z}_m$ -gauge witness.

Theorem 15.58 (Channel-root criterion for blocked-to-root reconstruction). Lean ✓ Stmt *If the blocked-to-root channel-root hypothesis holds, then the tensor-level blocked-to-root reconstruction hypothesis holds as well.*

Proof. ✓ Proof Choose a tensor A' realizing the bounded Kraus family of the root channel. Because the channel is trace-preserving, the Kraus operators of A' satisfy $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$. The identity $\mathcal{E}_C = (\mathcal{E}_{A'})^p$ is then equivalent to $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ by the blocking formula. $\square$

Theorem 15.59 (Transfer-map *p*-divisibility implies *p*-refinability). Lean ✓ Stmt *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the reverse transfer-map root hypothesis: there is a tensor A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ whenever $\mathcal{E}_B$ is *p*-divisible. Then *p*-divisibility of $\mathcal{E}_B$ implies *p*-refinability of B .*

Proof. ✓ Proof From the reverse transfer-map root hypothesis, extract A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. This matches two finite Kraus families of the same completely positive map: the blocked family $A^{[p]}$ with d^p operators and the original family B with d operators. The cardinality comparison $d \leq d^p$ holds whenever $p \geq 1$, so [Wol12, Theorem 2.18] supplies an isometry $V \in M_{d^p \times d}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $(A^{[p]})^\alpha = \sum_j V_{\alpha,j} B^j$. Expanding $\text{coeff}(A^{[p]})$ (ofFn τ) by linearity of matrix multiplication and of the trace, the contributions reorganise into the V -weighted sum over length- N words of B required by Definition 15.47, exhibiting V as the isometry W in the statement of *p*-refinability. $\square$

Remark 15.60 (Continuum-limit application). Not ready The original motivation for Theorem 15.55 is a continuum-limit construction in the spirit of [DICCSPG17]: refinability along arbitrarily large p is equivalent to infinite divisibility of the transfer map, which selects out continuum limits of translationally invariant MPS. The further consequences are explored elsewhere.

Chapter 16

The Algebraic Fundamental Theorem

This chapter develops the one-dimensional MPS results of [MGRPG⁺18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly with $M_D(\mathbb{C})$. Two preparatory lemmas—the virtual bond gauge (Lemma 16.13) and the proportionality lemma (Lemma 16.14)—carry the entire argument, yielding the main theorem and its corollaries for translation-invariant chains, non-translation-invariant descriptions of translation-invariant states, and blocked chains.

Throughout, D is the bond dimension and d is the physical dimension with $d \geq D^2$. The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ($2L + 1$ in place of $3L$) are not pursued in this chapter. The non-translation-invariant chain data, including same chain state, injectivity, and cyclic gauge equivalence, is set up below.

16.1 Alternative fixed-length non-TI setup

The algebraic proof of [MGRPG⁺18] keeps a fixed periodic chain length and allows the local tensors to vary from site to site. The first step is therefore a separate setup for finite non-translation-invariant chains.

Definition 16.1 (Periodic non-TI chain data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Fix a chain length n . A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \quad k = 0, \dots, n-1,$$

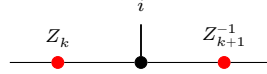
where each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \text{tr}(A_0^{\sigma_0} A_1^{\sigma_1} \dots A_{n-1}^{\sigma_{n-1}}).$$

Two such chains generate the *same chain state* if these coefficients agree for every σ . The chain is *injective* if each local tensor A_k is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site k and physical index i , with indices understood modulo n so that $Z_n = Z_0$. Locally, this gauge relation is represented by



where the black node denotes the site tensor named in the surrounding formula and the two red side nodes denote the bond gauges on the neighboring virtual legs.

Lemma 16.2 (Cyclic gauge equivalence is an equivalence relation). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

Proof. [Proof](#) Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. $\square$

16.2 One-sided inverse and decomposition

If A is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ is surjective and admits a right inverse.

Lemma 16.3 (Existence of a right inverse). [Lean](#) [Stmnt](#) *If A is injective, then there exists a $\mathbb{C}$ -linear map $\Psi : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \text{id}_{M_D(\mathbb{C})}$.*

Proof. [Proof](#) Surjectivity of Φ_A yields a linear section by choice. $\square$

Definition 16.4 (Decomposition map). [Lean](#) [Stmnt](#) Let A be an injective MPS tensor. The decomposition map $\Psi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ is a fixed choice of right inverse of Φ_A .

Lemma 16.5 (Decomposition property). [Lean](#) [Stmnt](#) *If A is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that*

$$X = \sum_{i=0}^{d-1} c_i A^i.$$

Proof. [Proof](#) Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. $\square$

Remark 16.6 (Non-uniqueness of the decomposition). When $d > D^2$, the map Φ_A has a nontrivial kernel, so the section Ψ_A is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \text{id}$, so the choice does not affect the conclusions.

16.3 Physical realisation of virtual insertions

For an injective tensor A , the physical realisation map encodes the effect of inserting a virtual matrix X on the bond as a $d \times d$ operation on the physical indices.

Definition 16.7 (Physical realisation map). [Lean](#) [Stmnt](#) Let A be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row i of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:

The map $X \mapsto O_A(X)$ is $\mathbb{C}$ -linear.

Lemma 16.8 (Defining property of the physical realisation map). [Lean](#) [✓ Stmt](#) For every $X \in M_D(\mathbb{C})$ and every physical index i ,

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Proof. [✓ Proof](#) By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying Φ_A gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \text{id}$. $\square$

Theorem 16.9 (Physical realisation is multiplicative). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor. For all $X, Y \in M_D(\mathbb{C})$,

$$O_A(XY) = O_A(X) O_A(Y).$$

Proof. [✓ Proof](#) Applying Lemma 16.8 to X yields

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Multiply by Y on the right and apply the linear map Ψ_A :

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij} \Psi_A(A^j Y).$$

Taking the k th component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij} O_A(Y)_{jk} = (O_A(X) O_A(Y))_{ik}.$$

Since this holds for all i and k , the matrices are equal. $\square$

Remark 16.10 (General injective case). When $d > D^2$, the decomposition map Ψ_A need not be unique. Nevertheless, Theorem 16.9 is already the general injective statement: for the fixed choice of Ψ_A in Definition 16.4, the associated physical realisation map O_A is multiplicative. This supersedes the earlier basis-only formulation.

16.4 Intermediate lemmas

Lemma 16.11 (Blocking preserves injectivity). [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor $(A_0 \otimes \dots \otimes A_{m-1})^{(i_0, \dots, i_{m-1})} := A_0^{i_0} A_1^{i_1} \dots A_{m-1}^{i_{m-1}}$ is again injective.

Proof. [✓ Proof](#) The span of the blocked matrices contains all products $A_0^{i_0} \dots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. $\square$

Lemma 16.12 (Centre of $M_D(\mathbb{C})$). [Lean](#) [✓ Stmt](#) If $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then $Z = \lambda 1_D$ for some $\lambda \in \mathbb{C}$.

Proof. [✓ Proof](#) Standard: Z commutes with every matrix unit E_{ij} , which forces all off-diagonal entries to vanish and all diagonal entries to agree. $\square$

16.5 Same state forces a virtual bond gauge

This section presents the 3-site virtual-bond statement used later in the proof of the PEPS fundamental theorem. Here the hypothesis is already the all-length MPV equality of the combined tensors, and the conclusion is the conjugacy of the middle-bond insertion data.

Lemma 16.13 (Virtual bond gauge – 3-site case). Lean ✓ Stmt *Let $A = (A_0, A_1, A_2)$ and $B = (B_0, B_1, B_2)$ be two 3-site injective periodic chains with common bond dimension $D \geq 1$ and $\mathcal{V}(A) = \mathcal{V}(B)$ (same MPV family). Then at the middle bond there exists an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ such that the virtual-insertion trace coefficients agree after conjugation:*

$$\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Z^{-1} X Z B_1^{i_1} B_2^{i_2})$$

for all $X \in M_D(\mathbb{C})$ and all physical indices i_0, i_1, i_2 .

In [MGRPG⁺18, Lemma 1], the statement is given for arbitrary chain length $n \geq 3$ and any bond position, with uniqueness of Z up to scalar. The chain-level generalization to arbitrary $n \geq 3$ and any bond position is not pursued here.

Proof. ✓ Proof The $\mathcal{V}(A) = \mathcal{V}(B)$ hypothesis gives, for every $X \in M_D(\mathbb{C})$, equality of the virtual-insertion trace coefficients: $\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Y B_1^{i_1} B_2^{i_2})$ for some Y depending on X . Since all three site tensors on each side are injective, the physical realisation maps O_A, O_B are algebra homomorphisms by Theorem 16.9. Setting $T(X) = Y$ defines a $\mathbb{C}$ -linear multiplicative map, hence a nonzero $\mathbb{C}$ -algebra endomorphism of $M_D(\mathbb{C})$, which is bijective by Theorem 3.8. Skolem–Noether (Theorem 3.9) gives $Z \in \text{GL}_D(\mathbb{C})$ with $T(X) = Z^{-1} X Z$. $\square$

16.6 Aligned gauges force proportionality

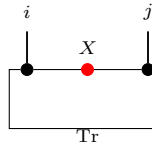
After applying Lemma 16.13 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG⁺18] shows that this agreement forces the local tensors to be proportional.

Lemma 16.14 (Aligned gauges force proportionality – Lemma 2 of [MGRPG⁺18]). Lean ✓ Stmt *Let A_1, A_2 and B_1, B_2 be injective MPS tensors with the same bond dimension D . Suppose that for all $X \in M_D(\mathbb{C})$ (on the internal bond), all $Y \in M_D(\mathbb{C})$ (on the external bond), and all physical indices i, j ,*

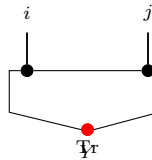
$$\text{tr}(A_1^i X A_2^j) = \text{tr}(B_1^i X B_2^j) \tag{16.1}$$

$$\text{tr}(A_2^j Y A_1^i) = \text{tr}(B_2^j Y B_1^i) \tag{16.2}$$

Diagrammatically, inserting X on the bond between the two sites (Eq. 16.1):



while the insertion Y on the outer trace bond (Eq. 16.2) is:



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that

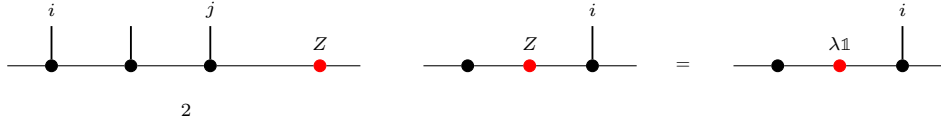
$$A_1^i = \lambda B_1^i \quad \text{and} \quad A_2^j = \lambda^{-1} B_2^j$$

for all physical indices i, j .

Proof. ✓ Proof By cyclicity and non-degeneracy of the trace pairing, the conditions in Eqs. (16.1) and (16.2) give, for all physical indices i, j ,

$$\begin{aligned} A_2^j A_1^i &= B_2^j B_1^i, \\ A_1^i A_2^j &= B_1^i B_2^j. \end{aligned}$$

Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map Ψ_{A_2} , and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity gives $A_1^i = Z B_1^i$, and the second gives $A_1^i = B_1^i Z$. Hence Z commutes with every B_1^i ; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, Z is central. By Lemma 16.12, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into the first product identity forces $A_2^j = \lambda^{-1} B_2^j$. $\square$

16.7 The Fundamental Theorem for injective chains

The chain result below combines the site and physical indices into a single physical index and applies the single-block Fundamental Theorem to the resulting tensor. The fixed-length same-state statement requires a separate same-state reduction hypothesis.

Theorem 16.15 (Fundamental Theorem for injective chains). Lean ✓ Stmt Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two n -site chains with common bond dimension D , and assume that $\mathbf{A}$ is injective. Suppose that, after combining the site index and physical index into a single MPS tensor, the resulting two tensors generate the same MPV family. Then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.

Proof. ✓ Proof Choose one site of $\mathbf{A}$; its injectivity implies injectivity of the combined tensor by Lemma 16.11. Apply the single-block Fundamental Theorem (Theorem 3.11) to the two combined tensors. This yields one invertible matrix X such that every combined matrix of $\mathbf{B}$ equals the corresponding combined matrix of $\mathbf{A}$ conjugated by X . Reading this identity back at each site gives $B_k^i = X A_k^i X^{-1}$ for all k and i , so the constant choice $Z_k = X$ is a cyclic gauge. $\square$

Lemma 16.16 (Uniform site rescaling rescales the combined tensor). Lean ✓ Stmt Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an n -site chain and let $\zeta \in \mathbb{C}$. If $\zeta \mathbf{A}$ denotes the chain obtained by replacing each site tensor A_k with ζA_k , and if $\widetilde{\mathbf{A}}$ denotes the combined tensor of $\mathbf{A}$, then

$$\widetilde{\zeta \mathbf{A}} = \zeta \widetilde{\mathbf{A}}.$$

Proof. ✓ Proof Combining the site and physical indices does not change which local matrix is read at the component (k, i) , so both sides evaluate to ζA_k^i there. $\square$

Theorem 16.17 (Injective-chain Fundamental Theorem up to a global phase). Lean ✓ Stmt Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two n -site chains with common bond dimension D , and assume that $\mathbf{A}$ is injective. Suppose that the combined tensors $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$ are gauge-phase equivalent. Then there exist invertible matrices $Z_0, \dots, Z_{n-1} \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\zeta \in \mathbb{C}^\times$ such that

$$B_k^i = \zeta Z_k A_k^i Z_{k+1}^{-1}$$

for all sites k and physical indices i , where $k+1$ is interpreted cyclically modulo n .

Proof. ✓ Proof If the combined tensor of $\mathbf{B}$ differs from that of $\mathbf{A}$ by a gauge and a nonzero scalar ζ , rescale each site of $\mathbf{B}$ by ζ^{-1} . Lemma 16.16 removes this common scalar from the combined tensor, so Theorem 16.15 applies to the rescaled chain. Multiplying the resulting sitewise relations by ζ gives the displayed conclusion. $\square$

16.7.1 From fixed-length chain states to combined-tensor MPV agreement

Definition 16.18 (Same-state reduction hypothesis). Lean ✓ Stmt A *same-state reduction hypothesis* for dimensions (d, D) is the assertion that whenever $\mathbf{A}$ is an injective n -site chain with $n \geq 3$ and $\mathbf{A}, \mathbf{B}$ generate the same chain state, then the combined tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.

Lemma 16.19 (Same state gives combined-tensor MPV agreement). Lean ✓ Stmt Assume a *same-state reduction hypothesis* for (d, D) . Let $\mathbf{A}$ and $\mathbf{B}$ be n -site chains with common bond dimension D , with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then the combined tensors of $\mathbf{A}$ and $\mathbf{B}$ generate the same MPV family.

Proof. ✓ Proof This is exactly the defining property of the same-state reduction hypothesis. $\square$

Theorem 16.20 (Fundamental Theorem for injective chains from same state). Lean ✓ Stmt Assume a *same-state reduction hypothesis* for (d, D) . Let $\mathbf{A}$ and $\mathbf{B}$ be n -site chains with common bond dimension D , with $n \geq 3$, and assume that $\mathbf{A}$ is injective. If $\mathbf{A}$ and $\mathbf{B}$ generate the same chain state, then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.

Proof. ✓ Proof Apply Lemma 16.19 to convert the fixed-length chain-state hypothesis into equality of the combined-tensor MPV families, then invoke Theorem 16.15. $\square$

Remark 16.21 (Relationship to the translation-invariant all-sizes theorem). Theorem 16.15 treats a fixed chain length and assumes equality of the combined-tensor MPV families. The single-block Fundamental Theorem (Theorem 3.11) instead studies one tensor at all system sizes and concludes exact gauge equivalence $B^i = X A^i X^{-1}$. The chain theorem is recovered by converting the chain into one combined tensor; the price is that the hypothesis is all-length MPV equality for that combined tensor rather than equality of the chain state at one fixed length.

16.7.2 Finite-length MPV agreement

Definition 16.22 (Finite-length MPV agreement). Lean ✓ Stmt Two tensors A and B of the same bond dimension satisfy *finite-length MPV agreement from N_0 onward* if they generate the same MPV family for all system sizes $N \geq N_0$, meaning that for every $N \geq N_0$ and $\sigma \in [d]^N$,

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma$$

Theorem 16.23 (Finite-length agreement implies full agreement). [Lean](#) [✓ Stmt](#) *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B generate the same MPV family for every system size $N \geq N_0$, then they generate the same MPV family for every system size.*

Proof. [✓ Proof](#) Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show that the length- N_0 word span of B is all of $M_D(\mathbb{C})$ via the linear relation between the length- N_0 word-evaluation maps of A and B induced by trace agreement at length $N_0 + 1$, together with a finrank transfer argument.
2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\text{tr}(M(S - 1)) = 0$.
3. Downward induction on $k = N_0 - |w|$: decompose $\text{tr}(w_A)$ and $\text{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

□

Theorem 16.24 (Strengthened single-block FT (finite-length)). [Lean](#) [✓ Stmt](#) *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold N_0), then A and B are gauge equivalent.*

Proof. [✓ Proof](#) Apply Theorem 16.23 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 3.11). □

16.8 Corollaries

16.8.1 Translation-invariant chains

Theorem 16.25 (Single gauge for translation-invariant chains). [Lean](#) [✓ Stmt](#) *Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that*

$$B^i = X A^i X^{-1}$$

for every physical index i .

Proof. [✓ Proof](#) Bundle the constant chains into one tensor. Injectivity of A implies injectivity of the combined tensor by Lemma 16.11, and the combined-tensor MPV equality is exactly the hypothesis needed for the single-block Fundamental Theorem. The resulting gauge matrix is independent of the site because every site carries the same local tensor. □

Corollary 16.26 (Translation-invariant collapse to one gauge). [Lean](#) [✓ Stmt](#) *Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exist an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that*

$$B^i = \lambda Z^{-1} A^i Z$$

for every physical index i .

Proof. [✓ Proof](#) Apply Theorem 16.25 to obtain $B^i = X A^i X^{-1}$. Set $Z = X^{-1}$ and $\lambda = 1$. Then $\lambda^n = 1$ and the displayed identity is exactly the same gauge relation rewritten in the form used by the corollary. □

16.8.2 Non-translation-invariant descriptions of translation-invariant states

Corollary 16.27 (Translation-invariant states from nonuniform chains). Not ready Assume a same-state reduction hypothesis for (d, D) . Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an injective non-translation-invariant chain of length $n \geq 3$. Suppose the state generated by $\mathbf{A}$ is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain $(A_1, \dots, A_{n-1}, A_0)$. Then $\mathbf{A}$ admits a translation-invariant description with the same bond dimension: there exists an injective tensor B such that the constant chain $(B, \dots, B)$ generates the same state as $\mathbf{A}$. Explicitly, $B^i = A_0^i Z_0^{-1}$ (right multiplication, not conjugation), where Z_0 is extracted from the cyclic shift.

Proof. Apply Theorem 16.20 to $\mathbf{A}$ and its cyclic shift. This gives invertible gauges $Z_0, \dots, Z_{n-1}$ with $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$. Iterating these identities yields $A_k^i = L_k^{-1} A_0^i R_k$, where the products L_k, R_k satisfy $R_k L_{k+1}^{-1} = Z_0^{-1}$. Substituting into the chain trace shows that the same state is generated by the constant tensor $B^i := A_0^i Z_0^{-1}$. Since A_0 is injective and Z_0 is invertible, B is injective. $\square$

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 14.

16.8.3 Blocked chains

Lemma 16.28 (Block injectivity and blocked-tensor injectivity). Lean ✓ Stmt A tensor A is L -block injective if and only if its L -blocked tensor $A^{[L]}$ is injective.

Proof. ✓ Proof The blocked physical alphabet is just a reindexing of words of length L . Decoding blocked indices therefore identifies the two spanning families, so the two injectivity conditions are equivalent. $\square$

Definition 16.29 (Constant blocked chain). Lean ✓ Stmt For an MPS tensor A , a blocking length L , and a chain length n , the *blocked chain* is the constant n -site chain whose local tensor is the blocked tensor $A^{[L]}$.

Lemma 16.30 (Blocked chains are injective). Lean ✓ Stmt If A is L -block injective, then the constant blocked chain built from $A^{[L]}$ is injective at every site.

Proof. ✓ Proof By Lemma 16.28, the local blocked tensor $A^{[L]}$ is injective, and therefore every site of the constant blocked chain is injective. $\square$

Theorem 16.31 (Fundamental Theorem for blocked chains). Lean ✓ Stmt Let A and B be MPS tensors of bond dimension D , let L be a blocking length, and let n be a chain length. Assume that A is L -block injective and that the combined tensors of the two constant blocked chains built from $A^{[L]}$ and $B^{[L]}$ generate the same MPV family. Then the two blocked chains are gauge equivalent.

Proof. ✓ Proof By Lemma 16.30, the blocked chain built from $A^{[L]}$ is injective. Apply Theorem 16.15 to the two constant blocked chains. $\square$

Remark 16.32. The theorem above is the blocked-chain result established here. Recovering site-wise gauges for the original unblocked tensors gives the normal-MPS corollary of [MGRPG⁺18]; that step is outside this blocked-chain statement.

16.9 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 16.15) handles non-translation-invariant periodic chains. A parallel line of argument addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions D_k and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k A_k^i X_k^{-1}$ for each k independently, and a global block-permutation plus inner-automorphism decomposition.

16.9.1 Per-block linear extension

Definition 16.33 (Per-block linear extension). [Lean] [✓ Stmt] Let A_k, B_k be MPS tensors of bond dimension D_k with A_k injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \dots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$ -linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index i .

Lemma 16.34 (Per-block multiplicativity). [Lean] [✓ Stmt] For each block k , the per-block linear extension T_k is multiplicative: $T_k(MN) = T_k(M)T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.

Proof. [✓ Proof] Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. □

Lemma 16.35 (Per-block bijectivity). [Lean] [✓ Stmt] Each T_k is bijective.

Proof. [✓ Proof] A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of T_k is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. □

16.9.2 Product algebra automorphism

Definition 16.36 (Product algebra automorphism). [Lean] [✓ Stmt] The *product algebra automorphism* is the $\mathbb{C}$ -algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each T_k is the per-block linear extension. It is an algebra automorphism since each T_k is a multiplicative, unital, bijective linear map. Diagrammatically, this is the componentwise map on the product algebra obtained by applying the corresponding extension to each summand.

Theorem 16.37 (Block-permutation decomposition). [Lean] [✓ Stmt] The product algebra automorphism T decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all k) and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that

$$(T(M))_{\sigma(k)} = X_k M X_k^{-1}$$

for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block k) and all k . Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.

Proof. ✓ Proof The automorphism T permutes the block ideals of $\prod_k M_{D_k}(\mathbb{C})$ (each block ideal is simple, and an automorphism sends simple ideals to simple ideals of the same dimension). This gives the permutation σ . On each block, the restricted map is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$; by Skolem–Noether (Theorem 3.9), it is inner. Thus T splits algebraically into the permutation of block ideals followed by inner conjugation on each surviving block. $\square$

16.9.3 Nondegeneracy of the product trace pairing

Lemma 16.38 (Nondegeneracy of the product trace pairing). Lean ✓ Stmt Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \text{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.

Proof. ✓ Proof Choose N supported on a single block k (all other components zero). The hypothesis reduces to $\text{tr}(M_k N_k) = 0$ for all N_k , so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since k was arbitrary, $M = 0$. $\square$

Lemma 16.39 (Injectivity of the product Gram map). Lean ✓ Stmt Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \text{tr}(M_k \cdot A_k^i))$ is injective.

Proof. ✓ Proof Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \text{tr}(M_k \cdot A_k^i))$, which holds by injectivity of A_k (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). $\square$

16.10 Gauge equivalence for direct sums

Definition 16.40 (Flattened block-diagonal gauge). Lean Lean ✓ Stmt Given per-block gauges $X_k \in \text{GL}_{D_k}(\mathbb{C})$, first form the dependent block-diagonal element $\bigoplus_k X_k$ and then reindex the dependent block-coordinate index to the standard index $\{0, \dots, \sum_k D_k - 1\}$ of the assembled tensor. The resulting element of $\text{GL}_{\sum_k D_k}(\mathbb{C})$ is the global gauge used in the conjugation formula for $\bigoplus_k \mu_k A_k$.

Lemma 16.41 (Explicit direct-sum conjugation by the flattened gauge). Lean Lean ✓ Stmt If each block satisfies $B_k^i = X_k A_k^i X_k^{-1}$, then the weighted direct sums satisfy, for every physical index i ,

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1},$$

where X is the flattened block-diagonal gauge of Definition 16.40. Consequently the two assembled tensors are gauge equivalent with this explicit witness.

Proof. ✓ Proof The block-diagonal matrix product is computed blockwise. The scalar weight μ_k commutes with the per-block conjugation, so each diagonal block of the right-hand side is $\mu_k B_k^i$. Reindexing from the dependent block coordinates to the standard bond index preserves multiplication and inverses. The gauge equivalence follows directly from the conjugation identity. $\square$

Lemma 16.42 (Gauge equivalence of direct sums from per-block gauge equivalence). Lean ✓ Stmt If each pair (A_k, B_k) is gauge equivalent, then the weighted block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.

Proof. ✓ Proof Choose per-block gauge matrices via the hypothesis; then apply Lemma 16.41. $\square$

Lemma 16.43 (Gauge equivalence of direct sums with phase-weight absorption). [Lean](#) [Stmt](#) Let A_k, B_k be block families, and let $X_k \in \text{GL}_{D_k}(\mathbb{C})$ and scalars $\zeta_k \in \mathbb{C}$ satisfy $B_k^i = \zeta_k \cdot X_k A_k^i X_k^{-1}$ for every block k and physical index i . If the weights satisfy $\mu_A(k) = \mu_B(k) \zeta_k$, then the weighted block-diagonal tensors $\bigoplus_k \mu_A(k) A_k$ and $\bigoplus_k \mu_B(k) B_k$ are gauge equivalent.

Proof. [Proof](#) Per-block, the weight identity $\mu_A(k) = \mu_B(k) \zeta_k$ rewrites the scaled relation as $\mu_B(k) B_k^i = X_k (\mu_A(k) A_k^i) X_k^{-1}$. Each weighted block then satisfies an ordinary conjugation, and the block-diagonal conjugation formula assembles the per-block X_k into the flattened global gauge $\bigoplus_k X_k$ relating $\bigoplus_k \mu_A(k) A_k$ to $\bigoplus_k \mu_B(k) B_k$. $\square$

Lemma 16.44 (Gauge equivalence of direct sums from per-block MPV equality). [Lean](#) [Stmt](#) Let $\{A_k\}_{k=0}^{r-1}$ and $\{B_k\}_{k=0}^{r-1}$ be two families of MPS tensors with block dimensions D_k , each A_k injective, and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k . Then:

1. Per-block gauge equivalence: $B_k^i = X_k A_k^i X_k^{-1}$ for some $X_k \in \text{GL}_{D_k}(\mathbb{C})$, for every k and i .
2. Global gauge equivalence: the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.

Explicitly, for $X = \bigoplus_k X_k$,

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1}.$$

Proof. [Proof](#) Per-block gauge equivalence follows from applying the single-block Fundamental Theorem to each block, using injectivity of A_k and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. With $X = \bigoplus_k X_k$, the displayed identity follows by multiplying block-diagonal matrices block by block. $\square$

Lemma 16.45 (Explicit per-block gauges). [Lean](#) [Stmt](#) Under the hypotheses of Lemma 16.44, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that

$$B_k^i = X_k A_k^i X_k^{-1}$$

for every block k and every physical index i .

Proof. [Proof](#) Extract the gauge matrix from the per-block part of Lemma 16.44 for each block. $\square$

Lemma 16.46 (Product-algebra decomposition of per-block MPV equality). [Lean](#) [Stmt](#) Assume that every block dimension is nonzero. Under the hypotheses of Lemma 16.44, the product-algebra automorphism determined by the per-block linear extensions decomposes into a block permutation together with inner conjugation on each block: there exist a permutation σ of the blocks, compatible block-dimension equalities, and invertible matrices X_k such that, after reindexing the target block $\sigma(k)$ back to block k , the action on the k th matrix block is $M \mapsto X_k M X_k^{-1}$.

Proof. [Proof](#) Apply Theorem 16.37 to the product-algebra automorphism built from the per-block linear extensions. $\square$

16.10.1 MPV invariance under reindexing

Lemma 16.47 (MPV invariance under block permutation). [Lean](#) [✓ Stmt](#) *Permuting the block index set and transporting the weights correspondingly does not change the SameMPV_2 family of the weighted block-diagonal tensor.*

Proof. [✓ Proof](#) The sum of per-block MPV contributions is invariant under rearrangement of the summands. $\square$

Lemma 16.48 (MPV invariance under dimension cast). [Lean](#) [✓ Stmt](#) *If two block-dimension families agree pointwise, the corresponding weighted block-diagonal tensors generate the same SameMPV_2 family after transporting the blocks across the dimension equalities.*

Proof. [✓ Proof](#) Substituting the equality of dimensions makes the two tensor families identical. $\square$

16.10.2 Converse: gauge equivalence implies same MPV

Lemma 16.49 (Gauge equivalence of direct sums implies same MPV). [Lean](#) [✓ Stmt](#) *If the weighted block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent, then they generate the same MPV family.*

Proof. [✓ Proof](#) Gauge equivalence preserves all traces of word evaluations, hence all MPV coefficients. $\square$

16.10.3 Canonical-form tensor as a weighted direct sum

Lemma 16.50 (Canonical-form tensor agrees with block sum). [Lean](#) [✓ Stmt](#) *For a canonical-form record C , the assembled tensor T_C is definitionally equal to the weighted block-diagonal tensor $\bigoplus_k \mu_k^C A_k^C$, where μ_k^C are the block weights and A_k^C are the block tensors stored in C .*

Proof. [✓ Proof](#) The equality holds by definition; T_C is defined as the weighted block-diagonal tensor formed from the weights and block tensors of C . $\square$

Theorem 16.51 (Fundamental theorem for canonical forms with the same block structure). [Lean](#) [✓ Stmt](#) *Let C be a canonical-form record with block weights μ_k^C and block tensors A_k^C . Let $\{B_k\}$ be a family of MPS tensors on the same block dimensions, and assume that each A_k^C is injective and generates the same MPV family as B_k . Then T_C and $\bigoplus_k \mu_k^C B_k$ are gauge equivalent.*

Proof. [✓ Proof](#) Rewrite T_C as a weighted block-diagonal tensor via Lemma 16.50, then apply the multi-block global fundamental theorem (Theorem 12.56). $\square$

Lemma 16.52 (Per-block equality of MPV families and per-block gauge equivalence). [Lean](#) [✓ Stmt](#) *Let $\{A_k\}$ be a family of injective block tensors. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k if and only if A_k and B_k are gauge equivalent for all k .*

Proof. [✓ Proof](#) The forward direction applies the single-block Fundamental Theorem to each block. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. $\square$

16.11 Block separation from ordered canonical-form data

Lemma 16.44 assumes $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k . In practice, one starts from the weaker global hypothesis that the block-diagonal tensors satisfy the same MPV family: $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$. Block separation extracts the per-block identities from the global one under canonical-form normalization plus a strict separation of the weight moduli. The first step is the weighted block-sum identity

$$\sum_k \mu_k^N V^{(N)}(A_k)_\sigma = \sum_k \mu_k^N V^{(N)}(B_k)_\sigma,$$

which is then combined with overlap estimates and the leading-block peel.

Definition 16.53 (Canonical-form data with ordered weights). [Lean](#) [✓ Stmt](#) A family of block tensors $\{A_k\}_{k=0}^{r-1}$ with scaling weights $\mu_0, \dots, \mu_{r-1} \in \mathbb{C}^\times$ has *canonical-form data with ordered weights* if:

1. each A_k is injective,
2. each A_k is left-canonical: $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}_{D_k}$,
3. the weights are non-increasing by norm and nonzero: $|\mu_0| \geq |\mu_1| \geq \dots \geq |\mu_{r-1}| > 0$,
4. the self-overlaps are normalised: $O_{A_k A_k}(N) \rightarrow 1$ as $N \rightarrow \infty$.

This records the normalized ordered data used before the final block-separation argument. Equal-modulus blocks are still allowed here, as in the canonical decompositions in arXiv:1606.00608. The strict inequality needed for the leading-block peeling argument is a separate hypothesis in the block-separation lemmas below, and at the BNT level it is obtained only after grouping repeated representatives.

Remark 16.54. See Definition 9.18 for the formalized normal canonical-form data statement.

Lemma 16.55 (Block separation from weighted word identities). [Lean](#) [✓ Stmt](#) Let $\{A_k\}$ and $\{B_k\}$ be block families with strictly ordered, nonzero weights μ_k . Assume that every A_k and B_k is injective and left-canonical, that each self-overlap $O_{A_k A_k}(N)$ tends to 1, and that for every system size N and every word σ one has the weighted identity

$$\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0.$$

Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k .

Proof. [✓ Proof](#) The proof is the peeling argument described below for canonical-form data equipped with a separate strict ordering of the weight moduli. The strict weight ordering isolates the leading block, the overlap bounds show that the tail decays geometrically after dividing by the dominant weight, and one then iterates on the remaining blocks. $\square$

Lemma 16.56 (Block separation from canonical-form data and strict weights). [Lean](#) [✓ Stmt](#) Let $\{A_k\}$ have canonical-form data with ordered weights $\{\mu_k\}$, and assume in addition that the weight norms are strictly decreasing. Let $\{B_k\}$ be another family with each B_k injective and left-canonical. If $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k)$, then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block k individually. The proof isolates the leading block and treats the remaining blocks as an exponentially decaying tail with $\rho = |\mu_1|/|\mu_0| < 1$.

Proof. ✓ Proof The proof proceeds by strong induction on the number of blocks r . For $r = 1$, the weighted sum degenerates and the result is immediate after cancelling $\mu_0^N \neq 0$.

For $r \geq 2$, the global identity $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ holds for all N and σ .

$$\mu_0^N (V^{(N)}(A_0)_\sigma - V^{(N)}(B_0)_\sigma) = - \sum_{k \geq 1} \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma).$$

Dividing by μ_0^N shows that the right-hand side is $O(\rho^N)$ with $\rho = |\mu_1|/|\mu_0| < 1$. Multiplying by the MPV components of A_0 and summing over σ converts this to an overlap identity. Dividing by $|\mu_0|^{2N}$ and using the strict norm ordering $|\mu_1|/|\mu_0| < 1$, the tail terms ($k \geq 1$) decay geometrically with ratio $\rho = |\mu_1|/|\mu_0|$. The leading term ($k = 0$) therefore converges: $O_{A_0 B_0}(N) \rightarrow 1$ as $N \rightarrow \infty$. Combined with the normalised self-overlap $O_{A_0 A_0}(N) \rightarrow 1$, this gives $\mathcal{V}(A_0) = \mathcal{V}(B_0)$ via the gauge-phase equivalence criterion. Subtracting the leading block and applying the inductive hypothesis yields $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all remaining k . $\square$

Lemma 16.57 (Block separation from separated canonical data). Lean ✓ Stmt *Suppose the weights are nonzero with strictly decreasing norms, the A -blocks are injective and left-canonical with normalized self-overlaps, the B -blocks are injective and left-canonical, and the block-diagonal tensors generate the same MPV family. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k .*

Proof. ✓ Proof Equality of the block-diagonal MPV families gives the weighted word identity required by Lemma 16.55. The separated hypotheses supply exactly the injectivity, left-canonicity, strict-weight, and self-overlap assumptions needed there. $\square$

Lemma 16.58 (Gauge comparison from separated canonical data). Lean ✓ Stmt *Under the hypotheses of Lemma 16.57, every block pair A_k, B_k is gauge equivalent, and the block-diagonal tensors are globally gauge equivalent.*

Proof. ✓ Proof First apply Lemma 16.57 to get $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for every block k . Then apply Lemma 16.44. $\square$

Lemma 16.59 (Explicit gauge from separated canonical data). Lean ✓ Stmt *Under the same hypotheses, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that*

$$B_k^i = X_k A_k^i X_k^{-1}$$

for every block k and physical index i .

Proof. ✓ Proof Combine Lemma 16.57 with the matrices X_k from Lemma 16.45. $\square$

Lemma 16.60 (Gauge comparison from canonical-form data and strict weights). Lean ✓ Stmt *Under the hypotheses of Lemma 16.56, every block pair A_k, B_k is gauge equivalent, and the block-diagonal tensors are globally gauge equivalent.*

Proof. ✓ Proof Lemma 16.56 gives per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. Lemma 16.44 converts this to per-block and global gauge equivalence. $\square$

Lemma 16.61 (Explicit gauge from canonical-form data and strict weights). Lean ✓ Stmt *Under the same hypotheses, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that $B_k^i = X_k A_k^i X_k^{-1}$ for every block k and physical index i .*

Proof. ✓ Proof Apply Lemma 16.59 with the strict-ordering data from the separated hypotheses. $\square$

16.12 Heterogeneous equal-case block matching

Lemma 16.62 (Heterogeneous equal-case block matching). [Lean](#) [✓ Stmt](#) *Let P and Q be two BNT sector decompositions (Definition 11.1), possibly with different block counts and different bond dimensions in corresponding blocks. If their block-diagonal tensors generate the same MPV family (Definition 2.11), then the basis sectors are bijectively matched by a permutation β , each matched basis pair is gauge-phase equivalent (Definition 2.21), the matched copy multiplicities agree, and the copy weights agree up to a unit-modulus phase ζ and a copy-index reindexing. This is the sector-data form of the equal-case theorem [CPGSV17a, Corollary II.2, lines 354–361 and appendix 1172–1192]; the corresponding global gauge corollary is recorded in Chapter 12.*

Proof. [✓ Proof](#) On the BNT canonical form, the proof proceeds in three steps: bijective block matching from non-decaying cross-overlaps, coefficient identity via the global-gauge phase ζ , and multiset comparison of the scalar power sums recovering matched copy multiplicities and weights up to ζ^{-1} . $\square$

16.13 Per-block to global proportional gauge

The block-permutation gauge-phase conclusion (see Definition 12.39) provides per-block gauge matrices X_k and phases ζ_k . Using the block-diagonal GL construction of Definition 16.40, one assembles these per-block gauges into a single global gauge matrix acting on the total bond space of the weighted direct sum. This is the *global proportionality matrix* of [CPGSV17a, Corollary II.2, lines 1155–1192].

Theorem 16.63 (Global conjugation from per-block gauge-phase data). [Lean](#) [Lean](#) [✓ Stmt](#) *Let the block weights absorb the per-block phases via $\mu_A(k) = \mu_B(\sigma(k)) \zeta_k$. Then the weighted direct sum of the permuted right-side blocks is conjugate to the weighted direct sum of the cast left-side blocks by the flattened block-diagonal gauge $X = \bigoplus_k X_k$. Consequently the two assembled tensors are gauge equivalent.*

Proof. [✓ Proof](#) Let $\tilde{A}_k$ denote the block A_k transported along the dimension equality $\dim_k^A = \dim_{\sigma(k)}^B$. Absorb each phase ζ_k into the corresponding weight via the identity $\mu_A(k) = \mu_B(\sigma(k)) \zeta_k$. After this absorption each weighted block satisfies an ordinary conjugation relation $\mu_B(\sigma(k)) B_{\sigma(k)}^i = X_k (\mu_A(k) \tilde{A}_k^i) X_k^{-1}$. Lemma 16.41 then assembles the per-block X_k into the global block-diagonal gauge $X = \bigoplus_k X_k$. $\square$

16.14 Translation-invariant reduction

Theorem 16.64 (Translation-invariant reduction corollary). [Lean](#) [✓ Stmt](#) *Let A and B be MPS tensors of bond dimension D with A injective. Consider the two constant translation-invariant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n \geq 1$. If the combined tensors of the two chains generate the same MPV family, then:*

1. *B is gauge equivalent to A : there exists $X \in \text{GL}_D(\mathbb{C})$ with $B^i = X A^i X^{-1}$ for all i .*
2. *B is injective.*

Proof. [✓ Proof](#) Theorem 16.25 gives a single gauge matrix X with $B^i = X A^i X^{-1}$ for all i . Since gauge equivalence preserves injectivity, B is injective as well. $\square$

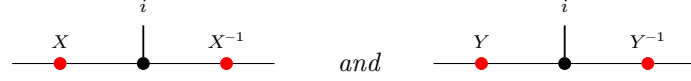
Theorem 16.65 (Translation-invariant reduction from same state). [Lean](#) [Stmt](#) Assume a same-state reduction hypothesis for (d, D) . Let A and B be MPS tensors of bond dimension D , and let $n \geq 3$. If the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same chain state, with A injective, then the same conclusion as in Theorem 16.64 holds: B is gauge equivalent to A , and B is injective.

Proof. [Proof](#) The same-state reduction hypothesis upgrades equality of the two constant chain states to equality of their combined-tensor MPV families. Then Theorem 16.64 applies. $\square$

16.14.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 14.15) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

Lemma 16.66 (Gauge ratio commutes with tensor matrices). [Lean](#) [Stmt](#) Diagrammatically, the hypothesis is that the two local gauges



produce the same target tensor B^i from the same source tensor A^i . If $X, Y \in \text{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = XA^iX^{-1} = YA^iY^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i .

Proof. [Proof](#) Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. $\square$

Lemma 16.67 (Gauge ratio is scalar for injective tensors). [Lean](#) [Stmt](#) Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.

Proof. [Proof](#) The same two-gauge picture shows that the ratio $Y^{-1}X$ acts invisibly on the local tensor. By Lemma 16.66, $Y^{-1}X$ commutes with all A^i . Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 16.12, it is a scalar matrix. $\square$

Theorem 16.68 (Projective multiplication law for gauge matrices). [Lean](#) [Stmt](#) Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \text{GL}_D(\mathbb{C})$ satisfy $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$ for all i . Then for all $g, h \in G$, there exists $c(g, h) \in \mathbb{C}$ such that

$$X(h) X(g) = c(g, h) X(g \cdot h).$$

In fact $c(g, h) \neq 0$ since both sides are invertible, so $c(g, h) \in \mathbb{C}^\times$. Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:



and both sides conjugate A^i to the same twisted tensor $\tilde{A}_{gh}^i$.

Proof. [Proof](#) By the composition law $\tilde{A}_{gh} = (\tilde{A}_h)_g$ (Lemma 14.5), both $X(h)X(g)$ and $X(gh)$ conjugate A to $\tilde{A}_{gh}$. Lemma 16.67 then gives the scalar factor. $\square$

16.15 Closing remarks

Remark 16.69. Appendix A of [MGRPG⁺18] strengthens the normal-MPS bound from $n \geq 3L$ to $n \geq 2L + 1$ by a more refined blocking argument. The injective PEPS theorem of Chapter 17 and the normal PEPS theorem on arbitrary graphs both rest on the same two core lemmas (Lemma 16.13 and Lemma 16.14); the graph geometry enters only through the blocking step that reduces to a three-site chain.

Chapter 17

Fundamental Theorems for Injective and Normal PEPS

The following definitions and results extend the Fundamental Theorem to injective and normal PEPS, following Theorems 2 and 3 of [MGRPG⁺18].

Definition 17.1 (Graph edge for PEPS). [Lean](#) [✓ Stmt](#) For a finite simple graph G , an *edge* is an ordered pair (u, v) with $u < v$ and u adjacent to v . The ordering is only used to avoid double counting.

Definition 17.2 (Incident edges at a vertex). [Lean](#) [✓ Stmt](#) For a vertex v , an *incident edge* is an edge of G having v as one of its endpoints.

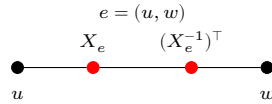
Definition 17.3 (PEPS tensor on a graph). [Lean](#) [✓ Stmt](#) A PEPS tensor on G with physical dimension d consists of a bond dimension D_e for every edge e , together with a local tensor at each vertex v whose virtual indices are indexed by the edges incident to v and whose physical index lies in $\{0, \dots, d - 1\}$.

Definition 17.4 (Virtual configuration). [Lean](#) [✓ Stmt](#) A *virtual configuration* assigns to every edge e one basis index in the corresponding bond space of dimension D_e .

Definition 17.5 (PEPS state coefficient). [Lean](#) [✓ Stmt](#) For a physical configuration σ on the vertices, the PEPS state coefficient is obtained by summing over all virtual configurations and, for each such configuration, multiplying the local tensor entries over all vertices.

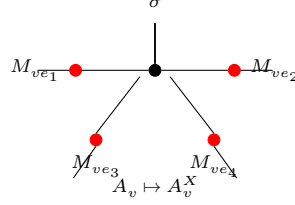
Definition 17.6 (Equality of PEPS states). [Lean](#) [✓ Stmt](#) Two PEPS tensors represent the same state if all their state coefficients agree for every physical configuration.

Definition 17.7 (Edge gauge at a vertex). [Lean](#) [✓ Stmt](#) For an edge $e = (u, w)$ with $u < w$ and gauge $X_e \in \text{GL}(D_e)$, the gauge matrix applied at vertex v is X_e if $v = u$ and $(X_e^{-1})^\top$ if $v = w$. Diagrammatically, the ordered edge carries opposite endpoint actions:



Definition 17.8 (Gauge vertex action). [Lean](#) [✓ Stmt](#) The gauged tensor at vertex v is A_v^X . For each incident edge ie , write M_{ie} for the matrix specified by Definition 17.7, applied at the vertex v

to the gauge on the underlying edge of ie . Then $A_v^X(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) A_v(\eta', \sigma)$. In tensor-network notation, one inserts the appropriate gauge matrix on each incident virtual leg of the local tensor:



Definition 17.9 (Apply gauge). [Lean](#) [Stmnt](#) The PEPS tensor obtained by applying edge-gauge matrices to A , with vertex tensors A_v^X .

Definition 17.10 (Gauge equivalence for PEPS). [Lean](#) [Stmnt](#) Two PEPS tensors A, B are gauge-equivalent if they have the same bond dimensions and B is obtained from A by applying invertible gauge matrices on each edge.

Theorem 17.11 (Gauge invariance of PEPS state). [Lean](#) [Stmnt](#) Applying a gauge to a PEPS tensor preserves the state coefficients. Along each internal edge, the two endpoint insertions cancel:

$$\begin{array}{ccc}
 X_e(j, i) ((X_e^{-1})^\top)(j, k) & \sum_j X_e(j, i) ((X_e^{-1})^\top)(j, k) = \delta(i, k) & \\
 \bullet \text{---} \bullet \text{---} \bullet & = & \bullet \text{---} \bullet
 \end{array}$$

Theorem 17.12 (Gauge equivalence implies same state). [Lean](#) [Stmnt](#) Gauge equivalence implies that the two PEPS tensors represent the same state.

Definition 17.13 (Local tensor evaluation). [Lean](#) [Stmnt](#) The local tensor evaluated at vertex v with virtual-index weighting f , computing $\sum_{\eta} (\prod_{ie} f(ie)(\eta(ie))) \cdot A_v(\eta, \sigma)$. This is multilinear in the components of f , not linear in the full tuple.

Definition 17.14 (Vertex injectivity). [Lean](#) [Stmnt](#) A PEPS tensor A is *vertex-injective* if, at every vertex v , the family of physical vectors

$$(A_v(\eta, \cdot) \in \mathbb{C}^d)_\eta$$

indexed by virtual configurations η on the edges incident to v is linearly independent in $\mathbb{C}^d$. Equivalently, extending A_v by linearity to a map from the free vector space on virtual configurations into the physical space $\mathbb{C}^d$ gives a linear map with trivial kernel. This matches the injectivity formulation of [MGRPG⁺18, Section 3], where the local tensor is interpreted as a map from virtual configurations to the physical space.

Definition 17.15 (Local tensor map). [Lean](#) [Stmnt](#) For a fixed vertex v , the family of local physical vectors $\eta \mapsto A_v(\eta, \cdot)$ extends by linearity to a map from the coefficient space on local virtual configurations at v into $\mathbb{C}^d$.

Definition 17.16 (Local left inverse). [Lean](#) [Stmnt](#) If A is vertex-injective, one may choose a linear map Ψ_v from $\mathbb{C}^d$ to the coefficient space on local virtual configurations at v such that $\Psi_v \circ \Phi_v = \mathbb{1}$.

Definition 17.17 (Physical realization of a local virtual operator). [Lean](#) [✓ Stmt](#) For an endomorphism T of the coefficient space on local virtual configurations at v , define the local physical operator

$$O_T = \Phi_v \circ T \circ \Psi_v.$$

Theorem 17.18 (Local virtual operations on the image). [Lean](#) [✓ Stmt](#) For every coefficient function c on the local virtual configurations at v ,

$$O_T(\Phi_v(c)) = \Phi_v(Tc).$$

Proof. [✓ Proof](#) This is immediate from $\Psi_v \circ \Phi_v = 1$. □

Definition 17.19 (Split at one incident edge). [Lean](#) [✓ Stmt](#) Choosing one incident edge at v identifies the set of local virtual configurations with the product of the bond index on that edge and the residual indices on the remaining incident edges.

Definition 17.20 (Middle region of an edge). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, the middle region is the complement $V \setminus \{u, v\}$.

Theorem 17.21 (Edge-centered vertex partition). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, the three regions $\{u\}$, $V \setminus \{u, v\}$, and $\{v\}$ cover the whole vertex set.

Proof. [✓ Proof](#) Every vertex is either u , v , or different from both. □

Theorem 17.22 (Product splitting at an edge). [Lean](#) [✓ Stmt](#) For any edge $e = (u, v)$ and any function f on the vertex set with values in a commutative monoid M ,

$$\prod_{x \in V} f(x) = f(u) \left(\prod_{x \in V \setminus \{u, v\}} f(x) \right) f(v).$$

Proof. [✓ Proof](#) Isolate the two distinguished endpoints in the product over V and group the remaining factors into the middle region. □

Theorem 17.23 (PEPS coefficient splitting at an edge). [Lean](#) [✓ Stmt](#) For any edge $e = (u, v)$, each summand in the PEPS coefficient factors into the tensor contribution at u , the product over the middle region $V \setminus \{u, v\}$, and the tensor contribution at v .

Proof. [✓ Proof](#) Expand the state coefficient and apply the product splitting at the chosen edge to the per-vertex product inside each summand. □

Definition 17.24 (Edge-centered boundary data). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, the boundary data consist of the virtual index on e , the remaining virtual indices incident to u , and the remaining virtual indices incident to v . These are the indices over which the endpoint tensors couple to the blocked middle region.

Definition 17.25 (Boundary data read from a virtual configuration). [Lean](#) [✓ Stmt](#) A global virtual configuration determines edge-centered boundary data by reading the index on the distinguished edge and the residual endpoint indices.

Definition 17.26 (Boundary compatibility at an edge). [Lean](#) [✓ Stmt](#) A global virtual configuration is compatible with fixed edge-centered boundary data if it has the prescribed index on the distinguished edge and the prescribed residual indices at both endpoint stars.

Definition 17.27 (Middle configurations over fixed boundary data). [Lean](#) [✓ Stmt](#) For fixed boundary data β , the middle configurations are the subtype of global virtual configurations satisfying the compatibility condition with β .

Definition 17.28 (Virtual configurations fibred by edge boundary data). [Lean](#) [✓ Stmt](#) A global virtual configuration is equivalently a choice of edge-centered boundary data together with a middle configuration in the corresponding fibre.

Definition 17.29 (Left endpoint local configuration from boundary data). [Lean](#) [✓ Stmt](#) The left endpoint local virtual configuration is reconstructed from the edge index and the residual data on the left endpoint star.

Definition 17.30 (Right endpoint local configuration from boundary data). [Lean](#) [✓ Stmt](#) The right endpoint local virtual configuration is reconstructed from the edge index and the residual data on the right endpoint star.

Theorem 17.31 (Left endpoint reconstruction for compatible boundary data). [Lean](#) [✓ Stmt](#) *If a global virtual configuration is compatible with boundary data, then its restriction to the left endpoint star is the local configuration reconstructed from those data.*

Theorem 17.32 (Right endpoint reconstruction for compatible boundary data). [Lean](#) [✓ Stmt](#) *If a global virtual configuration is compatible with boundary data, then its restriction to the right endpoint star is the local configuration reconstructed from those data.*

Definition 17.33 (Boundary data with an inserted edge matrix). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, inserted-edge boundary data consist of two independent indices on the bond e , together with the residual virtual indices incident to u and to v . These are the boundary indices used when a matrix is inserted on the distinguished bond.

Definition 17.34 (Diagonal inserted-edge boundary data). [Lean](#) [✓ Stmt](#) Ordinary edge-centered boundary data determine inserted-edge boundary data by using the same bond index at both endpoints. This is the diagonal boundary datum selected by the identity matrix in the edge-insertion coefficient.

Definition 17.35 (Diagonal inserted-boundary configurations). [Lean](#) [✓ Stmt](#) The diagonal inserted-boundary configurations are those inserted-edge boundary data for which the two distinguished bond indices agree. This is the support of the identity matrix inside the inserted-boundary sum.

Definition 17.36 (Diagonal boundary reindexing). [Lean](#) [✓ Stmt](#) Ordinary edge-boundary data are equivalent to diagonal inserted-boundary data. This is the finite reindexing map used after the identity matrix has removed all off-diagonal inserted-boundary summands.

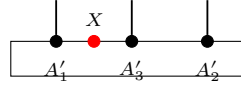
Definition 17.37 (Open middle tensor at an edge). [Lean](#) [✓ Stmt](#) Fixing the residual endpoint data but leaving the bond e out of the middle region, the open middle tensor is the contraction over all virtual indices away from e that are compatible with those endpoint residual data.

Definition 17.38 (Restricting middle configurations away from the edge). [Lean](#) [✓ Stmt](#) An ordinary middle configuration for the edge-blocked contraction restricts to a virtual configuration on all edges except the distinguished edge. This is the first reindexing map needed to compare the identity insertion with the ordinary edge-blocked middle tensor.

Definition 17.39 (Restoring the distinguished edge index). [Lean](#) [✓ Stmt](#) Conversely, a compatible open middle configuration becomes an ordinary middle configuration after the fixed edge-boundary index is restored on the distinguished edge. Together with the restriction map, this gives the finite reindexing underlying the identity-insertion specialization.

Definition 17.40 (Open-middle reindexing equivalence). [Lean](#) [✓ Stmt](#) For fixed edge-boundary data, ordinary middle configurations are equivalent to compatible open-middle configurations. This packages the finite reindexing used when the inserted matrix is the identity.

Definition 17.41 (Edge insertion coefficient). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$ and a matrix X acting on the bond space of e , the edge insertion coefficient is the contraction obtained by inserting X between the endpoint tensors and the open middle tensor:



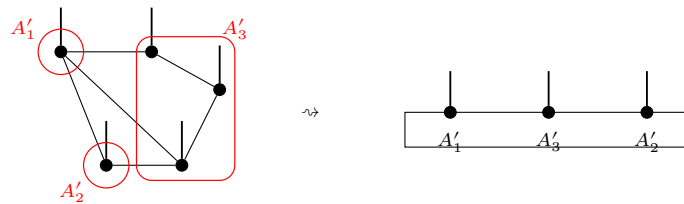
Theorem 17.42 (Middle tensor reindexing for an opened edge). [Lean](#) [✓ Stmt](#) Restoring the fixed distinguished-edge index identifies the ordinary blocked middle tensor with the open middle tensor. This is the finite reindexing used before the identity matrix collapses the two inserted bond indices to the diagonal.

Theorem 17.43 (Diagonal summand of the identity insertion). [Lean](#) [✓ Stmt](#) On the diagonal boundary datum selected by the identity matrix, the inserted-edge summand is the ordinary edge-blocked summand. Thus the remaining identity-insertion step is purely a finite summation statement: the off-diagonal inserted boundary data vanish and the diagonal data reindex the ordinary edge-boundary sum.

Theorem 17.44 (Off-diagonal summands of the identity insertion). [Lean](#) [✓ Stmt](#) If the two inserted bond indices are distinct, then the corresponding summand in the identity insertion is zero. This is the pointwise off-diagonal vanishing used before the inserted-boundary sum is restricted to its diagonal part.

Theorem 17.45 (Identity insertion recovers the edge-blocked coefficient). [Not ready](#) If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the ordinary edge-blocked coefficient. After the diagonal summand comparison, the remaining proof is the finite reindexing of the inserted-boundary sum along the diagonal and the off-diagonal vanishing of the identity matrix.

Definition 17.46 (Blocked middle tensor at an edge). [Lean](#) [✓ Stmt](#) Fixing the edge-centered boundary data, the blocked middle tensor is the sum over global virtual configurations compatible with those data of the product of local tensor entries over the middle region $V \setminus \{u, v\}$. The graph-local contraction is read as a three-site chain:



Definition 17.47 (Edge-blocked PEPS coefficient). [Lean](#) [✓ Stmt](#) The edge-blocked coefficient is the three-block contraction obtained from the left endpoint tensor, the blocked middle tensor, and the right endpoint tensor at the chosen edge.

Theorem 17.48 (Edge-blocked coefficient equals the PEPS coefficient). [Lean](#) [✓ Stmt](#) For every edge e , the edge-blocked three-block coefficient is exactly the original PEPS state coefficient.

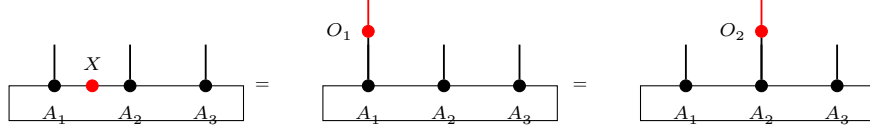
Proof. ✓ Proof Fibre the global virtual sum by the edge-centered boundary data. The remaining fibre sum is the blocked middle tensor, and the two endpoint factors are reconstructed from the same boundary data. $\square$

Theorem 17.49 (Same state gives the same edge-blocked coefficients). Lean ✓ Stmt *If two PEPS tensors have the same state, then their edge-blocked coefficients agree at every edge and every physical configuration.*

Proof. ✓ Proof Rewrite both edge-blocked coefficients as the corresponding PEPS state coefficients and apply the same-state hypothesis. $\square$

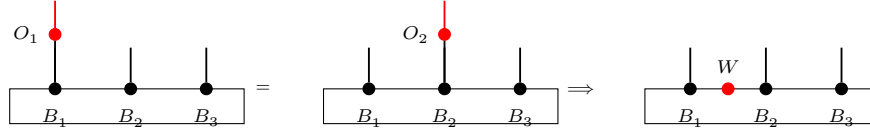
Theorem 17.50 (Edge blocking preserves injectivity). Not ready *If A is vertex-injective, then for every edge $e = (u, v)$, the two endpoint tensors together with the tensor obtained by blocking all vertices outside u and v form an injective three-site MPS. This is the formal version of the assertion after the diagram eq:block_to_mps in [MGRPG⁺ 18, Section 3].*

Theorem 17.51 (Virtual insertion realized physically on either endpoint). Not ready *In the edge-blocked three-site MPS, injectivity of the endpoint tensors lets an arbitrary virtual matrix X on the chosen bond be realized by a physical operation on either neighboring particle:*



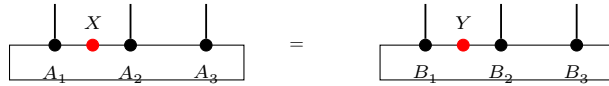
This is the diagrammatic step $X \mapsto O_1, O_2$ in the proof of Lemma inj_isomorph in [MGRPG⁺ 18, Section 3].

Theorem 17.52 (Equal neighboring physical actions recover a virtual insertion). Not ready *Conversely, if two physical operations on neighboring particles act in the same way on the three-site state, then injectivity of the adjacent tensors recovers a virtual operation on their shared bond:*



This is the $O_1, O_2 \mapsto W$ step in the proof of Lemma inj_isomorph.

Theorem 17.53 (Edge-blocked insertion algebra isomorphism). Not ready *Applying the three-site injective-chain theorem to the edge-blocked MPS gives an algebra isomorphism between matrix insertions on the chosen edge in the two PEPS tensors. Equivalently, for every matrix X inserted on that edge for the first tensor family, there is a uniquely determined matrix Y inserted on the corresponding edge for the second tensor family that gives the same edge-blocked state:*



Theorem 17.54 (Edge gauge from the insertion algebra isomorphism). Not ready *The insertion correspondence obtained from the three-site injective-chain theorem is an isomorphism between full matrix algebras on the chosen bond. Hence the two bond dimensions on that edge are equal, and the isomorphism is implemented by conjugation with an invertible edge matrix Z_e : $Y = Z_e X Z_e^{-1}$. The matrix Z_e is unique up to multiplication by a nonzero scalar. This is the finite-dimensional matrix-algebra step used in [MGRPG⁺ 18, Section 3] after Lemma inj_isomorph.*

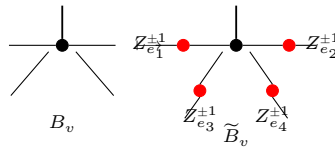
Definition 17.55 (Factorized local gauge datum). [Lean](#) [✓ Stmt](#) For fixed A , B , a bond-dimension equality, and a vertex v , this states the two local conclusions still needed from the blocked-middle reduction in [MGRPG⁺18, Section 3]: (i) each local component of B lies in the image of the local tensor map of A , and (ii) the canonical pullback through the chosen left inverse of A factorizes as one invertible matrix on each incident edge.

Definition 17.56 (Blocked-middle local gauge formula). [Lean](#) [✓ Stmt](#) This states the explicit local gauge formula that the edge-centered blocked-middle reduction is expected to produce: a family of invertible matrices on the incident edges of v whose product kernel reconstructs each local component of B from those of A .

Theorem 17.57 (Blocked-middle formula implies factorized local gauge). [Lean](#) [✓ Stmt](#) Any explicit local gauge formula from the blocked-middle reduction yields the factorized local gauge datum. The remaining open step is to derive this blocked-middle formula from equality of PEPS states.

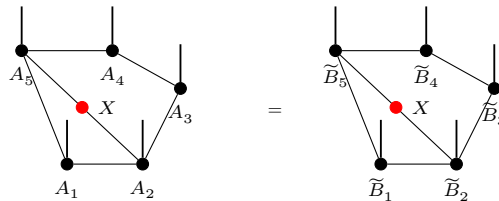
Proof. [✓ Proof](#) The explicit local gauge formula already places each local component of B in the image of the local tensor map of A . Applying the chosen left inverse of A identifies the canonical candidate with the same factorized coefficient kernel, so both parts of the factorized local gauge datum follow. $\square$

Theorem 17.58 (Absorbing edge gauges into the second PEPS). [Not ready](#) The edge gauges obtained by applying the three-site injective-chain theorem at each edge can be absorbed into the tensors of the second family. At a vertex v , each incident edge contributes the appropriate oriented gauge or inverse gauge, producing the modified tensor $\tilde{B}_v$:



After this absorption, arbitrary insertions of the same matrix on the same edge are compared in the first PEPS and the modified second PEPS.

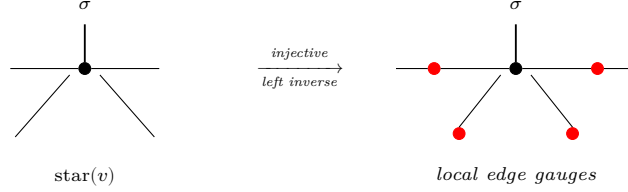
Theorem 17.59 (Post-absorption edge insertion equality). [Not ready](#) After the edge gauges have been absorbed into the second tensor family, for every edge and every matrix X , inserting X on that edge in the first PEPS gives the same state as inserting X on the same edge in the modified second PEPS:



This is the blueprint form of the displayed equation eq:inj_equal_edge in [MGRPG⁺18, Section 3].

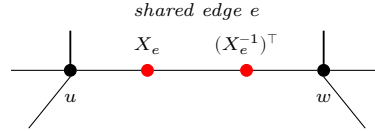
Theorem 17.60 (Local gauge extraction). [Lean](#) [✓ Stmt](#) Under the factorized local gauge hypothesis, the local gauge formula follows. What remains open is to upgrade equality of the edge-blocked

coefficients to the blocked-middle formula using the corresponding three-site injective-chain argument. The physical-realization and physical-to-virtual steps in that argument are recorded separately as Theorems 17.51 and 17.52. In the paper route, once the post-absorption insertion equality of Theorem 17.59 has supplied the factorized local gauge hypothesis, injectivity on the star neighborhood provides a left inverse that isolates the local edge gauges:



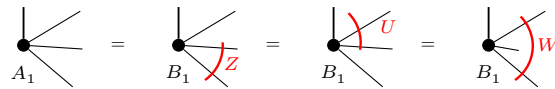
Proof. ✓ Proof The factorized local gauge datum already supplies the incident-edge invertible matrices and the factorized local coefficient identity. $\square$

Theorem 17.61 (Gauge consistency). Lean Not ready Local gauges extracted at adjacent vertices are consistent across shared edges, yielding a global gauge family. On a shared edge, the two neighboring local extractions match as one edge gauge:



Remark 17.62. The PEPS diagrams in this section record the gauge steps used in [MGRPG⁺18, Section 3]. The edge-orientation and vertex-action diagrams fix the endpoint convention for the edge gauges obtained by blocking the PEPS to a three-site injective MPS. The cancellation diagram is the converse contracted identity: inserting opposite endpoint actions on an internal edge leaves the PEPS coefficient unchanged. The edge-gauge absorption diagram records the formation of $\tilde{B}$ before the post-absorption insertion equality. The local-extraction diagram summarizes the comparison of one-site-enlarged injective regions with injective complements, after the edge gauges have been absorbed into $\tilde{B}$. The global-consistency diagram summarizes repeating the edge blocking over all edges, so the two endpoint actions on a shared edge are one edge gauge. The two-injective-tensor diagram records Lemma inj_equal_tensors_2 from the paper, and the final one-vertex-complement diagram records its application to one vertex and its complement. These pictures use the blueprint tensor-dot convention, but each one is attached to the same contraction, insertion, or blocking step as the corresponding argument in the paper.

Theorem 17.63 (Scalar reduction of the residual two-tensor gauges). Not ready In the proof of the generalized two-injective-tensor comparison, applying the inverse of the second injective tensor leaves three possible virtual operators on the exposed legs of the first tensor:

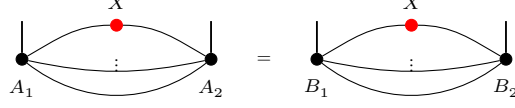


Inverting the first injective tensor then shows that the corresponding tensor decompositions

$$\sum_i I \otimes Z_i^{(1)} \otimes Z_i^{(2)} = \sum_i U_i^{(1)} \otimes U_i^{(2)} \otimes I = \sum_i W_i^{(1)} \otimes I \otimes W_i^{(2)}$$

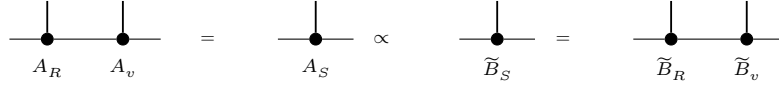
force the residual virtual operators to be scalar multiples of the identity. This is the scalar step in the proof of Lemma inj_equal_tensors_2.

Theorem 17.64 (Generalized two-injective-tensor comparison). Not ready Let A_1, A_2, B_1, B_2 be injective tensors with the same virtual boundary spaces. Suppose that inserting an arbitrary matrix X on any shared virtual bond gives the same two-tensor state for the A -pair and the B -pair:



Then there is a nonzero scalar λ such that $A_1 = \lambda B_1$ and $A_2 = \lambda^{-1} B_2$. This is the blueprint form of Lemma `inj_equal_tensors_2` in [MGRPG⁺ 18, Section 3].

Theorem 17.65 (One-vertex versus complement comparison). Not ready After the edge gauges have been absorbed into the second tensor family, the paper blocks one vertex v against its injective complement. Applying the generalized two-injective-tensor comparison to two regions that differ by exactly this vertex gives $A_v \propto \tilde{B}_v$:



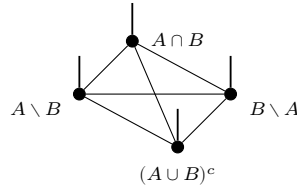
This is the final local comparison following the equation `eq:inj_equal_edge` in [MGRPG⁺ 18, Section 3].

Theorem 17.66 (Fundamental Theorem for injective PEPS, conditional on bond-dimension equality). Lean Not ready Given matching bond dimensions, two vertex-injective PEPS that generate the same state are gauge-equivalent. This conditional form isolates the bond-dimension obligation from the edge-centered consistency obligation of the global gauge construction.

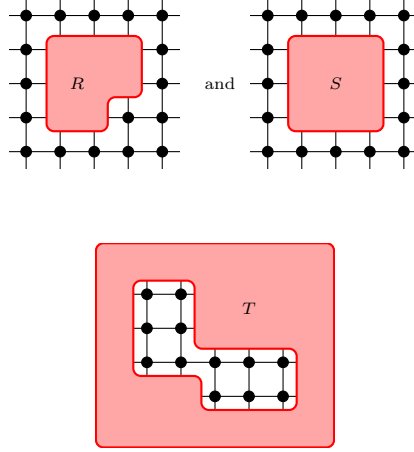
Theorem 17.67 (Fundamental Theorem for injective PEPS). Lean Not ready If two PEPS tensors on a simple graph are both vertex-injective and generate the same state, then they are gauge-equivalent (Theorem 2 of [MGRPG⁺ 18]).

17.1 Normal PEPS

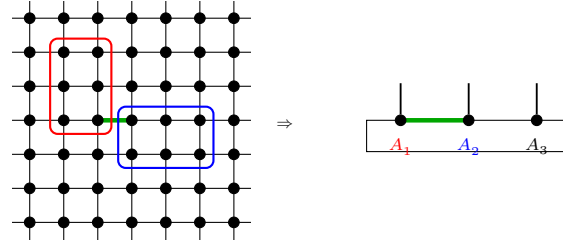
Lemma 17.68 (Union of injective regions). Not ready If two regions of a PEPS are injective, then their union is injective. The proof in [MGRPG⁺ 18, Section 3] blocks the network into four parts $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$, and applies the two available left inverses in sequence:



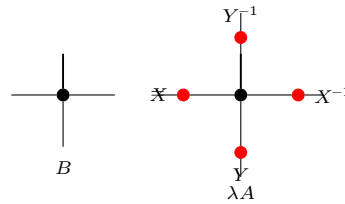
Definition 17.69 (Normal square-lattice blocking regions). Not ready In the translationally invariant square-lattice proof, the source paper assumes that every 2×3 and 3×2 region is injective. Lemma 17.68 then implies that the regions R , S , and T below are injective:



Theorem 17.70 (Normal edge blocking to a three-site injective chain). Not ready Under the square-lattice normality hypotheses of [MGRPG⁺ 18, Theorem 3], the tensor network may be blocked around each edge into three injective regions. Hence the same three-site injective-chain insertion argument used in the injective PEPS theorem applies to the blocked normal PEPS:



Theorem 17.71 (Translationally invariant normal gauge absorption). Not ready In the translationally invariant normal square-lattice case, the edge gauges obtained after blocking are represented by one horizontal matrix X and one vertical matrix Y , unique up to scalar. Absorbing these matrices into B gives a tensor $\tilde{B}$, equivalently the final local gauge formula of [MGRPG⁺ 18, Theorem 3]:



Theorem 17.72 (Fundamental Theorem for normal square-lattice PEPS). Not ready Let A and B be translationally invariant normal square-lattice PEPS tensors such that every 2×3 and 3×2 region is injective. If they generate the same state on an $n \times m$ region with $n, m \geq 7$, then they are related by horizontal and vertical gauges X, Y , up to a scalar λ satisfying $\lambda^{nm} = 1$. This is [MGRPG⁺ 18, Theorem 3].

Theorem 17.73 (Fundamental Theorem for normal PEPS). Not ready Suppose two normal PEPS generate the same state, can be blocked into three-site injective chains around every edge, and

for every site admit two injective regions with injective complements that differ exactly at that site. Then their defining tensors are related by a local gauge, and the gauges are unique up to the scalar freedom allowed by the graph. This is the general normal-PEPS theorem stated after [MGRPG⁺18, Theorem 3].

Definition 17.74 (Gauge equivalence modulo balanced edge scalars). Lean ✓ Stmt Two gauge families are equivalent modulo balanced edge scalars if there is a nonzero scalar c_e on each edge such that the corresponding endpoint actions rescale the oriented edge gauges, and the oriented product of those endpoint scalars is 1 at every vertex.

Theorem 17.75 (Balanced edge scalars preserve the local gauge action). Lean ✓ Stmt If two gauge families differ by a balanced edge-scalar reweighting, then they produce the same gauged tensor at every vertex.

Proof. ✓ Proof In the defining sum for the gauged tensor, each incident edge contributes its endpoint scalar factor in addition to the original gauge entry. The product of these endpoint scalars is 1 by the balancing condition, so every summand is unchanged. $\square$

Theorem 17.76 (Gauge uniqueness modulo balanced edge scalars). Lean Not ready If two gauge families relate the same injective PEPS tensor to the same target tensor, then they define the same class modulo balanced edge scalars. This replaces the earlier global-scalar claim, which is false on a triangle with bond dimension 1.

Remark 17.77. The local left inverse Ψ_v , the realization map $T \mapsto O_T$, the split at one incident edge, and the partition $V = \{u\} \sqcup (V \setminus \{u, v\}) \sqcup \{v\}$ isolate the local data needed for the edge-centered reduction of [MGRPG⁺18, Section 3]. The middle-region blocked coefficient is constructed by the preceding definitions; the remaining local-gauge step is the comparison with the corresponding three-site injective chain. The uniqueness statement has now been repaired to a quotient by balanced edge scalars, but its proof still requires the same blocking construction together with the corresponding tensor-factor uniqueness statement at a single vertex.

Chapter 18

Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Chapter 7].

Section 18.1 establishes definitions and the basic exponential form (Proposition 18.14). Section 18.2 gives the Duhamel formula and perturbation bound. Section 18.3 develops the generator theory and proves the GKSL theorem.

18.1 Dynamical semigroups and the exponential form

Definition 18.1 (Dynamical semigroup). [Lean](#) [✓ Stmt](#) A family of linear maps $T : \mathbb{R} \rightarrow (M_D(\mathbb{C}) \rightarrow_\ell M_D(\mathbb{C}))$ is a *dynamical semigroup* if

1. (semigroup law) $T_{t+s} = T_t \circ T_s$ for all $t, s \geq 0$; and
2. (initial condition) $T_0 = \text{id}$.

This is [Wol12, Equation (7.1)].

Definition 18.2 (Continuous dynamical semigroup). [Lean](#) [✓ Stmt](#) A dynamical semigroup T is *norm-continuous* if the map $t \mapsto T_t$ is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

Definition 18.3 (Exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) Given a generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$, the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

Theorem 18.4 (Semigroup law for the exponential). [Lean](#) [Lean](#) [✓ Stmt](#) For all $t, s \in \mathbb{R}$, $e^{(t+s)L} = e^{tL} \cdot e^{sL}$.

Proof. [✓ Proof](#) Since $(t \cdot L)$ commutes with $(s \cdot L)$, this follows from the exponential addition formula for commuting elements in a Banach algebra. $\square$

Theorem 18.5 (Exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) The continuous-linear-map exponential satisfies $e^{0L} = \mathbb{1}$.

Proof. [✓ Proof](#) The exponential of the zero endomorphism is the identity. $\square$

Theorem 18.6 (Continuity of the exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) *The map $t \mapsto e^{tL}$ is continuous in the operator norm topology.*

Proof. [✓ Proof](#) The exponential is analytic (convergence radius = ∞), hence continuous. $\square$

Theorem 18.7 (Derivative of the exponential semigroup). [Lean](#) [✓ Stmt](#) *For all $t \in \mathbb{R}$,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

This is [Wol12, Equation (7.2)].

Proof. [✓ Proof](#) Chain rule applied to the composition $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \text{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \text{End}(M_D(\mathbb{C}))$. $\square$

Theorem 18.8 (Derivative at the origin). [Lean](#) [✓ Stmt](#) *One has*

$$\left. \frac{d}{dt}e^{tL} \right|_{t=0} = L.$$

Proof. [✓ Proof](#) This is Theorem 18.7 at $t = 0$ together with Theorem 18.5. $\square$

Theorem 18.9 (Generator uniqueness). [Lean](#) [✓ Stmt](#) *If $e^{tL} = e^{tL'}$ for all $t \geq 0$, then $L = L'$.*

Proof. [✓ Proof](#) Both semigroups agree on $[0, \infty)$, so their derivatives within $[0, \infty)$ at $t = 0$ agree. Since $[0, \infty)$ has unique differentials at 0, $L = L'$. $\square$

Theorem 18.10 (CLM and linear-map forms coincide). [Lean](#) [✓ Stmt](#) *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

Proof. [✓ Proof](#) This is a direct unpacking of the definition of the linear-map exponential semigroup. $\square$

Theorem 18.11 (Pointwise derivative formula). [Lean](#) [✓ Stmt](#) *For every $X \in M_D(\mathbb{C})$ and every $t \in \mathbb{R}$,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

Proof. [✓ Proof](#) Evaluate the derivative of the continuous-linear-map semigroup at X and transport the statement through Theorem 18.10. $\square$

Theorem 18.12 (Linear-map exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) *The linear-map exponential semigroup satisfies $e^{0L} = \mathbb{1}$.*

Proof. [✓ Proof](#) Transport Theorem 18.5 through Theorem 18.10. $\square$

Theorem 18.13 (Exponential semigroup as a dynamical semigroup). [Lean](#) [✓ Stmt](#) *The family $t \mapsto e^{tL}$ satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

Proof. [✓ Proof](#) Transport the continuous-linear-map semigroup law from Theorem 18.4 through Theorem 18.10; the initial condition is Theorem 18.12. $\square$

Theorem 18.14 (Every continuous semigroup is exponential). [Lean](#) [✓ Stmt](#) Every norm-continuous dynamical semigroup T on $M_D(\mathbb{C})$ is of the form

$$T_t = e^{tL}$$

for some generator $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ and every $t \geq 0$. This is [\[Wol12, Proposition 7.1\]](#).

Proof. [✓ Proof](#) Since $T_0 = \mathbb{1}$ and T is continuous, the integral $M_\varepsilon = \int_0^\varepsilon T_s ds$ is invertible for small $\varepsilon > 0$. Using the semigroup property, $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$ is differentiable for $t \geq 0$. ODE uniqueness then gives $T_t = e^{tL}$ for all $t \geq 0$. $\square$

18.2 Perturbation theory

Theorem 18.15 (Duhamel formula). [Lean](#) [✓ Stmt](#) Let $\Delta = L' - L$. For $t \geq 0$,

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta e^{sL'} ds.$$

This is [\[Wol12, Lem. 7.1\]](#).

Proof. [✓ Proof](#) Define $f(s) = e^{(t-s)L} e^{sL'}$. Then $f'(s) = e^{(t-s)L} \Delta e^{sL'}$ (Theorem 18.7), and $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) ds$. $\square$

Theorem 18.16 (Perturbation bound). [Lean](#) [✓ Stmt](#) For $t \geq 0$,

$$\|e^{tL'} - e^{tL}\| \leq t \|\Delta\| \cdot \sup_{s \in [0, t]} \|e^{sL}\| \cdot \sup_{s \in [0, t]} \|e^{sL'}\|,$$

where $\Delta = L' - L$. This is [\[Wol12, Corollary 7.1\]](#).

Proof. [✓ Proof](#) Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 18.15). $\square$

Corollary 18.17 (Perturbation bound under unit-norm control). [Lean](#) [✓ Stmt](#) For $t \geq 0$, if $\|e^{sL}\| \leq 1$ and $\|e^{sL'}\| \leq 1$ for every $s \in [0, t]$, then

$$\|e^{tL'} - e^{tL}\| \leq t \|L' - L\|.$$

Proof. [✓ Proof](#) Apply Theorem 18.16 and bound both suprema by 1. $\square$

Lemma 18.18 (Dyson-series summability from factorial majorant). [Lean](#) [✓ Stmt](#) Assume a factorial majorant of Dyson–Phillips iterates:

$$\|D_n(t)\| \leq M \frac{(t \|L' - L\| M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

Proof. [✓ Proof](#) This is the Weierstrass M-test step, using the scalar convergence of $\sum_n a^n/n!$ and lifting via norm-bounded summability. $\square$

Theorem 18.19 (Factorial norm bound for Dyson iterates). [Lean](#) [✓ Stmt](#) For $s \in [0, t]$ and $M = \sup_{u \in [0, t]} \|e^{uL}\|$,

$$\|\tilde{T}_s^{(n)}\| \leq M \frac{(s \|\Delta\| M)^n}{n!}.$$

Proof. ✓ Proof Induction on n . The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$ and integrates $\int_0^s u^n du = s^{n+1}/(n+1)$, recovering the factorial. $\square$

Corollary 18.20 (Dyson series summability). Lean ✓ Stmt *The Dyson–Phillips series $\sum_n \tilde{T}_t^{(n)}$ converges in operator norm for every $t \geq 0$.*

Proof. ✓ Proof Compose the factorial norm bound (Theorem 18.19) with the Weierstrass M-test step (Lemma 18.18). $\square$

Corollary 18.21 (Factorial norm bound at the final time). Lean ✓ Stmt *For $t \geq 0$, setting $s = t$ in Theorem 18.19 gives*

$$\|\tilde{T}_t^{(n)}\| \leq M \frac{(t\|\Delta\|M)^n}{n!},$$

where $M = \sup_{u \in [0, t]} \|e^{uL}\|$.

Proof. ✓ Proof This is the special case $s = t$ of Theorem 18.19. $\square$

Theorem 18.22 (Continuity of Dyson iterates). Lean ✓ Stmt *Each Dyson iterate $t \mapsto \tilde{T}_t^{(n)}$ is continuous in the time parameter.*

Proof. ✓ Proof By induction on n . The base case is continuity of the matrix exponential. For the inductive step, the parametric integral $t \mapsto \int_0^t e^{(t-s)L} \Delta \tilde{T}_s^{(n)} ds$ is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of $\mathbb{R}$ since $\tilde{T}_t^{(n+1)} = 0$ for $t \leq 0$. $\square$

Theorem 18.23 (Factorial bound on the Dyson remainder). Lean ✓ Stmt *For $s \in [0, t]$, $M = \sup_{u \in [0, t]} \|e^{uL}\|$, and $M' = \sup_{u \in [0, t]} \|e^{uL'}\|$:*

$$\|T'_s - \sum_{n < N} \tilde{T}_s^{(n)}\| \leq M' \frac{(s\|\Delta\|M)^N}{N!}.$$

Proof. ✓ Proof Induction on N . The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity $R_{N+1}(s) = \int_0^s e^{(s-u)L} \Delta R_N(u) du$ (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial. $\square$

Theorem 18.24 (Dyson series identity [Wol12, Equation (7.13)]). Lean ✓ Stmt *For $t \geq 0$, the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}.$$

Proof. ✓ Proof The partial-sum remainder $\|e^{tL'} - \sum_{n < N} \tilde{T}_t^{(n)}\| \leq M' (t\|\Delta\|M)^N/N!$ tends to zero as $N \rightarrow \infty$, since $c^N/N! \rightarrow 0$ for any constant c . Together with the summability of the series (Corollary 18.20), this identifies the sum via uniqueness of limits. $\square$

Corollary 18.25 (Tsum form of the Dyson series). Lean ✓ Stmt $\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}$ (tsum form).

Proof. ✓ Proof Immediate from Theorem 18.24. $\square$

18.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 18.54), which shows that such generators have the standard Lindblad form.

18.3.1 Generator decomposition and conditional complete positivity

Definition 18.26 (Generator decomposition). [Lean](#) [✓ Stmt](#) A *generator decomposition* consists of a completely positive map $\phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ and a matrix $\kappa \in M_d(\mathbb{C})$, defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Equation (7.14)].

Definition 18.27 (Conditional complete positivity). [Lean](#) [✓ Stmt](#) A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map ϕ and a matrix κ such that $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. This is condition 1 of [Wol12, Proposition 7.2].

Definition 18.28 (Trace-annihilating map). [Lean](#) [✓ Stmt](#) A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is *trace-annihilating* if $\text{tr}(L(\rho)) = 0$ for all $\rho \in M_d(\mathbb{C})$. This is the infinitesimal version of trace preservation.

Definition 18.29 (Trace constraint). [Lean](#) [✓ Stmt](#) A generator decomposition (ϕ, κ) satisfies the *trace constraint* if $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, where ϕ^* is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Equation (7.20)].

Theorem 18.30 (Trace constraint implies trace-annihilating). [Lean](#) [✓ Stmt](#) If (ϕ, κ) satisfies $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$, then $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ is trace-annihilating.

Proof. [✓ Proof](#) Let $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$. By the cyclic property of trace, $\text{tr}(L(\rho)) = \text{tr}((\sum_i K_i^\dagger K_i)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = \text{tr}((\kappa + \kappa^\dagger)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = 0$. $\square$

18.3.2 The Lindblad form

Definition 18.31 (Lindblad form). [Lean](#) [✓ Stmt](#) A *Lindblad form* consists of a Hermitian matrix $H = H^\dagger$ (the Hamiltonian) and a family of matrices $\{L_j\}_{j=0}^{r-1}$ (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\}_+ \right),$$

where $[A, B] = AB - BA$ and $\{A, B\}_+ = AB + BA$. This is [Wol12, Equation (7.21)].

Theorem 18.32 (Lindblad form is trace-annihilating). [Lean](#) [✓ Stmt](#) Every Lindblad form defines a trace-annihilating linear map.

Proof. [✓ Proof](#) The Hamiltonian part satisfies $\text{tr}(i[\rho, H]) = 0$ by the cyclic property. Each dissipator term satisfies $\text{tr}(L_j \rho L_j^\dagger) = \text{tr}(L_j^\dagger L_j \rho) = \text{tr}(\rho L_j^\dagger L_j)$, so the three terms sum to zero. $\square$

Theorem 18.33 (Lindblad form equals generator decomposition). [Lean](#) [✓ Stmt](#) A Lindblad form with Hamiltonian H and operators $\{L_j\}$ defines the same linear map as the generator decomposition (ϕ, κ) with $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$ and $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$. This is [Wol12, Equation (7.24)].

Proof. ✓ Proof Compute $\kappa^\dagger = -iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ using $H^\dagger = H$ and $(A^\dagger A)^\dagger = A^\dagger A$. Setting $S = \sum_j L_j^\dagger L_j$, expand $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$, which is $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$. $\square$

Theorem 18.34 (Lindblad form is CCP). Lean ✓ Stmt *Every Lindblad form defines a conditionally completely positive map.*

Proof. ✓ Proof By Theorem 18.33, the Lindblad form equals a generator decomposition with ϕ CP. $\square$

Definition 18.35 (Commutator form of Lindblad equation). Lean ✓ Stmt *The commutator form of the Lindblad equation writes the generator as*

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho, L_j^\dagger]),$$

where $[A, B] = AB - BA$. This is [Wol12, Equation (7.22)].

Theorem 18.36 (Commutator form equals standard Lindblad form). Lean ✓ Stmt *The commutator form (Definition 18.35) defines the same linear map as the standard Lindblad form (Definition 18.31).*

Proof. ✓ Proof Expanding the commutator brackets $[L_j, \rho L_j^\dagger]$ and $[L_j \rho, L_j^\dagger]$ yields the standard dissipator terms. $\square$

18.3.3 Characterization of conditional complete positivity

Theorem 18.37 (Conditional complete positivity implies projected Choi positivity). Lean ✓ Stmt *If L is CCP and $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then*

$$P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

This is [Wol12, Proposition 7.2].

Proof. ✓ Proof Write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$. The left- and right-multiplication terms are annihilated by P , so the projected Choi matrix reduces to the projected Choi matrix of ϕ , which is positive. $\square$

Theorem 18.38 (Projected Choi positivity implies conditional complete positivity). Lean ✓ Stmt *If L is Hermiticity-preserving and $P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0$ where $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$, then L is CCP. This is [Wol12, Proposition 7.2].*

Proof. ✓ Proof Decompose the Hermitian Choi matrix as $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$ with $Q \geq 0$ supported on $|\Omega\rangle^\perp$. The Choi–Jamiołkowski isomorphism applied to Q gives the CP map ϕ , and $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$ defines κ . $\square$

18.3.4 Completely positive semigroups and conditionally completely positive generators

Theorem 18.39 (CP semigroups have CCP generators). Lean ✓ Stmt *If e^{tL} is CP for all $t \geq 0$, then L is CCP. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof From $(e^{tL} \otimes \text{id})(|\Omega\rangle\langle\Omega|) \geq 0$, expand at infinitesimal t and project onto P to get $P(L \otimes \text{id})(|\Omega\rangle\langle\Omega|)P \geq 0$. Then Theorem 18.38 gives CCP. □

Theorem 18.40 (CCP generators yield CP semigroups). Lean ✓ Stmt *If L is CCP, then e^{tL} is CP for all $t \geq 0$. This is [Wol12, Proposition 7.3].*

Proof. ✓ Proof Approximate e^{tL} by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \quad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to e^{tL} , and the cone of CP maps is closed. □

Theorem 18.41 (CP semigroups iff CCP generators). Lean ✓ Stmt *e^{tL} is CP for all $t \geq 0$ if and only if L is CCP.*

Proof. ✓ Proof Combine Theorems 18.39 and 18.40. □

18.3.5 Freedom in generator representation

Theorem 18.42 (Traceless Kraus operators exist). Lean ✓ Stmt *Assume $d > 0$. Given any Kraus operators $\{K_j\}$, there exist $c_j \in \mathbb{C}$ such that $K'_j = K_j + c_j\mathbb{1}$ is traceless. This is part of [Wol12, Proposition 7.4].*

Proof. ✓ Proof Set $c_j = -\text{tr}(K_j)/d$. □

Theorem 18.43 (Shift invariance of generators). Lean ✓ Stmt *If $K'_j = K_j + c_j\mathbb{1}$ and κ' is adjusted per [Wol12, Equation (7.19)], then (ϕ', κ') and (ϕ, κ) define the same generator.*

Proof. ✓ Proof Direct algebraic expansion. □

18.3.6 Uniqueness of the traceless Lindblad form

Definition 18.44 (Traceless Kraus operators). Lean ✓ Stmt A Lindblad form has *traceless Kraus operators* if $\text{tr}(L_j) = 0$ for every j .

Theorem 18.45 (Uniqueness of the completely positive part). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their CP parts agree: $\phi = \phi'$. This is part of [Wol12, Proposition 7.4, item 2].*

Proof. ✓ Proof Compare projected Choi matrices and use Choi injectivity. □

Theorem 18.46 (Uniqueness of the drift term modulo phase). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar: $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$. This is part of [Wol12, Proposition 7.4, item 2].*

Proof. ✓ Proof After identifying $\phi = \phi'$ (Theorem 18.45), the generator equality forces $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$ for all ρ . Setting $\rho = \mathbb{1}$ gives $\Delta\kappa + (\Delta\kappa)^\dagger = 0$ (skew-Hermiticity), which converts the equation to $[\Delta\kappa, \rho] = 0$ for all ρ . The scalar commutant lemma gives $\Delta\kappa = c \cdot \mathbb{1}$, and skew-Hermiticity forces $c = i\lambda$ for some $\lambda \in \mathbb{R}$. □

Theorem 18.47 (Combined uniqueness of the traceless Lindblad form). Lean ✓ Stmt *If two Lindblad forms F, F' induce the same generator and both have traceless Kraus operators, then $\phi = \phi'$ and $\kappa' = \kappa + i\lambda\mathbb{1}$ for some $\lambda \in \mathbb{R}$.*

Proof. ✓ Proof Combine Theorems 18.45 and 18.46. □

18.3.7 Trace-annihilation and trace preservation

Theorem 18.48 (Trace-annihilating generators yield trace-preserving semigroups). [Lean](#) [✓ Stmt](#)
If L is trace-annihilating, then e^{tL} is trace-preserving for every $t \in \mathbb{R}$.

Proof. [✓ Proof](#) For fixed ρ , the function $f(t) = \text{tr}(e^{tL}(\rho))$ has derivative $f'(t) = \text{tr}(L(e^{tL}(\rho))) = 0$, so f is constant and $\text{tr}(e^{tL}(\rho)) = f(t) = f(0) = \text{tr}(\rho)$. $\square$

Theorem 18.49 (Trace-preserving semigroups have trace-annihilating generators). [Lean](#) [✓ Stmt](#)
If e^{tL} is trace-preserving for every $t \geq 0$, then L is trace-annihilating.

Proof. [✓ Proof](#) Differentiate the identity $\text{tr}(e^{tL}(\rho)) = \text{tr}(\rho)$ at $t = 0$ from the right. $\square$

18.3.8 The GKSL/Lindblad theorem

Definition 18.50 (GKSL generator). [Lean](#) [✓ Stmt](#) A linear map $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ is a *GKSL generator* if e^{tL} is a quantum channel (CPTP) for all $t \geq 0$.

Theorem 18.51 (A trace-constrained generator decomposition gives a GKSL generator). [Lean](#) [✓ Stmt](#)
If $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with ϕ CP and $\phi^(\mathbb{1}) = \kappa + \kappa^\dagger$, then L is a GKSL generator. This is [Wol12, Theorem 7.1, Equation (7.20)].*

Proof. [✓ Proof](#) CP follows from Theorem 18.40. The trace constraint implies trace-annihilation by Theorem 18.30, so Theorem 18.48 gives trace preservation. $\square$

Theorem 18.52 (GKSL generators admit a trace-constrained generator decomposition). [Lean](#) [✓ Stmt](#)
If L is a GKSL generator, then there exist a CP map ϕ and a matrix κ such that

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$.

Proof. [✓ Proof](#) Apply Theorem 18.39 to obtain a CCP decomposition. Then apply Theorem 18.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$. $\square$

Theorem 18.53 (GKSL iff conditional complete positivity and trace annihilation). [Lean](#) [✓ Stmt](#)
 L is a GKSL generator if and only if L is CCP and trace-annihilating.

Proof. [✓ Proof](#) Forward: apply Theorem 18.39 to the CP part and Theorem 18.49 to the TP part. Reverse: combine Theorem 18.40 with Theorem 18.48. $\square$

Theorem 18.54 (GKSL iff Lindblad form). [Lean](#) [✓ Stmt](#) *L is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \}_+ \right)$$

with $H = H^\dagger$. This is [Wol12, Theorem 7.1].

Proof. [✓ Proof](#) Forward: apply Theorem 18.52 to write $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ with trace constraint. Then choose traceless Kraus operators by Theorem 18.42 and rewrite κ as $iH + \frac{1}{2}\phi^*(\mathbb{1})$; Theorem 18.33 gives the Lindblad formula. Reverse: Theorems 18.34 and 18.32 show that every Lindblad form satisfies the right-hand side of Theorem 18.53. $\square$

18.3.9 Kossakowski matrix form

Definition 18.55 (Kossakowski form). [Lean](#) [✓ Stmt](#) A *Kossakowski form* consists of $H = H^\dagger$, a finite family of matrices $\{F_k\}_{k=0}^{n-1}$, and a positive semidefinite matrix $C \in M_n(\mathbb{C})$, defining

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_{k,l} C_{lk} ([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes $\{F_k\}$ to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

Theorem 18.56 (Kossakowski form iff Lindblad form). [Lean](#) [✓ Stmt](#) A linear map admits a Kossakowski form iff it admits a Lindblad form.

Proof. [✓ Proof](#) Forward: diagonalize $C = B^\dagger B$ and set $L_j = \sum_k B_{jk} F_k$. Reverse: keep the same Hamiltonian and family $F_k = L_k$, and take $C = \mathbb{1}$. $\square$

18.4 Primitivity and irreducibility of QDS

18.4.1 Auxiliary spectral and semigroup lemmas

Definition 18.57 (Quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every $t \geq 0$.

Theorem 18.58 (Semigroup power identity). [Lean](#) [✓ Stmt](#) For $t \geq 0$ and $n \in \mathbb{N}$, one has $T_{nt} = (T_t)^n$.

Theorem 18.59 (Eigenvector propagation under powers). [Lean](#) [✓ Stmt](#) If $E(v) = \mu v$, then $E^n(v) = \mu^n v$ for all $n \in \mathbb{N}$.

Lemma 18.60 (Density matrices are nonzero). [Lean](#) [✓ Stmt](#) Every density matrix is nonzero.

Theorem 18.61 (Compression invariance is stable under powers). [Lean](#) [✓ Stmt](#) If a corner compression is invariant under E , it remains invariant under every power E^n .

Theorem 18.62 (Generator eigenvectors along the exponential semigroup). [Lean](#) [✓ Stmt](#) If $L(X) = \mu X$, then $e^{tL}(X) = e^{t\mu} X$ for all $t \in \mathbb{R}$.

Theorem 18.63 (Spectral mapping for semigroup generators). [Lean](#) [✓ Stmt](#) If μ is in the spectrum of L , then $e^{t\mu}$ is in the spectrum of e^{tL} .

Theorem 18.64 (Root-of-unity rigidity for phases). [Lean](#) [✓ Stmt](#) If $e^{it\theta}$ is a root of unity for every $t > 0$, then $\theta = 0$.

Theorem 18.65 (Peripheral generator eigenvalues are purely imaginary). [Lean](#) [✓ Stmt](#) Peripheral spectral points of the generator have vanishing real part.

Theorem 18.66 (Peripheral cardinality bound). [Lean](#) [✓ Stmt](#) The number of peripheral eigenvalues is bounded by $\dim(M_D(\mathbb{C}))$.

Theorem 18.67 (Bounded order under peripheral power-closure). [Lean](#) [✓ Stmt](#) If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding $\dim(M_D(\mathbb{C}))$ equals 1.

Theorem 18.68 (Peripheral powers stay peripheral under irreducibility). [Lean](#) [✓ Stmt](#) *For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.*

Theorem 18.69 (Continuous-linear-map power evaluation). [Lean](#) [✓ Stmt](#) *Evaluating powers of a linear map agrees with powers in its continuous linear realization.*

Theorem 18.70 (Spectral-radius criterion from eigenvalue bounds). [Lean](#) [✓ Stmt](#) *If all eigenvalues satisfy $|\lambda| < 1$, then the spectral radius is < 1 .*

Theorem 18.71 (Primitive powers annihilate trace-zero matrices). [Lean](#) [✓ Stmt](#) *If E is an irreducible primitive channel with faithful fixed density σ , then $E^n(X) \rightarrow 0$ for every matrix X with $\text{tr}(X) = 0$.*

Proof. [✓ Proof](#) Decompose E^n into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus < 1 , so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component. $\square$

Theorem 18.72 (Fixed points of an irreducible channel are trace multiples). [Lean](#) [✓ Stmt](#) *If E is an irreducible channel with faithful fixed density σ , then every nonzero fixed point V of E satisfies $V = \text{tr}(V)\sigma$.*

Proof. [✓ Proof](#) Subtract the trace multiple $\text{tr}(V)\sigma$ and apply the vanishing theorem for trace-zero fixed points of an irreducible channel. $\square$

Corollary 18.73 (Nonzero fixed points have nonzero trace). [Lean](#) [✓ Stmt](#) *Every nonzero fixed point of an irreducible channel has nonzero trace.*

Proof. [✓ Proof](#) Otherwise the preceding theorem would force the fixed point to vanish. $\square$

Theorem 18.74 (Peripheral eigenvectors become periodic fixed points). [Lean](#) [✓ Stmt](#) *For every positive time t , a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue μ of T_t admits a nonzero eigenvector V and a positive integer p such that $T_t(V) = \mu V$ and $T_{pt}(V) = V$.*

Proof. [✓ Proof](#) Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives $\mu^p = 1$, and the semigroup power identity turns the eigenvector equation for T_t into a fixed-point equation for T_{pt} . $\square$

Theorem 18.75 (Peripheral eigenvectors with nonzero trace). [Lean](#) [✓ Stmt](#) *For every positive time t , under the same hypotheses every peripheral eigenvalue of T_t has a nonzero eigenvector with nonzero trace.*

Proof. [✓ Proof](#) Apply Theorem 18.74 and then use the previous corollary at the irreducible time pt . $\square$

Theorem 18.76 (Trace-preserving eigenvectors force eigenvalue 1). [Lean](#) [✓ Stmt](#) *If E is trace-preserving, $E(V) = \mu V$, and $\text{tr}(V) \neq 0$, then $\mu = 1$.*

Proof. [✓ Proof](#) Take traces in $E(V) = \mu V$ and use trace preservation. $\square$

Theorem 18.77 (Peripheral collapse under irreducibility at all times). [Lean](#) [✓ Stmt](#) *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

Proof. [✓ Proof](#) Combine Theorem 18.75 with Theorem 18.76. □

Theorem 18.78 (Primitivity from irreducibility at all times). [Lean](#) [✓ Stmt](#) *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

Proof. [✓ Proof](#) The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 18.77 collapses the peripheral set to $\{1\}$. □

Remark 18.79 (Semigroup irreducible/primitive equivalence). [✓ Stmt](#) The semigroup analogue of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 18.95 and 18.96).

Remark 18.80 (Reduction to a smaller time step for semigroups). [✓ Stmt](#) Starting from one irreducible time t_0 , we construct a positive time step $u = t_0/N$ with $T_{t_0} = (T_u)^N$, prove peripheral eigenvalue collapse at time u , and deduce primitivity of T_u .

Theorem 18.81 (The stationary density is fixed for all times). [Lean](#) [✓ Stmt](#) *If $t_0 > 0$, T_{t_0} is irreducible, and σ is its unique fixed density matrix, then $T_u(\sigma) = \sigma$ for every $u \geq 0$.*

Proof. [✓ Proof](#) By semigroup commutativity, $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$, and uniqueness of the fixed density at time t_0 identifies $T_u(\sigma)$ with σ . □

Theorem 18.82 (Fixed points at the irreducible time are trace multiples). [Lean](#) [✓ Stmt](#) *If T_{t_0} is irreducible with faithful fixed density σ , then every fixed point V of T_{t_0} satisfies $V = \text{tr}(V)\sigma$.*

Proof. [✓ Proof](#) This is the fixed-point trace-scalar theorem for the irreducible channel T_{t_0} . □

Definition 18.83 (Residual slice index). [Lean](#) [✓ Stmt](#) For $u \geq 0$ and $s > 0$, the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

Definition 18.84 (Residual slice time). [Lean](#) [✓ Stmt](#) For $u \geq 0$ and $s > 0$, the residual slice time is the remainder

$$r_n = s \text{ fract} \left(\frac{nu}{s} \right).$$

Theorem 18.85 (Residual slice decomposition). [Lean](#) [Lean](#) [✓ Stmt](#) *For $u \geq 0$ and $s > 0$, one has $r_n \in [0, s]$ and*

$$nu = m_n s + r_n.$$

Proof. [✓ Proof](#) This is the floor-plus-fraction decomposition of nu/s , multiplied by s . □

Theorem 18.86 (A residual slice can vanish). [Lean](#) [✓ Stmt](#) *Suppose $u \geq 0$ and $s > 0$, and that $T_s(\delta) = \delta$ and $T_{nu}(\delta) \rightarrow 0$ as $n \rightarrow \infty$. Then there exists $a \in [0, s]$ such that $T_a(\delta) = 0$.*

Proof. [✓ Proof](#) The residual slice times lie in the compact interval $[0, s]$, so a subsequence converges to some $a \in [0, s]$. The residual decomposition rewrites $T_{nu}(\delta)$ in terms of $T_{r_n}(\delta)$, and continuity of the semigroup identifies the limit with $T_a(\delta)$. □

Theorem 18.87 (Reduced-time eigenvector has nonzero trace). [Lean](#) [✓ Stmt](#) *Under the reduced-time hypotheses $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$, if μ is a peripheral eigenvalue of T_u , then T_u admits a μ -eigenvector with nonzero trace.*

Theorem 18.88 (Peripheral collapse at the reduced time step). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, every peripheral eigenvalue of T_u equals 1.*

Theorem 18.89 (Primitivity at the reduced time step). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, T_u is primitive.*

Theorem 18.90 (Irreducibility descends to a reduced time step). [Lean](#) [✓ Stmt](#) *If $T_{t_0} = (T_u)^N$ and T_{t_0} is irreducible, then T_u is irreducible.*

Proof. [✓ Proof](#) Any invariant compression for T_u remains invariant under powers, hence also for T_{t_0} ; irreducibility of T_{t_0} rules this out. $\square$

Theorem 18.91 (Existence of an irreducible reduced time step). [Lean](#) [✓ Stmt](#) *If $t_0 > 0$, T_{t_0} is irreducible, and σ is fixed for all times, then there exists a positive time step u such that*

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

the slice T_u is a channel, T_u is irreducible, and $T_u(\sigma) = \sigma$.

Proof. [✓ Proof](#) Take $u = t_0/N$ with $N = (\dim M_D(\mathbb{C}))!$. The semigroup power identity gives the factorization of T_{t_0} , channelity comes from the semigroup hypotheses, and Theorem 18.90 gives irreducibility of T_u . $\square$

Theorem 18.92 (Fixed points of the primitive reduced time step are trace-scalars). [Lean](#) [✓ Stmt](#) *If T_u is primitive and σ is the common fixed density, then every fixed point of any positive-time slice T_s has the form $\tau = \text{tr}(\tau)\sigma$.*

Theorem 18.93 (Existence of a primitive reduced time step). [Lean](#) [✓ Stmt](#) *If T_{t_0} is irreducible for some $t_0 > 0$, then there exists a positive reduced time step u such that T_u is primitive.*

Theorem 18.94 (Irreducibility propagates to all positive times). [Lean](#) [✓ Stmt](#) *If T_{t_0} is irreducible for some $t_0 > 0$, then every positive-time slice T_s is irreducible.*

Proof. [✓ Proof](#) Let σ be the unique faithful fixed density of T_{t_0} . By Theorem 18.81, it is fixed for all times, and Theorem 18.82 makes the fixed-point space of T_{t_0} one-dimensional. Choose a primitive reduced time step by Theorem 18.93; then the trace-scalar description of fixed points from Theorem 18.92 gives the uniqueness criterion for irreducibility at every positive time. $\square$

Theorem 18.95 (An irreducible semigroup is primitive). [Lean](#) [✓ Stmt](#) *Let $T_t = e^{tL}$ be a quantum dynamical semigroup. If T_{t_0} is irreducible for some $t_0 > 0$, then T_t is primitive (and irreducible) for every $t > 0$. This is [Wol12, Proposition 7.5].*

Proof. [✓ Proof](#) First apply Theorem 18.94 to propagate irreducibility from the single time t_0 to every positive time. Then apply Theorem 18.78 to conclude primitivity of every positive-time slice. $\square$

Theorem 18.96 (Irreducibility iff primitivity for a quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) *For a quantum dynamical semigroup $T_t = e^{tL}$:*

$$(\exists t_0 > 0, T_{t_0} \text{ irreducible}) \iff (\forall t > 0, T_t \text{ primitive and irreducible}).$$

This is [Wol12, Proposition 7.5].

Proof. [✓ Proof](#) Forward: apply Theorem 18.95. Reverse: take $t_0 = 1$ (or any positive time); the right-hand side already includes irreducibility of T_t for every $t > 0$. $\square$

18.5 Kernel of the adjoint Liouvillian

Definition 18.97 (Faithful stationary state). [Lean](#) [✓ Stmt](#) A generator L has a faithful stationary state if there exists a positive definite density matrix ρ_0 with $L(\rho_0) = 0$.

Definition 18.98 (Adjoint dissipator). [Lean](#) [✓ Stmt](#) For a Lindblad operator L_j , the adjoint dissipator on observables is

$$\mathcal{D}_{L_j}^*(A) = L_j^\dagger A L_j - \frac{1}{2} L_j^\dagger L_j A - \frac{1}{2} A L_j^\dagger L_j.$$

Definition 18.99 (Adjoint generator). [Lean](#) [✓ Stmt](#) The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}_{L_j}^*(A).$$

Theorem 18.100 (Trace-pairing characterization of the adjoint generator). [Lean](#) [✓ Stmt](#) For every density matrix ρ and every observable A ,

$$\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A).$$

Proof. [✓ Proof](#) Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term. $\square$

Definition 18.101 (Commutant of the Lindblad data). [Lean](#) [✓ Stmt](#) The commutant of a Lindblad form is the set of matrices commuting with H , with every Lindblad operator L_j , and with every adjoint $L_j^\dagger$.

Definition 18.102 (Adjoint kernel). [Lean](#) [✓ Stmt](#) The adjoint kernel is the set of observables A satisfying $L^*(A) = 0$.

Theorem 18.103 (Commutant lies in the adjoint kernel). [Lean](#) [Lean](#) [✓ Stmt](#) If an observable commutes with H , with every L_j , and with every $L_j^\dagger$, then it lies in $\ker(L^*)$.

Proof. [✓ Proof](#) The Hamiltonian commutator term vanishes, and each adjoint dissipator $\mathcal{D}_{L_j}^*(A)$ is zero under the commutation hypotheses. $\square$

Theorem 18.104 (Adjoint kernel lies in the commutant). [Lean](#) [✓ Stmt](#) Let L be a GKSL generator in Lindblad form with Hamiltonian H and Lindblad operators $\{L_j\}$. Define the adjoint generator L^* by $\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A)$. If L has a faithful stationary state $\rho_0 > 0$ with $L(\rho_0) = 0$, then for any A :

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \quad \forall j.$$

That is, $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$. This is [Wol12, Theorem 7.2].

Proof. [✓ Proof](#) From $L^*(A) = 0$ and $L^*(A^\dagger) = 0$, compute $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$. Faithfulness of ρ_0 and $\text{tr}(\rho_0 L^*(A^\dagger A)) = 0$ force $[A, L_j] = 0$ for all j . Similarly $[A, L_j^\dagger] = 0$, and then $L^*(A) = 0$ directly gives $[A, H] = 0$. $\square$

Theorem 18.105 (Adjoint kernel equals the commutant). [Lean](#) [✓ Stmt](#) Under the same hypotheses as Theorem 18.104,

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of $\{H, L_j, L_j^\dagger\}$. This is [Wol12, Theorem 7.2].

Proof. [✓ Proof](#) The inclusion $\supseteq$ is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse $\subseteq$ is Theorem 18.104. $\square$

18.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Proposition 7.6] are as follows.

Definition 18.106 (Nontrivial projection). [Lean](#) [✓ Stmt](#) A projection is nontrivial if it is orthogonal and neither 0 nor $\mathbb{1}$.

Definition 18.107 (Rank-deficient fixed density). [Lean](#) [✓ Stmt](#) Condition (1): there exists a density matrix ρ_0 with nontrivial kernel such that $e^{tL}(\rho_0) = \rho_0$ for all $t \geq 0$.

Definition 18.108 (Rank-deficient kernel element). [Lean](#) [✓ Stmt](#) Condition (2): there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$.

Definition 18.109 (Invariant compression). [Lean](#) [✓ Stmt](#) Condition (3): there exists a nontrivial orthogonal projector P such that $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ for all $t \geq 0$.

Definition 18.110 (Block-upper-triangular Lindblad form). [Lean](#) [✓ Stmt](#) Condition (4): there exists a nontrivial projector P and a Lindblad form $(H, \{L_j\})$ for L such that $(1 - P)L_jP = 0$ and $(1 - P)\kappa P = 0$ for all j , where $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$.

Definition 18.111 (Generator-preserved compression). [Lean](#) [✓ Stmt](#) For a projection P , the generator preserves the compression $PM_d(\mathbb{C})P$ if

$$P L(PXP) P = L(PXP)$$

for every $X \in M_d(\mathbb{C})$.

Definition 18.112 (Reducible quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

Theorem 18.113 (Semigroup invariance implies generator invariance). [Lean](#) [✓ Stmt](#) *If the semigroup preserves $PM_d(\mathbb{C})P$ for all nonnegative times, then the generator preserves the same compression.*

Proof. [✓ Proof](#) Differentiate the identity $Pe^{tL}(PXP)P = e^{tL}(PXP)$ at $t = 0$. □

Theorem 18.114 (Generator invariance implies semigroup invariance). [Lean](#) [✓ Stmt](#) *If the generator preserves $PM_d(\mathbb{C})P$, then so does the exponential semigroup e^{tL} for every $t \geq 0$.*

Proof. [✓ Proof](#) Apply the compression map termwise to the exponential series. Every iterate of L preserves the compression, so the whole series does as well. □

Theorem 18.115 (Reducibility as a generator-level criterion). [Lean](#) [✓ Stmt](#) *A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.*

Proof. [✓ Proof](#) Combine Theorems 18.113 and 18.114. □

Theorem 18.116 (Rank-deficient fixed densities and kernel elements). [Lean](#) [✓ Stmt](#) *A rank-deficient density matrix is a fixed point of e^{tL} for all $t \geq 0$ if and only if it lies in $\ker L$. This is [Wol12, Proposition 7.6, (1)↔(2)].*

Proof. [✓ Proof](#) The equivalence $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$ for all $t \geq 0$ follows from the generator-semigroup correspondence: differentiate at $t = 0$ for the forward direction, and exponentiate for the reverse. □

Theorem 18.117 (A rank-deficient fixed density yields an invariant compression). [Lean](#) [✓ Stmt](#) *If L is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Proposition 7.6, (1)→(3)].*

Proof. [✓ Proof](#) Let ρ_0 be a rank-deficient fixed density. Let P be the support projection of ρ_0 . Since ρ_0 is rank-deficient, $P \neq \mathbb{1}$; since $\rho_0 \neq 0$, $P \neq 0$. Channel positivity and trace preservation force e^{tL} to preserve $PM_D(\mathbb{C})P$. $\square$

Theorem 18.118 (Invariant compression yields block-upper-triangular Lindblad data). [Lean](#) [✓ Stmt](#) *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

Proof. [✓ Proof](#) First pass from semigroup invariance to generator invariance by Theorem 18.113. The generator-level vanishing condition forces $(1 - P)L_j P = 0$ and $(1 - P)\kappa P = 0$ through the sum-of-squares identity. $\square$

Theorem 18.119 (Block-upper-triangular Lindblad data yield invariant compression). [Lean](#) [✓ Stmt](#) *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

Proof. [✓ Proof](#) The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 18.114. $\square$

Theorem 18.120 (Invariant compression and triangular Lindblad data). [Lean](#) [Lean](#) [✓ Stmt](#) *For a GKSL generator L , the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Proposition 7.6, (3)↔(4)].*

Proof. [✓ Proof](#) Combine Theorems 18.119 and 18.118. $\square$

Theorem 18.121 (Triangular Lindblad data give deficient kernels). [Lean](#) [✓ Stmt](#) *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector P , then there exists a density matrix ρ_0 with nontrivial kernel satisfying $L(\rho_0) = 0$. This is [Wol12, Proposition 7.6, (4)→(2)].*

Proof. [✓ Proof](#) The block-upper-triangular structure implies L preserves the compression $PM_d(\mathbb{C})P$. Therefore the semigroup e^{tL} also preserves $PM_d(\mathbb{C})P$ for all $t \geq 0$. By the Brouwer fixed-point theorem, $e^{(1/m)L}$ has a fixed density matrix ρ_m supported in $PM_d(\mathbb{C})P$ for each m . Passing to a subsequential limit gives a density matrix ρ_0 with $P\rho_0 P = \rho_0$ and $L(\rho_0) = 0$. Since $P \neq \mathbb{1}$, the matrix ρ_0 has nontrivial kernel. $\square$

Theorem 18.122 (Equivalent formulations of reducibility). [Lean](#) [✓ Stmt](#) *For a GKSL generator $L : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the four conditions*

1. rank-deficient fixed density,
2. rank-deficient kernel element,
3. invariant compression, and
4. block-upper-triangular Lindblad form

are equivalent. This is [Wol12, Proposition 7.6].

Proof. [✓ Proof](#) The cycle $(1) \leftrightarrow (2)$, $(1) \rightarrow (3)$, $(3) \leftrightarrow (4)$, and $(4) \rightarrow (2)$ closes the equivalence chain. $\square$

18.6.1 Sufficient conditions for non-reducibility

Definition 18.123 (Lindblad span). [Lean](#) [✓ Stmt](#) The $\mathbb{C}$ -linear span of the Lindblad operators $\{L_j\}$.

Definition 18.124 (Hermitian-closed Lindblad span). [Lean](#) [✓ Stmt](#) The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a $*$ -subspace of $M_d(\mathbb{C})$.

Definition 18.125 (Trivial Lindblad commutant). [Lean](#) [✓ Stmt](#) The commutant of the Lindblad family $\{L_j\}'$ equals $\mathbb{C} \cdot \mathbb{1}$, i.e. the Lindblad operators act irreducibly on $M_d(\mathbb{C})$.

Definition 18.126 (Kossakowski rank). [Lean](#) [✓ Stmt](#) The minimal number of Lindblad operators across all GKSL representations of L , which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

Theorem 18.127 (κ block vanishing from generator compression). [Lean](#) [✓ Stmt](#) Let $(H, \{L_j\})$ be a Lindblad form with $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$, and let P be an orthogonal projection such that the generator preserves the compression $PM_d(\mathbb{C})P$. If $(1 - P)L_jP = 0$ for every j , then $(1 - P)\kappa P = 0$.

Proof. [✓ Proof](#) From $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$ and $(1 - P)L(P) = 0$: the $(1 - P)\varphi(P)$ term vanishes because each $(1 - P)L_jP = 0$, and the $(1 - P)(P\kappa^\dagger)$ term vanishes because $(1 - P)P = 0$. What remains is $(1 - P)\kappa P = 0$. $\square$

Theorem 18.128 (Full algebra generation forbids block-upper-triangular form). [Lean](#) [✓ Stmt](#) Let $(H, \{L_j\})$ be a Lindblad form. If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the Lindblad data do not admit a block-upper-triangular decomposition.

Proof. [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P . By Theorem 18.127, $(1 - P)\kappa P = 0$. Induction on elements of the subalgebra shows $(1 - P)AP = 0$ for every generator. Since the generated algebra is all of $M_d(\mathbb{C})$, every matrix satisfies $(1 - P)AP = 0$, forcing $P = 0$ or $P = \mathbb{1}$ — a contradiction. $\square$

Theorem 18.129 (Hermitian span with trivial commutant forbids triangular form). [Lean](#) [✓ Stmt](#) If the Lindblad span is closed under Hermitian conjugation and its commutant is $\mathbb{C} \cdot \mathbb{1}$, then the Lindblad data do not admit a block-upper-triangular decomposition.

Proof. [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection P . The block vanishing $(1 - P)L_jP = 0$ and Hermitian closure give $PL_j(1 - P) = 0$ as well, so $[P, L_j] = 0$ for all j . But then P lies in the commutant of the Lindblad span, contradicting triviality. $\square$

Theorem 18.130 (Large Kossakowski rank forbids block-upper-triangular form). [Lean](#) [✓ Stmt](#) If the Kossakowski rank satisfies

$$\text{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

Proof. [✓ Proof](#) A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most $d^2 - d$. The formal rank minimization argument then contradicts the large-rank bound. $\square$

Theorem 18.131 (No block-upper-triangular Lindblad form implies non-reducibility). [Lean](#) [✓ Stmt](#) *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

Proof. [✓ Proof](#) A reducible semigroup would have an invariant compression, and Theorem 18.118 would then produce a block-upper-triangular Lindblad form. $\square$

Theorem 18.132 (Full algebra generation implies non-reducibility). [Lean](#) [✓ Stmt](#) *If the algebra generated by $\{L_j\}$ and κ is the entire matrix algebra $M_d(\mathbb{C})$, then the QDS is not reducible. This is [Wol12, Corollary 7.2, condition 1].*

Proof. [✓ Proof](#) By Theorem 18.128, no block-upper-triangular decomposition exists. Then apply Theorem 18.131. $\square$

Theorem 18.133 (Hermitian span and trivial commutant imply non-reducibility). [Lean](#) [✓ Stmt](#) *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Corollary 7.2, condition 2].*

Proof. [✓ Proof](#) By Theorem 18.129, no block-upper-triangular decomposition exists. Then apply Theorem 18.131. $\square$

Theorem 18.134 (Large Kossakowski rank implies non-reducibility). [Lean](#) [✓ Stmt](#) *If $\text{rank}(C) > d^2 - d$, then the QDS is not reducible. This is [Wol12, Corollary 7.2, condition 3].*

Proof. [✓ Proof](#) The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 18.130. Then apply Theorem 18.131. $\square$

Chapter 19

Operator Convexity and Jensen Inequalities

This chapter collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis ([Bha97, Chapter V]; [Wol12, Theorem 5.1]; Hansen–Pedersen’s Jensen inequality [HP82]; and Lieb’s concavity theorem [Lie73]). They are provided here as cited inputs, since the corresponding matrix-analysis proofs for real powers $x \mapsto x^p$ on the Loewner order and for the matrix logarithm are not developed elsewhere in this document.

19.1 Diagonal Jensen inequality

Lemma 19.1 (Diagonal Jensen inequality). Lean ✓ Stmt *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex on $[0, \infty)$, let A be a positive semidefinite matrix on a finite-dimensional space indexed by n , and let $v \in \mathbb{C}^n$ satisfy $\langle v, v \rangle = 1$. Then*

$$f(\Re \langle v, Av \rangle) \leq \Re \langle v, f(A)v \rangle,$$

where $f(A)$ is defined by the Hermitian continuous functional calculus.

Proof. ✓ Proof Write $A = U \operatorname{diag}(\mu) U^*$ by the spectral theorem and set $w = U^*v$. Since U is unitary, $p_i := |w_i|^2$ satisfies $\sum_i p_i = \|v\|^2 = 1$, so (p_i) is a probability distribution on n . The eigenvalues μ_i lie in $[0, \infty)$ because A is positive semidefinite. A direct computation yields $\Re \langle v, Av \rangle = \sum_i p_i \mu_i$ and $\Re \langle v, f(A)v \rangle = \sum_i p_i f(\mu_i)$, so the scalar Jensen inequality applied to the weights (p_i) and points (μ_i) gives the conclusion. See [Bha97, Chapter V]. □

19.2 Trace concavity and convexity of matrix powers

Theorem 19.2 (Trace concavity of real powers). Lean Lean ✓ Stmt *For $p \in [0, 1]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$:*

$$t \Re \operatorname{tr}(A_1^p) + (1 - t) \Re \operatorname{tr}(A_2^p) \leq \Re \operatorname{tr}((tA_1 + (1 - t)A_2)^p).$$

Follows from operator concavity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Let $A = tA_1 + (1-t)A_2$ and let $\{\psi_j\}$ with eigenvalues $\mu_j \geq 0$ be an orthonormal eigenbasis of A . A direct computation gives $\Re \operatorname{tr}(A^p) = \sum_j \mu_j^p$ and $\mu_j = t a_j + (1-t) b_j$ where $a_j = \Re \langle \psi_j, A_1 \psi_j \rangle$, $b_j = \Re \langle \psi_j, A_2 \psi_j \rangle$. Scalar concavity of $x \mapsto x^p$ on $[0, \infty)$ yields $\mu_j^p \geq t a_j^p + (1-t) b_j^p$, and the diagonal Jensen inequality (Lemma 19.1, concave form) gives $a_j^p \leq \Re \langle \psi_j, A_1^p \psi_j \rangle$ (and similarly for b_j). Summing over j and using $\sum_j \Re \langle \psi_j, M \psi_j \rangle = \Re \operatorname{tr} M$ yields the conclusion. See [Bha97, Chapter V]. □

Theorem 19.3 (Trace convexity of real powers). Lean Lean ✓ Stmt For $p \in [1, 2]$, PSD matrices A_1, A_2 , and $t \in [0, 1]$:

$$\Re \operatorname{tr}((tA_1 + (1-t)A_2)^p) \leq t \Re \operatorname{tr}(A_1^p) + (1-t) \Re \operatorname{tr}(A_2^p).$$

Follows from operator convexity of $x \mapsto x^p$ composed with trace monotonicity on the Loewner order.

Proof. ✓ Proof Identical eigenbasis reduction as in Theorem 19.2, with scalar convexity (rather than concavity) of $x \mapsto x^p$ on $[0, \infty)$ for $p \geq 1$ and the convex form of the diagonal Jensen inequality. □

19.3 Operator Jensen inequality for positive maps

Definition 19.4 (Finite-POVM dilation construction). Lean Lean ✓ Stmt For a finite family of matrices $\{C_i\}_{i \in \iota}$ and a defect block S , one forms an isometric dilation whose rows are the adjoints of the matrices C_i together with the adjoint of S . One also forms the scalar block-diagonal matrix whose ι -blocks carry prescribed weights and whose defect block carries a single scalar.

Lemma 19.5 (Compression auxiliary for Jensen). Lean ✓ Stmt Let Y be positive definite and let W be an isometry. Then

$$(W^* Y W)^{-1} \leq W^* Y^{-1} W.$$

Proof. ✓ Proof Apply the Schur-complement criterion to the block matrix $\begin{pmatrix} Y^{-1} & W \\ W^* & W^* Y W \end{pmatrix}$. The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by W gives the stated inequality. □

Lemma 19.6 (Finite-POVM compression identities). Lean Lean Lean Lean Lean Lean ✓ Stmt If the defect block satisfies

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*,$$

then the dilation is an isometry, compressing the weighted scalar block-diagonal matrix gives

$$\sum_{i \in \iota} w_i C_i C_i^* + t SS^*,$$

the defect relation rewrites as

$$\sum_{i \in \iota} C_i C_i^* + SS^* = \mathbb{1},$$

and strict positivity of all weights together with $t > 0$ implies positive definiteness of the scalar block-diagonal matrix. If all weights are nonzero and the defect scalar is nonzero, the inverse diagonal has reciprocal weights, and compressing this inverse gives

$$\sum_{i \in \iota} w_i^{-1} C_i C_i^* + t^{-1} SS^*.$$

Proof. ✓ Proof Expand the block matrix multiplication entrywise. The defect identity gives the isometry relation after summing the ι -blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positive definiteness uses strict positivity of every diagonal entry, while the inverse formula uses the nonvanishing of every diagonal entry. $\square$

Lemma 19.7 (Finite-POVM resolvent inequality). Lean ✓ Stmt Let $w_i \geq 0$ and $t > 0$. If the defect block satisfies

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*,$$

then

$$\left(\sum_{i \in \iota} w_i C_i C_i^* + t \mathbb{1} \right)^{-1} \leq \sum_{i \in \iota} (w_i + t)^{-1} C_i C_i^* + t^{-1} SS^*.$$

Proof. ✓ Proof Apply the compression inequality to the positive definite scalar block-diagonal matrix with entries $w_i + t$ and defect entry t . The compression identities rewrite the compressed matrix as $\sum_i w_i C_i C_i^* + t \mathbb{1}$ and its inverse compression as the displayed upper bound. $\square$

Theorem 19.8 (Jensen for concave real powers). Lean Not ready For a positive subunital map T , a positive semidefinite matrix A , and $p \in [0, 1]$:

$$T(A^p) \leq (TA)^p.$$

This is [Wol12, Theorem 5.1] for concave $f(x) = x^p$.

Theorem 19.9 (Map form of Jensen for concave real powers). Lean ✓ Stmt For a positive subunital map T , a positive semidefinite matrix A , and $p \in [0, 1]$, the inequality $T(A^p) \leq (TA)^p$ holds.

Proof. ✓ Proof Direct application of the axiom. $\square$

Theorem 19.10 (Jensen for convex real powers). Lean Not ready For a positive subunital map T , a positive semidefinite matrix A , and $p \in [1, 2]$:

$$(TA)^p \leq T(A^p).$$

This is [Wol12, Theorem 5.1] for convex $f(x) = x^p$.

Theorem 19.11 (Map form of Jensen for convex real powers). Lean ✓ Stmt For a positive subunital map T , a positive semidefinite matrix A , and $p \in [1, 2]$, the inequality $(TA)^p \leq T(A^p)$ holds.

Proof. ✓ Proof Direct application of the axiom. $\square$

Theorem 19.12 (Jensen for concave log). Lean Not ready For a positive unital map T and positive-definite A :

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Theorem 5.1] for $f = \log$. Requires unitality ($T(\mathbb{1}) = \mathbb{1}$), not merely subunitality.

Theorem 19.13 (Map form of Jensen for the concave logarithm). Lean ✓ Stmt For a positive unital map T and a positive-definite matrix A , the inequality $T(\log A) \leq \log(TA)$ holds.

Proof. ✓ Proof Direct application of the axiom. $\square$

19.4 Lieb concavity theorem

Theorem 19.14 (Lieb concavity). [Lean](#) [Not ready](#) For $s \in [0, 1]$, any matrix K , and positive-definite matrices A_1, A_2, B_1, B_2 , the map $(A, B) \mapsto \text{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave:

$$t \Re \text{tr}(K^\dagger A_1^s K B_1^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A_2^s K B_2^{1-s}) \leq \Re \text{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K (tB_1 + (1-t)B_2)^{1-s}).$$

See [\[Lie73, And79\]](#).

Theorem 19.15 (Map form of Lieb concavity). [Lean](#) [✓ Stmt](#) For $s \in [0, 1]$, any matrix K , and positive-definite matrices A_1, A_2, B_1, B_2 , the function $(A, B) \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$ is jointly concave, with the same inequality as Theorem 19.14.

Proof. [✓ Proof](#) Direct application of the axiom. □

Corollary 19.16 (Lieb concavity, $K = \mathbb{1}$ case). [Lean](#) [✓ Stmt](#) For $s \in [0, 1]$, $t \in [0, 1]$, and positive-definite matrices A_1, A_2, B_1, B_2 , the map $(A, B) \mapsto \Re \text{tr}(A^s B^{1-s})$ is jointly concave:

$$t \Re \text{tr}(A_1^s B_1^{1-s}) + (1-t) \Re \text{tr}(A_2^s B_2^{1-s}) \leq \Re \text{tr}((tA_1 + (1-t)A_2)^s (tB_1 + (1-t)B_2)^{1-s}).$$

Obtained from Theorem 19.15 by taking $K = \mathbb{1}$.

Proof. [✓ Proof](#) Specialize Theorem 19.15 at $K = \mathbb{1}$ and simplify $\mathbb{1}^\dagger = \mathbb{1}$ together with $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$. □

Corollary 19.17 (Concavity of Lieb's map in the first argument). [Lean](#) [✓ Stmt](#) For $s \in [0, 1]$, $t \in [0, 1]$, matrix K , and positive-definite matrices A_1, A_2, B , the map $A \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$ is concave:

$$t \Re \text{tr}(K^\dagger A_1^s K B^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A_2^s K B^{1-s}) \leq \Re \text{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K B^{1-s}).$$

Proof. [✓ Proof](#) Specialize Theorem 19.15 at $B_1 = B_2 = B$, so that the convex combination collapses: $tB + (1-t)B = B$. □

Corollary 19.18 (Concavity of Lieb's map in the second argument). [Lean](#) [✓ Stmt](#) For $s \in [0, 1]$, $t \in [0, 1]$, matrix K , and positive-definite matrices A, B_1, B_2 , the map $B \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$ is concave:

$$t \Re \text{tr}(K^\dagger A^s K B_1^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A^s K B_2^{1-s}) \leq \Re \text{tr}(K^\dagger A^s K (tB_1 + (1-t)B_2)^{1-s}).$$

Proof. [✓ Proof](#) Specialize Theorem 19.15 at $A_1 = A_2 = A$, so that the convex combination collapses: $tA + (1-t)A = A$. □

Chapter 20

Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

20.1 Von Neumann entropy

Definition 20.1 (Von Neumann entropy). [Lean](#) [✓ Stmt](#) For a Hermitian matrix $\rho \in M_D(\mathbb{C})$ with eigenvalues $\lambda_0, \dots, \lambda_{D-1}$, the *von Neumann entropy* is

$$S(\rho) = - \sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where $0 \log 0 := 0$.

Theorem 20.2 (Entropy is non-negative). [Lean](#) [✓ Stmt](#) For any density matrix ρ , $S(\rho) \geq 0$.

Proof. [✓ Proof](#) Each eigenvalue λ_i of a density matrix satisfies $0 \leq \lambda_i \leq 1$, and $-x \log x \geq 0$ on $[0, 1]$. $\square$

Lemma 20.3 (Eigenvalue sum). [Lean](#) [✓ Stmt](#) The eigenvalues of a density matrix sum to 1.

Proof. [✓ Proof](#) Follows from $\text{tr}(\rho) = \sum_i \lambda_i = 1$. $\square$

Theorem 20.4 (Eigenvalue bound). [Lean](#) [✓ Stmt](#) Each eigenvalue of a density matrix lies in $[0, 1]$.

Proof. [✓ Proof](#) Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. $\square$

Theorem 20.5 (Entropy upper bound). [Lean](#) [✓ Stmt](#) For a density matrix $\rho \in M_D(\mathbb{C})$ with $D \geq 1$,

$$S(\rho) \leq \log D.$$

Proof. [✓ Proof](#) By Jensen's inequality applied to the concave function $-x \log x$, the entropy is maximized when all eigenvalues equal $1/D$. $\square$

20.2 Tripartite partial traces

Definition 20.6 (Partial trace over A). [Lean](#) [✓ Stmt](#) For a tripartite matrix ρ_{ABC} on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$, the partial trace over A is

$$(\mathrm{tr}_A \rho_{ABC})_{(b_1, c_1)(b_2, c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a, b_1, c_1)(a, b_2, c_2)}.$$

Definition 20.7 (Partial trace over C). [Lean](#) [✓ Stmt](#) The partial trace over C :

$$(\mathrm{tr}_C \rho_{ABC})_{(a_1, b_1)(a_2, b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1, b_1, c)(a_2, b_2, c)}.$$

Definition 20.8 (Partial trace over AC). [Lean](#) [✓ Stmt](#) The partial trace over A and C :

$$(\mathrm{tr}_{AC} \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a, b_1, c)(a, b_2, c)}.$$

Lemma 20.9 (Partial trace preserves Hermiticity). [Lean](#) [✓ Stmt](#) If ρ_{ABC} is Hermitian, then $\mathrm{tr}_A(\rho_{ABC})$, $\mathrm{tr}_C(\rho_{ABC})$, and $\mathrm{tr}_{AC}(\rho_{ABC})$ are all Hermitian. The same holds for bipartite partial traces $\mathrm{tr}_A(\rho_{AB})$ and $\mathrm{tr}_B(\rho_{AB})$.

Proof. [✓ Proof](#) Follows from $\overline{\rho_{ji}} = \rho_{ij}$ applied entry-wise inside the summation defining each partial trace. $\square$

20.3 Strong subadditivity

Theorem 20.10 (Strong subadditivity). [Lean](#) [Not ready](#) For a tripartite density matrix ρ_{ABC} ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 20.11 (SSA equality). [Lean](#) [✓ Stmt](#) A tripartite density matrix ρ_{ABC} satisfies SSA equality if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

Definition 20.12 (Quantum Markov decomposition). [Lean](#) [✓ Stmt](#) A Hayashi Markov decomposition of a tripartite state ρ_{ABC} consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on B , a probability vector $(p_j)_j$, and density matrices $\rho_{AB_j^L}$ and $\rho_{B_j^R C}$ such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

Theorem 20.13 (Hayashi SSA-equality characterization). [Lean](#) [Not ready](#) For a tripartite density matrix ρ_{ABC} ,

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .

This equality criterion is cited here as an input for the later MPDO arguments. It is the equality criterion needed by Appendix C of *arXiv:1606.00608*. We cite Hayashi, J. Phys. A 37 (2004) L205–L208, and Ruskai, J. Math. Phys. 43 (2002) 4358; the structural block-decomposition formulation used for MPDOs also appears in Hayden–Jozsa–Petz–Winter, Commun. Math. Phys. 246 (2004) 359–374.

20.4 Mutual information

Definition 20.14 (Mutual information). [Lean](#) [✓ Stmt](#) The *quantum mutual information* of a bipartite state ρ_{AB} is

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

Theorem 20.15 (Mutual information is non-negative). [Lean](#) [✓ Stmt](#) For any bipartite density matrix ρ_{AB} , $I(A:B) \geq 0$.

Proof. [✓ Proof](#) Apply strong subadditivity (Theorem 20.10) with trivial B (one-dimensional middle system). The SSA inequality $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ reduces to subadditivity $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$, which gives $I(A:C) \geq 0$. $\square$

20.5 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

Definition 20.16 (Entropy formulation of von Neumann entropy). [Lean](#) [✓ Stmt](#) This formulation has the same value $S(\rho) = -\sum_i \lambda_i \log \lambda_i$ as in Definition 20.1.

Theorem 20.17 (Entropy formulation of strong subadditivity). [Lean](#) [✓ Stmt](#) For any tripartite density matrix ρ_{ABC} on $A \otimes B \otimes C$,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 20.10. The underlying axiom's proof (Lieb–Ruskai 1973) is deferred to the umbrella entropy issue.

Definition 20.18 (Entropy formulation of quantum Markov decomposition). [Lean](#) [✓ Stmt](#) This is the entropy formulation of the Hayashi Markov decomposition from Definition 20.12.

Theorem 20.19 (SSA equality iff quantum Markov decomposition). [Lean](#) [✓ Stmt](#) For any tripartite density matrix ρ_{ABC} , equality in strong subadditivity holds if and only if ρ_{ABC} admits a quantum Markov decomposition on the middle subsystem B .

This formulation introduces no new axiom: it is the same equality criterion as Theorem 20.13.

Definition 20.20 (Entropy formulation of mutual information). [Lean](#) [✓ Stmt](#) This formulation has the same value $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ as in Definition 20.14.

Theorem 20.21 (Subadditivity from trivial-middle SSA). [Lean](#) [✓ Stmt](#) For a tripartite density matrix ρ_{ABC} with $\dim B = 1$,

$$S(\rho_{ABC}) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

Proof. [✓ Proof](#) Apply Theorem 20.17 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 20.24, hence its entropy vanishes by Theorem 20.25. $\square$

20.6 Trivial-factor corollaries

The partial traces and von Neumann entropy are introduced in Chapter 20. This supplement collects elementary trace-preservation identities and the dimension-1 entropy bound that follow directly from those definitions. They are listed as separate results because each proof requires only unfolding definitions and re-indexing finite sums.

Theorem 20.22 (Partial trace over A preserves the full trace). [Lean](#) [✓ Stmt](#) *For any tripartite matrix ρ_{ABC} ,*

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_A(\rho_{ABC})).$$

Proof. [✓ Proof](#) Unfolding the definitions,

$$\mathrm{tr}(\mathrm{tr}_A(\rho_{ABC})) = \sum_{b,c} \sum_a (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

Theorem 20.23 (Partial trace over C preserves the full trace). [Lean](#) [✓ Stmt](#) *For any tripartite matrix ρ_{ABC} ,*

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_C(\rho_{ABC})).$$

Proof. [✓ Proof](#) Unfolding the definitions,

$$\mathrm{tr}(\mathrm{tr}_C(\rho_{ABC})) = \sum_{a,b} \sum_c (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

Theorem 20.24 (Partial trace over AC preserves the full trace). [Lean](#) [✓ Stmt](#) *For any tripartite matrix ρ_{ABC} ,*

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})).$$

Proof. [✓ Proof](#) Unfolding the definitions and interchanging the outer sums,

$$\mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})) = \sum_b \sum_{a,c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

Theorem 20.25 (Entropy vanishes in dimension 1). [Lean](#) [✓ Stmt](#) *If $\rho \in M_1(\mathbb{C})$ is Hermitian and $\mathrm{tr}(\rho) = 1$, then*

$$S(\rho) = 0.$$

Proof. [✓ Proof](#) The unique eigenvalue equals the trace, hence equals 1, and $-1 \cdot \log 1 = 0$:

$$\lambda_0 = \mathrm{tr}(\rho) = 1, \quad S(\rho) = -\lambda_0 \log \lambda_0 = -1 \cdot \log 1 = 0.$$

□

Chapter 21

Matrix Product Operators and Density Operators

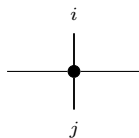
Fix a physical dimension d and a bond dimension D . An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

21.1 MPO tensors

Definition 21.1 (MPO tensor). [Lean](#) [✓ Stmt](#) An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \quad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index i and a bra index j . Diagrammatically,



The black node denotes the tensor M , the horizontal legs are virtual, the upper leg is the ket index i , and the lower leg is the bra index j .

Definition 21.2 (Doubled-index MPS view). [Lean](#) [✓ Stmt](#) By combining the pair (i, j) into a single physical index in $\{0, \dots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension d^2 and bond dimension D :

$$M_{(i,j)}^{\text{MPS}} = M^{ij},$$

identifying $\{0, \dots, d-1\} \times \{0, \dots, d-1\}$ with $\{0, \dots, d^2-1\}$ via the product encoding.

Definition 21.3 (Word evaluation). [Lean](#) [✓ Stmt](#) For words $u = (i_1, \dots, i_N)$ and $v = (j_1, \dots, j_N)$ in $\{0, \dots, d-1\}^N$, the *word evaluation* is

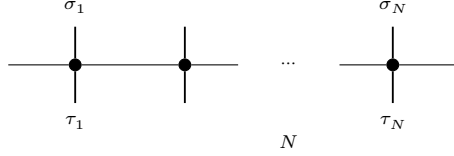
$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to $\mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

Definition 21.4 (MPO operator family). [Lean](#) [✓ Stmt](#) The operator generated by M on N sites is the matrix $\rho^{(N)}(M)$ indexed by configurations $\sigma, \tau \in \{0, \dots, d-1\}^N$ and given by

$$\rho^{(N)}(M)_{\sigma, \tau} = \text{tr}(M^{\sigma, \tau}).$$

Diagrammatically,



each black node denotes the same local tensor M , and the matrix entry $\rho^{(N)}(M)_{\sigma, \tau}$ is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

Definition 21.5 (Hermitian MPO tensor). [Lean](#) [✓ Stmt](#) An MPO tensor is *Hermitian* if, for all $i, j \in \{0, \dots, d-1\}$,

$$M^{ij} = (M^{ji})^\dagger$$

21.2 Transfer maps

Definition 21.6 (MPO transfer map). [Lean](#) [✓ Stmt](#) The *transfer map* associated to M is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

Lemma 21.7 (Identification with the doubled-index MPS transfer map). [Lean](#) [✓ Stmt](#) The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view.

Proof. [✓ Proof](#) After identifying the single physical index of the doubled tensor with a pair (i, j) , the single sum defining the MPS transfer map is exactly the double sum defining $\mathcal{E}_M$. $\square$

Theorem 21.8 (Positivity of the MPO transfer map). [Lean](#) [✓ Stmt](#) If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.

Proof. [✓ Proof](#) By Lemma 21.7, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. $\square$

21.3 Zero correlation length

Definition 21.9 (Zero correlation length for MPO tensors). [Lean](#) [✓ Stmt](#) An MPO tensor M has *zero correlation length* when its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

This is the mixed-state analogue of the renormalization fixed-point condition for MPS tensors; see [CPGSV21, Definition 4.2].

Theorem 21.10 (MPO ZCL iff doubled-index MPS RFP). [Lean](#) [✓ Stmt](#) An MPO tensor has zero correlation length if and only if its doubled-index MPS view is a renormalization fixed point.

Proof. [✓ Proof](#) By Lemma 21.7, the transfer map of M agrees with the transfer map of its doubled-index MPS view. Thus $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ holds exactly when the transfer map of that doubled-index MPS tensor is idempotent, which is the RFP condition. $\square$

21.4 MPDOs and LPDOs

Definition 21.11 (MPDO). [Lean](#) [✓ Stmt](#) An MPO tensor is an *MPDO* if, for every $N \in \mathbb{N}$,

$$\rho^{(N)}(M) \geq 0$$

This is the global positivity condition of [\[CPGSV17a, Section 4.1\]](#).

Definition 21.12 (LPDO). [Lean](#) [✓ Stmt](#) An MPO tensor is an *LPDO* if there exist a Kraus dimension d_K , an inner bond dimension D' , a bond-space identification $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ (equivalently, D and $(D')^2$ have the same cardinality), and matrices $A^{(i,k)} \in M_{D'}(\mathbb{C})$ such that, for all physical indices i, j ,

$$M^{ij} = \left(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}$$

where $\otimes$ denotes the Kronecker product, $\overline{(\cdot)}$ denotes entrywise complex conjugation, and $(\cdot)_{e,e}$ denotes reindexing along e back to $\{0, \dots, D-1\}$. See [\[CPGSV17a, Section 4.3\]](#).

Theorem 21.13 (LPDOs are MPDOs). [Lean](#) [✓ Stmt](#) Every LPDO tensor is an MPDO.

Proof. [✓ Proof](#) By the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, the N -site product decomposes as a sum of Kronecker products after transporting along the bond-space identification e :

$$M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N} = \left(\sum_{\kappa} P_{\kappa} \otimes \overline{Q_{\kappa}} \right)_e,$$

where $P_{\kappa} = A^{(\sigma_1, \kappa_1)} \dots A^{(\sigma_N, \kappa_N)}$ and $Q_{\kappa} = A^{(\tau_1, \kappa_1)} \dots A^{(\tau_N, \kappa_N)}$. Since the trace is invariant under the reindexing by e , we apply $\text{tr}(X \otimes Y) = \text{tr}(X) \text{tr}(Y)$ to obtain

$$\rho^{(N)}(M)_{\sigma, \tau} = \sum_{\kappa} \text{tr}(P_{\kappa}) \overline{\text{tr}(Q_{\kappa})} = \sum_{\kappa} \psi_{\kappa}(\sigma) \overline{\psi_{\kappa}(\tau)},$$

where $\psi_{\kappa}(\sigma) = \text{tr}(P_{\kappa})$. Hence $\rho^{(N)}(M) = \sum_{\kappa} |\psi_{\kappa}\rangle \langle \psi_{\kappa}| \geq 0$. □

21.5 Purification RFP

Definition 21.14 (Purifying MPS tensor). [Lean](#) [✓ Stmt](#) Given a purifying family $A^{(i,k)} \in M_{D'}(\mathbb{C})$ for $i \in \{0, \dots, d-1\}$ and $k \in \{0, \dots, d_K-1\}$, the *purifying MPS tensor* is

$$A_{(i,k)}^{\text{pur}} = A^{(i,k)},$$

an MPS tensor with physical dimension $d \cdot d_K$ and bond dimension D' .

Definition 21.15 (Purification RFP). [Lean](#) [✓ Stmt](#) An MPO tensor M has a *purification RFP* if it admits an LPDO witness whose purifying MPS tensor is a pure-state renormalization fixed point. This is [\[CPGSV21, Definition 4.3\]](#).

Remark 21.16 (PRFP does not imply MPO ZCL). [Lean](#) [✓ Stmt](#) Purification RFP is strictly weaker than MPO zero correlation length: there exists an MPO tensor with purification RFP whose transfer map is not idempotent. Equivalently, purification RFP does not imply MPO zero correlation length.

Theorem 21.17 (Purification RFP implies LPDO). [Lean](#) [✓ Stmt](#) A purification-RFP tensor is, in particular, an LPDO.

Proof. [✓ Proof](#) The PRFP condition provides an LPDO witness by construction. □

21.6 Pure-state recovery inside the MPO formalism

Definition 21.18 (MPDO renormalization fixed point). [Lean](#) [✓ Stmt](#) An MPO tensor M is an *MPDO renormalization fixed point* if its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

Theorem 21.19 (MPDO RFP iff MPO ZCL). [Lean](#) [✓ Stmt](#) *MPDO renormalization fixed points are exactly MPO tensors with zero correlation length.*

Proof. [✓ Proof](#) Both conditions assert $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$; they are definitionally identical. $\square$

Theorem 21.20 (MPDO RFP iff doubled-index MPS RFP). [Lean](#) [✓ Stmt](#) *The MPDO RFP condition is equivalent to the pure-state RFP condition for the doubled-index MPS tensor.*

Proof. [✓ Proof](#) Unfold the MPDO RFP condition to its definitional equivalent, MPO ZCL, then apply Theorem 21.10. $\square$

Definition 21.21 (Diagonal pure-state embedding as an MPO). [Lean](#) [✓ Stmt](#) An MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ determines an MPO tensor by placing A^i on the diagonal in the bra and ket indices:

$$M^{ij} = \delta_{ij} A^i.$$

Equivalently, $M^{ii} = A^i$ and $M^{ij} = 0$ for $i \neq j$.

Theorem 21.22 (Transfer map of the diagonal pure-state embedding). [Lean](#) [✓ Stmt](#) *The transfer map of the diagonal MPO associated to A agrees with the original MPS transfer map:*

$$\mathcal{E}_{A^{\text{MPO}}} = \mathcal{E}_A.$$

Proof. [✓ Proof](#) In the double sum defining $\mathcal{E}_{A^{\text{MPO}}}$, all off-diagonal terms vanish because $M^{ij} = 0$ for $i \neq j$, while the diagonal terms are exactly $A^i X(A^i)^\dagger$. Thus only the sum over i remains, which is the defining formula for $\mathcal{E}_A$. $\square$

Theorem 21.23 (Pure-state recovery of the RFP condition). [Lean](#) [✓ Stmt](#) *For an MPS tensor A , its diagonal MPO embedding is MPDO-RFP if and only if A is an RFP:*

$$A^{\text{MPO}} \text{ is MPDO-RFP} \iff A \text{ is RFP}.$$

Proof. [✓ Proof](#) Both conditions assert idempotence of the corresponding transfer map. Theorem 21.22 identifies those transfer maps, so the two predicates are equivalent. $\square$

Theorem 21.24 (Pure-state recovery of zero correlation length). [Lean](#) [✓ Stmt](#) *For an MPS tensor A , the diagonal MPO embedding is MPDO-RFP if and only if A has zero correlation length.*

Proof. [✓ Proof](#) Combine Theorem 21.23 with the characterization of zero correlation length by idempotence of the MPS transfer map (Theorem 23.28). $\square$

21.7 Simple local structure from SAL and ZCL

For the simple MPDO case in Appendix C.2 of [CPGSV17a], the strong-area-law hypothesis is used locally through the equality case of strong subadditivity for a normalized three-site reduced state.

Definition 21.25 (Local η -structure for a three-site reduced state). [Lean](#) [✓ Stmt](#) For a three-site density operator ρ_{ABC} , the local η -structure is the quantum Markov decomposition on the middle subsystem B produced by equality in strong subadditivity. Concretely, after a unitary change of basis on B , one has a finite direct-sum decomposition

$$B = \bigoplus_k (b_1^{(k)} \otimes b_2^{(k)})$$

such that

$$\rho_{ABC} = \bigoplus_k \rho_{Ab_1}^{(k)} \otimes \rho_{b_2C}^{(k)}.$$

Theorem 21.26 (SAL implies local η -structure). [Lean](#) [✓ Stmt](#) *If a normalized three-site reduced state satisfies the equality case of strong subadditivity — the local entropy form of the simple-MPDO strong area law — then it admits the local η -structure of Definition 21.25.*

Proof. [✓ Proof](#) This is exactly the forward implication of the Hayashi equality characterization: equality in strong subadditivity yields a quantum Markov decomposition on the middle subsystem. $\square$

Definition 21.27 (Injective inverse-map layer for a simple MPO tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A simple MPO tensor K is called injective if its doubled-index MPS tensor is injective. For such a tensor, the chosen decomposition map of the doubled-index MPS furnishes a concrete inverse tensor K^{-1} together with right- and left-physical realization maps that turn virtual bond insertions back into local physical operators.

Theorem 21.28 (Inverse-map identities for an injective simple MPO tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For an injective simple MPO tensor, contracting K^{-1} with K recovers the matrix units on the virtual bond space. Likewise, every right virtual insertion admits a physical realization, this realization is multiplicative, and the analogous left-insertion statement also holds.*

Definition 21.29 (Explicit neighboring η -operator data). [Lean](#) [Lean](#) [✓ Stmt](#) Fix a Hayashi decomposition witness h_η . An explicit neighboring η -family is a dependent family $(\eta_{k,h})$ indexed by sector pairs, where $\eta_{k,h}$ acts on the neighboring bond space $B_k^R \otimes B_h^L$. Requiring positivity for each pair gives the corresponding positive η -data.

Definition 21.30 (Trace matrices of explicit η -data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Every explicit neighboring η -family determines a sector-by-sector trace matrix. We give both its complex-valued form and the real-part version, which is the direct data for the real Perron–Frobenius matrix T used in the rank-one step.

Theorem 21.31 (Positivity of bipartite partial traces). [Lean](#) [Lean](#) [✓ Stmt](#) *If ρ is positive semidefinite on a bipartite space $A \otimes B$, then both reduced operators $\text{tr}_A \rho$ and $\text{tr}_B \rho$ are positive semidefinite.*

Proof. [✓ Proof](#) Write each partial trace as the finite sum, over the traced index, of a principal submatrix of ρ . Principal submatrices of a positive semidefinite matrix are positive semidefinite, and finite sums of positive semidefinite matrices remain positive semidefinite. $\square$

Definition 21.32 (Hayashi–Markov extraction of explicit η -operators). [Lean](#) [✓ Stmt](#) Every Hayashi decomposition witness h_η canonically produces an explicit neighboring η -family by Kronecker-multiplying the sector-reduced states on the neighboring bond spaces:

$$\eta_{k,h} := \text{tr}_C(\rho_{b_2C}^{(k)}) \otimes \text{tr}_A(\rho_{Ab_1}^{(h)}).$$

Positive semidefiniteness of each $\eta_{k,h}$ follows from the fact that partial traces preserve positivity and that positive semidefiniteness is closed under Kronecker products. This is the sector-reduced extraction layer; the remaining connection to Appendix C.2 is the identification of this family with the inverse-map construction in the original simple-MPDO tensor coordinates.

Theorem 21.33 (Trace matrix of the Hayashi–Markov η -operators). [Lean](#) [Lean](#) [✓ Stmt](#) *The trace matrix induced by the extracted family is entrywise $T_{k,h} = 1$ in both its complex-valued form and its real-valued form, matching the normalization $T_{k,h} = a_k b_h$ used in $a_k = b_h = 1$.*

Proof. [✓ Proof](#) Each $\rho_{b_2C}^{(k)}$ is a density matrix, hence has unit trace; the partial trace tr_C preserves the trace. The analogous statement holds for $\text{tr}_A \rho_{Ab_1}^{(h)}$. Multiplicativity of the trace under Kronecker products gives $T_{k,h} = 1 \cdot 1 = 1$, and taking real parts gives the real-valued statement. $\square$

Theorem 21.34 (Entrywise nonnegativity of the real trace matrix). [Lean](#) [✓ Stmt](#) *Positivity of the neighboring operators gives entrywise nonnegativity of the real trace matrix. Matrix-level positive semidefiniteness of T is a separate structural condition.*

Proof. [✓ Proof](#) The trace of a positive semidefinite neighboring operator is nonnegative in the complex order. Taking real parts gives the stated entrywise nonnegativity. $\square$

Definition 21.35 (Auxiliary matrix predicates for the rank-one step). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The auxiliary predicates state three matrix-theoretic ingredients used in the rank-one step of Appendix C.2, Lemma C.4: constant traces of all positive powers, existence of a rank-one factorization $T = ab^\top$, and the conditional Perron–Frobenius implication from primitivity together with constant trace powers to such a factorization.

This conditional implication is not a globally provable theorem. As a universal statement over primitive nonnegative matrices it is false; a concrete 3×3 counterexample appears in the archive. The corrected criterion proved here adds positive semidefiniteness and trace normalization, which provide the diagonalizability needed to turn the trace-power condition into a rank-one factorization.

Theorem 21.36 (Trace square of a Hermitian matrix). [Lean](#) [✓ Stmt](#) *Let T be a finite Hermitian matrix over a real or complex scalar field, with Hermitian eigenvalues λ_i . Then*

$$\text{tr}(T^2) = \sum_i \lambda_i^2.$$

Proof. [✓ Proof](#) The spectral theorem writes T as a unitary conjugate of the diagonal matrix of its Hermitian eigenvalues. Squaring commutes with this conjugation, and cyclicity of trace reduces the identity to the trace of the squared diagonal matrix. $\square$

Theorem 21.37 (PSD trace-power rank-one criterion). [Lean](#) [Lean](#) [✓ Stmt](#) *Let T be a finite real square matrix. If T is positive semidefinite, normalized by $\text{tr}(T) = 1$, and has constant traces on all positive powers, then T has a rank-one factorization. Consequently, under the same positive-semidefinite and trace-normalization hypotheses, the conditional rank-one criterion used in Lemma C.4 follows.*

span the full product algebra of block matrices. The corresponding finite-length separation property asserts that there exists a length L such that if, for every word w of length L ,

$$\sum_k \text{tr}(\Delta_k A_k^w) = 0$$

then these pairings vanish on a spanning set for the product algebra, so the induced linear functional on the product algebra is zero; by nondegeneracy of the product trace pairing, every Δ_k vanishes. In other words, a single finite blocking length already separates the blocks by their word evaluations. Combined with injectivity, left-canonicity, and nonzero block weights, this finite-length separation yields the horizontal canonical-form data of the block-diagonal tensor.

The entrywise scalar family of block-matrix coefficients encodes the same finite-length data. Linear independence of that scalar family implies the product-algebra spanning statement, hence a block-injective canonical-form witness. The same criterion therefore gives horizontal canonical-form data directly. The duplicate-scalar counterexample shows that this finite-length witness is strictly stronger than blockwise injectivity, left-canonicity, and nonzero weights alone: two identical scalar blocks satisfy those hypotheses, but the block-injective canonical-form separation condition fails and no blocking length makes the entrywise family linearly independent.

[CPGSV21, Proposition IV.3] gives a concrete sufficient criterion for the spanning condition: after a common blocking length each block is block-injective, and after a further finite blocking length one can form linear combinations of word evaluations which equal the identity on one chosen block and vanish on all others. Concatenating an injective prefix with this finite block-separation data produces the product-algebra spanning condition, hence the horizontal canonical-form data.

Theorem 21.43 (Finite block-separation criterion). Lean ✓ Stmt *If a block family admits the finite-length data described in [CPGSV21, Proposition IV.3], then it is in block-injective canonical form.*

Proof. ✓ Proof This is the direct implication from the finite-length data to the finite-length trace-separation statement. What remains open in the full [CPGSV21, Proposition IV.3] is the derivation of this finite witness from separated canonical-form / BNT data. □

Lemma 21.44 (Blockwise equality from agreement on the MPV family). Lean ✓ Stmt *Let $\bigoplus_k \mu_k A_k$ be a block-diagonal tensor whose blocks are injective, left-canonical, weighted by nonzero scalars, and jointly block-injective in the sense that for some blocking length L the tuple of trace pairings against length- L block words is faithful across all blocks simultaneously. If two physical operators induce the same first-site action on every MPV generated by this tensor, then their first-site insertions agree on each block A_k .*

Proof. ✓ Proof This is Lemma L from [CPGSV17a, Appendix A], using the finite criterion from Theorem 21.43. Expanding the hypothesis over a word of length $L + 1$ beginning with the fixed site and folding each block contribution into a trace pairing against the corresponding first-site insertion tensor yields, upon taking the Y - Z difference, a tuple of block matrices whose trace pairings against every length- L block word vanish. The block-injective canonical-form hypothesis then forces each block difference to be zero, and dividing by the nonzero weight μ_k^{L+1} concludes the blockwise equality of the two first-site insertions. □

Theorem 21.45 (Horizontal canonical form implies vertical canonical form). Not ready *Every MPDO in horizontal canonical form is also in vertical canonical form.*

Proof. Starting from the horizontal canonical-form decomposition, one uses the MPDO positivity condition to compare the first-site action of the doubled-index tensor on the diagonal sector. Lemma 21.44 then shows that the diagonal tensor splits along the same normal blocks, with the coefficients rearranged into positive scalar weights. Flattening those positive diagonal coefficients produces the required vertical canonical-form decomposition. $\square$

21.9 Fusion isometries

Following [CPGSV17a, Section 4.5], the next definitions describe the transfer-map content of the fusion-isometry picture. For each blocked size $n \geq 1$, the doubled-index blocked tensor of an MPO has a blocked transfer map E_n acting on bond-space matrices, and the support algebra is modeled by a subspace through which E_n factors as a retract.

Definition 21.46 (Transfer-map fusion-isometry datum). Lean ✓ Stmt A *transfer-map fusion-isometry datum* at blocked size n for an MPO tensor consists of a subspace $\mathcal{A}_n \subseteq M_D(\mathbb{C})$ together with linear maps

$$T_n : M_D(\mathbb{C}) \rightarrow \mathcal{A}_n, \quad S_n : \mathcal{A}_n \rightarrow M_D(\mathbb{C})$$

such that

$$T_n \circ S_n = \text{id}_{\mathcal{A}_n}, \quad S_n \circ T_n = E_n,$$

where E_n denotes the blocked transfer map.

Definition 21.47 (Fusion-isometry RFP predicate). Lean ✓ Stmt An MPO tensor M satisfies the *fusion-isometry formulation* of the renormalization fixed-point condition when for every blocked size $n \geq 1$ there exists transfer-map fusion-isometry data for the blocked transfer map E_n .

Theorem 21.48 (Fusion-isometry data imply the MPDO RFP condition). Lean ✓ Stmt *The fusion-isometry formulation implies the MPDO RFP condition.*

Proof. ✓ Proof Apply the definition at blocked size $n = 1$. If $E_1 = S_1 \circ T_1$ and $T_1 \circ S_1 = \text{id}$, then

$$E_1^2 = S_1 T_1 S_1 T_1 = S_1 (T_1 S_1) T_1 = S_1 T_1 = E_1.$$

Since E_1 is the original transfer map, this is exactly the MPDO RFP condition. $\square$

Theorem 21.49 (Transfer-map fusion criterion for MPDO RFP). Lean ✓ Stmt *The fusion-isometry formulation is equivalent to the MPDO RFP condition.*

Proof. ✓ Proof The forward implication is Theorem 21.48. Conversely, if the transfer map E is idempotent, then every positive blocked transfer map equals E and therefore factors through its range. This range-factorization gives the required fusion-isometry data. $\square$

Corollary 21.50 (Pure-state recovery). Lean ✓ Stmt *For the diagonal MPO associated to a pure MPS tensor, the fusion-isometry formulation reduces to the usual pure-state RFP condition.*

Proof. ✓ Proof Combine Theorem 21.49 with the diagonal pure-state recovery theorem for the MPO predicate. $\square$

21.10 Algebra structure

Definition 21.51 (Algebra-structure data for an MPDO). [Lean](#) [✓ Stmt](#) A family of support $*$ -algebras $\{\mathcal{A}_n\}_{n \geq 0}$, one at every blocking size, equipped with blocking maps $m_n : \mathcal{A}_n \otimes \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$ and inclusion maps $\iota_n : \mathcal{A}_n \rightarrow \mathcal{A}_{n+1}$ such that, after viewing all elements inside the ambient bond-space matrix algebra, m_n is ordinary matrix multiplication and ι_n is the ambient inclusion.

Definition 21.52 (Algebra-structure RFP predicate). [Lean](#) [✓ Stmt](#) An MPO tensor M satisfies the *algebra-structure formulation* used here when there exist algebra-structure data whose support algebra at every positive blocking size coincides with the fixed-point algebra of the adjoint blocked transfer map. This is a nontrivial algebraic condition, but it is still weaker than the full coefficient formulation of [CPGSV17a, Theorem IV.13 (ii)]; that formulation also includes the coefficient family $c_{\alpha,\beta,\gamma}^{(L)}$.

Theorem 21.53 (RFP yields a stationary algebra tower under a faithful fixed point). [Lean](#) [✓ Stmt](#) Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer map is idempotent, then there exists a stationary family of support algebras, all equal to the fixed-point algebra of the adjoint transfer map, with multiplication given by matrix multiplication and inclusions given by the identity.

Proof. [✓ Proof](#) Under the trace-preserving normalization and the faithful fixed point, Wolf's fixed-point-algebra theorem gives a $*$ -subalgebra of adjoint fixed points. Idempotence identifies every positive blocked transfer map with the original one, so the same fixed-point algebra works at every blocking size. $\square$

Theorem 21.54 (Fusion yields a stationary algebra tower under a faithful fixed point). [Lean](#) [✓ Stmt](#) Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer-map fusion criterion holds, then the MPO satisfies the algebra-structure formulation of the renormalization fixed-point condition.

Proof. [✓ Proof](#) Combine Theorem 21.48 with Theorem 21.53. $\square$

Theorem 21.55 (Stabilized blocked adjoint fixed points yield algebra data). [Lean](#) [✓ Stmt](#) Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If, for every positive blocked size n , the adjoint fixed-point space of the blocked transfer map agrees with the adjoint fixed-point space of the original transfer map, then the algebra-structure formulation holds.

Proof. [✓ Proof](#) Use the adjoint fixed-point algebra of the original transfer map as a constant support-algebra tower. The stabilization hypothesis identifies that same algebra with the adjoint fixed-point space of every positive blocked transfer map, so the stationary family satisfies the definition. $\square$

This does not give the full converse direction of [CPGSV17a, Theorem IV.13]: the compatibility condition above only sees stabilized adjoint fixed-point algebras, not the coefficient/BNT data. In particular, Theorem IV.13(ii) also requires the special diagonal matrices $\chi_{\alpha,\beta,\gamma}$. The coefficient data attached to the algebra tower are: a chosen basis of each support algebra, coordinate maps into a finite scalar family, and the corresponding multiplication / inclusion coefficient arrays.

Definition 21.56 (Blocked coefficients for the algebra tower). [Lean](#) [✓ Stmt](#) For each blocked size n , once a basis of $\mathcal{A}_n$ is chosen, every element of $\mathcal{A}_n$ is identified with a finite coefficient family on that basis.

Theorem 21.57 (Reconstruction from blocked coefficients). [Lean](#) [✓ Stmt](#) *Reconstructing a coefficient family gives the corresponding basis expansion $\sum_i a_i b_i$ in $\mathcal{A}_n$.*

Proof. [✓ Proof](#) This is exactly the finite-basis reconstruction formula for the chosen basis of $\mathcal{A}_n$. $\square$

Definition 21.58 (Blocked multiplication coefficients). [Lean](#) [✓ Stmt](#) The blocking multiplication map m_n determines a coefficient family for the product of any two chosen basis elements of $\mathcal{A}_n$, now viewed in the basis of $\mathcal{A}_{2n}$.

Theorem 21.59 (Blocked multiplication reconstructs from its coefficients). [Lean](#) [✓ Stmt](#) *In the ambient bond-space matrix algebra, the product of two chosen basis elements of $\mathcal{A}_n$ is the sum of the basis elements of $\mathcal{A}_{2n}$ weighted by these extracted coefficients.*

Proof. [✓ Proof](#) Apply the reconstruction formula of Theorem 21.57 to the element $m_n(b_i, b_j) \in \mathcal{A}_{2n}$. $\square$

Theorem 21.60 (Compatible adjoint fixed points reconstruct from blocked coefficients). [Lean](#) [✓ Stmt](#) *If the algebra tower is compatible with an MPO tensor M , then every adjoint blocked fixed point of the blocked transfer map E_n belongs to $\mathcal{A}_n$ and is recovered exactly from its extracted coefficient family.*

Proof. [✓ Proof](#) Compatibility identifies $\mathcal{A}_n$ with the adjoint fixed-point algebra of E_n , so the fixed point first becomes an element of $\mathcal{A}_n$ and then Theorem 21.57 reconstructs it. $\square$

Theorem 21.61 (Adjoint fixed points descend through the algebra tower). [Lean](#) [✓ Stmt](#) *If an MPO tensor satisfies the algebra-structure formulation, then every adjoint fixed point of the blocked transfer map E_n at a positive blocking size n is already an adjoint fixed point of the unblocked transfer map E_1 .*

Proof. [✓ Proof](#) Compatibility at size n places X in $\mathcal{A}_n$. The inclusion map ι_n lifts X into $\mathcal{A}_{n+1}$, and since ι_n is just an inclusion it does not change the underlying operator, so we continue to denote the lifted element by X . Thus compatibility at size $n + 1$ yields $E_{n+1}^\dagger X = X$. Rewriting $E_{n+1} = E_n \circ E_1$ and applying $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$ together with the hypothesis $E_n^\dagger X = X$ extracts $E_1^\dagger X = X$. $\square$

Lemma 21.62 (Reverse inclusion of adjoint fixed points). [Lean](#) [✓ Stmt](#) *Every adjoint fixed point of the unblocked transfer map E_1 is an adjoint fixed point of every blocked transfer map E_n .*

Proof. [✓ Proof](#) Induction on n using $E_{n+1} = E_n \circ E_1$ and $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$; the base case is $E_0 = \text{id}$. This step does not require the algebra-structure data. Combined with Theorem 21.61 this yields the fixed-point equality $\text{Fix}(E_n^\dagger) = \text{Fix}(E_1^\dagger)$ at every positive blocking size under the algebra-structure predicate. $\square$

Remark 21.63 (Why algebra data do not imply fusion data). The converse implication 21.52 $\Rightarrow$ 21.47 requires a stronger hypothesis than the algebra-structure formulation above. Theorem 21.61 already gives the strongest available consequence of the algebra-structure data – exclusion of *finite order* (root-of-unity) peripheral eigenvalues of the adjoint transfer map, since any such eigenvalue would produce a new adjoint fixed point at some positive blocking size. This does *not*, however, exclude unit-modulus eigenvalues of irrational phase, nor force idempotence of the transfer map, so ergodic quantum channels with a strict spectral gap on the 1-complement still satisfy the predicate without being RFPs. For instance, on $M_D(\mathbb{C})$ with $0 < \varepsilon < 1$ take

$$E(X) = \varepsilon X + (1 - \varepsilon) \Pi_{\text{diag}}(X),$$

where Π_{diag} is the projection that zeroes the off-diagonal entries of X ; then $\Pi_{\text{diag}} \circ \Pi_{\text{diag}} = \Pi_{\text{diag}}$, the channel E has peripheral spectrum $\{1\}$ and diagonal fixed-point algebra at every blocking size, yet $E \circ E \neq E$. Closing the converse therefore requires strengthening the predicate to the coefficient formulation of [CPGSV17a, Theorem IV.13 (ii)]: the positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ of Appendix C.3 and the coefficient identity $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ of Appendix C.4. The scalar Newton–Girard power-sum identity needed for the latter step is already available as Theorem 11.25.

21.10.1 Diagonal χ -matrices and the trace-power formula

The following statements isolate the form of [CPGSV17a, Theorem IV.13 (ii)]: the structure coefficients of the length- L operator algebra are trace-powers of positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$. They state the coefficient identity and the elementary trace-power identity for diagonal matrices.

Definition 21.64 (Diagonal χ family). Lean ✓ Stmt A family of diagonal complex matrices $\chi_{\alpha,\beta,\gamma}$ indexed by ordered triples drawn from a common index type, each of a specified size. This is the matrix $\chi_{\alpha,\beta,\gamma}$ of [CPGSV17a, Theorem IV.13(ii)].

Theorem 21.65 (Trace of χ^L equals the sum of L -th powers of entries). Lean ✓ Stmt For every triple (α, β, γ) , if $\chi_{\alpha,\beta,\gamma}$ has size $r_{\alpha,\beta,\gamma}$, then for every $L \geq 0$,

$$\text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi_{\alpha,\beta,\gamma,k}^L,$$

where $\chi_{\alpha,\beta,\gamma,k}$ are the diagonal entries.

Proof. ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. □

Definition 21.66 (Trace-power form of a structure-coefficient family). Lean ✓ Stmt A binary compatibility predicate between an abstract structure-coefficient family $c_{\alpha,\beta,\gamma}^{(L)}$ and a diagonal χ family: the pair (c, χ) is said to be in *trace-power form* when

$$c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi_{\alpha,\beta,\gamma,k}^L$$

for every $L \geq 0$ and every triple (α, β, γ) , where $r_{\alpha,\beta,\gamma}$ is the size of $\chi_{\alpha,\beta,\gamma}$. This is the formal analogue of the target identity of [CPGSV17a, Theorem IV.13 (ii)]; the existential version—“there exists χ for which (c, χ) has trace-power form”—is obtained by quantifying over χ .

21.11 Commuting form and GSNNCH

Definition 21.67 (Commuting-form data for an MPDO). Lean ✓ Stmt For a fixed chain length $N \geq 2$, this consists of a positive semidefinite two-site operator B such that its translated copies on the periodic chain commute pairwise.

Definition 21.68 (Commuting-form property). Lean ✓ Stmt An MPO tensor has the commuting-form property if for every $N \geq 2$ its N -site operator is a positive scalar multiple of the product of the translated copies of some commuting two-site operator B .

Definition 21.69 (GSNNCH). [Lean](#) [✓ Stmt](#) An MPO tensor is a Gibbs state of a nearest-neighbor commuting Hamiltonian if every finite-chain operator admits the equivalent positive-operator presentation $\rho^{(N)} = c \prod_i B_{i,i+1}$ with $c > 0$ and commuting nearest-neighbor factors. The projector-limit convention in [\[CPGSV17a\]](#) identifies this with the exponential Gibbs form.

Theorem 21.70 (GSNNCH iff commuting form). [Lean](#) [✓ Stmt](#) For an MPO tensor, the GSNNCH condition is equivalent to the existence of commuting-form data on every finite chain.

Proof. [✓ Proof](#) A positive normalization constant converts commuting-form data into a GSNNCH witness, and every GSNNCH witness remembers its underlying commuting-form data. $\square$

Definition 21.71 (GSNNCH with zero correlation length). [Lean](#) [✓ Stmt](#) This is the conjunction of the GSNNCH condition with MPO zero correlation length, matching item (iii) of the simple-MPDO equivalence in [\[CPGSV17a\]](#).

Theorem 21.72 (GSNNCH–ZCL iff commuting form with ZCL). [Lean](#) [✓ Stmt](#) The GSNNCH–ZCL branch is equivalent to the conjunction of the commuting-form property with MPO zero correlation length.

Proof. [✓ Proof](#) Unfold the definition of GSNNCH with zero correlation length and apply Theorem 21.70. $\square$

Definition 21.73 (η -local commuting-form data). [Lean](#) [✓ Stmt](#) This defines the post-extraction form of Appendix C.2: a single translation-invariant positive two-site bond whose translated copies commute on every periodic chain and realize the MPO at every finite length.

Theorem 21.74 (η -local structure implies commuting form). [Lean](#) [✓ Stmt](#) Once the explicit neighboring operators $\eta_{k,h}$ have been assembled into the data above, the global commuting-form property follows.

Proof. [✓ Proof](#) For each $N \geq 2$, take the commuting-form datum given by these data. $\square$

Remark 21.75 (Entropy-side construction). The entropy-side theorem still to be supplied constructs the local data needed for Theorem 21.74 from the SAL and ZCL hypotheses. In particular, one must extract explicit neighboring operators $\eta_{k,h}$ from the Hayashi Markov decomposition. Once these operators are available, Theorem 21.74 yields the commuting-form conclusion and hence the GSNNCH branch of the simple-MPDO equivalence.

21.12 Blocked-RFP construction for simple MPDOs

Definition 21.76 (Local simple-MPDO structure data). [Lean](#) [✓ Stmt](#) This structure collects the local Appendix C.2 data: a normalized three-site reduced state with equality in strong sub-additivity, the resulting local η -structure, and the rank-one conclusion for the auxiliary primitive matrix T .

Definition 21.77 (Simple-MPDO blocked-RFP data). [Lean](#) [✓ Stmt](#) For an MPO tensor K , this consists of the local Appendix C.2 data together with a compatibility relation attaching those local data to K , the commuting-form property, and zero correlation length. The local component consists of the hypotheses used in the entropy-side route to commuting form, while the blocked-RFP consequences below use only commuting form and zero correlation length.

Theorem 21.78 (Zero correlation length gives a blocked fusion-isometry witness). Lean ✓ Stmt
If an MPO tensor K has zero correlation length, then it has a blocked fusion-isometry witness at size 2 in the sense of Definition 21.46.

Proof. ✓ Proof Zero correlation length is the MPDO RFP condition. Applying the converse direction of Theorem 21.49 to that RFP witness yields fusion-isometry data at every positive blocked size; evaluating it at size 2 produces the blocked witness. $\square$

Theorem 21.79 (Simple-MPDO RFP chain from commuting form and ZCL). Lean ✓ Stmt
From the commuting-form property and zero correlation length one simultaneously recovers the GSNNCH-with-ZCL criterion, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of MPDO RFP.

Proof. ✓ Proof The commuting-form and zero-correlation-length hypotheses give the GSNNCH–ZCL branch via Theorem 21.72. The blocked witness is Theorem 21.78. The full transfer-map fusion formulation follows by applying the converse direction of Theorem 21.49 to zero correlation length, which is the MPDO RFP condition. $\square$

Remark 21.80 (SAL–ZCL implication for the GSNNCH–ZCL criterion). Not ready The implication from SAL and ZCL alone to the conclusion of Appendix C.2 requires a derivation of the commuting-form property from the local simple-MPDO structure of Lemmas C.3–C.4. Once that derivation is available, Theorem 21.79 gives the GSNNCH–ZCL branch and the blocked fusion-isometry witness.

Chapter 22

Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07], with the N -site Hilbert space modelled as coefficient functions

$$\{0, \dots, d-1\}^N \rightarrow \mathbb{C},$$

rather than as an explicit tensor product.

Definition 22.1 (N -site coefficient space). [Lean](#) [✓ Stmt](#) For local dimension d and chain length N , the ambient coefficient space is

$$\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \rightarrow \mathbb{C}.$$

This is the concrete model used throughout the chapter for periodic-chain states.

22.1 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \dots A^{i_L}$. The local ground-space map is

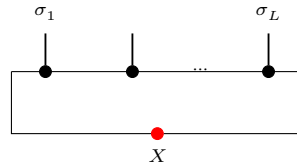
$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}), \quad \Gamma_L(X)(\sigma) := \text{tr}(A^\sigma X),$$

where A^σ abbreviates $A^{\sigma(0)} \dots A^{\sigma(L-1)}$.

Definition 22.2 (Ground-space map). [Lean](#) [✓ Stmt](#) The map Γ_L above is a $\mathbb{C}$ -linear map

$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the lower red node denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The chain length L is fixed by the surrounding formula, not by an extra label inside the diagram.

Lemma 22.3 (Evaluation formula). Lean ✓ Stmt For every boundary matrix $X \in M_D(\mathbb{C})$ and every length- L word σ ,

$$\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X).$$

Definition 22.4 (Local ground space). Lean ✓ Stmt The *local ground space* is the image

$$G_L(A) := \text{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

Lemma 22.5 (Dimension bound). Lean ✓ Stmt For every L ,

$$\dim G_L(A) \leq D^2.$$

Proof. ✓ Proof Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$,

$$\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2.$$

□

Lemma 22.6 (Ambient-space dimension). Lean ✓ Stmt The coefficient-function model satisfies

$$\dim(\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) = d^L.$$

Lemma 22.7 (Nontriviality criterion). Lean ✓ Stmt If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space:

$$G_L(A) \neq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

Proof. ✓ Proof If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 22.6; Lemma 22.5 gives the contradiction $d^L \leq D^2$. □

22.2 Parent interaction and chain Hamiltonian

The parent interaction at range L is the orthogonal projector onto $G_L(A)^\perp$ in the L -site Euclidean space.

Definition 22.8 (Euclidean transport of the local ground space). Lean ✓ Stmt Via the canonical identification between coefficient functions and the Euclidean space $\ell^2(\{0, \dots, d-1\}^L)$, one transports the local ground space to a Hilbert-space subspace

$$G_L^{\text{ES}}(A) \subseteq \ell^2(\{0, \dots, d-1\}^L).$$

Lemma 22.9 (Membership in the transported local ground space). Lean ✓ Stmt A Euclidean vector lies in $G_L^{\text{ES}}(A)$ exactly when transporting it back through the canonical coefficient-space equivalence gives an element of the original local ground space $G_L(A)$.

Proof. ✓ Proof This is just the definition of $G_L^{\text{ES}}(A)$ as the image of $G_L(A)$ under the inverse of the canonical equivalence. □

Definition 22.10 (Parent interaction). [Lean](#) [✓ Stmt](#) The parent interaction at range L is the orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp},$$

on the local L -site space.

Lemma 22.11 (Parent interaction kills ground-space elements). [Lean](#) [✓ Stmt](#) For any $v \in G_L(A)$, we have $h_L(A)v = 0$.

Proof. [✓ Proof](#) Since $h_L(A)$ is the orthogonal projector onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. $\square$

Definition 22.12 (Periodic window extraction). [Lean](#) [✓ Stmt](#) For a configuration $\sigma \in \{0, \dots, d-1\}^N$ on the periodic chain and a starting site i , the extracted word

$$\text{extractWindow}_{L,i}(\sigma) \in \{0, \dots, d-1\}^L$$

reads the entries of σ on the cyclic interval $i, i+1, \dots, i+L-1$ modulo N .

Definition 22.13 (Periodic window replacement). [Lean](#) [✓ Stmt](#) If $L \leq N$, the configuration

$$\text{replaceWindow}_{L,i}(\sigma, \tau)$$

is obtained from σ by replacing the cyclic window starting at i by the length- L word τ .

Lemma 22.14 (Extracting a replaced window). [Lean](#) [✓ Stmt](#) If $L \leq N$, then extracting the same cyclic window after replacement recovers the inserted word:

$$\text{extractWindow}_{L,i}(\text{replaceWindow}_{L,i}(\sigma, \tau)) = \tau.$$

Lemma 22.15 (Replacing an extracted window). [Lean](#) [✓ Stmt](#) If $L \leq N$, then replacing the cyclic window at i in σ by the values extracted from that same σ leaves σ unchanged:

$$\text{replaceWindow}_{L,i}(\sigma, \text{extractWindow}_{L,i}(\sigma)) = \sigma.$$

Proof. [✓ Proof](#) Case-split on whether the offset $(k-i+N) \bmod N$ lies in $[0, L)$: inside the window the value comes from extractWindow , which returns σ at the same position; outside the window both sides already agree with σ . $\square$

For a periodic chain of length N , the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 22.16 (Translated local term). [Lean](#) [✓ Stmt](#) For each site index $i \in \{0, \dots, N-1\}$, the translated local term h_i embeds the parent interaction into the full N -site space at position i (with periodic boundary conditions). Thus h_i acts on the cyclic interval $[i, i+L-1]$ and as the identity outside that interval.

Definition 22.17 (Parent Hamiltonian). [Lean](#) [✓ Stmt](#) The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

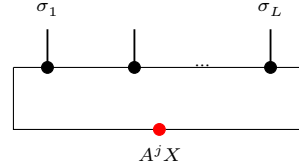
Definition 22.18 (Frustration-free condition). [Lean](#) [✓ Stmt](#) A state ψ is frustration-free if each local term annihilates it:

$$\forall i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

22.3.2 Forward direction: ground space restricts

Lemma 22.28 (Left restriction). [Lean](#) [✓ Stmt](#) If $\psi \in G_{L+1}(A)$, i.e. $\psi(\sigma) = \text{tr}(A^\sigma X)$ for some X , then for each j , the state obtained by fixing the last index to j lies in $G_L(A)$ with witness $A^j X$.

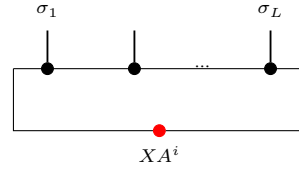
Proof. [✓ Proof](#) The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation: $\psi(\sigma_1, \dots, \sigma_L, j) = \text{tr}(A^{\sigma_1} \dots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$. $\square$

Lemma 22.29 (Right restriction). [Lean](#) [✓ Stmt](#) If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \text{tr}(A^\sigma X)$, then for each i , the state obtained by fixing the first index to i lies in $G_L(A)$ with witness $X A^i$.

Proof. [✓ Proof](#) On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity: $\psi(i, \sigma_1, \dots, \sigma_L) = \text{tr}(A^i A^\sigma X) = \text{tr}(A^\sigma \cdot X A^i) = \Gamma_L(X A^i)(\sigma)$. $\square$

22.3.3 Injectivity of the ground-space map

Theorem 22.30 (Ground-space map injectivity). [Lean](#) [✓ Stmt](#) If A is injective and $L \geq 1$, then Γ_L is injective.

Proof. [✓ Proof](#) If $\Gamma_L(X) = 0$ then $\text{tr}(A^\sigma X) = 0$ for every length- L word σ . Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. $\square$

Theorem 22.31 (Ground-space map injectivity from word span). [Lean](#) [✓ Stmt](#) If the length- L products $\{A^\sigma : |\sigma| = L\}$ span $M_D(\mathbb{C})$, then Γ_L is injective.

Proof. [✓ Proof](#) If $\Gamma_L(X) = 0$, then $\text{tr}(A^\sigma X) = 0$ for every word σ of length L . The spanning hypothesis turns this into $\text{tr}(MX) = 0$ for every $M \in M_D(\mathbb{C})$, and the trace pairing gives $X = 0$. $\square$

Lemma 22.32 (Ground-space map injectivity for block-injective tensors). [Lean](#) [✓ Stmt](#) If A is L_0 -block-injective, then Γ_{L_0} is injective.

Proof. [✓ Proof](#) By definition, L_0 -block injectivity says precisely that the length- L_0 products span $M_D(\mathbb{C})$. Apply Theorem 22.31. $\square$

Theorem 22.33 (Ground-space dimension). [Lean](#) [✓ Stmt](#) If A is injective and $L \geq 1$, then $\dim G_L(A) = D^2$.

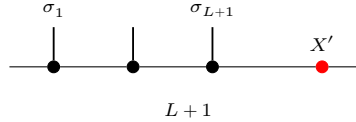
Proof. [✓ Proof](#) The upper bound $\dim G_L(A) \leq D^2$ is Lemma 22.5. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. $\square$

22.3.4 The intersection property

Theorem 22.34 (Intersection property). [Lean](#) [✓ Stmt](#) *If A is injective and $L \geq 2$, then a state ψ on $L + 1$ sites satisfying both the left and right ground conditions lies in $G_{L+1}(A)$:*

$$\psi \in G_L^{\text{left}} \cap G_L^{\text{right}} \implies \psi \in G_{L+1}(A).$$

Proof. [✓ Proof](#) The “invert-and-regrow” argument compares the left and right restrictions on their common $(L - 1)$ -site overlap and then reconstructs the full $(L + 1)$ -site word from a single boundary matrix:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$ (by injectivity of Γ_L) such that $\psi(i, \sigma) = \text{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each j there exists a unique Z_j with $\psi(\sigma, j) = \text{tr}(A^\sigma Z_j)$.
3. Matching on the $(L - 1)$ -site overlap and using nondegeneracy of the trace pairing: $A^j Y_i = Z_j A^i$ for all i, j .
4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.26), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Then $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$, where $X' := \sum_j c_j Z_j$.
5. By trace cyclicity, $\psi(\sigma) = \text{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

□

Theorem 22.35 (Intersection characterization). [Lean](#) [✓ Stmt](#) *For injective A and $L \geq 2$: $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$.*

Proof. [✓ Proof](#) Immediate from the two forward-direction lemmas and Theorem 22.34. □

22.3.5 Unique ground state

Contiguous and cyclic window operators

Remark 22.36 (Contiguous window operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a non-wrapping interval $[s, s + M - 1] \subseteq \{0, \dots, N - 1\}$, one can insert a length- M word into that interval and keep the complementary sites fixed. The associated restriction map evaluates an N -site state on those configurations. The auxiliary lemmas state the full-window case $s = 0$, $M = N$, the compatibility of this restriction with deleting the last or first site, and the fact that fixing the first K entries of a contiguous $(K + L)$ -window leaves the contiguous L -window beginning at $s + K$.

Proof. ✓ Proof For $K = 1$, the hypothesis gives $Z_j A_i = A_j Y_i$ for every letter i . Every blocked word of length L_0 is nonempty, so this identity extends to the coefficients of the blocked tensor $A^{[L_0]}$. Since $A^{[L_0]}$ is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields $Z_j = A_j X$ for a common matrix X . For larger K , strip the first letter: for each i , the matrices $Z_j A_i$ satisfy the same compatibility hypothesis with words of length $K - 1$, so the induction hypothesis reduces the problem to the case $K = 1$. $\square$

Theorem 22.44 (Open-chain right extension at the injectivity length). Lean ✓ Stmt Assume A is L_0 -block injective with $L_0 > 0$. If an element ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every prefix of length K , then $\psi \in G_{K+L_0+1}(A)$.

Proof. ✓ Proof Lemma 22.42 produces matrices $(Z_j)_j$ and $(Y_u)_u$ with $Z_j A^u = A_j Y_u$ for every prefix word u of length K . Theorem 22.43 gives a common right factor X such that $Z_j = A_j X$ for all j . Therefore each last-site restriction of ψ agrees with the corresponding last-site restriction of the ground-space vector determined by X . Since every configuration is obtained by adjoining its final letter to its initial $(K + L_0)$ -tuple, the two states coincide, so ψ lies in $G_{K+L_0+1}(A)$. $\square$

Theorem 22.45 (Open-chain normal range reduction). Lean ✓ Stmt Let A be L_0 -block injective with $D \geq 1$ and $L_0 > 0$. If an N -site state satisfies the ground-space constraint on every non-wrapping contiguous window of length $L_0 + 1$, for every choice τ of the complementary outside configuration, with $N \geq L_0 + 1$, then it lies in $G_N(A)$.

Proof. ✓ Proof Induct on the extra length beyond $L_0 + 1$. The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 22.39 and the contiguous-window transport above identify every fixed-prefix suffix with one of the assumed length- $L_0 + 1$ windows. Theorem 22.44 then regrows the rightmost site. $\square$

Lemma 22.46 (Cyclic-window range antitonicity). Lean Lean ✓ Stmt On a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if $L' \leq L \leq N$, then $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$.

Proof. ✓ Proof Fix a cyclic window of length $L + 1$ and restrict its last site. The resulting length- L vector is again in the ground space, so one-site peeling gives the inclusion for consecutive ranges. Iteration gives the general antitonicity. $\square$

Theorem 22.47 (Cyclic-window normal open-chain reduction). Lean ✓ Stmt Let A be L_0 -block injective with $L_0 > 0$ and $D \geq 1$. If an N -site periodic-chain state satisfies all cyclic ground-space constraints at some range $L_0 + 1 \leq L \leq N$, then it lies in the open-chain ground space $G_N(A)$.

Proof. ✓ Proof Lemma 22.46 reduces the cyclic hypotheses to range $L_0 + 1$. On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 22.45 applies. $\square$

Lemma 22.48 (Wrapped-window one-sided compatibility). Lean ✓ Stmt Assume A is L_0 -block injective with $L_0 > 0$ and $M \geq L_0$. If the reduced wrapped window at the last site has boundary matrices Y_τ , then for every physical letter j one has

$$C_\tau^+ A^j X = Y_\tau A^j,$$

where C_τ^+ is the complementary word seen by that wrapped window.

Proof. ✓ Proof The trace identity for the wrapped window is tested against all words of length L_0 . Injectivity of Γ_{L_0} , which follows from L_0 -block injectivity, strips the test word and gives the displayed matrix identity. $\square$

Lemma 22.49 (Mirror wrapped-window one-sided compatibility). Lean ✓ Stmt Assume A is L_0 -block injective with $L_0 > 0$ and $M \geq L_0$. If the opposite reduced wrapped window has boundary matrices Y_τ , then

$$XA^j C_\tau^- = A^j Y_\tau$$

for every physical letter j , where C_τ^- is the complementary word seen from the opposite boundary-crossing position.

Proof. ✓ Proof Factor the cyclic word at the opposite wrapped position, rotate the trace, and use injectivity of Γ_{L_0} to remove the length- L_0 test word. $\square$

Theorem 22.50 (Reduced wrapped-boundary compatibilities). Lean ✓ Stmt Let A be L_0 -block injective with $L_0 > 0$ and $D \geq 1$. Suppose $\psi \in \mathcal{G}_{N,L}(A)$, $N \geq 2$, $L_0 < L \leq N$, and $\psi = \Gamma_N(X)$ for a boundary matrix X . Then there are two families of boundary matrices Y_τ^+ and Y_τ^- , indexed by outside configurations τ , such that for every physical letter j ,

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \quad X A^j C_\tau^- = A^j Y_\tau^-,$$

where C_τ^+ and C_τ^- are the complementary words exposed by the two opposite wrapped windows of reduced length $L_0 + 1$.

Proof. ✓ Proof First use Lemma 22.46 to reduce the cyclic constraints from range L to range $L_0 + 1$. Then apply the wrapped window and mirror wrapped window compatibility lemmas at the two boundary-crossing positions to obtain the displayed one-sided identities. $\square$

Lemma 22.51 (Middle-word backgrounds for the two wrapped complements). Lean Lean Lean ✓ Stmt Fix a dummy physical letter η and a middle word μ of length $N - (L_0 + 1)$. Write $\tau_\eta^+(\mu)$ for the last-site wrapped background and $\tau_\eta^-(\mu)$ for the opposite wrapped background. Their exposed complements are both exactly μ .

Proof. ✓ Proof With the same dummy letter η on the remaining sites, fill the physical sites $L_0, \dots, N - 2$ with μ for the last-site wrapped window, and fill the sites $1, \dots, N - L_0 - 1$ with μ for the opposite wrapped window. The two complement-extraction identities are then immediate from the index arithmetic. $\square$

Lemma 22.52 (Same-witness identities from compared wrapped witnesses). Lean ✓ Stmt Fix a dummy physical letter η . Suppose the wrapped and mirror one-sided witnesses Y^+ and Y^- have been extracted. If, for every middle word μ , the witnesses agree on the backgrounds built from η ,

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-,$$

then there is a single family Y_μ satisfying

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu$$

for all letters a, b .

Proof. ✓ Proof Define Y_μ to be the wrapped witness evaluated on the background $\tau_\eta^+(\mu)$. Lemma 22.51 rewrites the wrapped complement as μ , while the assumed comparison at $\tau_\eta^-(\mu)$ rewrites the mirror witness to the same Y_μ . $\square$

Lemma 22.53 (Same-witness common-middle commutation). [Lean](#) [✓ Stmt](#) Suppose a single family of boundary matrices Y_μ , indexed by middle words μ , satisfies the two one-sided identities

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu$$

for every pair of physical letters a, b . Then X commutes with every word product $A^a A^\mu A^b$ obtained by adjoining one letter on each side of the common middle.

Proof. [✓ Proof](#) Multiply the second identity on the right by A^b and the first identity on the left by A^a :

$$X A^a A^\mu A^b = A^a Y_\mu A^b = A^a A^\mu A^b X.$$

□

Theorem 22.54 (Boundary matrix commutation from the wrapping window). [Lean](#) [✓ Stmt](#) Let A be injective with $D \geq 1$, and let $\psi = \Gamma_N(X)$ for some boundary matrix $X \in M_D(\mathbb{C})$. If every cyclic restriction of length L of ψ lies in $G_L(A)$, with $N \geq 2$ and $1 < L \leq N$, then, for every $j = 0, \dots, d-1$,

$$X A^j = A^j X$$

Proof. [✓ Proof](#) The wrapping window across the boundary produces matrix identities of the form $X M W = M Y$ for arbitrary test matrices M and complementary words W . Spanning first over the length- $(L-1)$ tails and then over the complementary words, injectivity forces $(X M - M X) W = 0$ for every W , hence $X M = M X$ for every matrix M . In particular, X commutes with each letter A^j . □

Definition 22.55 (MPV submodule). [Lean](#) [✓ Stmt](#) The subspace spanned by the MPV state $V^{(N)}(A)(\sigma) = \text{tr}(A^{\sigma_0} \dots A^{\sigma_{N-1}})$.

Lemma 22.56 (The MPV is the ground-space map at the identity). [Lean](#) [✓ Stmt](#) For every chain length N ,

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

Lemma 22.57 (MPV lies in the ground space). [Lean](#) [✓ Stmt](#) $V^{(N)}(A) \in G_N(A)$ for every N , with boundary matrix $X = I$.

Proof. [✓ Proof](#) $V^{(N)}(A)(\sigma) = \text{tr}(A^\sigma) = \text{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$. □

Definition 22.58 (Unique ground state). [Lean](#) [✓ Stmt](#) A submodule S has a unique ground state if $\dim S = 1$.

Theorem 22.59 (One-dimensionality criterion). [Lean](#) [✓ Stmt](#) For finite-dimensional S , $\dim S = 1$ iff there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c \psi_0$.

Proof. [✓ Proof](#) Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. □

Definition 22.60 (Periodic chain ground space). [Lean](#) [✓ Stmt](#) For $N > 0$ and $L \leq N$, define the periodic chain ground space

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}) : \forall i \in \{0, \dots, N-1\}, \psi|_{[i, i+L-1]} \in G_L(A)\},$$

where the condition $\psi|_{[i, i+L-1]} \in G_L(A)$ means that for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \top$ by convention.

Lemma 22.61 (The MPV satisfies every cyclic window constraint). Lean ✓ Stmt If $0 < N$ and $L \leq N$, then

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

Theorem 22.62 (MPV nonvanishing for block-injective tensors). Lean ✓ Stmt If A is L_0 -block-injective with $L_0 > 0$, $D \geq 1$, and $N \geq L_0 + 1$, then $V^{(N)}(A) \neq 0$ in the N -site space.

Proof. ✓ Proof Suppose for contradiction that $V^{(N)}(A) = 0$, i.e. $\text{tr}(A^\sigma) = 0$ for every word σ of length N . For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\text{tr}(A^w A^u) = 0$. Since A is L_0 -block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeating the argument with words of length $N - 2L_0$ (since $|w'| + L_0 = N - L_0$, the product $A^{w'} A^{u_1}$ has length $N - L_0$, so $A^{w'} A^{u_1} = 0$ by the previous step) and iterating, we eventually reach $\text{tr}(A^u) = 0$ for $|u| = L_0$, which gives $I = 0$, a contradiction. $\square$

22.3.6 Block-word stripping

Theorem 22.63 (Common right factor from block-word compatibility). Lean ✓ Stmt Assume A is L_0 -block-injective with $L_0 > 0$. Let $\{Z_u\}_{|u|=L_0}$ be a family indexed by words of length L_0 . If for some $K \geq 0$ and every suffix word w of length K there exists $Y_w \in M_D(\mathbb{C})$ such that, for every word u of length L_0 ,

$$Z_u A^w = A^u Y_w$$

then there exists $X \in M_D(\mathbb{C})$ with $Z_u = A^u X$ for every word u of length L_0 .

Proof. ✓ Proof Induct on K . For $K = 0$ the empty suffix gives $Z_u = A^u Y_\emptyset$. For the successor step, fix the first physical index and apply the induction hypothesis to the shortened suffix. This reduces the statement to letter compatibility on the right. Extending that letter compatibility to all length- L_0 words and decomposing $\mathbb{1}$ in the spanning family $\{A^u : |u| = L_0\}$ yields a common factor X . $\square$

Theorem 22.64 (Block-word commutation from long-word commutation). Lean ✓ Stmt If A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m product A^w , then X already commutes with every length- L_0 product A^u .

Proof. ✓ Proof Write each length- m word as a concatenation uw with $|u| = L_0$ and $|w| = m - L_0$. The hypothesis gives

$$X A^u A^w = A^u A^w X = A^u (A^w X).$$

Thus the family $Z_u := X A^u$ satisfies the hypothesis of Theorem 22.63. One obtains $X A^u = A^u X$ for all $|u| = L_0$. Decomposing $\mathbb{1}$ in the span of the length- L_0 words shows $X = X$, hence $X A^u = A^u X$. $\square$

Theorem 22.65 (Centrality from long-word commutation). Lean ✓ Stmt If A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m word product, then X commutes with every matrix in $M_D(\mathbb{C})$.

Proof. ✓ Proof By Theorem 22.64, X commutes with every length- L_0 word product. These products span $M_D(\mathbb{C})$ by block injectivity, so linearity gives $X M = M X$ for every $M \in M_D(\mathbb{C})$. $\square$

Lemma 22.66 (Amplifying fixed-length word commutation). Lean ✓ Stmt If a boundary matrix X commutes with every word product of length m , then it commutes with every word product whose length is a multiple $q m$.

Proof. ✓ Proof Proceed by induction on q . For $q = 0$, the claim is trivial. For the inductive step, let a word of length $(q + 1)m$ be given and write it as a length- m prefix followed by a length- q suffix. The prefix commutes with X by hypothesis, and the suffix commutes by the induction hypothesis. $\square$

Lemma 22.67 (Padded complement annihilation). Lean ✓ Stmt *If A is L_0 -block-injective and $q \geq 1$, then an operator Z that annihilates every length- k word product on the left is zero whenever $k \leq qL_0$:*

$$ZA^w = 0 \quad (|w| = k) \quad \implies \quad Z = 0.$$

Proof. ✓ Proof The hypothesis $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$ implies the same spanning statement at every positive multiple qL_0 . Each length- qL_0 word factors as a length- k prefix followed by padding, so Z annihilates all generators of the full span and hence annihilates the identity. $\square$

Lemma 22.68 (Generator commutation from long-word commutation). Lean ✓ Stmt *Under the hypotheses of Theorem 22.65, one has $XA^i = A^iX$ for every physical index i .*

Proof. ✓ Proof Apply Theorem 22.65 with $M = A^i$. $\square$

Theorem 22.69 (Boundary contraction in the MPV span from long-word commutation). Lean ✓ Stmt *Let A be L_0 -block-injective with $L_0 > 0$ and let $m \geq L_0$. If a boundary matrix X commutes with every length- m word product, then $\Gamma_N(X)$ lies in the MPV line $\text{span}\{V^{(N)}(A)\}$.*

Proof. ✓ Proof Theorem 22.65 makes X commute with the whole matrix algebra. The center of $M_D(\mathbb{C})$ consists of scalar matrices, so $X = c\mathbf{1}$, and therefore $\Gamma_N(X) = cV^{(N)}(A)$. $\square$

Theorem 22.70 (Boundary contraction in the MPV span from positive-length commutation). Lean ✓ Stmt *Let A be L_0 -block-injective with $L_0 > 0$. If X commutes with every word product of some positive length m , then $\Gamma_N(X)$ lies in the MPV line.*

Proof. ✓ Proof Lemma 22.66 amplifies the length- m commutation hypothesis to length $L_0 m \geq L_0$. Then Theorem 22.69 applies. $\square$

Theorem 22.71 (Boundary contraction in the MPV span from a common-middle witness). Lean ✓ Stmt *Let A be L_0 -block-injective with $L_0 > 0$. If the common-middle same-witness identities of Lemma 22.53 hold for X , then $\Gamma_N(X)$ lies in the MPV line.*

Proof. ✓ Proof The common-middle identities give commutation with all words of length $|\mu| + 2$, a positive length. The positive-length containment theorem then applies. $\square$

Theorem 22.72 (Boundary contraction in the MPV span from wrapped-witness comparison). Lean ✓ Stmt *Let A be L_0 -block-injective with $L_0 > 0$, and fix a dummy physical letter η . If the wrapped and mirror one-sided witnesses satisfy*

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$$

for every middle word μ , then $\Gamma_N(X)$ lies in the MPV line.

Proof. ✓ Proof Lemma 22.52 uses the comparison at the two η -backgrounds to turn the wrapped witnesses into a single same-witness common-middle family. Then Theorem 22.71 applies. $\square$

Theorem 22.73 (Conditional normal-range reduction from witness comparison). Lean ✓ Stmt
Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$. Assume $N \geq 2$ and $L_0 < L \leq N$. Suppose that, for every dummy letter η and every boundary matrix obtained from an N -site chain ground state, the wrapped and mirror witnesses supplied by Theorem 22.50 agree at $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$ for every middle word μ . Then the range- L chain ground space is contained in the MPV line.

Proof. ✓ Proof The cyclic-to-open-chain reduction writes any chain ground state as $\Gamma_N(X)$. The reduced wrapped-boundary compatibility theorem supplies the two one-sided witness families for this X . The comparison hypothesis, applied to a dummy letter η , says that these witnesses agree at $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$ for every middle word μ . Theorem 22.72 finishes. $\square$

Theorem 22.74 (Chain ground space equals the MPV line in the injective case). Lean ✓ Stmt
If A is injective, $D \geq 1$, $N \geq 2$, and $1 < L \leq N$, then

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

Proof. ✓ Proof Let $\psi \in \mathcal{G}_{N,L}(A)$. The non-wrapping window conditions and Lemma 22.38 imply $\psi \in G_N(A)$, so $\psi = \Gamma_N(X)$ for some boundary matrix X . The wrapping window condition now applies Theorem 22.54 and forces X to commute with every letter A^j . Since A is injective, the letters generate the full matrix algebra, hence X is scalar. Therefore ψ is proportional to $V^{(N)}(A)$, while the reverse inclusion is Lemma 22.61. $\square$

Theorem 22.75 (Normal-range reduction: containment in the MPV line). Lean Not ready If A is normal, $D \geq 1$, L_0 -block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then

$$\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}.$$

Proof. Not ready The closure property needed here is the one from [CPGSV21, Section 4.3, lines 2049–2094]. Theorem 22.50 supplies the two one-sided wrapped-boundary identities for the open-chain boundary matrix X . The padded-annihilation lemma states that exact complement-span length mismatches are no longer the obstruction. The new conditional reduction theorem shows that the proof would finish as soon as those actual wrapped-window witnesses are compared after reindexing their complements to a common middle word. What is still missing is precisely that comparison of the witnesses themselves. $\square$

Theorem 22.76 (Chain ground space equals the MPV line in the normal case). Lean Not ready
If A is normal, $D \geq 1$, L_0 -block-injective, with $N \geq 2$ and $L_0 < L \leq N$, then

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

Theorem 22.77 (Unique ground state on the periodic chain). Lean ✓ Stmt For an injective tensor A with $D \geq 1$, if $N \geq 2$ and $1 < L \leq N$, then the periodic chain ground space is one-dimensional.

Proof. ✓ Proof By Theorem 22.74, the ground space is the line $\text{span}\{V^{(N)}(A)\}$. It remains to show that the MPV is nonzero. If $V^{(N)}(A) = 0$, then Lemma 22.56 gives

$$\Gamma_N(\mathbb{1}) = \Gamma_N(0).$$

Since A is injective and $N > 0$, Theorem 22.30 forces $\mathbb{1} = 0$, a contradiction. Therefore the spanning line is one-dimensional. $\square$

22.5 Decorrelation and commuting parent Hamiltonians

This section develops the abstract operator-algebraic formulation of decorrelation and its connection to commuting parent Hamiltonians, following [CPGSV16, Appendix D, Section D.2].

Definition 22.86 (Decorrelation). Lean ✓ Stmt Given an idempotent endomorphism P_K and sets of observables $\mathcal{O}_A, \mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A, O_B \in \mathcal{O}_B$.

Definition 22.87 (Commuting parent Hamiltonian structure). Lean ✓ Stmt A subspace K (given by its idempotent endomorphism P_K) has a *commuting parent Hamiltonian structure* if there exist idempotent endomorphisms P_{AX} and P_{XB} that commute and satisfy $P_{AX} \circ P_{XB} = P_K$.

Theorem 22.88 (Idempotence gives a trivial commuting parent Hamiltonian). Lean ✓ Stmt If

$$P_K \circ P_K = P_K,$$

then P_K has a commuting parent Hamiltonian structure.

Proof. ✓ Proof Take $P_{AX} = P_{XB} = P_K$. Their commutativity is immediate, and the product identity is exactly the assumed idempotence of P_K . □

Lemma 22.89 (Cancellation for commuting idempotents). Lean ✓ Stmt If P and Q are commuting idempotents, then

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

Proof. ✓ Proof Expand the middle factor, use commutativity to rewrite $QPQP$ as $QPPQ$, and then apply idempotence of P and Q . □

Theorem 22.90 (Idempotence of the product projector). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K \circ P_K = P_K.$$

Proof. ✓ Proof Since P_{AX} and P_{XB} are idempotent and commute, their product $P_{AX} \circ P_{XB}$ is idempotent. Substituting $P_K = P_{AX} \circ P_{XB}$ gives the claim. □

Theorem 22.91 (Absorption by the left projector). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_{AX} \circ P_K = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{AX} ,

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 22.92 (Absorption by the right projector). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K \circ P_{XB} = P_K.$$

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{XB} ,

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 22.93 (Cross-absorption by the right projector). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_{XB} \circ P_K = P_K.$$

Proof. ✓ Proof By commutativity, $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$, and idempotence of P_{XB} yields $P_{XB} \circ P_K = P_K$. □

Theorem 22.94 (Cross-absorption by the left projector). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K \circ P_{AX} = P_K.$$

Proof. ✓ Proof By commutativity, $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$, and idempotence of P_{AX} yields $P_K \circ P_{AX} = P_K$. □

Theorem 22.95 (Reverse product identity). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_{XB} \circ P_{AX} = P_K.$$

Proof. ✓ Proof This is immediate from commutativity of P_{AX} and P_{XB} together with the identity $P_{AX} \circ P_{XB} = P_K$. □

Theorem 22.96 (Commuting complementary projectors). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition and

$$Q_{AX} = 1 - P_{AX}, \quad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

Proof. ✓ Proof Expanding both products and using $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ shows that the complementary projectors commute. □

Theorem 22.97 (Commuting parent Hamiltonian implies decorrelation). Lean ✓ Stmt If a subspace K has a commuting parent Hamiltonian decomposition $P_K = P_{AX} \circ P_{XB}$ with $[P_{AX}, P_{XB}] = 0$, and A -observables commute with P_{XB} while B -observables commute with P_{AX} , then regions A and B are decorrelated w.r.t. K . This is the backward direction of [CPGSV16, Appendix D, Section D.2, Proposition D.3].

Proof. ✓ Proof The key identity is $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Using the commutation hypotheses to slide O_A past P_{XB} and O_B past P_{AX} , the five-fold composition factors through this zero term. □

Theorem 22.98 (Commuting parent Hamiltonian implies decorrelation). Lean ✓ Stmt Corollary of Theorem 22.97: if P_K has a commuting parent Hamiltonian decomposition and observables respect locality, then regions A and B are decorrelated w.r.t. K .

Lemma 22.99 (Frustration-free identity). [Lean](#) [✓ Stmt](#) For any endomorphisms P and Q ,

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

Proof. [✓ Proof](#) Expanding the left-hand side and cancelling gives the result. $\square$

Theorem 22.100 (Commuting parent Hamiltonian identity). [Lean](#) [✓ Stmt](#) If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where $Q_{AX} = 1 - P_{AX}$ and $Q_{XB} = 1 - P_{XB}$ are the complementary projectors. This is the frustration-free form of the parent Hamiltonian from [\[CPGSV16, Appendix D, Section D.2\]](#).

Proof. [✓ Proof](#) Applying Lemma 22.99 to P_{AX} and P_{XB} , and substituting $P_{AX} \circ P_{XB} = P_K$. $\square$

Theorem 22.101 (Ground-space membership). [Lean](#) [✓ Stmt](#) If $P_K = P_{AX} \circ P_{XB}$ is a commuting parent Hamiltonian decomposition, then

$$P_K v = v \iff P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic form of equation (D.2) from [\[CPGSV16\]](#): $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Proof. [✓ Proof](#) ($\Rightarrow$) If $P_K v = v$, then $P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$ by left absorption, and similarly for P_{XB} .

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then $P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v$. $\square$

22.6 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length N . For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [\[KL18\]](#).

Definition 22.102 (Transported parent ground space). [Lean](#) [✓ Stmt](#) Denote by $\mathcal{G}_{N,L}^{\text{ES}}(A)$ the parent ground space after identifying the coefficient space with its Euclidean ℓ^2 -model.

Definition 22.103 (Transported parent Hamiltonian). [Lean](#) [✓ Stmt](#) Denote by $H_{N,L}^{\text{ES}}(A)$ the conjugate of $H_N(A, L)$ by the canonical identification between the coefficient space and its Euclidean avatar.

Theorem 22.104 (Euclidean transport of the parent ground space). [Lean](#) [✓ Stmt](#) Under the canonical ℓ^2 transport of the N -site state space, the transported parent ground space is exactly the kernel of the transported parent Hamiltonian.

Proof. [✓ Proof](#) This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and its Euclidean-space avatar, so the transported ground space is precisely the transported kernel. $\square$

Lemma 22.105 (Membership in the transported parent ground space). [Lean](#) [✓ Stmt](#) A vector belongs to the transported parent ground space exactly when the transported parent Hamiltonian annihilates it.

Remark 22.106 (Euclidean local projector ingredients). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $P_L^{\text{ES}}(A)$ denote the orthogonal projector onto $(G_L^{\text{ES}}(A))^\perp$ on the Euclidean L -site space. For a cyclic window starting at i and an outside configuration τ , the transported restriction map $R_{i,\tau}$ evaluates an N -site Euclidean vector on that window. The transported local term $h_{i,\text{ES}}$ is the corresponding N -site local projector, and $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$ is the basic positive summand in its cyclic-averaging formula.

Lemma 22.107 (Positivity of the Euclidean local summands). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The Euclidean local projector $P_L^{\text{ES}}(A)$ is positive. Consequently, every summand $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$ is positive, and so is the finite sum over τ .*

Proof. [✓ Proof](#) Orthogonal projectors are positive. Conjugation by $R_{i,\tau}$ preserves positivity, and finite sums of positive operators remain positive. $\square$

Lemma 22.108 (Cyclic averaging for the transported ES local terms). [Lean](#) [Lean](#) [✓ Stmt](#) *The transported parent Hamiltonian is the sum of the transported local terms $H_{N,\text{ES}} = \sum_i h_{i,\text{ES}}$, and each transported local term admits the exact cyclic-averaging formula*

$$h_{i,\text{ES}} = d^{-L} \sum_{\tau} R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}.$$

Proof. [✓ Proof](#) Unfold the transported local term and compare it configuration-by-configuration with the cyclic restrictions. The d^{-L} factor compensates for the d^L choices of the window coordinates in the ambient configuration parameter τ . $\square$

Lemma 22.109 (Positivity of the transported ES Hamiltonian). [Lean](#) [Lean](#) [✓ Stmt](#) *Each transported local term $h_{i,\text{ES}}$ is positive, hence the full transported parent Hamiltonian $H_{N,\text{ES}}$ is positive.*

Proof. [✓ Proof](#) Apply the averaging formula above: $h_{i,\text{ES}}$ is a nonnegative scalar multiple of a finite sum of positive conjugates of $P_L^{\text{ES}}(A)$, hence is itself positive. Summing over i yields positivity of $H_{N,\text{ES}}$. $\square$

Lemma 22.110 (Symmetric-projection structure of transported ES local terms). [Lean](#) [Lean](#) [✓ Stmt](#) *The Euclidean local projector $P_L^{\text{ES}}(A)$ is a symmetric projection, and every transported local term $h_{i,\text{ES}}$ on the N -site chain is a symmetric projection.*

Proof. [✓ Proof](#) The local operator $P_L^{\text{ES}}(A)$ is an orthogonal projection onto $(G_L^{\text{ES}}(A))^\perp$. For $L \leq N$, restricting $h_{i,\text{ES}} v$ to a cyclic window gives $P_L^{\text{ES}}(A)$ applied to the corresponding restriction of v ; applying $h_{i,\text{ES}}$ a second time therefore applies $(P_L^{\text{ES}}(A))^2 = P_L^{\text{ES}}(A)$ on that window. The positivity result above supplies symmetry, and the case $L > N$ is the zero projection by definition. $\square$

Lemma 22.111 (Kernel of the Euclidean parent interaction). [Lean](#) [✓ Stmt](#) *The Euclidean local interaction $P_L^{\text{ES}}(A)$ annihilates exactly the Euclidean local ground space:*

$$P_L^{\text{ES}}(A)v = 0 \iff v \in G_L^{\text{ES}}(A).$$

Proof. [✓ Proof](#) Since $P_L^{\text{ES}}(A)$ is the orthogonal projector onto $(G_L^{\text{ES}}(A))^\perp$, its kernel is the original subspace $G_L^{\text{ES}}(A)$. $\square$

Lemma 22.112 (Local-term kernel via cyclic restrictions). [Lean](#) [Lean](#) [✓ Stmt](#) *For $L \leq N$, a transported local term $h_{i,\text{ES}}$ annihilates a vector v iff every boundary-filled restriction of v to the cyclic L -window starting at i lies in $G_L^{\text{ES}}(A)$.*

Proof. [✓ Proof](#) Restricting $h_{i,\text{ES}}v$ to the same cyclic window gives $P_L^{\text{ES}}(A)$ applied to the corresponding restriction of v . Lemma 22.111 identifies the vanishing of this local projector with membership in $G_L^{\text{ES}}(A)$. $\square$

Theorem 22.113 (Adjacent local kernels realize the intersection property). [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) *If A is injective and $L \geq 2$, then on an $(L+1)$ -site chain the kernels of the two adjacent transported L -site local terms are exactly the transported $(L+1)$ -site MPS ground space:*

$$h_{0,\text{ES}}v = 0 \quad \text{and} \quad h_{1,\text{ES}}v = 0 \quad \Longleftrightarrow \quad v \in G_{L+1}^{\text{ES}}(A).$$

Proof. [✓ Proof](#) Lemma 22.112 translates the two local kernel assumptions into the left and right L -site ground conditions. The open-chain intersection property, Theorem 22.34, then regrows the shared $(L-1)$ -site overlap to the $(L+1)$ -site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel–restriction characterization. $\square$

Lemma 22.114 (Non-overlap positivity for transported ES local terms). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) *Let $L \leq N$ and let $h_{i,\text{ES}}, h_{j,\text{ES}}$ be transported local terms whose cyclic L -windows are site-disjoint: no site of the periodic chain has cyclic offset $< L$ from both i and j . Then the terms commute pointwise: for every vector v ,*

$$h_{i,\text{ES}}(h_{j,\text{ES}}v) = h_{j,\text{ES}}(h_{i,\text{ES}}v),$$

and their ordered cross term is nonnegative:

$$0 \leq \text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

Proof. [✓ Proof](#) In coordinates, $h_{i,\text{ES}}$ applies $P_L^{\text{ES}}(A)$ only to the i -window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the i -window does not change the extracted j -window, and the i - and j -window replacements commute. Therefore the two embedded local operators commute. The preceding projection-geometry identity for commuting symmetric projections gives the nonnegative ordered cross term. $\square$

Lemma 22.115 (Quadratic form to norm bound). [Lean](#) [✓ Stmt](#) *Let H be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all v ,*

$$\gamma \langle v, Hv \rangle \leq \langle Hv, Hv \rangle$$

i.e. $H^2 \geq \gamma H$ as a quadratic form, for some $\gamma > 0$. Then for every v in the orthogonal complement of $\ker H$,

$$\gamma \|v\| \leq \|Hv\|.$$

In particular, every positive eigenvalue of H is at least γ . The positive-semidefiniteness hypothesis is essential: without it, $H = -\text{Id}$ with $\gamma = 2$ satisfies the quadratic-form inequality vacuously but violates the norm bound.

Proof. [✓ Proof](#) By the finite-dimensional spectral theorem, H diagonalises in an orthonormal eigenbasis with real eigenvalues λ_k . For a unit eigenvector ψ_k with eigenvalue λ_k the hypothesis gives $\gamma \lambda_k \leq \lambda_k^2$, whence $\lambda_k \leq 0$ or $\lambda_k \geq \gamma$. Since H is positive semidefinite (its quadratic form is ≥ 0), the negative case is excluded and every nonzero eigenvalue satisfies $\lambda_k \geq \gamma$. Writing $v = \sum_k c_k \psi_k$ and projecting onto $(\ker H)^\perp$ keeps only terms with $\lambda_k \geq \gamma$, so $\|Hv\|^2 = \sum_k |c_k|^2 \lambda_k^2 \geq \gamma^2 \sum_k |c_k|^2 = \gamma^2 \|v\|^2$. $\square$

Lemma 22.116 (Parent-Hamiltonian quadratic-form reduction). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $H_{N,L}^{\text{ES}}(A)$ be the transported parent Hamiltonian. If a constant $\gamma > 0$ satisfies the uniform quadratic-form estimate*

$$\gamma \operatorname{Re}\langle H_{N,L}^{\text{ES}}(A)v, v \rangle \leq \operatorname{Re}\langle H_{N,L}^{\text{ES}}(A)v, H_{N,L}^{\text{ES}}(A)v \rangle$$

for every $N \geq 2L$ and every vector v , then

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$$

for every $v \in (G_{N,L}^{\text{ES}}(A))^{\perp}$. In particular, with $L > 1$ and $\gamma = 1/(4L)$, the explicit gap-bound conclusion follows once this same quadratic-form estimate is supplied.

Proof. ✓ Proof The transported Hamiltonian is positive by Lemma 22.109, and its transported ground space is its kernel by Theorem 22.104. Therefore the abstract spectral-theorem conversion of Lemma 22.115 applies directly. The explicit constant is positive because $L > 1$ implies $4L > 0$. □

Remark 22.117 (Projection-geometry identities for row-sum arguments). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity $\text{Re}\langle Pv, Pv \rangle = \text{Re}\langle Pv, v \rangle$ for symmetric projections; nonnegativity of this quadratic form; nonnegativity of $\text{Re}\langle Pv, Qv \rangle$ for commuting symmetric projections P, Q ; the Cauchy–Schwarz conversion from $\|PQv\| \leq c\|Pv\|$ to $\text{Re}\langle Pv, Qv \rangle \geq -c \text{Re}\langle Pv, v \rangle$; finite-sum expansions of $\text{Re}\langle (\sum_i P_i)v, v \rangle$ and $\text{Re}\langle (\sum_i P_i)v, (\sum_i P_i)v \rangle$; the diagonal/off-diagonal split of the ordered double sum; and the aggregate off-diagonal estimate obtained from row sums $\sum_{i \neq j} c_{ij} \leq 1$.

Lemma 22.118 (Ordered projection row-sum quadratic-form reduction). [Lean] [Stm] Let P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. If the ordered off-diagonal forms obey

$$\operatorname{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) c_{ij} \operatorname{Re}\langle P_i v, v \rangle \quad (i \neq j),$$

with row sums $\sum_{j \neq i} c_{ij} \leq 1$ and $\gamma \leq 1$, then $H^2 \geq \gamma H$ as a quadratic form.

Proof.  **Proof** Expand $\text{Re}\langle Hv, Hv \rangle$ as an ordered double sum. Symmetric idempotence identifies each diagonal term with $\text{Re}\langle P_i v, v \rangle$, while the row-summable ordered cross-term bounds control the off-diagonal contribution by $-(1 - \gamma) \sum_i \text{Re}\langle P_i v, v \rangle$. $\square$

Lemma 22.119 (Finite-overlap projection row-sum reduction). [Lean] [✓ Stmt] Let P_i be a finite family of symmetric projections and let $\gamma \leq 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row for some positive integer m . If noninteracting ordered cross terms are nonnegative and, on each interacting pair,

$$\operatorname{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) \frac{1}{m} \operatorname{Re}\langle P_i v, v \rangle,$$

then $H = \sum_i P_i$ satisfies $H^2 \geq \gamma H$ as a quadratic form.

Proof.  Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row cardinality bound gives $\sum_{j \neq i} c_{ij} \leq 1$; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 22.118. $\square$

Lemma 22.120 (Finite-overlap projection norm-compression reduction). [Lean](#) [Lean](#) [✓ Stmt](#)
 Let P_i be a finite family of symmetric projections and let $\gamma \leq 1$. Suppose an interaction predicate has at most $m > 0$ off-diagonal neighbours in each row, noninteracting ordered cross terms are nonnegative, and interacting pairs satisfy the norm-compression estimate

$$\|P_i P_j v\| \leq (1 - \gamma) \frac{1}{m} \|P_i v\|.$$

Then $H = \sum_i P_i$ satisfies $H^2 \geq \gamma H$ as a quadratic form. The same conclusion holds from a separate compression coefficient η whenever $\eta \leq (1 - \gamma)/m$.

Proof. [✓ Proof](#) Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into $\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma)m^{-1} \text{Re}\langle P_i v, v \rangle$. If a separate coefficient η is used, the assumption $\eta \leq (1 - \gamma)/m$ first weakens that estimate to the displayed one. The finite-overlap row-sum reduction then applies. $\square$

Lemma 22.121 (Parent-Hamiltonian ordered row-sum quadratic-form reduction). [Lean](#) [✓ Stmt](#)
 For a fixed chain length N , suppose the transported local terms $h_{i,\text{ES}}$ satisfy the ordered row-sum estimate

$$\text{Re}\langle h_{i,\text{ES}} v, h_{j,\text{ES}} v \rangle \geq -(1 - \gamma) c_{ij} \text{Re}\langle h_{i,\text{ES}} v, v \rangle \quad (i \neq j),$$

with $\sum_{j \neq i} c_{ij} \leq 1$ and $\gamma \leq 1$. Then the transported parent Hamiltonian satisfies

$$\gamma \text{Re}\langle H_{N,L}^{\text{ES}} v, v \rangle \leq \text{Re}\langle H_{N,L}^{\text{ES}} v, H_{N,L}^{\text{ES}} v \rangle$$

for every vector v .

Proof. [✓ Proof](#) Lemma 22.110 supplies the symmetric-projection hypothesis for the family $P_i = h_{i,\text{ES}}$. Apply Lemma 22.118 to this family and use $H_{N,L}^{\text{ES}} = \sum_i h_{i,\text{ES}}$. $\square$

Lemma 22.122 (Parent-Hamiltonian gap from ordered local row bounds). [Lean](#) [✓ Stmt](#) Assume uniformly for every $N \geq 2L$ that the transported local terms satisfy the ordered row-sum estimate above with $\gamma = 1/(4L)$. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$,

$$\frac{1}{4L} \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

Proof. [✓ Proof](#) The fixed-chain row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N . The positivity and kernel identification of the transported Hamiltonian then allow Lemma 22.116 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. $\square$

Lemma 22.123 (Parent-Hamiltonian finite-overlap Friedrichs quadratic reduction). [Lean](#) [✓ Stmt](#)
 For a fixed chain length N , let m be a positive integer and let $\gamma \leq 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row. If non-overlapping local terms have nonnegative ordered cross terms, overlapping local terms satisfy the finite-overlap Friedrichs estimate

$$\text{Re}\langle h_{i,\text{ES}} v, h_{j,\text{ES}} v \rangle \geq -(1 - \gamma) \frac{1}{m} \text{Re}\langle h_{i,\text{ES}} v, v \rangle,$$

and the transported local terms are symmetric projections, then the transported Hamiltonian satisfies $H_{N,L}^{\text{ES}^2} \geq \gamma H_{N,L}^{\text{ES}}$ as a quadratic form.

Proof. [✓ Proof](#) Apply the finite-overlap projection reduction with $P_i = h_{i,\text{ES}}$. $\square$

Lemma 22.124 (Finite-overlap norm-compression reduction). [Lean](#) [Lean](#) [✓ Stmt](#) For a fixed chain length N , let $m > 0$ and $\gamma \leq 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row, non-overlapping local terms have nonnegative ordered cross terms, and overlapping local terms satisfy

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq (1 - \gamma) \frac{1}{m} \|h_{i,\text{ES}}v\|.$$

Then $H_{N,L}^{\text{ES}^2} \geq \gamma H_{N,L}^{\text{ES}}$ as a quadratic form. The same conclusion holds from a separate coefficient η satisfying $\eta \leq (1 - \gamma)/m$.

Proof. [✓ Proof](#) Apply the finite-overlap norm-compression projection reduction to the family $P_i = h_{i,\text{ES}}$. If a separate coefficient η is used, first weaken the compression estimate using $\eta \leq (1 - \gamma)/m$. $\square$

Definition 22.125 (Cyclic-window support). [Lean](#) [✓ Stmt](#) For a length- L cyclic window on $\{0, \dots, N-1\}$ starting at i , its support is

$$S_i = \{i, i+1, \dots, i+L-1\} \pmod{N},$$

with repeated visits counted once.

Lemma 22.126 (Cyclic-window support offsets). [Lean](#) [✓ Stmt](#) If $L \leq N$, then a site k belongs to the cyclic support S_i exactly when its cyclic offset from i is below L :

$$k \in S_i \iff (k - i) \bmod N < L.$$

Proof. [✓ Proof](#) Write $k = i + r \pmod{N}$ for a representative offset r . Membership in S_i gives an offset $r < L$, and conversely an offset below L gives the corresponding point of the support. $\square$

Definition 22.127 (Cyclic-window overlap). [Lean](#) [✓ Stmt](#) Two length- L cyclic windows overlap, written $i \sim_L j$, when their supports meet: $S_i \cap S_j \neq \emptyset$.

Lemma 22.128 (Concrete non-overlap positivity for cyclic supports). [Lean](#) [Lean](#) [✓ Stmt](#) If $L \leq N$ and the cyclic supports S_i and S_j do not meet, then the transported local ES terms have nonnegative ordered cross term:

$$0 \leq \text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

Proof. [✓ Proof](#) The support-offset characterization converts failure of $S_i \cap S_j \neq \emptyset$ into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies. $\square$

Lemma 22.129 (Cyclic-window overlap row cardinality). [Lean](#) [✓ Stmt](#) If $2L \leq N$ and $L > 1$, then for every $i \in \{0, \dots, N-1\}$,

$$\#\{j \neq i : i \sim_L j\} \leq 2(L-1).$$

Proof. [✓ Proof](#) If S_i and S_j meet, choose offsets $a, b < L$ with $i + a \equiv j + b \pmod{N}$. Since $2L \leq N$, the offset from i to j is either the clockwise distance $a - b \in \{1, \dots, L-1\}$ or the counterclockwise distance $b - a \in \{1, \dots, L-1\}$; the zero distance is excluded by $j \neq i$. Thus every overlapping off-diagonal start lies among the $L-1$ clockwise starts or the $L-1$ counterclockwise starts. $\square$

Lemma 22.130 (Parent-Hamiltonian gap from finite-overlap Friedrichs data). [Lean](#) [✓ Stmt](#) Fix $L > 1$ and set $m = 2(L - 1)$. Suppose uniformly for every $N \geq 2L$ that the hypotheses of the fixed-chain finite-overlap Friedrichs reduction hold with $\gamma = 1/(4L)$ and this value of m . Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$.

Proof. [✓ Proof](#) The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible N . Lemma 22.116 converts that estimate to the norm lower bound. $\square$

Lemma 22.131 (Parent-Hamiltonian gap from cyclic-window Friedrichs data). [Lean](#) [✓ Stmt](#) Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length- L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \frac{1}{4L}) \frac{1}{2(L-1)} \text{Re}\langle h_{i,\text{ES}}v, v \rangle.$$

Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$.

Proof. [✓ Proof](#) Lemma 22.129 gives the row-cardinality bound $m = 2(L - 1)$ for the concrete cyclic-window overlap relation, and Lemma 22.110 supplies the symmetric projection structure of each transported local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 22.128 gives the required nonnegative ordered cross term. Apply Lemma 22.130 with these facts and the displayed overlapping-window estimate. $\square$

Lemma 22.132 (Parent-Hamiltonian gap from overlapping cyclic Friedrichs data). [Lean](#) [✓ Stmt](#) Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length- L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \frac{1}{4L}) \frac{1}{2(L-1)} \text{Re}\langle h_{i,\text{ES}}v, v \rangle.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$.

Proof. [✓ Proof](#) Apply Lemma 22.131. Its non-overlap positivity and finite-row hypotheses are already internal to that cyclic-window reduction; the displayed overlapping estimate supplies its remaining ordered cross-term condition. $\square$

Lemma 22.133 (Parent-Hamiltonian gap from overlapping cyclic norm-compression data). [Lean](#) [✓ Stmt](#) Fix $L > 1$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq (1 - \frac{1}{4L}) \frac{1}{2(L-1)} \|h_{i,\text{ES}}v\|.$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$.

Proof. [✓ Proof](#) The norm-compression condition is converted to the ordered Friedrichs lower bound by the projection-geometry Cauchy-Schwarz lemma. The preceding overlapping-cyclic Friedrichs reduction then gives the norm lower bound. $\square$

Lemma 22.134 (Parent-Hamiltonian gap from a cyclic overlap compression constant). [Lean](#) [✓ Stmt](#) Fix $L > 1$, and let $m = 2(L - 1)$. Suppose uniformly for every $N \geq 2L$ and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{i,\text{ES}}v\|.$$

If $\gamma > 0$, $\gamma \leq 1$, and $\eta \leq (1 - \gamma)/m$, then $0 < \gamma$ and, for every admissible N and every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$,

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

The assertion $0 < \gamma$ repeats the hypothesis, in the same shape as the strict statement below. In particular, if $\eta \geq 0$ and $\eta m < 1$, then $0 < 1 - \eta m$ and, for every admissible N and every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$,

$$(1 - \eta m) \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

Proof. ✓ Proof The cyclic-window cardinality lemma gives at most $m = 2(L - 1)$ overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The finite-overlap norm-compression reduction gives the quadratic-form estimate with the chosen γ , and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with $\gamma = 1 - \eta m$. $\square$

Theorem 22.135 (Martingale criterion for spectral gap). Not ready Consider a frustration-free Hamiltonian

$$H = \sum_{i=1}^n h_i$$

with projector interaction terms ($h_i^2 = h_i$). Suppose that for every pair of overlapping terms h_i, h_j there exist constants $c_{ij} \geq 0$ and $\gamma > 0$ satisfying

$$h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j),$$

and suppose further that $c_{ij} = c_{ji}$ for every overlapping pair (i, j) and that $\sum_{j: j \neq i} c_{ij} \leq 1$ for every i . Then the smallest positive eigenvalue of H satisfies $\lambda_{\min}^+(H) \geq \gamma$.

Proof. Not ready The relevant geometry is a family of translated length- L projector windows on the ring; two terms h_i and h_j interact only when their supports overlap, equivalently when the cyclic distance $d_N(i, j)$ is smaller than L . From $h_i^2 = h_i$,

$$H^2 = \sum_i h_i + \sum_{\substack{i < j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i < j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ($h_i h_j = h_j h_i$), so their product $h_i h_j$ is itself a projector and hence PSD; thus the last sum is ≥ 0 . For the overlapping sum, the martingale condition gives $h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j)$. Summing over unordered pairs $\{i, j\}$:

$$\sum_{\{i, j\}, \text{overlap}} (h_i h_j + h_j h_i) \geq -(1 - \gamma) \sum_{\{i, j\}, \text{overlap}} c_{ij} (h_i + h_j).$$

Collecting terms for each h_i and using the symmetry assumption $c_{ij} = c_{ji}$:

$$\sum_{\{i, j\}, \text{overlap}} c_{ij} (h_i + h_j) = \sum_i \left(\sum_{\substack{j \neq i \\ \text{overlap}}} c_{ij} \right) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis $\sum_{j \neq i} c_{ij} \leq 1$. Combining, $H^2 \geq H - (1 - \gamma)H = \gamma H$. For any eigenvector with $H\psi = \lambda\psi$ and $\lambda > 0$, this gives $\lambda^2 \geq \gamma\lambda$, hence $\lambda \geq \gamma$ [KL18]. $\square$

Lemma 22.136 (Martingale condition for MPS). Not ready Let A be injective and fix an interaction range $L \geq 2$ as in Definition 22.10. Let $h_L(A) := \Pi_{G_L(A)^\perp}$ be the L -site parent interaction, and for each chain length and site i let h_i denote its translate as in Definition 22.16. Then for every pair of overlapping terms h_i and h_j with $0 < d_N(i, j) < L$, where $d_N(i, j) := \min(|i - j|, N - |i - j|)$ is the cyclic distance on the ring (i.e. whose L -site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 22.135 holds with constants c_{ij} and $\gamma > 0$ depending only on A (not on N). Note: The proof below fully establishes the martingale constants only for $L = 2$; for $L > 2$ the existence of suitable constants is asserted without complete proof (see the remark at the end of the proof). Only the $L = 2$ case is used in the spectral-gap theorem below. (When $L = 1$ the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale argument.)

Proof. Not ready Two terms h_i, h_j with $0 < d_N(i, j) < L$ have supports overlapping in $L - d_N(i, j)$ sites (where d_N is the cyclic distance on the N -site ring). Since A is injective, the intersection property (Theorem 22.34) gives the qualitative identity $\ker(h_i) \cap \ker(h_j) = G_{[i, j+L-1]}(A)$. Because the local Hilbert space is finite-dimensional, the subspaces $\ker(h_i)$ and $\ker(h_j)$ live in a fixed finite-dimensional space. The intersection identity shows that $\ker(h_i) \cap \ker(h_j)$ is exactly $G_{[i, j+L-1]}(A)$; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle $\theta_{d_N(i, j)}$ between $\ker(h_i)$ and $\ker(h_j)$ is strictly positive (for the fixed injective tensor A). Because the angle depends only on the overlap pattern $d_N(i, j) \in \{1, \dots, L-1\}$ and not on the absolute position, the number of distinct angles is at most $L-1$. Let $\alpha := \max_{1 \leq s < L} \cos \theta_s < 1$ be the worst-case cosine of the Friedrichs angles.

Row-sum bound. Since $L \geq 2$, each site i overlaps with at most $2(L-1)$ neighbours (those j with $0 < d_N(i, j) < L$). Set $c_{ij} := \frac{1}{2(L-1)}$ for every overlapping pair and $c_{ij} := 0$ otherwise. These constants depend only on the cyclic distance $d_N(i, j)$ and therefore satisfy $c_{ij} = c_{ji}$. Then $\sum_{j \neq i} c_{ij} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$, satisfying the row-sum hypothesis of Theorem 22.135.

Positivity of γ . The Friedrichs angle bound gives, for each overlapping pair, $h_i h_j + h_j h_i \geq -\alpha(h_i + h_j)$. The martingale inequality holds with $\gamma = 1 - 2(L-1)\alpha$, which requires $c_{ij}(1-\gamma) = \frac{1}{2(L-1)} \cdot 2(L-1)\alpha = \alpha$ (matching the Friedrichs bound exactly). Since $\alpha < 1$ (the Friedrichs angle is strictly positive), it remains to show $\gamma > 0$, i.e. $\alpha < \frac{1}{2(L-1)}$. In the blocked picture used in the spectral-gap theorem below, $L = 2$ and each site overlaps with at most 2 neighbours, so the condition reduces to $\alpha < \frac{1}{2}$. For $L = 2$ with an injective tensor, the two projectors h_i and h_{i+1} act on a finite-dimensional space, and the intersection property (Theorem 22.34) guarantees that $\ker(h_i) \cap \ker(h_{i+1})$ is exactly $G_{[i, i+2]}(A)$; hence $\|P_i P_{i+1}\| < 1$ (the Friedrichs cosine is strictly less than 1). On a finite-dimensional space with two projectors, the Friedrichs cosine satisfies the quantitative bound $\alpha \leq 1 - c_{\min}$ where $c_{\min} > 0$ depends on the minimal principal angle between the two kernels. The bound $\alpha < \frac{1}{2}$ then follows from the Nachtergaele finite-size criterion [Nac96], which establishes that the overlap $\|P_i P_{i+1}\|$ for injective MPS is bounded away from 1 by a constant depending only on the tensor A . For general $L > 2$, the uniform constants $c_{ij} = \frac{1}{2(L-1)}$ make the condition $\alpha < \frac{1}{2(L-1)}$ increasingly restrictive as L grows. A more refined choice of separation-dependent constants $c_{ij} = c_{d_N(i, j)}$ that weight small-separation (large-overlap) pairs more heavily can exploit the exponential decay of the Friedrichs cosine α_s in s to satisfy the row-sum constraint. However, this refinement is not needed for the main result: the spectral-gap argument below is applied only with $L = 2$ in the blocked picture. In all cases, the constants depend only on A and are independent of N [FNW92, KL18]. $\square$

Theorem 22.137 (Explicit Friedrichs-angle gap bound). Lean Not ready For an injective tensor A and a range $L > 1$, the theorem asserts the analytic estimate with the explicit constant

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, it asserts that if $N \geq 2L$ and $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$, then

$$\gamma_L \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

Proof. Not ready The local averaging, positivity, and symmetric-projection statements are now established in Lemmas 22.108, 22.109, and 22.110. Lemma 22.116 reduces the present theorem to the remaining MPS-specific task: deriving the uniform quadratic-form estimate. Lemma 22.122 establishes the finite-sum algebra from ordered cross-term row bounds, and Lemmas 22.123 and 22.130 state the common finite-overlap specialization with $m = 2(L - 1)$. Lemma 22.114 supplies the non-overlap positivity statement for site-disjoint windows, and Lemma 22.128 translates this to the concrete support-disjoint cyclic windows. Lemma 22.129 gives the finite-overlap row-cardinality bound. Lemma 22.131 combines these ingredients with the finite-overlap reduction, so the remaining analytic task is the Friedrichs-angle estimate for overlapping cyclic windows. Lemma 22.132 states the same overlap-only formulation, and Lemma 22.133 states a sufficient norm-compression formulation. $\square$

Theorem 22.138 (Spectral gap for MPS parent Hamiltonians). Lean ✓ Stmt For an injective tensor A and a range $L > 1$, there exists $\gamma > 0$ such that for every $N \geq 2L$ and every vector $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ one has

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

By Lemma 22.115, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

Proof. ✓ Proof This follows immediately from Theorem 22.137 by taking as γ the explicit constant produced there, which yields the required existential spectral-gap constant. $\square$

22.7 Ground space of non-injective MPS

For a general (non-injective) MPS tensor in block-diagonal canonical form, the ground space is no longer one-dimensional. Instead, it is spanned by the MPVs of the individual normal blocks — the basis of normal tensors (BNT) from Chapter 11 [FNW92, PGVWC07].

Definition 22.139 (Assembled parent ground space). Lean ✓ Stmt For a family of blocks $A = (A_j)_{j=1}^r$ and weights μ , write $\text{ParentGround}_{N,L}(A)$ for the periodic chain ground space of the block-diagonal tensor assembled from the blocks.

Lemma 22.140 (Definition by assembled chain ground space). Lean ✓ Stmt By definition, this subspace is exactly the chain ground space of the assembled tensor.

Definition 22.141 (Span of BNT block states). Lean ✓ Stmt The BNT span is the linear span of the periodic MPV states of the individual blocks:

$$\text{BNTSpan}_N(A) := \text{span}\{V^{(N)}(A_j) : j = 1, \dots, r\}.$$

Definition 22.142 (Route B use of the product-word span condition). Lean ✓ Stmt For a fixed length m , the Route B product-word span condition is the finite-length spanning condition already stated in Remark 21.42: the simultaneous block-word evaluations

$$\omega \mapsto (A_j^\omega)_j$$

span the whole product algebra $\prod_j M_{D_j}(\mathbb{C})$. The remaining mathematical step is to derive this condition for some positive length from the separated canonical-form/BNT hypotheses. This is the block-injective span requirement in [CPGSV21, Section 4.3, lines 2049–2094]: after blocking, the parent ground space is generated by the basis of normal tensors, with the improved bound coming from block-diagonal boundary conditions.

Lemma 22.143 (Each BNT MPV lies in the assembled parent ground space). Lean ✓ Stmt *If $1 < L$ and $N \geq L + 1$, then for each block A_j of a canonical-form/BNT decomposition, the corresponding MPV state lies in the assembled parent ground space.*

Proof. ✓ Proof One first applies Lemma 22.61 to the individual block. The proof then inserts the resulting boundary matrix into the j -th diagonal block of the assembled tensor, with the compensating scalar factor coming from the nonzero weight μ_j . □

Lemma 22.144 (Periodic block lift). Lean ✓ Stmt *For each block index j with $\mu_j \neq 0$, the chain ground space of the block A_j embeds into the chain ground space of the assembled tensor:*

$$\text{ChainGround}_{N,L}(A_j) \subseteq \text{ChainGround}_{N,L}(\text{Assemble}(\mu, A)).$$

Proof. ✓ Proof The cyclic-window definition of the chain ground space reduces the inclusion to the local block-embedding statement that each ground vector of A_j on the window of length L lifts to the assembled tensor by inserting the block-diagonal boundary matrix with the compensating scalar factor μ_j^{-L} . □

Lemma 22.145 (Periodic block decomposition: forward inclusion). Lean ✓ Stmt *Assume $\mu_j \neq 0$ for every block index j . The supremum of the blockwise periodic-chain ground spaces is contained in the assembled tensor's periodic-chain ground space:*

$$\bigsqcup_j \text{ChainGround}_{N,L}(A_j) \subseteq \text{ChainGround}_{N,L}(\text{Assemble}(\mu, A)),$$

where j ranges over all block indices.

Proof. ✓ Proof Aggregate the block lift over all block indices. □

Lemma 22.146 (Forward inclusion for assembled parent ground spaces). Lean ✓ Stmt *For a canonical-form/BNT family, the supremum of the blockwise periodic-chain ground spaces is contained in the assembled parent ground space:*

$$\bigsqcup_j \text{ChainGround}_{N,L}(A_j) \subseteq \text{ParentGround}_{N,L}(A).$$

Proof. ✓ Proof The canonical-form/BNT hypothesis supplies $\mu_j \neq 0$ for every j , and the assembled parent ground space is the chain ground space of the assembled tensor by definition. □

Lemma 22.147 (Block-diagonal commutant reduction from span extension). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Suppose the virtual sector projections P_j of a dependent direct-sum bond space lie in the span of a family of matrices. Any boundary matrix commuting with that family commutes with each P_j , hence is block diagonal. Applied after pulling an assembled tensor back along the direct-sum reindexing, length- m word commutation plus the corresponding projection-span hypothesis forces all off-block entries of the pulled-back boundary matrix to vanish.

Proof. [✓ Proof](#) Commutation is linear in the matrix from the spanning family, so it extends from the generators to their span. Since each P_j lies in that span, the boundary matrix commutes with every sector projection. Lemma 2.36 then gives block diagonality, and Lemma 2.35 gives the entrywise off-block-zero formulation. $\square$

Lemma 22.148 (Block projections from product-algebra span). [Lean](#) [✓ Stmt](#) Let $T_a = (T_a(i))_i$ be a family of tuples spanning the full product algebra $\prod_i M_{n_i}(\mathbb{C})$, and let $c_i \neq 0$ be nonzero scalars. Then, for every sector k , the sector projection P_k lies in the span of the scaled dependent block-diagonal matrices

$$\bigoplus_i c_i T_a(i).$$

Proof. [✓ Proof](#) Apply the linear map $(M_i)_i \mapsto \bigoplus_i c_i M_i$ to the full-span product-algebra element with $(c_k)^{-1}I$ in the k -th component and zero in every other component. Its image is exactly P_k . $\square$

Lemma 22.149 (Projection span from product-word span). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If the simultaneous length- m block-word tuples

$$\omega \mapsto (A_j^\omega)_j$$

span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$ and every μ_j is nonzero, then the pulled-back length- m word products of $\text{Assemble}(\mu, A)$ span every virtual sector projection P_j . Consequently, any boundary matrix commuting with all those assembled words has block-diagonal pulled-back form, and its off-block entries vanish.

Proof. [✓ Proof](#) First embed the product algebra linearly as dependent block-diagonal matrices, scaling the j -th block by μ_j^m . Since $\mu_j^m \neq 0$, the product-algebra span contains the tuple with $(\mu_j^m)^{-1}I$ in one component and zero elsewhere; its scaled diagonal image is exactly P_j . The word-evaluation formula for the assembled tensor identifies these scaled diagonal matrices with the pulled-back assembled word products, and the commutant conclusion follows from Lemma 22.147. $\square$

Definition 22.150 (Pairwise block-separating word polynomials). [Lean](#) [Lean](#) [✓ Stmt](#) A length- S partial selector for block k on a finite target set T is a tuple in the span of simultaneous length- S block-word evaluations which is I on block k and 0 on every block in T . The pairwise version asks for such a partial selector for each ordered pair of distinct blocks, with $T = \{j\}$.

Definition 22.151 (Pair trace-separation criterion). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For two blocks A and B , the length- S pair word tuple is $w \mapsto (A^w, B^w)$. The pair product-span condition says that these tuples span $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$. Equivalently, in the trace-dual formulation, the only pair of test matrices (Δ_A, Δ_B) whose trace pairing with every length- S tuple vanishes is $(0, 0)$. The cumulative variant replaces one fixed length by all word lengths at most S , and the all-length variant allows every finite word.

that length once the boundary-to-chain maps at the three-block length are injective. Combining the equal-dimension and unequal-dimension branches over all ordered block pairs gives one common three-block trace-separation length, and hence the positive-length product-word span required by the selector construction. Since canonical-form/BNT blocks are one-site injective, one may take $L = 2$; injectivity at lengths 2 and 6 then follows by passing from a fixed injective length to positive multiples. This length-6 trace separation is equivalently a full homogeneous pair span; concatenating one-letter words propagates fullness to all later homogeneous lengths, so the direct-sum witness also gives a uniform finite pair-span period window. Combining this window with the conditional period-window homogenization theorem gives the selector-format homogeneous word span without separately assuming homogeneous identity pairs. The same conclusion applies to normal-CF-BNT data once one-site injectivity is supplied explicitly, since the normal-CF-BNT predicate carries the irreducibility and normalization hypotheses used in the direct-sum comparison.

Proof. ✓ **Proof** Block injectivity makes the boundary-to-chain map injective, so $\dim \mathcal{G}_L^A = D_A^2$ and $\dim \mathcal{G}_L^B = D_B^2$. Inclusion gives $D_A^2 \leq D_B^2$, while $D_B \leq D_A$ gives the reverse inequality. Hence $D_A = D_B$, and equality of dimensions inside the inclusion gives equality of the image spaces. In the strict-size branch this contradicts $D_B < D_A$. For the conditional directness statement, a vector in the intersection of the two three-block image spaces has two boundary representations. If the first boundary matrix is zero, the vector is zero. Otherwise the difference of the two coefficient formulas is exactly the homogeneous three-block trace relation, so the dimension step gives the forbidden collapse. In the equal-size branch, equal local image spaces impose the same cyclic local constraints on a periodic chain. The injective uniqueness theorem then identifies both cyclic ground spaces with the spans of their MPV states, contradicting non-proportionality of those states. $\square$

Lemma 22.155 (All-length pair trace separation has finite cumulative length).
If no nonzero trace functional vanishes on all finite pair words, then there is a finite S such that vanishing on every word of length at most S already forces the test pair to be zero. For a finite family of ordered block pairs, these bounds can be replaced by one common bound.

Proof. ✓ **Proof** The trace condition is dual to the span of all finite pair words. If that span is the whole product algebra, every vector lies in some cumulative span. Since the product algebra is finite-dimensional, the increasing sequence of cumulative spans stabilizes; at the stabilization index it is already the whole product algebra. Applying the finite trace-duality criterion gives the stated finite bound. This proves the cumulative finite-dimensional step; the remaining BNT step is to convert it to a single homogeneous length. Cumulative span-top alone is not identity padding, because the length-zero word already contributes the identity. $\square$

Lemma 22.156 (Pair algebra projections are full). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) If one of the two blocks is injective, then the corresponding coordinate projection of the pair algebra generated by the simultaneous one-letter pairs is the full matrix algebra. Moreover, every finite pair word lies in this pair algebra, and the all-word linear span is the submodule generated by the pair algebra. If the pair algebra is the graph of an algebra automorphism of the matrix algebra, Skolem–Noether turns this graph case into gauge-phase equivalence, so under non-gauge-equivalence the graph branch is impossible. For two blocks of unequal bond dimensions, the graph branch is also impossible because it would identify matrix algebras with different finite dimensions over $\mathbb{C}$.

Proof. ✓ **Proof** Projecting the pair generators to one coordinate gives exactly the one-block generator family. Injectivity says that the linear span of this family is the full matrix algebra, and the

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let (A_j) be a finite family of block-injective tensors. Suppose there are length- S block-selector words: for each sector k , a linear combination of length- S words evaluates to I on A_k and to 0 on every other block. Then the simultaneous length- $(1 + S)$ block-word evaluations span $\prod_j M_{D_j}(\mathbb{C})$, with positive length. In particular, a canonical form with BNT separation supplies the required block injectivity, but the selector-word conclusion does not require the full canonical-form/BNT data. It is enough, by Lemma 22.159, to construct pairwise separators for every ordered pair of distinct blocks. Lemma 22.158, a common pair trace-separation length is one sufficient finite-dimensional witness. This witness remains homogeneous: either it is supplied directly as fixed-length pair separation, or it is obtained from an explicit period-window hypothesis as in Lemma 22.157. If the assembly weights are all nonzero, then the pulled-back words at the resulting length for $\text{Assemble}(\mu, A)$ span every virtual sector projection, and the corresponding commutant reduction is available.

Proof. [✓ Proof](#) Block injectivity writes each target sector matrix as a fixed-length linear combination of word evaluations. For a canonical-form/BNT family, Definition 11.1 gives one-site injectivity of each block, hence 1-block injectivity by Lemma 2.28. For a target tuple $(M_j)_j$, write M_k as a one-letter linear combination in block k , multiply by the selector for k , and sum over k . The selector kills all off-target blocks, while it is the identity on block k , so the length- $(1 + S)$ word tuples span the full product algebra. If one starts with pairwise separators instead, first use Lemma 22.159 to multiply them into full selectors; if one starts from the pair trace-separation criterion, first use Lemma 22.158. The projection-span and commutant consequences then follow from Lemma 22.149. $\square$

Lemma 22.161 (Conditional boundary-matrix block split from projection span). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be in canonical-form/BNT, and assume $1 < L$, $N \geq L + 1$, and $0 < m$. Assume every virtual sector projection lies in the pulled-back span of the length- m word products of the assembled tensor. If a boundary matrix for the assembled tensor commutes with all those length- m words, then the vector obtained by applying the assembled ground-space map to that boundary matrix lies in $\bigsqcup_j \mathcal{G}_{N,L}(A_j)$. Consequently, if every assembled periodic-chain ground vector admits such a commuting boundary matrix representation (the result of the wrapping-window comparison), then

$$\mathcal{G}_{N,L}(\text{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j),$$

and together with the already-proved forward inclusion this gives the corresponding conditional equality. Composing the reverse inclusion with blockwise injective uniqueness gives the conditional containment of the assembled parent ground space in the BNT span; this lemma keeps the projection-span and boundary-representation hypotheses explicit.

Proof. [✓ Proof](#) The commutant reduction makes the pulled-back boundary matrix block diagonal. Restricting the same word-commutation identities to a diagonal block and using the injectivity of that block shows, by the scalar-commutant lemma, that each diagonal block is scalar. Expanding the assembled ground-space map on a block-diagonal boundary matrix gives a finite sum of scalar multiples of the block MPV states, and each block MPV lies in its own periodic chain ground space. The reverse-inclusion, equality, and BNT-span forms are then reformulations of this boundary-matrix calculation, combined with the forward inclusion and the blockwise uniqueness theorem. $\square$

Lemma 22.162 (Conditional Route B block split from product-word span). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be in canonical-form/BNT, and assume $1 < L$, $N \geq L + 1$, and $0 < m$. If the

length- m product-word tuples span $\prod_j M_{D_j}(\mathbb{C})$ and every assembled periodic-chain ground vector comes with a boundary matrix representation commuting with all length- m assembled words, then

$$\mathcal{G}_{N,L}(\text{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Combined with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\text{BNTSpan}_N(A)$. The product-word span and boundary representation remain hypotheses here; they are not derived by this composition lemma.

Proof. ✓ Proof The product-word span witness gives the sector-projection span by Lemma 22.149; the nonzero weights required there are supplied by the canonical-form/BNT hypothesis. The conditional boundary-matrix block split from Lemma 22.161 then gives the reverse inclusion. The equality and BNT-span forms are the corresponding projection-span results after this product-word-to-projection-span reduction, followed by the blockwise uniqueness argument already built into the projection-span theorem. $\square$

Lemma 22.163 (Conditional Route B block split from BNT selector words). Lean Lean Lean

✓ Stmt Let A be in canonical-form/BNT, assume $1 < L$ and $N \geq L + 1$, and suppose length- S block-selector words are given. If every assembled periodic-chain ground vector has a boundary matrix representation commuting with all length- $(1 + S)$ assembled words, then

$$\mathcal{G}_{N,L}(\text{Assemble}(\mu, A)) \subseteq \bigsqcup_j \mathcal{G}_{N,L}(A_j).$$

Together with the forward inclusion, this gives the corresponding conditional block-splitting equality. In the parent-ground-space formulation, the same hypotheses imply containment in $\text{BNTSpan}_N(A)$. The selector-word hypothesis and the boundary representation remain explicit: this lemma does not derive selector existence from BNT separation, nor the wrapping/open-chain boundary representation. It is the formal composition matching the block-injective ground-space route of [CPGSV21, Section 4.3, lines 2049–2094], where the regrowing argument is restricted to block-diagonal boundary conditions.

Proof. ✓ Proof Lemma 22.160 turns the canonical-form/BNT hypotheses plus length- S selector words into the product-word span condition at the positive length $1 + S$. Lemma 22.162 then applies Route B with the boundary matrix commuting with the same length- $(1 + S)$ assembled words. The equality and BNT-span containment are exactly the corresponding product-word-span consequences after this selector-to-product-span replacement. $\square$

Theorem 22.164 (Containment in the BNT span by block decomposition). Lean Not ready If A is in canonical-form/BNT and $1 < L$ and $N \geq L + 1$, every vector in the assembled parent ground space should decompose into blockwise chain ground states, and therefore lie in the BNT span.

Proof. Not ready The remaining structural step is the periodic block decomposition of the assembled chain ground space. Lemma 22.159 reduces finite selector words to pairwise block-separating word polynomials. Lemma 22.163 then gives the conditional Route B composition once those selector words and the wrapping-window commuting boundary matrix are available. What remains open is to derive the needed separation data and to supply the compatible boundary representation. With the block decomposition available, one applies the blockwise injective uniqueness theorem to each block separately and recovers the BNT span. $\square$

Theorem 22.165 (Ground space equals span of BNT). Not ready Lean If $1 < L$ and $N \geq L + 1$, then for a canonical-form/BNT family the assembled parent ground space equals the BNT span:

$$\text{ParentGround}_{N,L}(A) = \text{BNTSpan}_N(A).$$

Proof. Not ready We prove the two inclusions separately. The inclusion

$$\text{ParentGround}_{N,L}(A) \subseteq \text{BNTSpan}_N(A)$$

is exactly Theorem 22.164. For the reverse inclusion, Lemma 22.143 shows that each BNT MPV lies in $\text{ParentGround}_{N,L}(A)$. Since the parent ground space is a linear subspace, it therefore contains every linear combination of such vectors, and hence contains $\text{BNTSpan}_N(A)$. Thus

$$\text{BNTSpan}_N(A) \subseteq \text{ParentGround}_{N,L}(A),$$

and the two spaces are equal. □

Remark. A separate kernel-identification statement for these assembled block families would convert the chain-ground-space statement into the corresponding statement for $\ker(H_N)$.

Chapter 23

Exponential Decay of Correlations

This chapter derives the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 22), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound $\rho(F_{AB}) \leq 1$, the strict spectral gap $\rho(F_{AB}) < 1$ for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 7. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

23.1 Connected correlators from the transfer map

Definition 23.1 (One-point expectation). [Lean](#) [✓ Stmt](#) Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R . Under the hypotheses of Theorem 6.10, that fixed point exists and is unique up to scaling; we normalize it so that $\text{tr}(\rho^R) = 1$. For a local observable $X \in M_D(\mathbb{C})$ define

$$\langle X_0 \rangle := \text{tr}(X \rho^R).$$

Definition 23.2 (Two-point expectation at distance n). [Lean](#) [✓ Stmt](#) For observables $X, Y \in M_D(\mathbb{C})$ and $n \geq 0$,

$$\langle X_0 Y_n \rangle := \text{tr}(Y \mathcal{E}_A^n(X \rho^R)).$$

Equivalently: insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.

Definition 23.3 (Connected correlator). [Lean](#) [✓ Stmt](#) The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

23.2 Spectral expansion and exponential decay

Theorem 23.4 (Sum of exponentials (spectral expansion interface)). [Lean](#) [✓ Stmt](#) If coefficients $c_j \in \mathbb{C}$ and eigenvalues $\lambda_j \in \mathbb{C}$ ($j = 1, \dots, D^2 - 1$) satisfy, for all $n \geq 0$,

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \lambda_j^n,$$

then the connected correlator equals the stated sum of exponentials. The source [CPGSV21, Sec. 2.3] states that the connected correlator is a sum of $D^2 - 1$ pure exponentials $C(X, Y; n) = \sum_{j \geq 2}^{D^2} c_{XY}(j) \lambda_j^n$, which always exists for a normal MPS via the transfer-map spectral decomposition.

Proof. [✓ Proof](#) Given the expansion hypothesis, the equality is immediate. The source [CPGSV21, Sec. 2.3] derives this expansion from the spectral decomposition of $\mathcal{E}_A^n$ restricted to the complement of the fixed-point eigenspace: the leading eigenvalue $\lambda_1 = 1$ contributes $\langle X_0 \rangle \langle Y_0 \rangle$, which is subtracted in the connected correlator, leaving the sum over subleading eigenvalues. $\square$

Theorem 23.5 (Exponential decay bound (interface)). [Lean](#) [✓ Stmt](#) If a constant $C_{XY} \in \mathbb{R}$ and $\lambda_2 \in \mathbb{C}$ satisfy, for all $n \geq 0$,

$$|C(X, Y; n)| \leq C_{XY} |\lambda_2|^n,$$

then the connected correlator satisfies that exponential decay bound. The source [CPGSV21, Sec. 2.3] obtains this by combining the sum-of-exponentials expansion with the triangle inequality, and Sec. 4 provides the spectral gap condition $|\lambda_j| < 1$ for the subleading eigenvalues.

Proof. [✓ Proof](#) Given the bound hypothesis, the estimate is immediate. The source [CPGSV21, Sec. 2.3] obtains this bound by applying the triangle inequality to the sum-of-exponentials expansion and using the spectral gap condition $|\lambda_j| \leq |\lambda_2| < 1$ for all subleading eigenvalues. $\square$

Remark 23.6 (Spectral hypothesis and the spectral gap). The hypothesis “ $|\lambda_2| < 1$ ” in Theorem 23.5 is the source-paper *spectral gap* condition for the transfer map $\mathcal{E}_A$. The complementary spectral gap $\rho(\mathcal{E}_A - P) < 1$ is established for primitive channels (Theorem 4.51, Chapter 4), for irreducible trace-preserving tensors (Theorem 7.29), and quantitatively for injective tensors (Theorem 23.11). Thus the analytical input needed to make Theorems 23.4 and 23.5 unconditional is already present; what remains is extracting the spectral expansion coefficients c_j, λ_j from the eigendecomposition of $\mathcal{E}_A$ (Issue #1447).

Definition 23.7 (Correlation length). [Lean](#) [✓ Stmt](#) The correlation length associated with a subleading eigenvalue λ_2 of the transfer map ($|\lambda_2| < 1$) is

$$\xi := -\frac{1}{\log |\lambda_2|}.$$

Lemma 23.8 (Correlation length is positive). [Lean](#) [✓ Stmt](#) If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.

Proof. [✓ Proof](#) Since $0 < |\lambda_2| < 1$, we have $\log |\lambda_2| < 0$, so $\xi = -1/\log |\lambda_2| > 0$. $\square$

23.3 Quantitative spectral gap bounds

The spectral gap theorems in Chapter 4 establish $\rho(\mathcal{E}_A - P) < 1$ for primitive channels qualitatively. This section gives the *quantitative* refinements with explicit constants and rates for the single-tensor transfer map $\mathcal{E}_A$.

Theorem 23.9 (Exponential convergence of injective primitive channels). Lean ✓ Stmt *Fix an injective, trace-preserving MPS tensor A with positive definite fixed point ρ . Let*

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \leq 1$ such that for all $n \geq 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|.$$

The spectral gap δ exists by primitivity (Theorem 4.51, the complementary spectral gap).

Proof. ✓ Proof Injectivity implies primitivity of the transfer map. The complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ then follows from the primitive spectral gap theorem. The Gelfand formula yields a geometric bound $\|(\mathcal{E}_A - P)^n\| \leq C(1 - \delta)^n$ for suitable constants, giving the claimed estimate. □

Theorem 23.10 (Correlation length bound). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$ (i.e. $\text{tr}(X) = 0$),*

$$\|\mathcal{E}_A^n(X)\| \leq C e^{-n/\xi} \|X\|.$$

Traceless matrices lie in $\ker P$, where P is the fixed-point projection. Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate $\xi = -1/\log \rho(\mathcal{E}_A - P)$.

Proof. ✓ Proof For traceless X we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. The exponential convergence theorem gives a geometric bound $C(1 - \delta)^n \|X\|$. Rewriting $(1 - \delta)^n = e^{-n/\xi}$ with $\xi = -1/\log(1 - \delta)$ yields the claimed estimate. □

Theorem 23.11 (Spectral gap from injectivity). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

Proof. ✓ Proof Injectivity implies eventual full Kraus rank. This yields primitivity, hence a spectral gap for the complementary map. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. □

23.4 Relation to parent Hamiltonians

Remark 23.12. When the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 22.17) whose ground state is the matrix product vector (Lemma 22.21), the exponential decay bound (Theorem 23.5) shows that ground-state correlations decay with a finite correlation length ξ .

23.5 Renormalization fixed points and zero correlation length

Definition 23.13 (Renormalization fixed point). [Lean](#) [✓ Stmt](#) An MPS tensor A is a *renormalization fixed point* (RFP) when its transfer map is idempotent, $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. See [CPGSV21, Definition 3.2].

Theorem 23.14 (Kraus-isometry criterion for RFP). [Lean](#) [✓ Stmt](#) Suppose there exists an isometry $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1, i_2), j}$, such that

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. Then A is a renormalization fixed point. This is the backward direction of [CPGSV21, Theorem 3.1].

Proof. [✓ Proof](#) Expanding $\mathcal{E}_A^2$ gives a double Kraus sum indexed by pairs (i_1, i_2) . Substituting the decomposition above and using the isometry relation $V^\dagger V = 1$ collapses the double sum to the single Kraus family $\{A^j\}$, hence $\mathcal{E}_A^2 = \mathcal{E}_A$. $\square$

Definition 23.15 (Local orthogonality). [Lean](#) [✓ Stmt](#) A single BNT block is *locally orthogonal* when its self-transfer map is idempotent. For a single tensor this coincides with the RFP condition (Definition 23.13); in the full multi-block BNT setting, local orthogonality additionally requires that mixed transfer operators vanish. See [CPGSV21, Definition 3.5].

Theorem 23.16 (RFP normal tensor is injective). [Lean](#) [✓ Stmt](#) If A is a nonzero normal RFP tensor, then A is injective.

Proof. [✓ Proof](#) The RFP condition $\mathcal{E}_A^2 = \mathcal{E}_A$ implies that the cumulative span $\text{span}\{A^w : |w| \leq n\}$ stabilises after one step. Since A is normal (hence has an injective blocked form), the one-step span already equals $M_D(\mathbb{C})$, so A is injective. $\square$

Theorem 23.17 (Left-canonical normal RFP tensor is injective). [Lean](#) [✓ Stmt](#) Assume $D > 0$. Let A be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then A is injective. This is the left-canonical injectivity step in [CPGSV21, Appendix B].

Proof. [✓ Proof](#) The normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ forces at least one matrix A^i to be nonzero, since otherwise the left-hand side would vanish. Hence A is a nonzero normal renormalization fixed point, and Theorem 23.16 applies. $\square$

Theorem 23.18 (Unitary diagonal fixed point for a left-canonical normal RFP). [Lean](#) [✓ Stmt](#) Assume $D > 0$. Let A be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [CPGSV21, Appendix B].

Proof. ✓ Proof By Theorem 23.17, the tensor A is injective. Therefore its transfer map is irreducible, and hence A is irreducible as a tensor. Applying Theorem 9.13 gives a unitary conjugate B for which the trace-preserving normalization remains valid and a diagonal positive definite matrix Λ with $\mathcal{E}_B(\Lambda) = \Lambda$. □

Theorem 23.19 (Rank-one classification for RFP transfer maps). Lean ✓ Stmt *Let A be an injective left-canonical RFP tensor with positive-definite fixed point ρ of the transfer map $\mathcal{E}_A$. Then $\mathcal{E}_A = P_\rho$, where $P_\rho(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$ is the rank-one fixed-point projection.*

Proof. ✓ Proof The exponential convergence bound gives $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1 - \delta)^n \|X\|$. Idempotence of $\mathcal{E}_A$ implies $\mathcal{E}_A^{n+1} = \mathcal{E}_A$ for all $n \geq 0$, so $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1 - \delta)^{n+1} \|X\|$ for all n . Taking $n \rightarrow \infty$ gives $\mathcal{E}_A = P_\rho$. □

Theorem 23.20 (Full structural form for RFP tensors). Lean ✓ Stmt *A normal tensor A in canonical form Π that is a renormalization fixed point admits a decomposition $A^i = X\Lambda U^i X^{-1}$, where Λ is diagonal positive and the family $U = (U^i)$ is an isometry in the physical indices.*

Proof. ✓ Proof Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a canonical Kraus form, applies Kraus freedom to identify the physical-index family with an isometry U , and transports the decomposition back through the conjugation. □

Theorem 23.21 (Canonical-form blocks are injective). Lean ✓ Stmt *For a family of tensors in canonical form, every block is injective. This is a direct projection of the injectivity field of the canonical-form predicate; see [CPGSV16, Section 2.3].*

Proof. ✓ Proof Immediate from the `block_injective` field of `IsCanonicalForm`. □

Theorem 23.22 (BNT basis blocks are injective). Lean ✓ Stmt *Let P be a sector decomposition in BNT canonical form. Then every basis block P_j is injective as an MPS tensor. This is the basis-block injectivity part of [CPGSV16, Section III] (*thm:charact-MPS*, lines 543–555; basis-block normality established at lines 271–301 and 317–344).*

Proof. ✓ Proof Direct projection of the `basis_injective` field of `IsBNTCanonicalForm`. The result holds for arbitrary copy multiplicities, since the statement concerns only BNT basis blocks and not copy weights. □

Theorem 23.23 (RG flow convergence for canonical-form tensors). Lean ✓ Stmt *For a tensor in canonical form with blocks (A_k) and weights (μ_k) , the iterated blocking $\mathcal{E}_{A_k}^{2^n}$ converges pointwise to an idempotent linear map $\mathcal{E}_\infty$ as $n \rightarrow \infty$. That is, there exists $\mathcal{E}_\infty$ with $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \rightarrow \mathcal{E}_\infty(\rho)$ for all density matrices ρ . This is [CPGSV21, Appendix B].*

Proof. ✓ Proof Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0 . The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|$ (Theorem 23.9) then gives pointwise convergence $\mathcal{E}^n(X) \rightarrow P(X)$, and composing with the subsequence $2^n \rightarrow \infty$ yields the result. The witness $\mathcal{E}_\infty = P$ is the rank-one fixed-point projection, which is idempotent. □

Remark 23.24. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \geq 1$. Informally, this corresponds to zero correlation length ($\xi = 0$). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

Definition 23.25 (Correlations independent of distance). [Lean](#) [✓ Stmt](#) For a positive definite right fixed point $\rho^R > 0$, an MPS tensor has *correlations independent of distance* (CID) when for all local observables X and Y and all separations $n, m \geq 1$,

$$C(X, Y; n) = C(X, Y; m).$$

Definition 23.26 (Zero correlation length). [Lean](#) [✓ Stmt](#) An MPS tensor has *zero correlation length* when it is both locally orthogonal and correlations are independent of distance. See [\[CPGSV21, Definition 3.6\]](#).

Theorem 23.27 (CID implies RFP). [Lean](#) [✓ Stmt](#) If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [\[CPGSV21, Theorem 3.8\]](#).

Proof. [✓ Proof](#) Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations $n = 2$ and $n = 1$. $\square$

Theorem 23.28 (Zero correlation length iff idempotent transfer). [Lean](#) [✓ Stmt](#) An MPS tensor has zero correlation length (i.e. is both locally orthogonal and CID) if and only if its transfer map is idempotent. This is [\[CPGSV21, Theorem 3.8\]](#).

Proof. [✓ Proof](#) The forward direction extracts local orthogonality (which equals idempotence by definition). The reverse uses $\mathcal{E}^n = \mathcal{E}$ for $n \geq 1$ to show the connected correlator is separation-independent. $\square$

Theorem 23.29 (Canonical-form blocks satisfy RFP iff ZCL). [Lean](#) [✓ Stmt](#) For a tensor in canonical form, each block is a renormalization fixed point if and only if it has zero correlation length. This is the RFP–ZCL part of [\[CPGSV21, Theorem 3.10\]](#).

Proof. [✓ Proof](#) Apply Theorem [23.28](#) to the chosen canonical-form block. $\square$

Chapter 24

Concrete Examples

This chapter collects concrete MPS tensors from [CPGSV21, Section III]. Each example is specified by its physical dimension d , bond dimension D , and explicit matrix family $\{A^i\}$. Key structural properties (injectivity, RFP status, symmetry) are stated for each tensor.

24.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

Definition 24.1 (GHZ tensor). [Lean](#) [✓ Stmt](#) Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \text{diag}(1, 0), \quad A^1 = |1\rangle\langle 1| = \text{diag}(0, 1).$$

For a system of N sites the resulting MPV is the GHZ state

$$|\text{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Theorem 24.2 (GHZ transfer map is idempotent). [Lean](#) [✓ Stmt](#) *The transfer map of the GHZ tensor satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$.*

Proof. [✓ Proof](#) The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij}X_{ij}$ (extraction of the diagonal). This operation is idempotent. $\square$

Theorem 24.3 (GHZ tensor is an RFP). [Lean](#) [✓ Stmt](#) *The GHZ tensor is a renormalization fixed point.*

Proof. [✓ Proof](#) Immediate from Theorem 24.2. $\square$

Theorem 24.4 (GHZ tensor is not injective). [Lean](#) [✓ Stmt](#) *The GHZ tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ is the two-dimensional space of diagonal matrices in $M_2(\mathbb{C})$, which does not equal $M_2(\mathbb{C})$.*

Proof. [✓ Proof](#) Every element of $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. $\square$

Theorem 24.5 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor). [Lean](#) [✓ Stmt](#) Let $\mathbb{Z}_2 = \{1, \sigma_x\}$ act on $\mathbb{C}^2$ by the Pauli-X matrix σ_x . The GHZ tensor is on-site symmetric under this action: for each $g \in \mathbb{Z}_2$, the twisted tensor $\tilde{A}_g$ defines the same MPV family as A , i.e. $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$.

Proof. [✓ Proof](#) The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\tilde{A}_g^i = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. $\square$

24.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

Definition 24.6 (AKLT tensor). [Lean](#) [✓ Stmt](#) Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}} \sigma_z, \quad A^1 = \frac{\sqrt{2}}{\sqrt{3}} |0\rangle\langle 1|, \quad A^2 = -\frac{\sqrt{2}}{\sqrt{3}} |1\rangle\langle 0|,$$

where $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}} \sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}} \sigma_-$ for $\sigma_+ = \sqrt{2} |0\rangle\langle 1|$ and $\sigma_- = \sqrt{2} |1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin- $\frac{1}{2}$ to spin- $\frac{1}{2}$.

Theorem 24.7 (AKLT tensor is not injective). [Lean](#) [✓ Stmt](#) The AKLT tensor is not injective at a single site: $\text{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ is the three-dimensional space of traceless matrices $\mathfrak{sl}(2, \mathbb{C})$, which is a proper subspace of $M_2(\mathbb{C})$.

Theorem 24.8 (AKLT tensor is normal). [Lean](#) [✓ Stmt](#) The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.

Theorem 24.9 ($\text{SO}(3)$ on-site symmetry of the AKLT tensor). [Not ready](#) The AKLT tensor is on-site symmetric under the spin-1 representation of $\text{SO}(3)$ (the spin-1 representation of $\text{SU}(2)$ factors through $\text{SO}(3)$), with the virtual representation being the projective spin- $\frac{1}{2}$ representation. The projective phase is the non-trivial element of $H^2(\text{SO}(3), \text{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase.

24.3 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

Definition 24.10 (Cluster-state tensor). [Not ready](#) Set $d = D = 2$. The cluster-state tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

Theorem 24.11 (Cluster-state tensor is not injective). Not ready *The cluster-state tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\} = \text{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ has dimension 2, not $\dim M_2(\mathbb{C}) = 4$.*

Proof. Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. $\square$

Remark 24.12. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\frac{1}{\sqrt{2}}|-\rangle\langle 1|.$$

Since $\{|+\rangle, |-\rangle\}$ and $\{|0\rangle, |1\rangle\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 24.13 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor). Not ready *The length-2 blocked cluster-state tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$. The two generators act on the 4-dimensional blocked physical space $(\mathbb{C}^2)^{\otimes 2}$ by $\sigma_x \otimes I$ and $I \otimes \sigma_x$, which commute and each square to the identity, defining a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$. The corresponding virtual matrices are $V_1 = \sigma_z$ and $V_2 = \sigma_x$, which anticommute ($V_1 V_2 = -V_2 V_1$). This projective phase is the non-trivial element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$, witnessing a non-trivial SPT phase.*

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